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(54) **TAM RECEPTOR-BINDING FUSION
MOLECULE HAVING
NON-INFLAMMATORY PHAGOCYTOSIS
INDUCING ACTIVITY**

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(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

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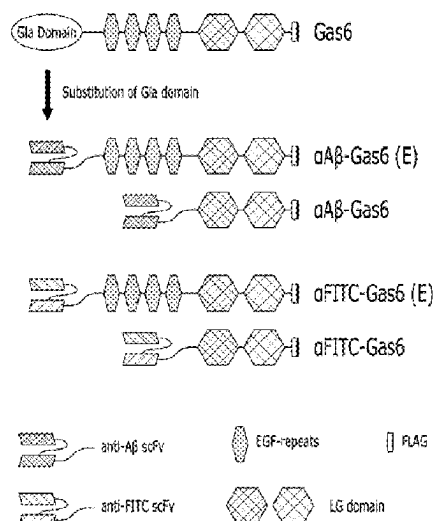
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(57) **ABSTRACT**
A fusion molecule having phagocytosis-inducing activity is
disclosed. The fusion molecule contains a first region
capable of binding a TAM receptor and a second region
capable of binding to a target substance of which aberrant
accumulation is associated with or characteristic of diseases.
The fusion molecule effectively clears and/or reduces and/or
suppresses accumulated abnormal proteins, such as beta-
amyloid, tau, alpha-synuclein, huntingtin, or prion, or the
like. Uses of the fusion molecule are disclosed. The fusion
molecule can be used for prevention or treatment of pro-
teinosis caused by the abnormal accumulation of substances.

13 Claims, 35 Drawing Sheets

Specification includes a Sequence Listing.



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FIG. 1A
--Prior Art--

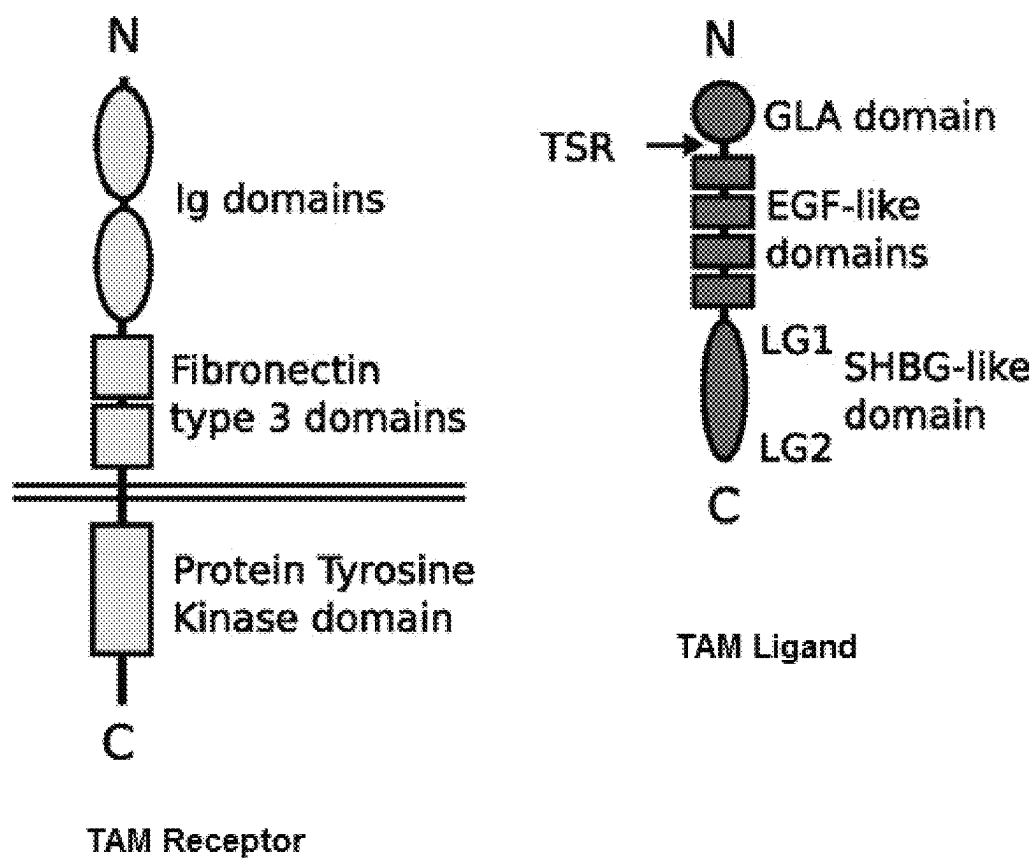


FIG. 1B

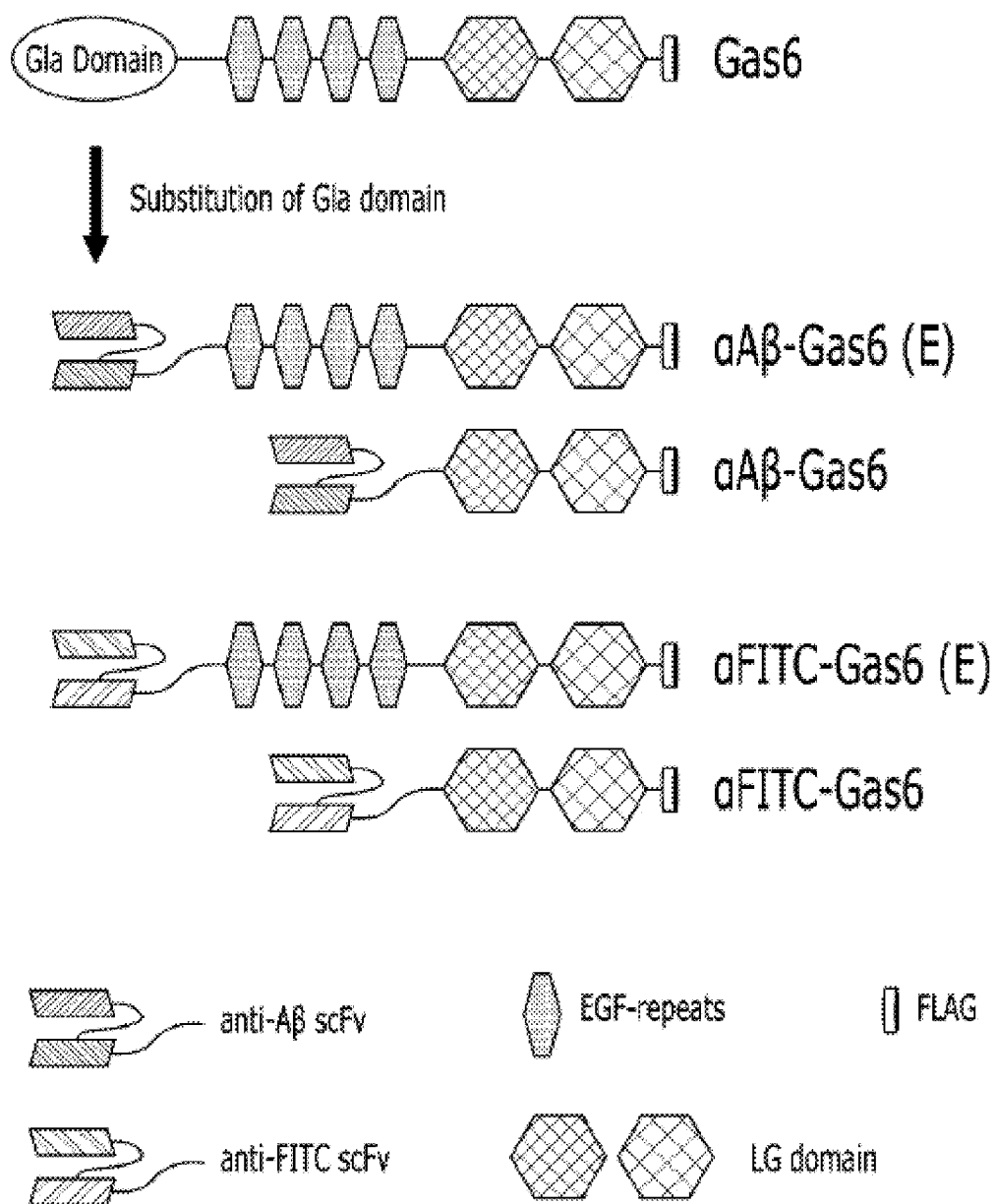


FIG. 1C

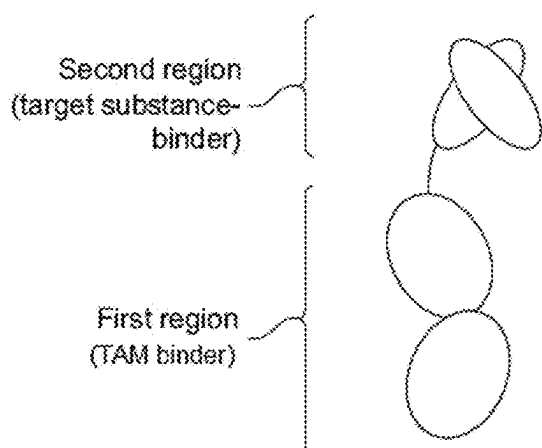


FIG. 1D

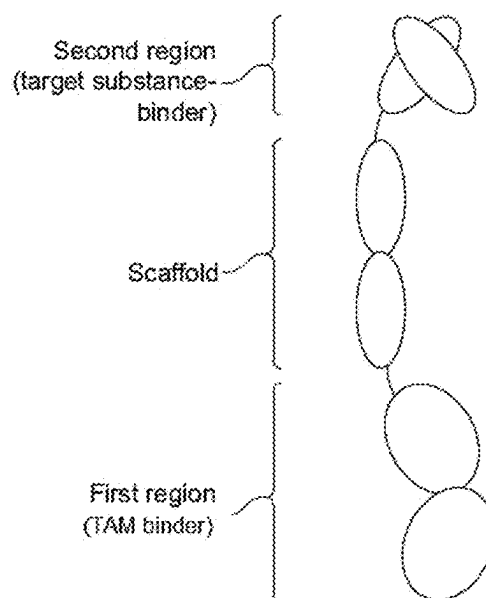


FIG. 1E

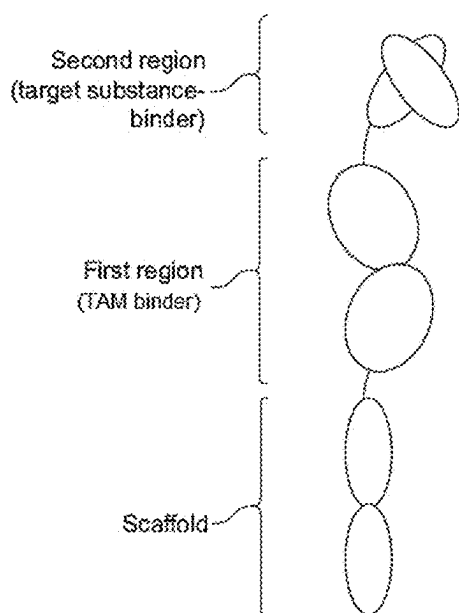


FIG. 1F

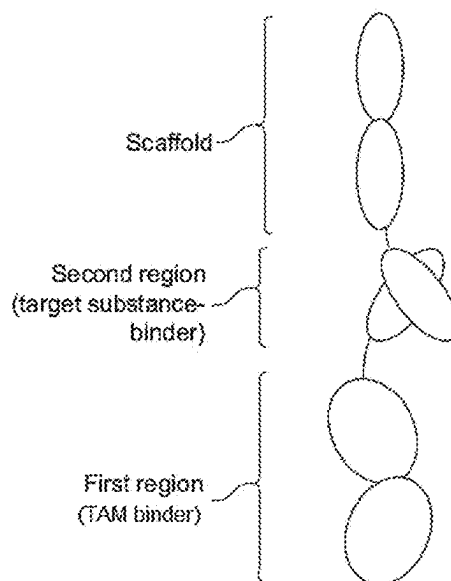


FIG. 1G

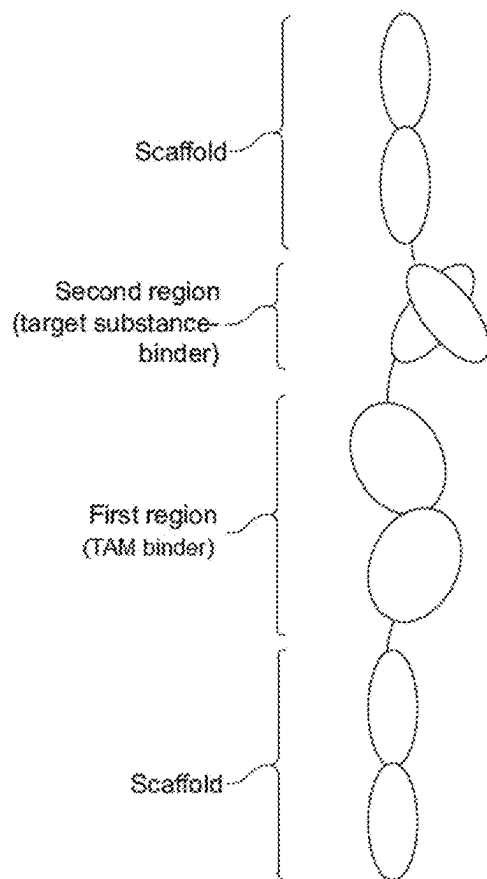


FIG. 1H

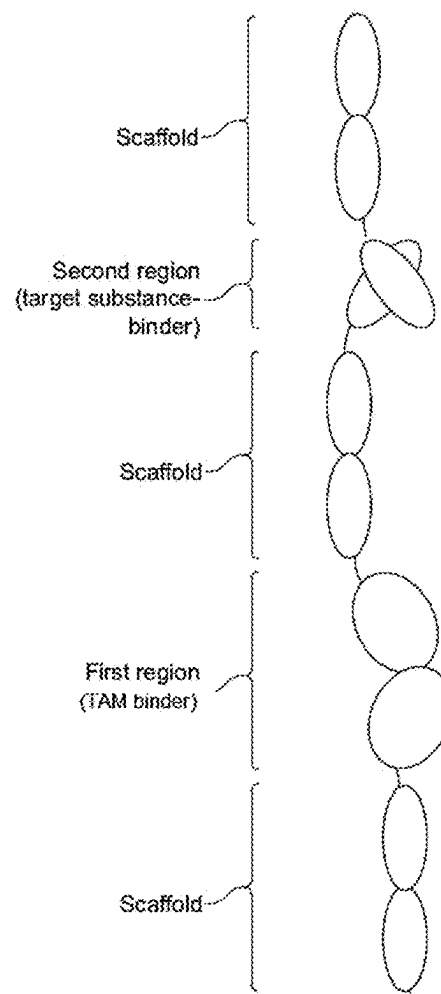


FIG. 1I

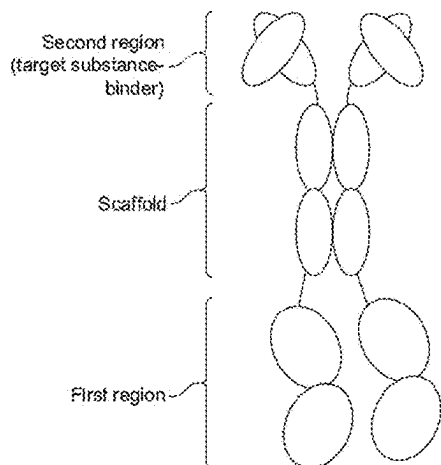


FIG. 1J

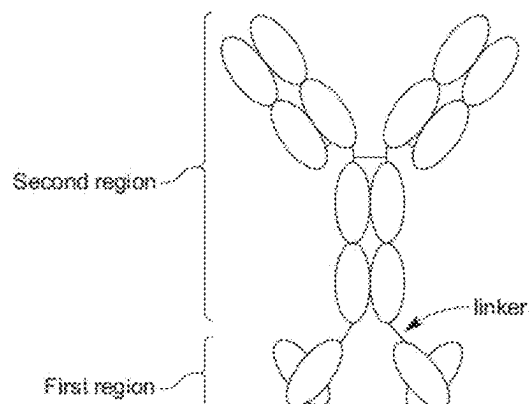


FIG. 1K

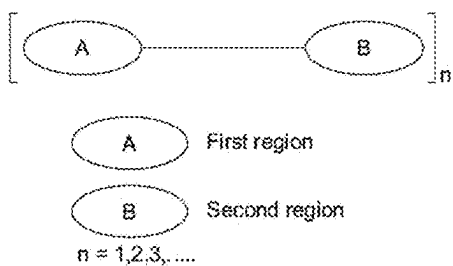


FIG. 1L

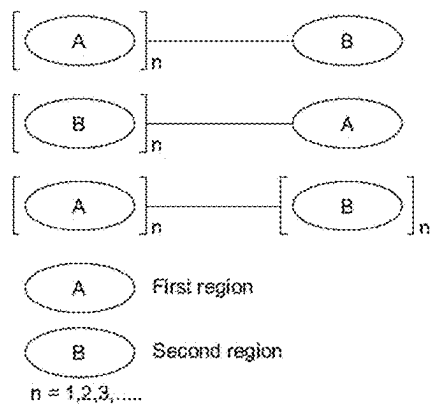


FIG. 1M

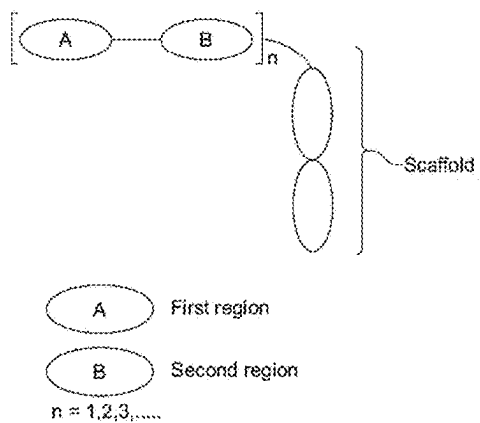


FIG. 2

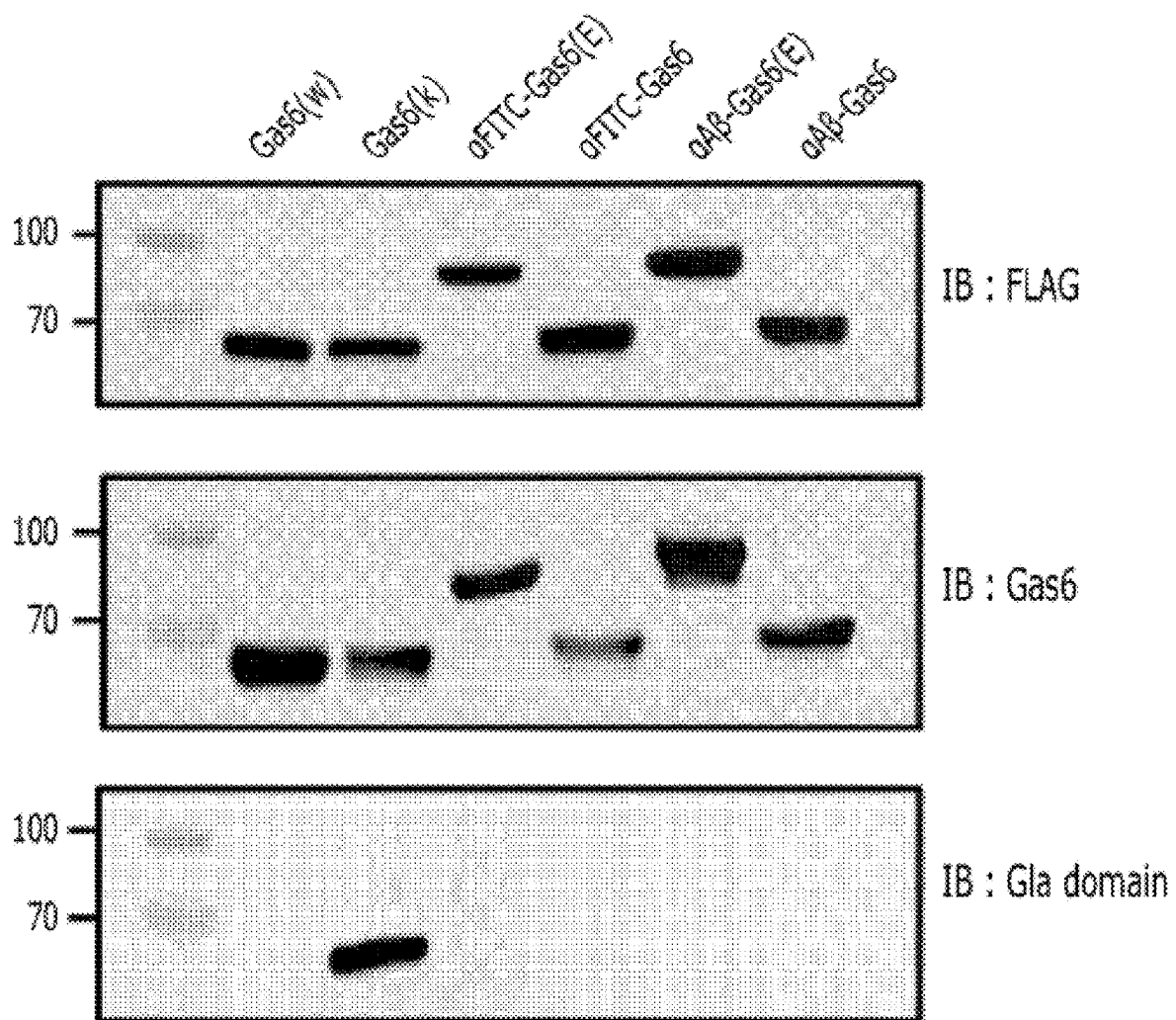


FIG. 3

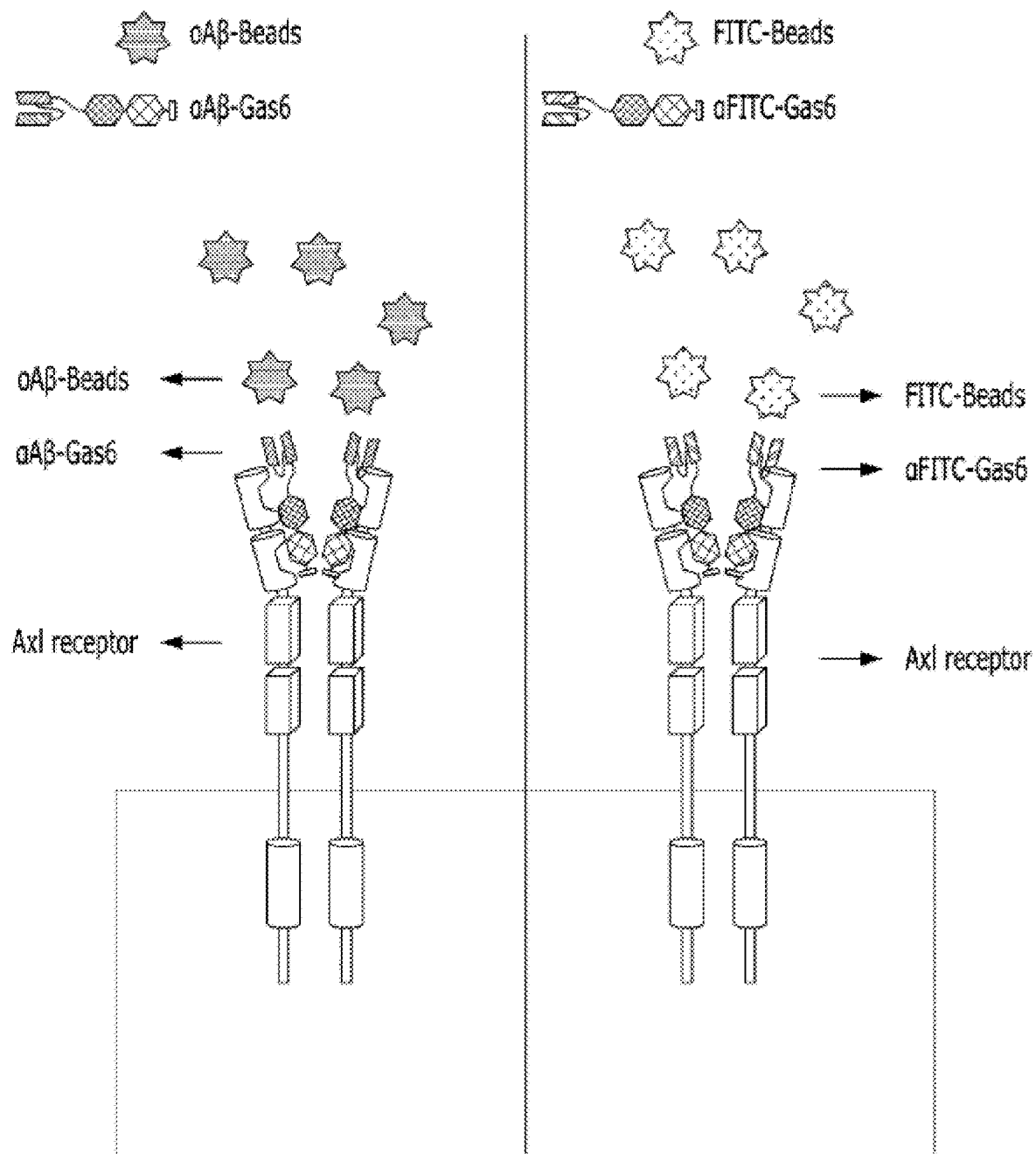


FIG. 4

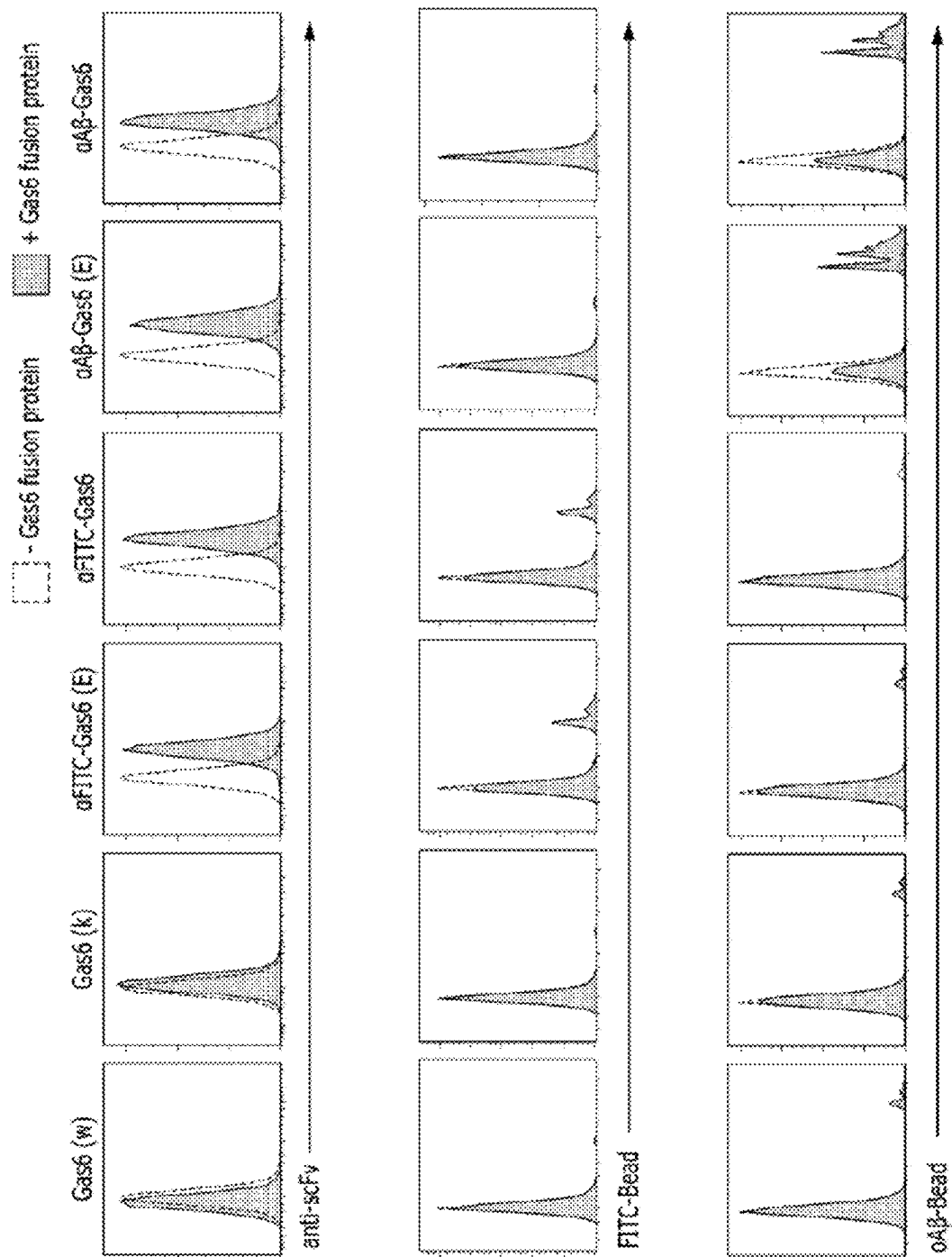


FIG. 5

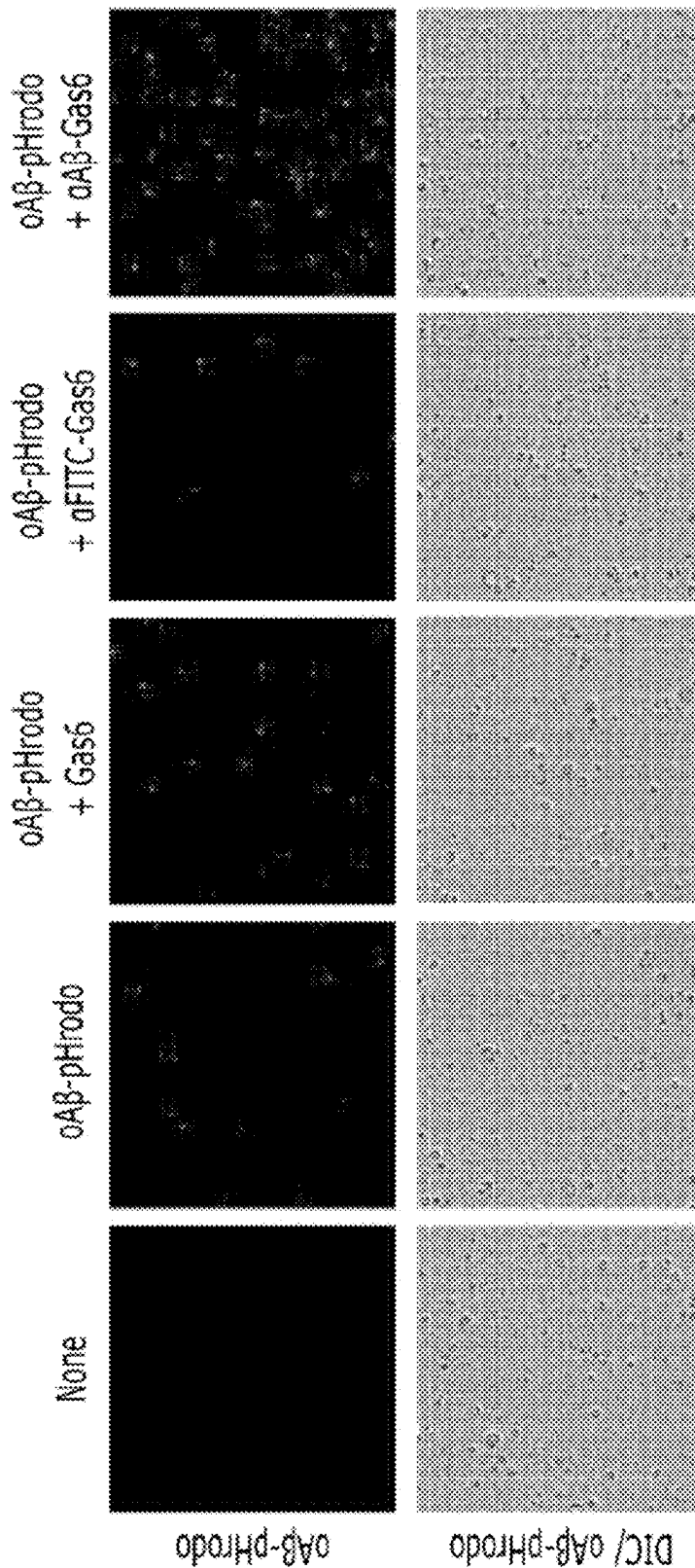


FIG. 6

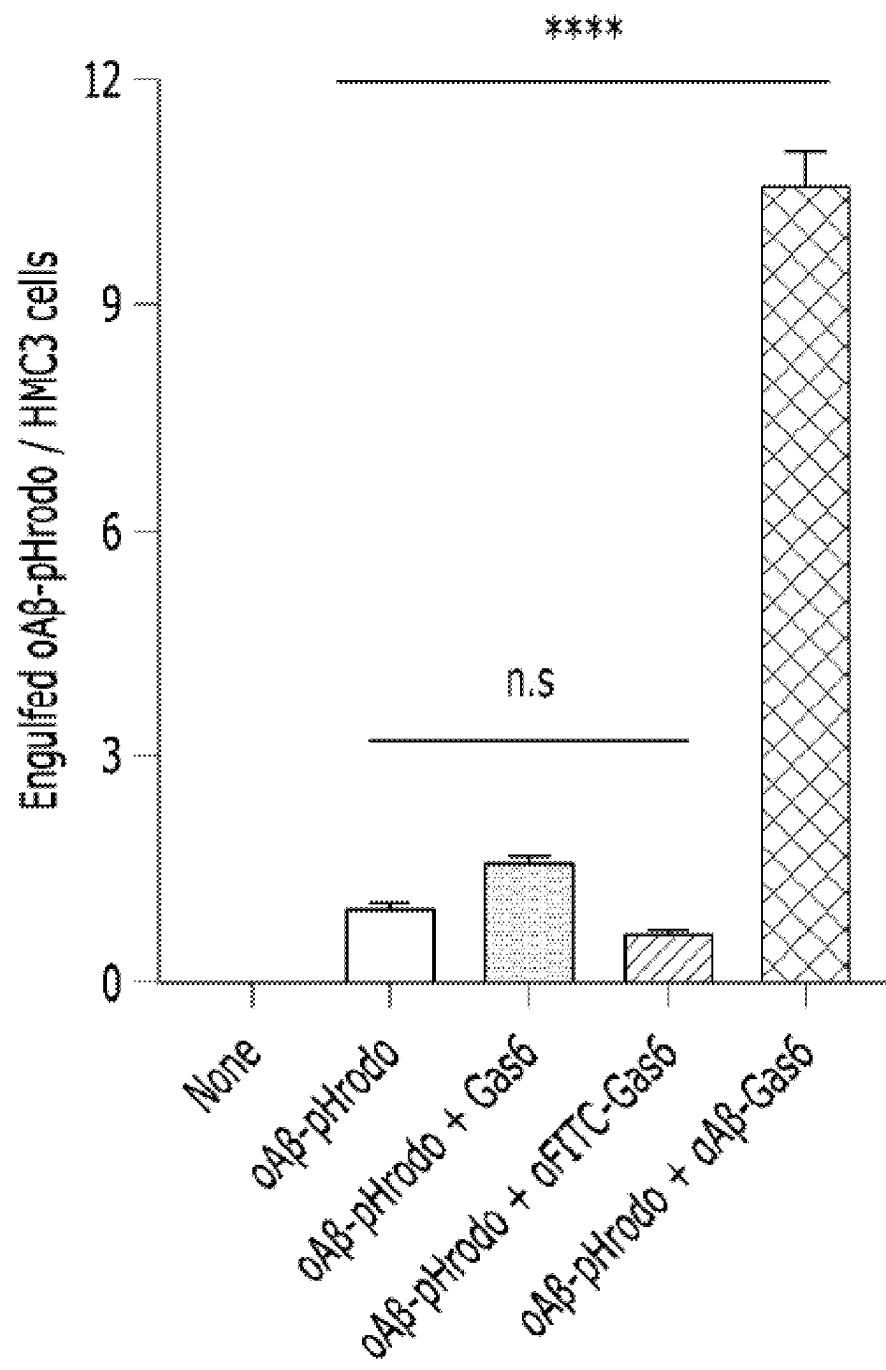


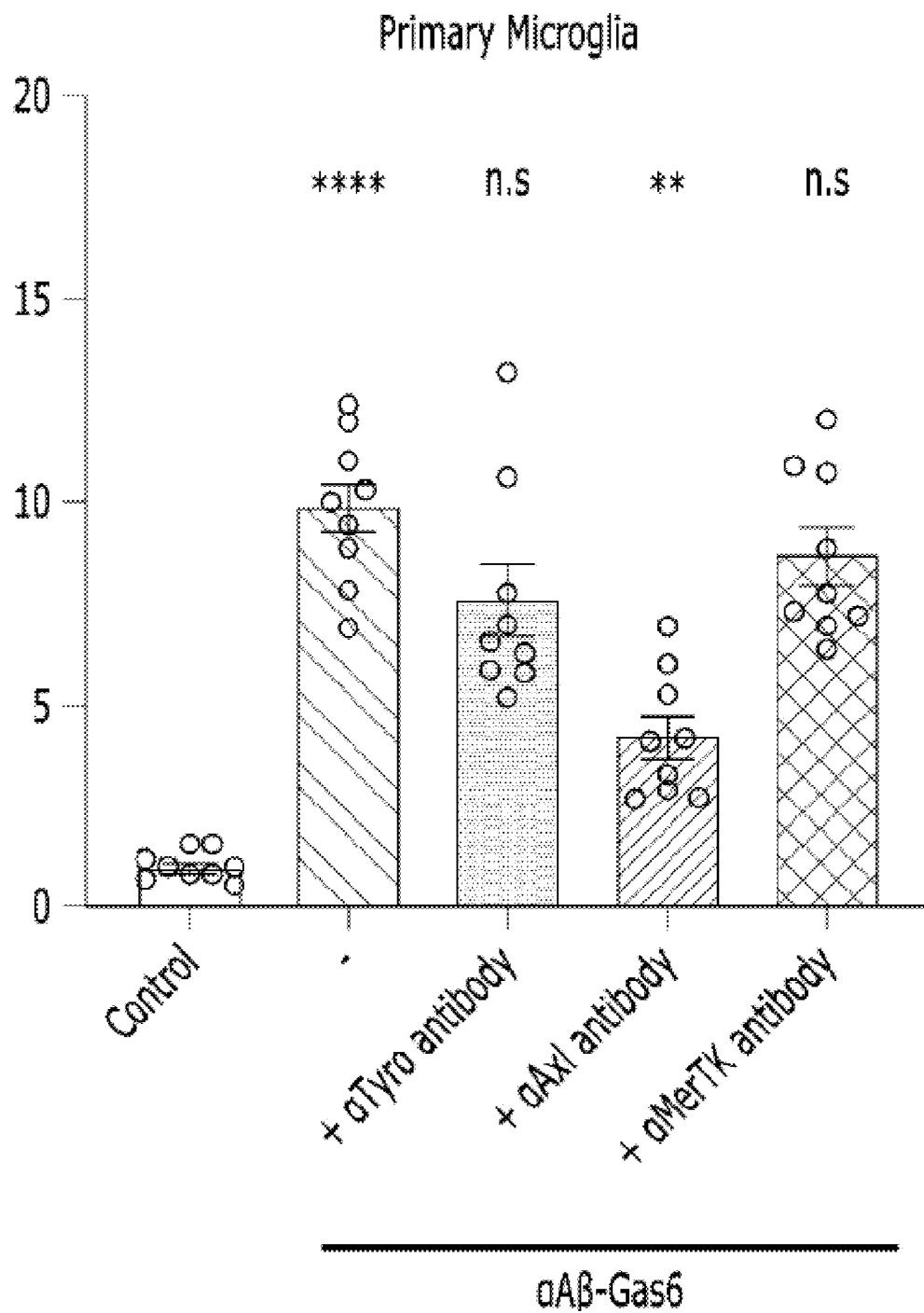
FIG. 7

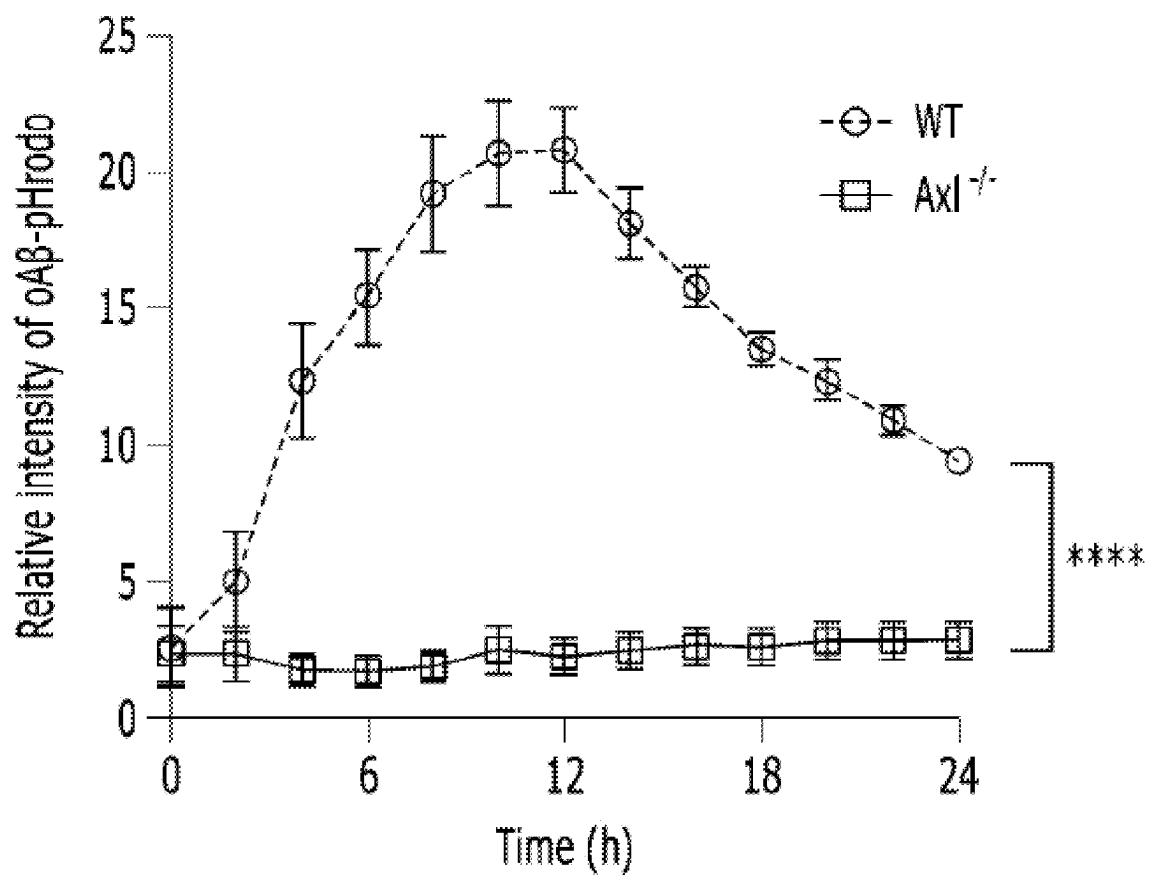
FIG. 8 α A β -Gas6

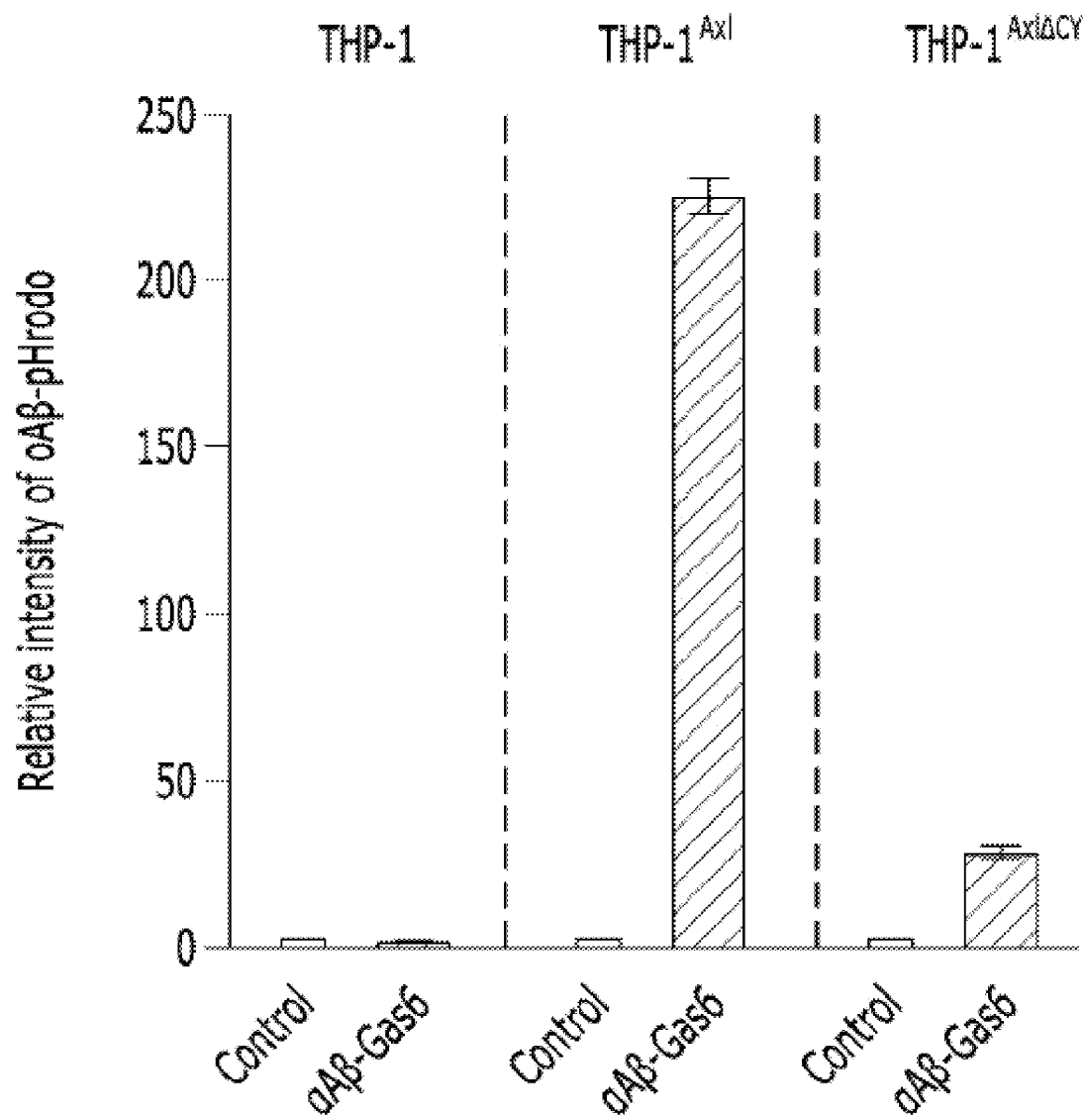
FIG. 9

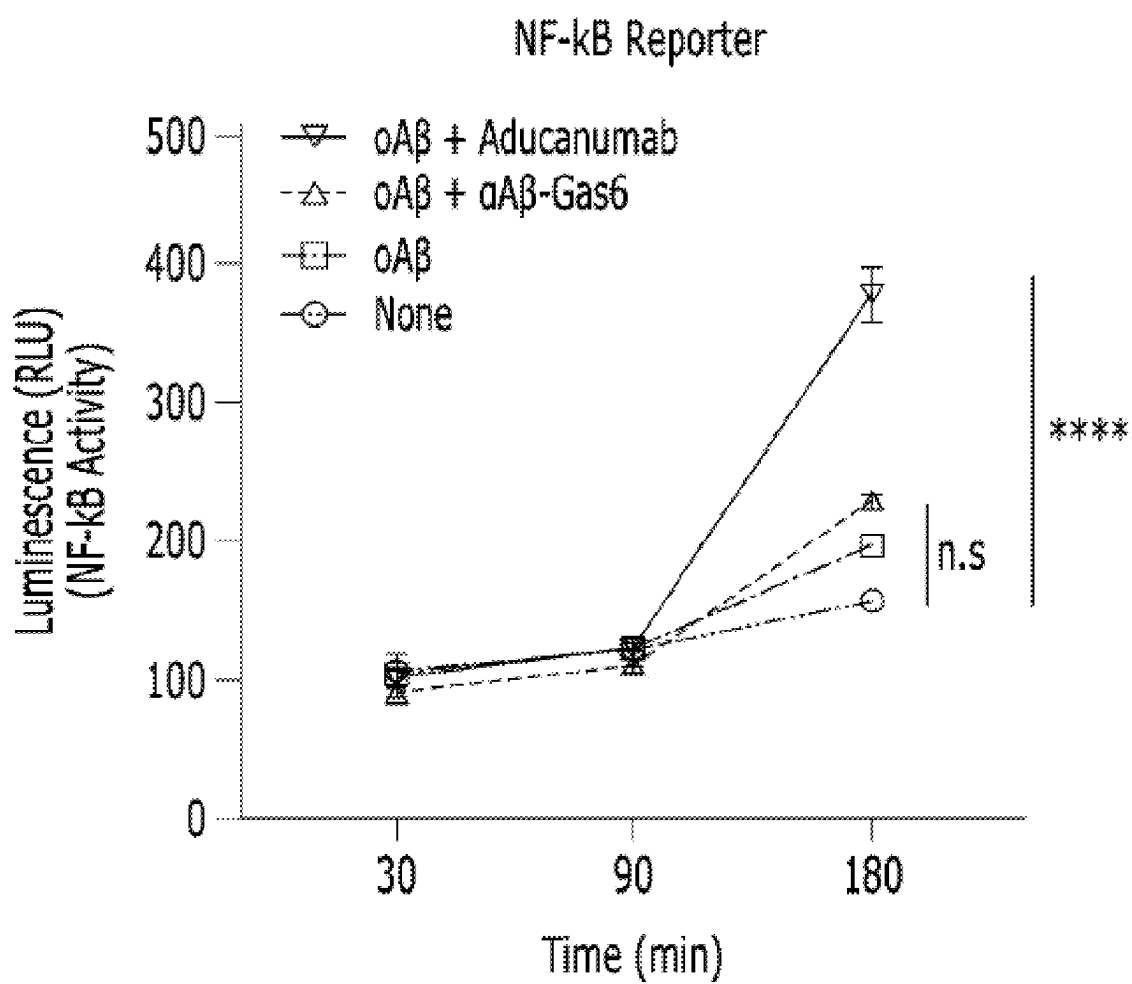
FIG. 10

FIG. 11

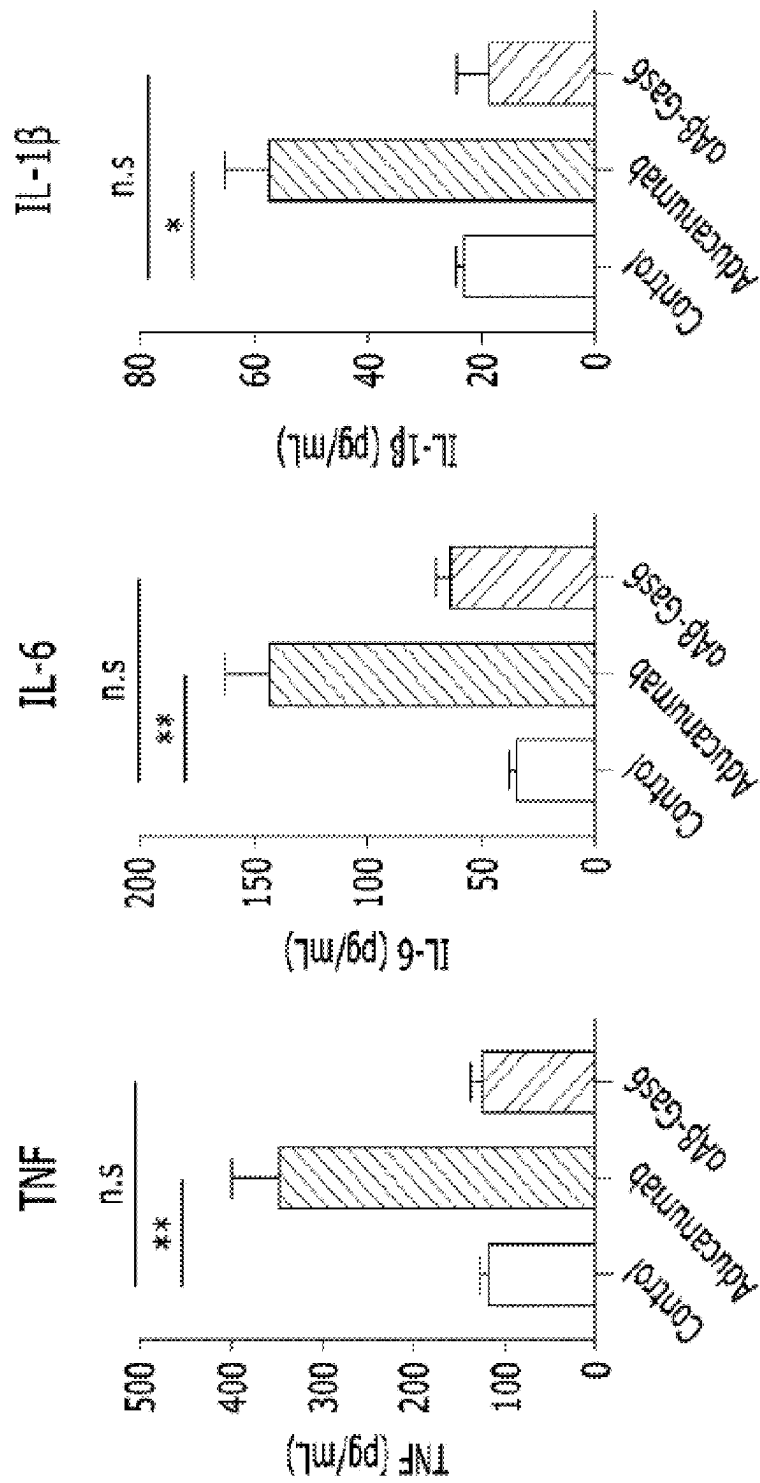


FIG. 12

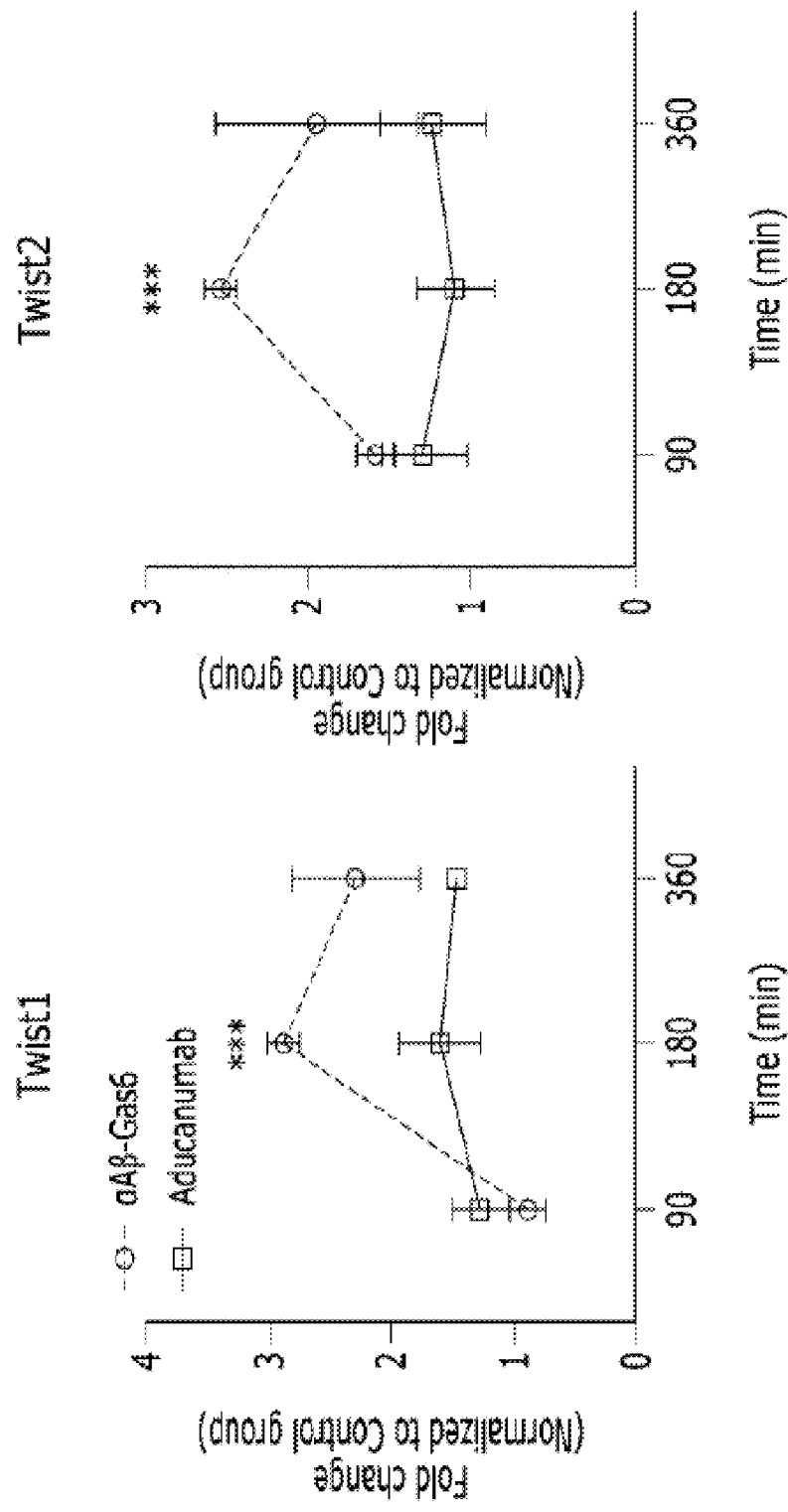


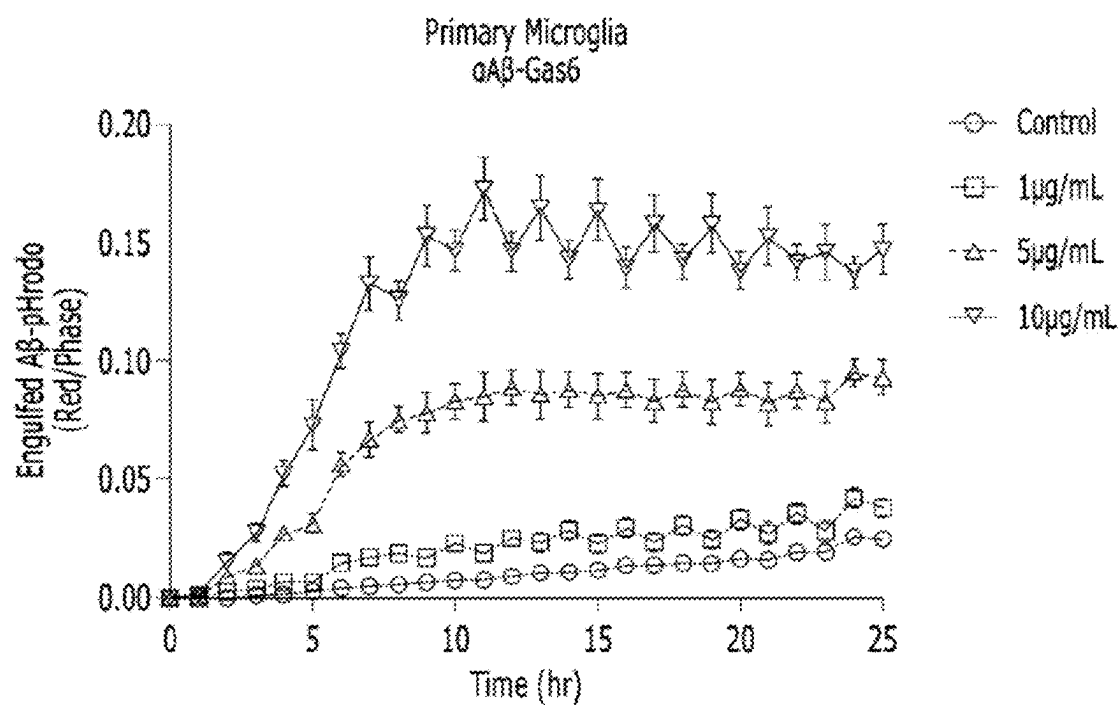
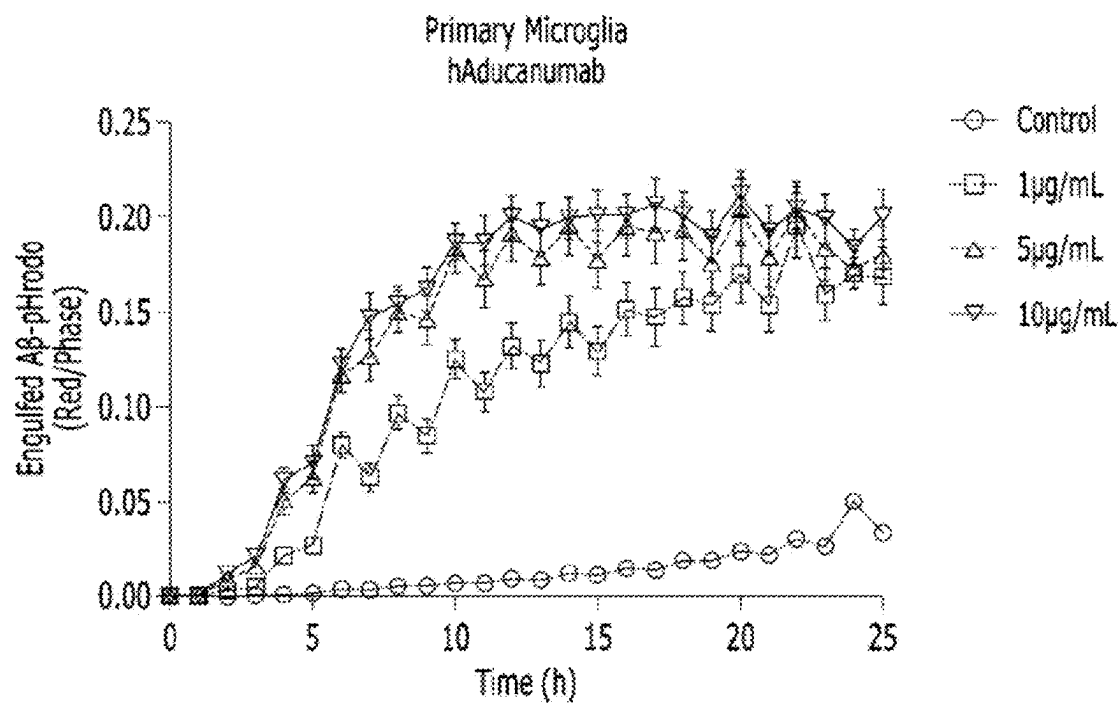
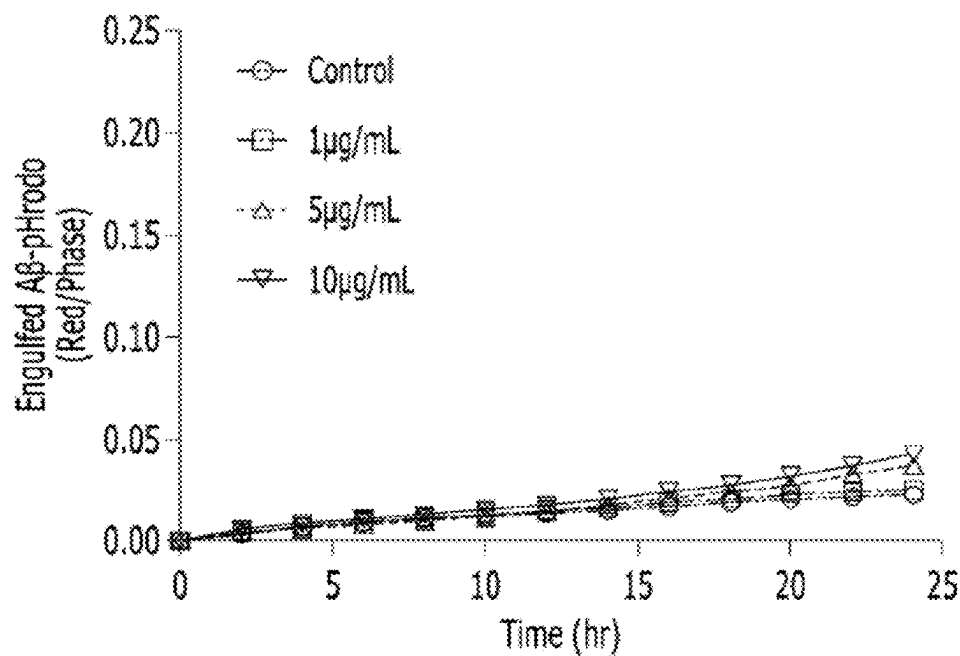
FIG. 13

FIG. 14

Primary Astrocyte
hAducanumab



Primary Astrocyte
 α A β -Gas6

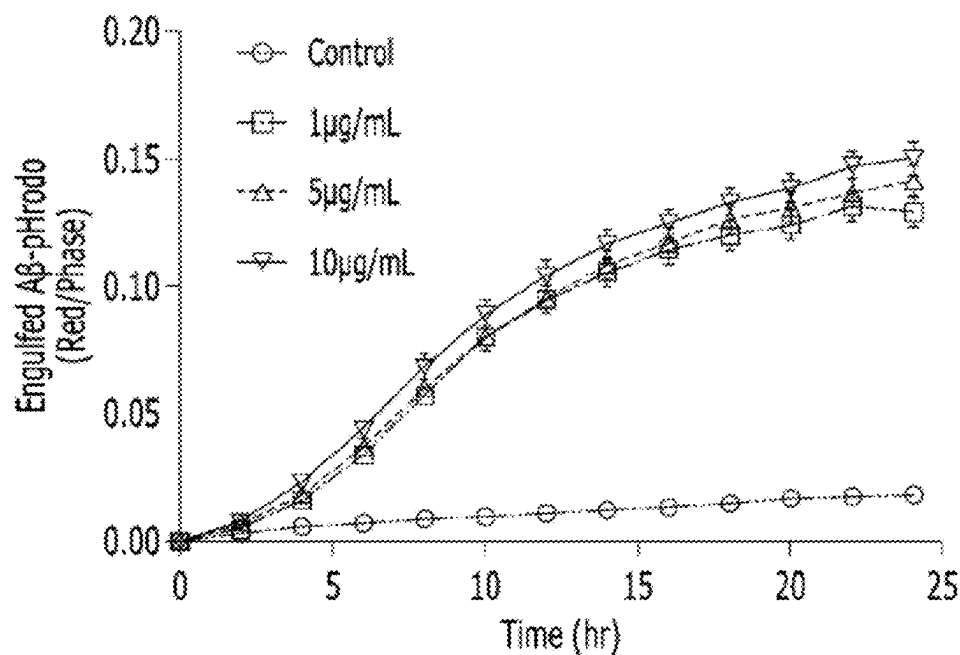


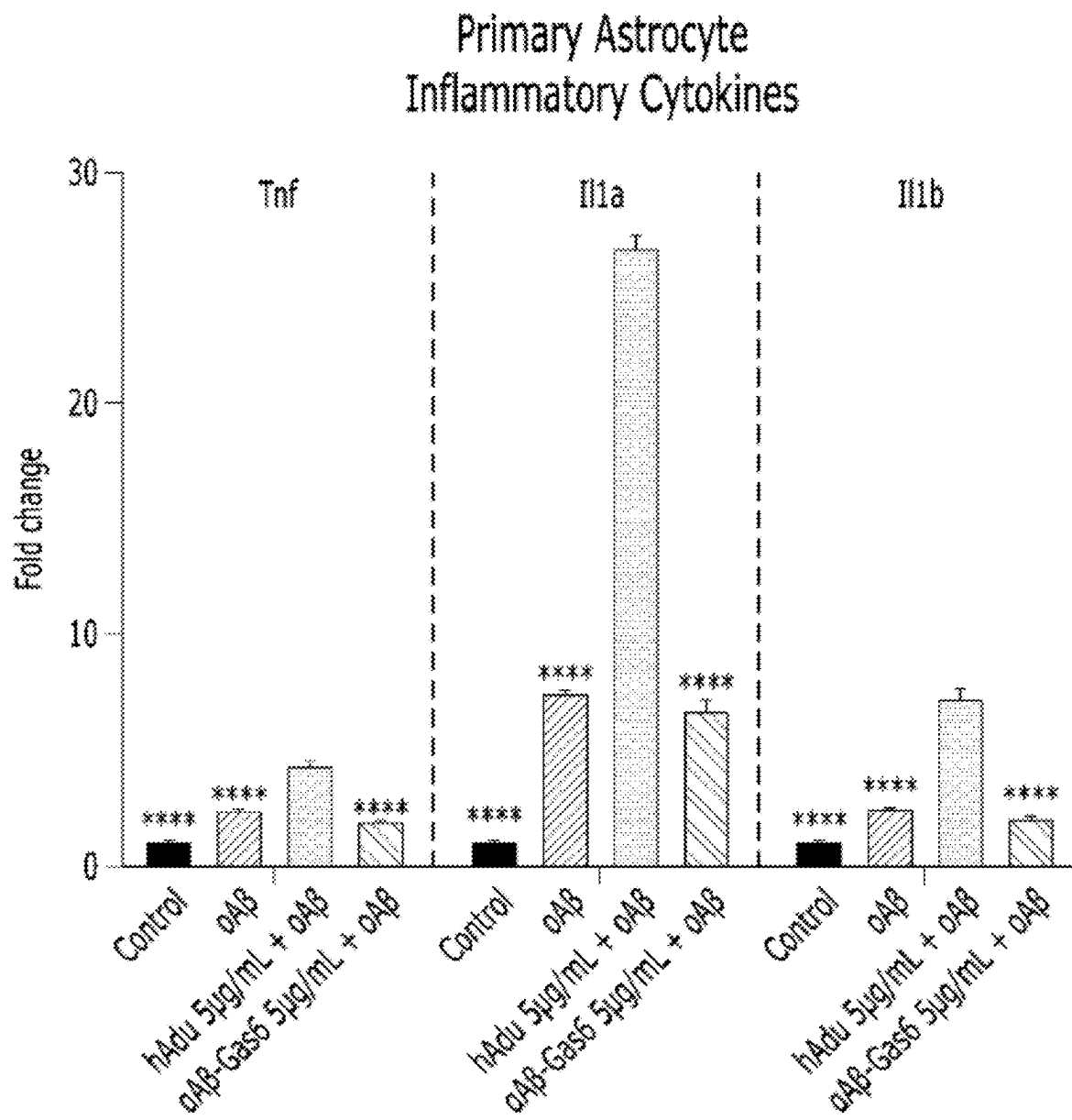
FIG. 15

FIG. 16

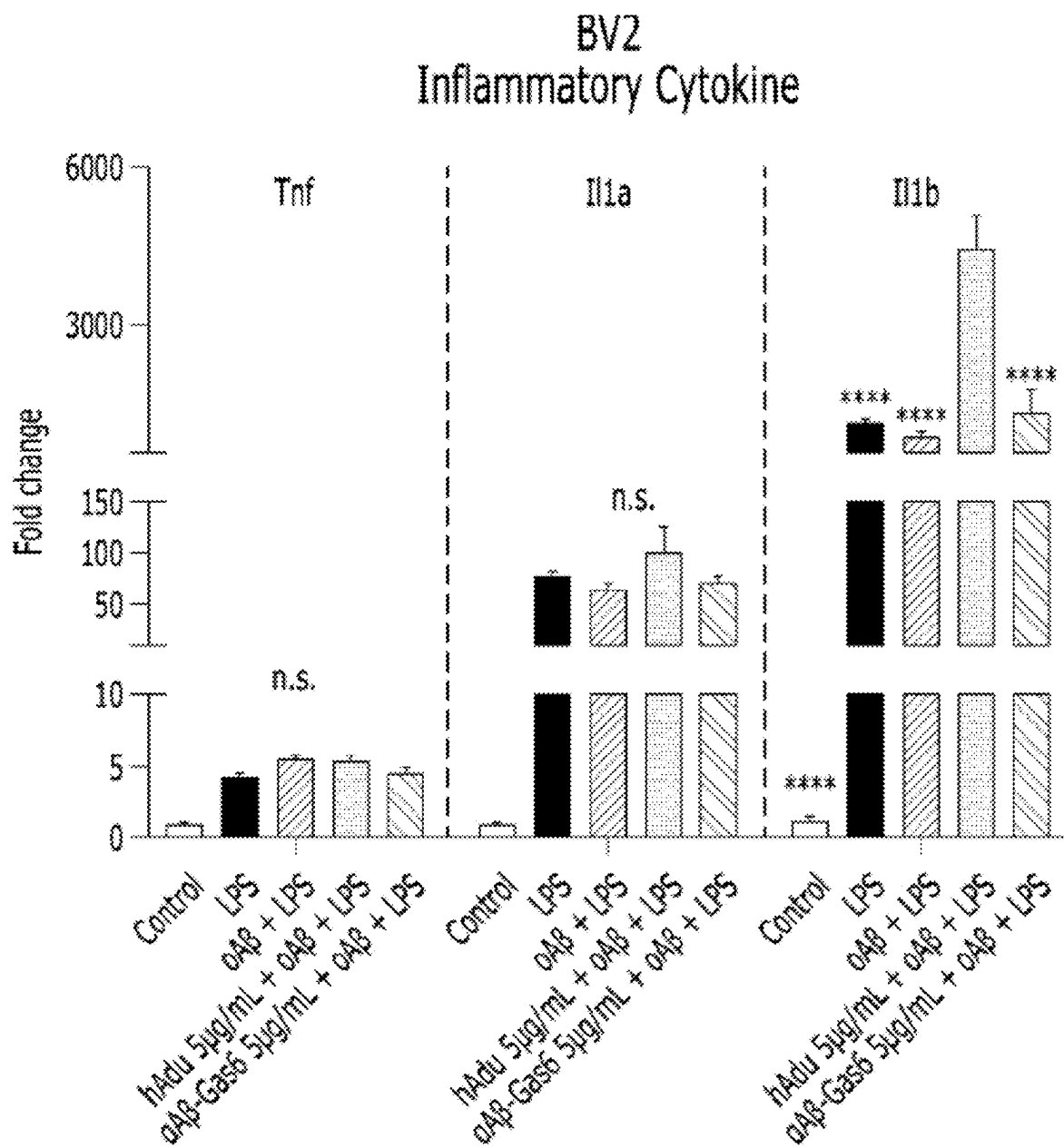


FIG. 17

Administration of
 α A β -Gas6 Fusion Protein

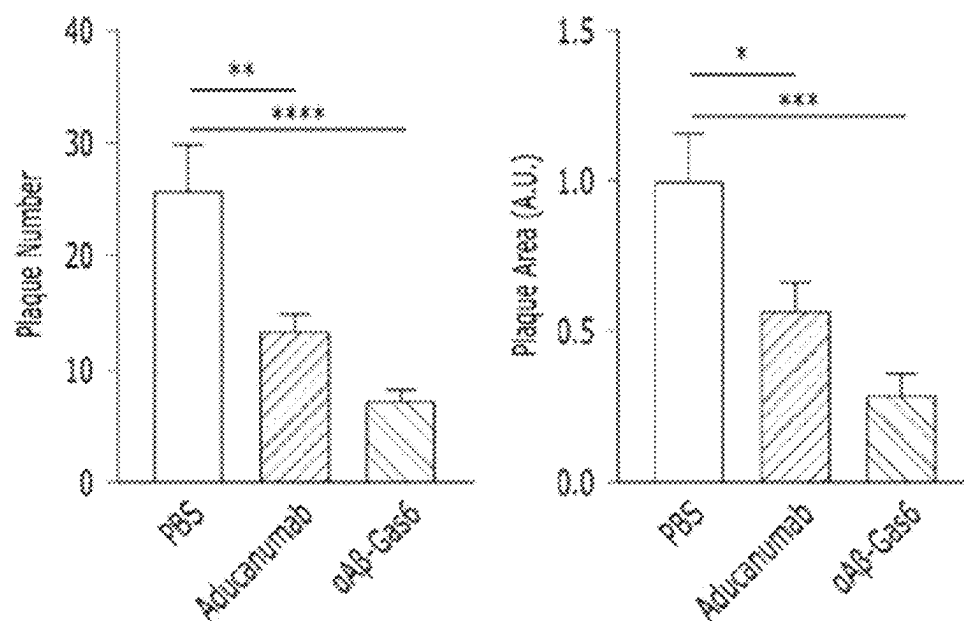
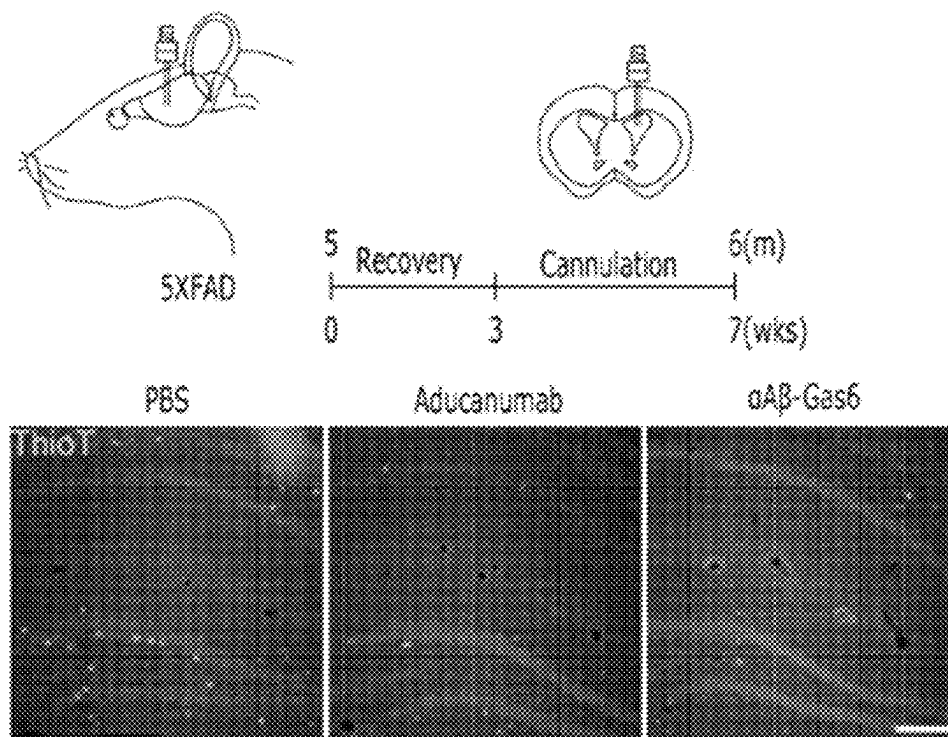


FIG. 18

Administration of virus carrying
 α A β -Gas6 - encoding gene

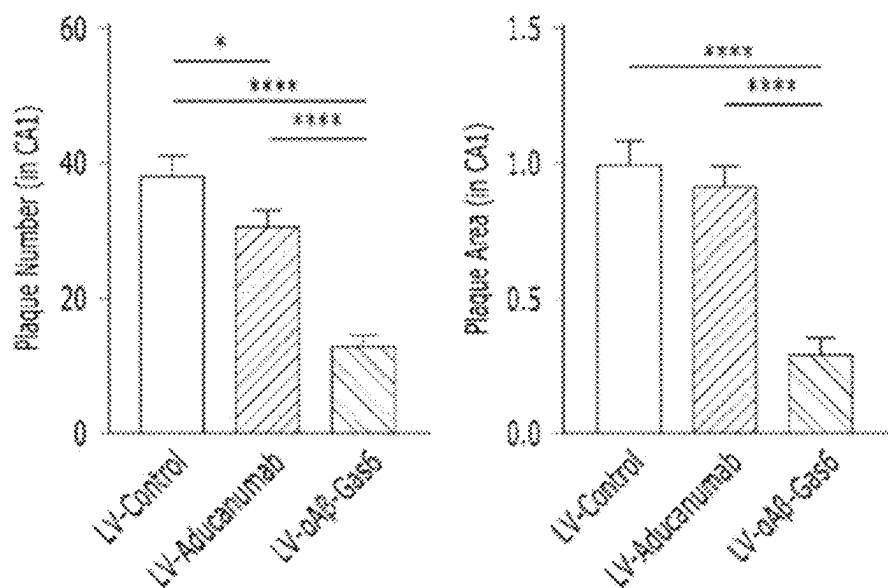
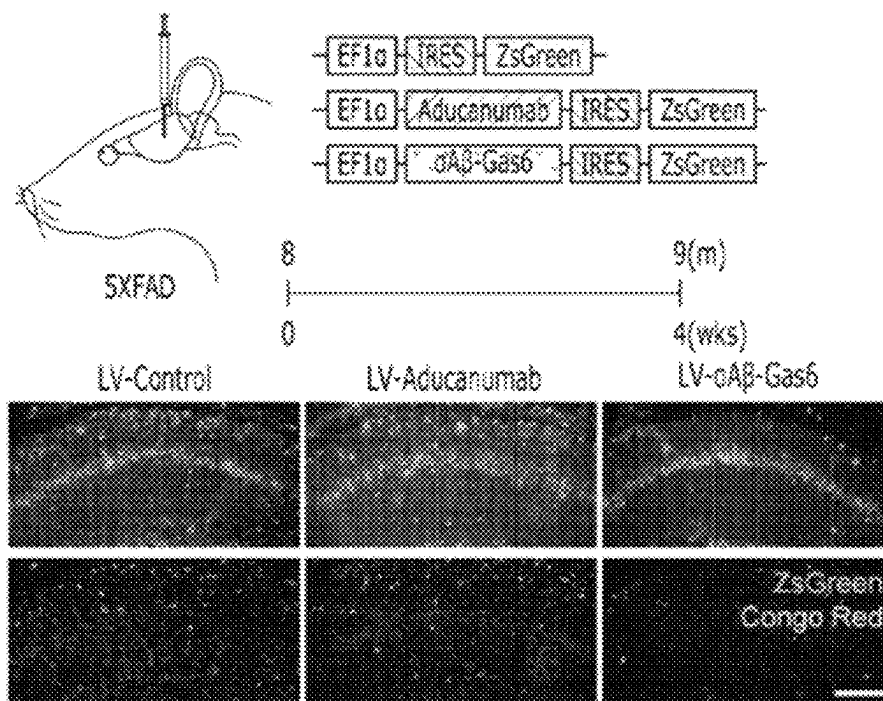


FIG. 19

Administration of
cA β -Gs6 Fusion Protein

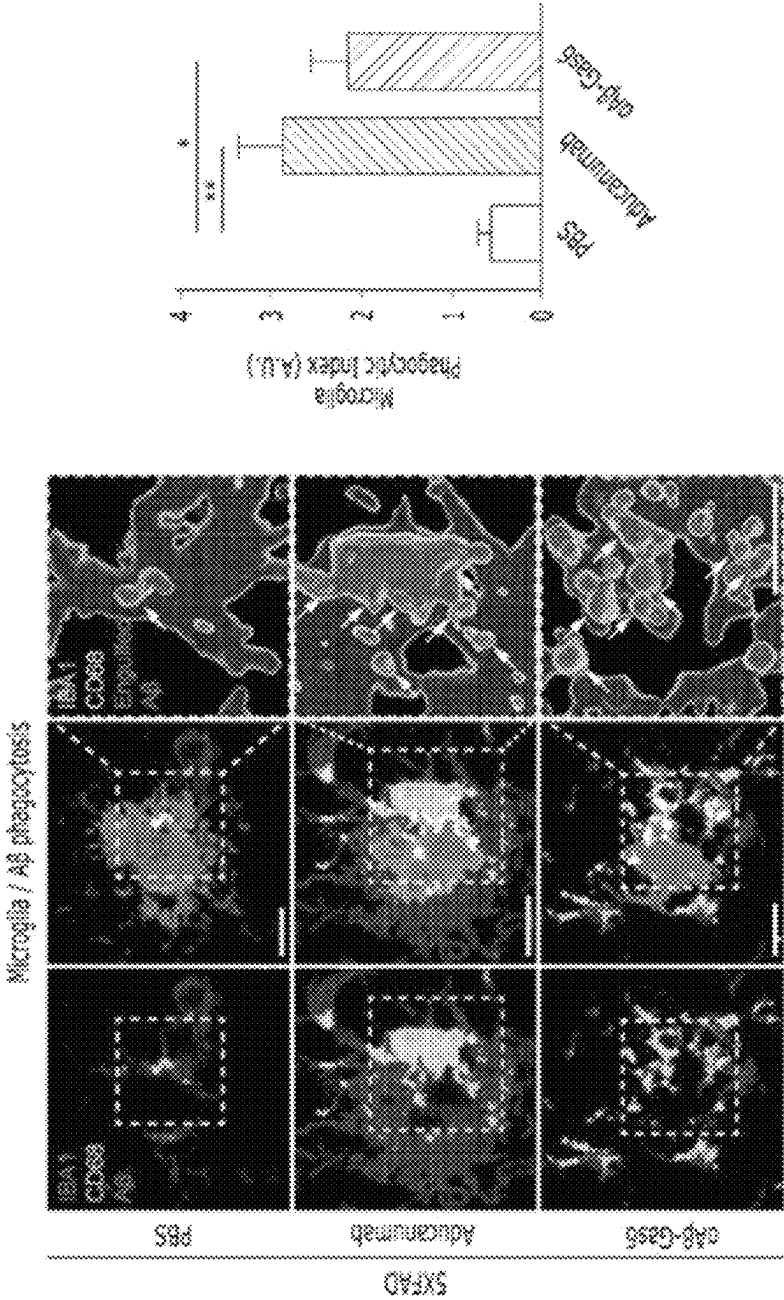


FIG. 20

Administration of
αAβ-Ga56 Fusion Protein

Astrocyte / Aβ phagocytosis

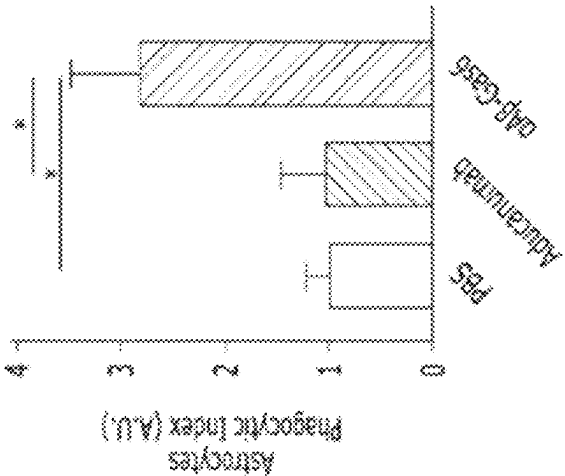
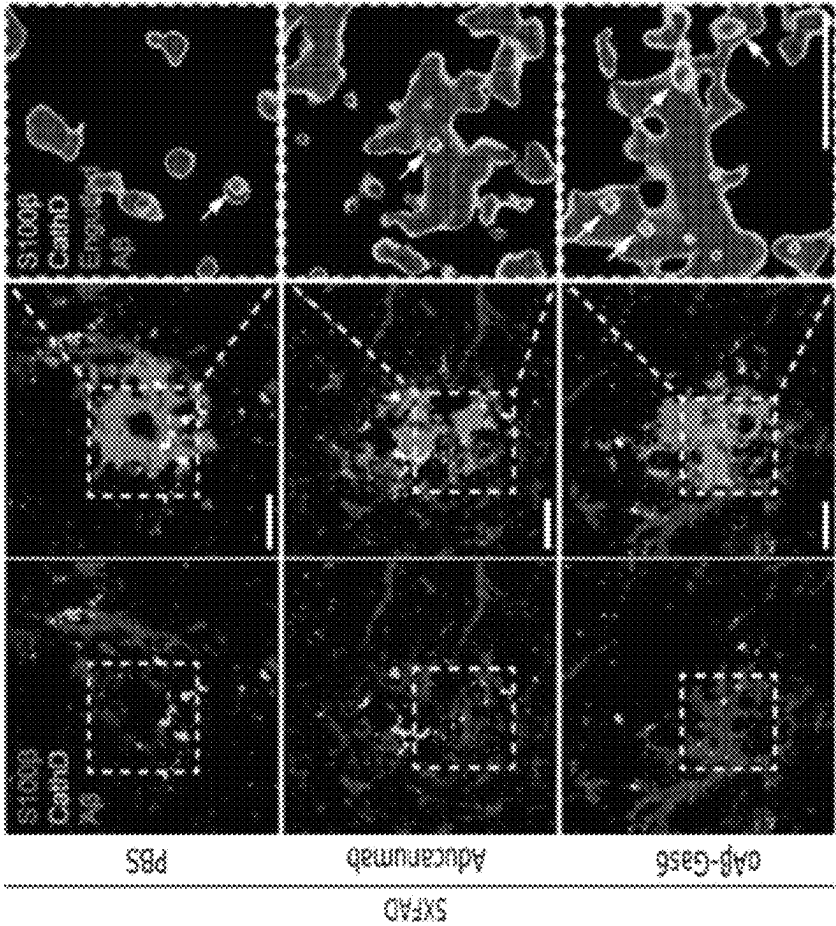


FIG. 21

Administration of vector carrying
 $\alpha\beta$ -Gas6 -encoding gene

Microglia / A β phagocytosis

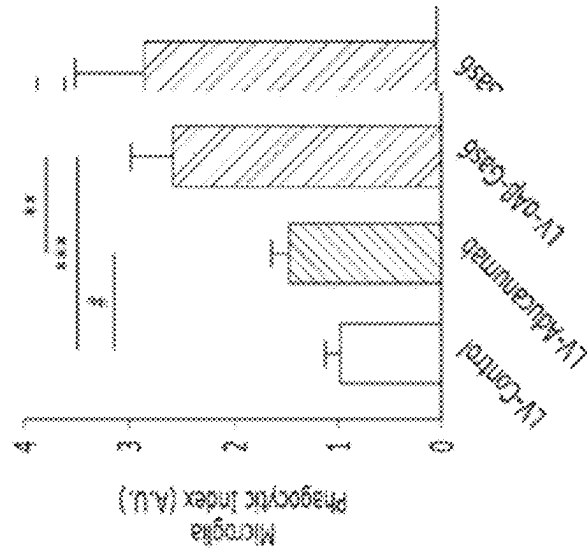
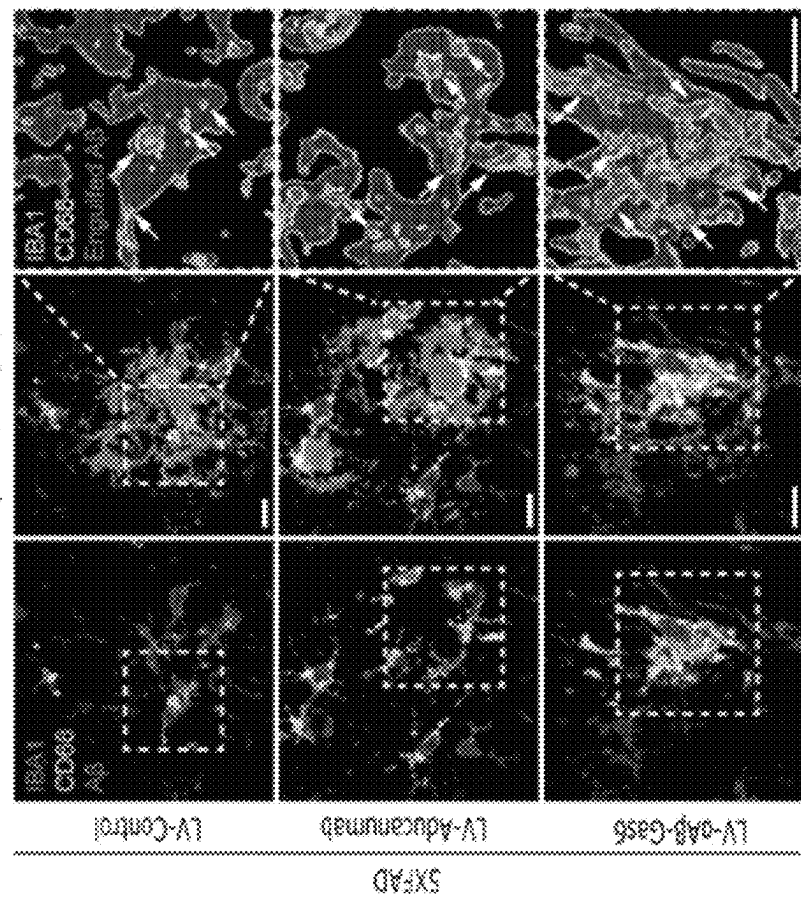


FIG. 22

Administration of vector carrying
 $\alpha A\beta$ -Gas6 -encoding gene

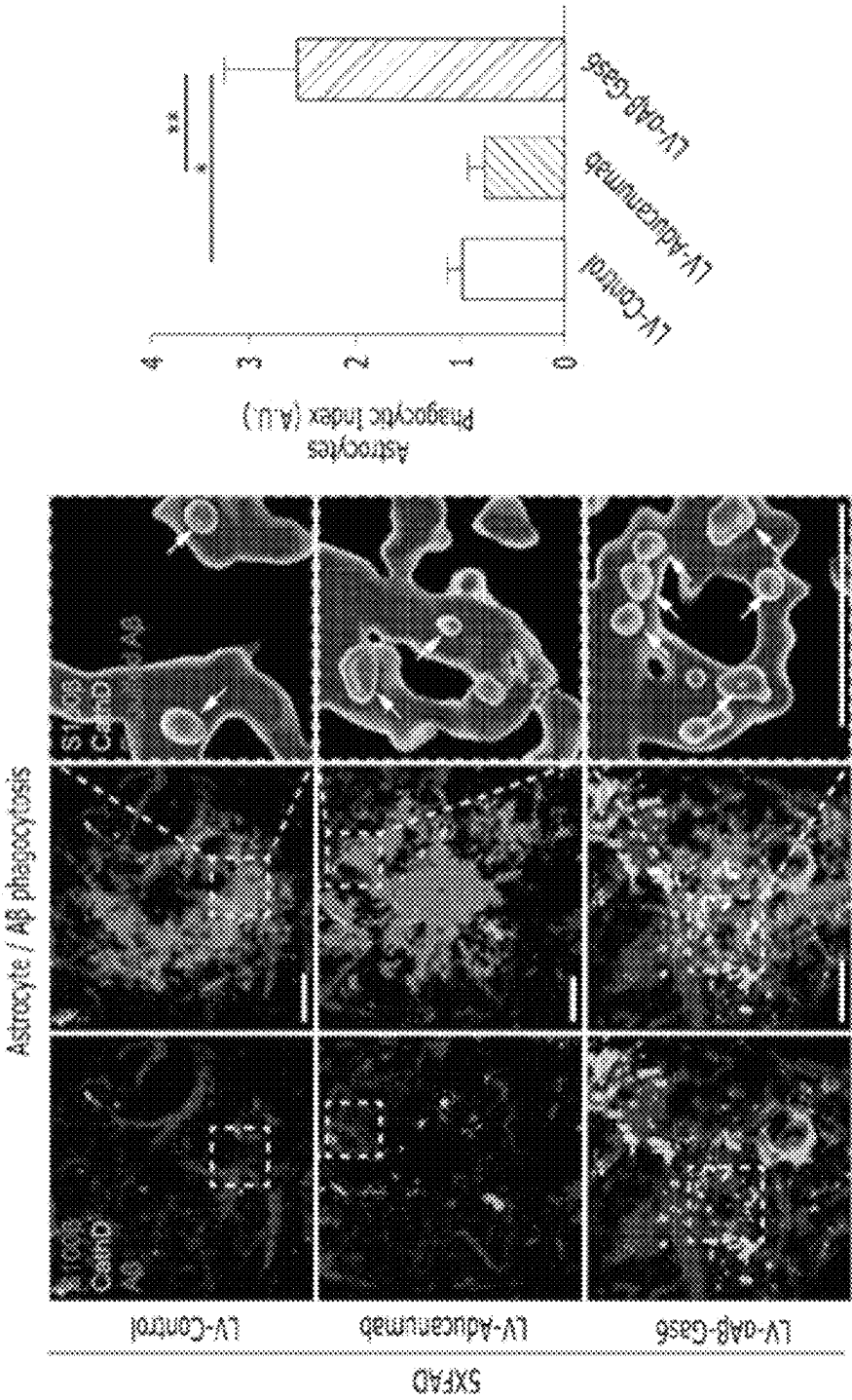


FIG. 23

Administration of vector carrying
 $\alpha\text{A}\beta\text{-Gas6}$ -encoding gene

Microglia / PSD95⁺ Synapse Engulfment

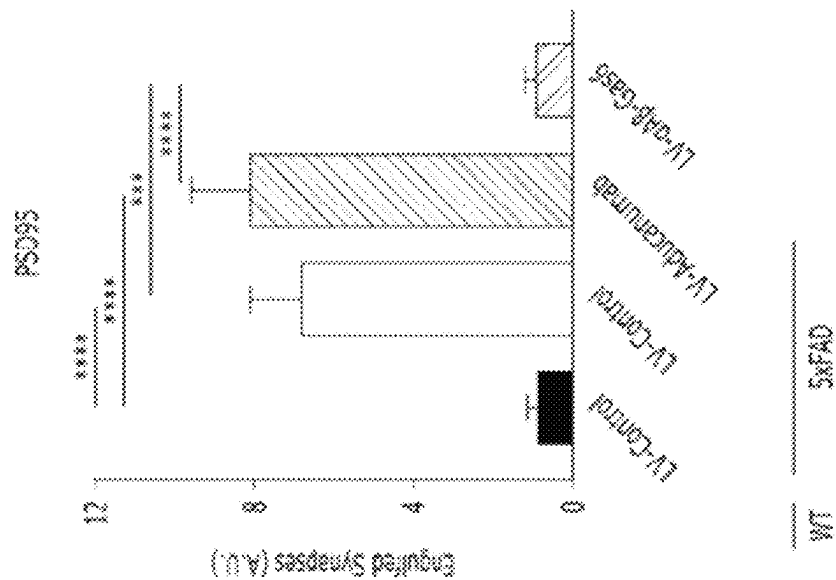
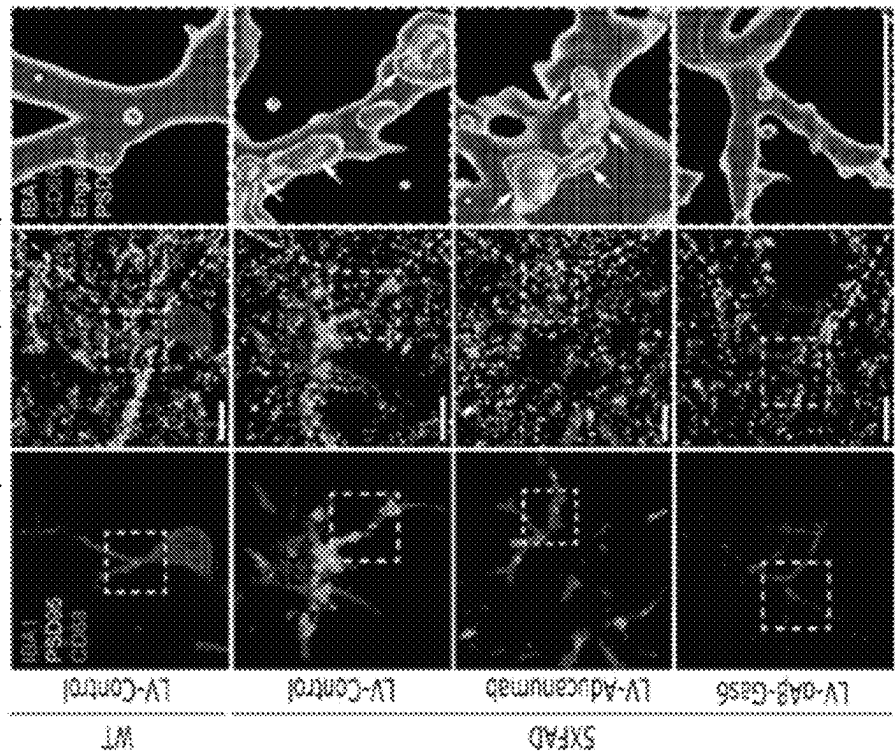


FIG. 24
Administration of vector carrying
 α A β -Gas6 -encoding gene

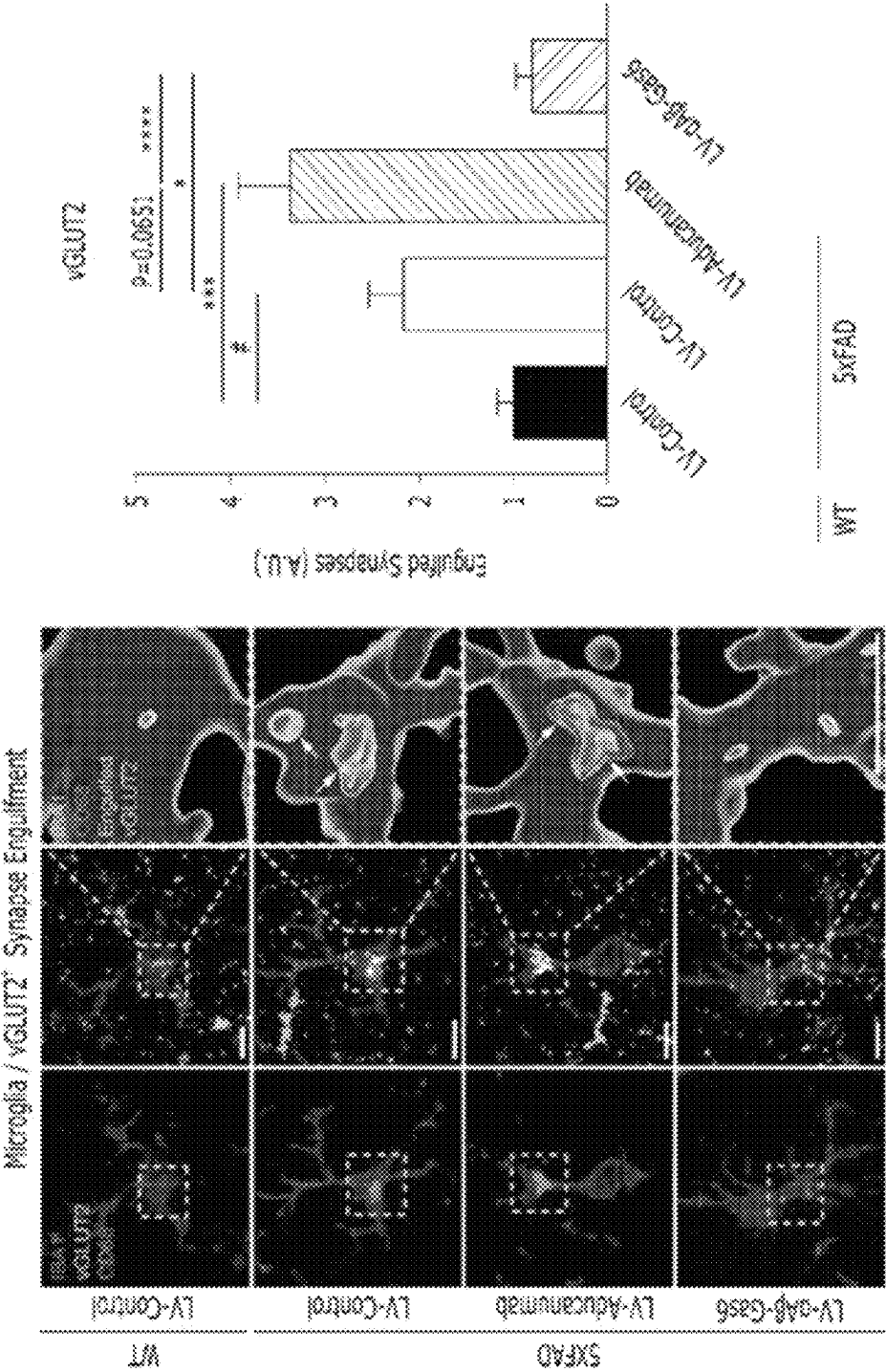
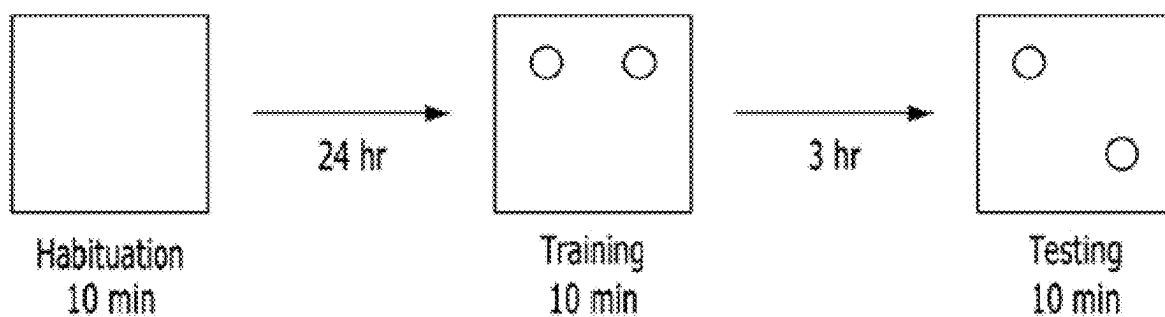


FIG. 25

Administration of vector carrying
 $\alpha\beta$ -Gas6 -encoding gene

Novel Object Location Test



Novel Object Recognition Test

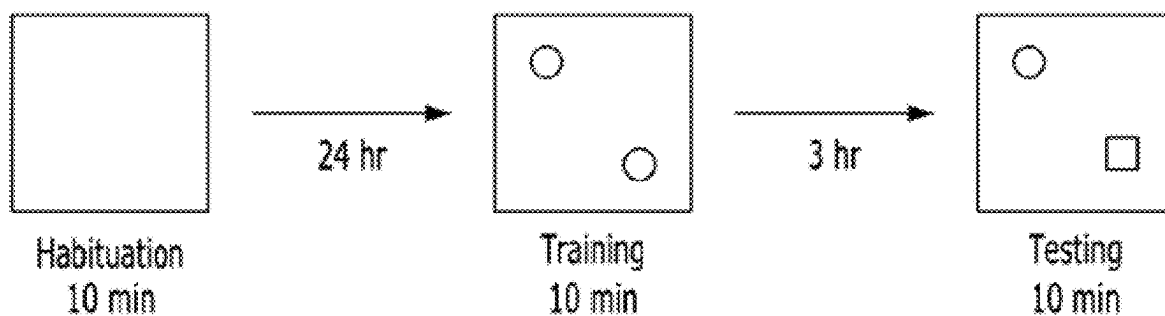


FIG. 26

Administration of vector carrying
 $\alpha\text{A}\beta$ -Gas6 -encoding gene

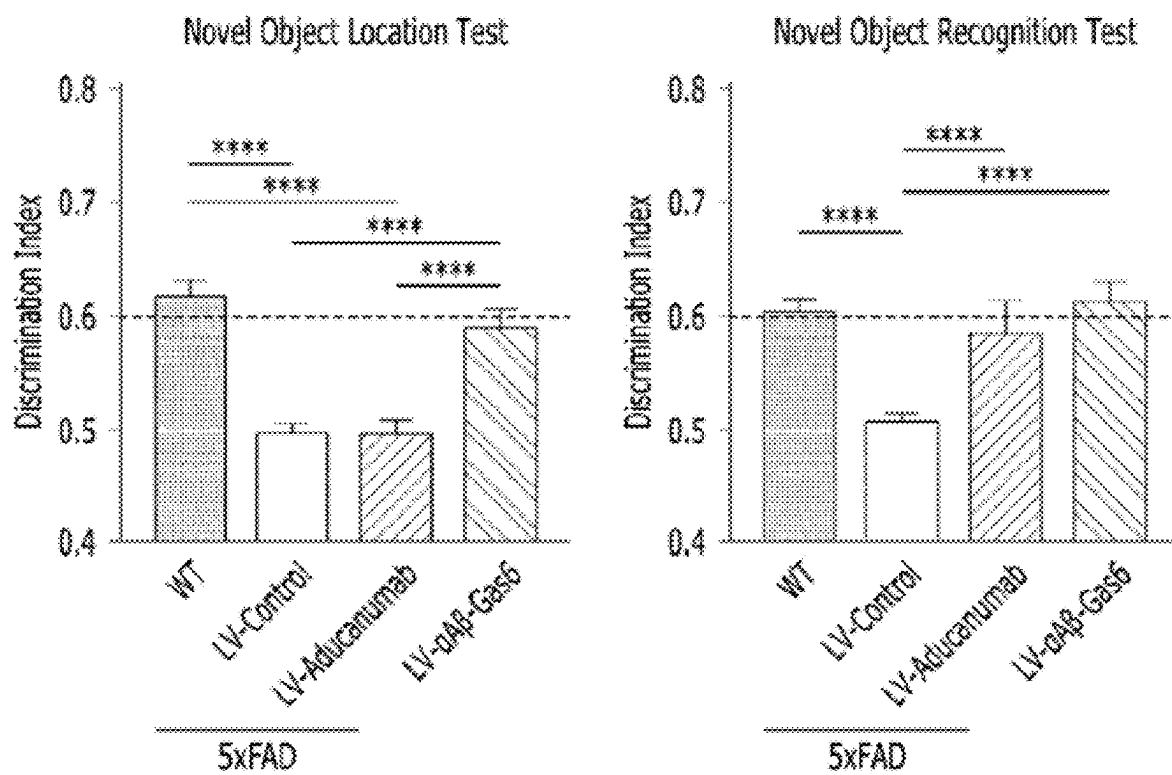


FIG. 27

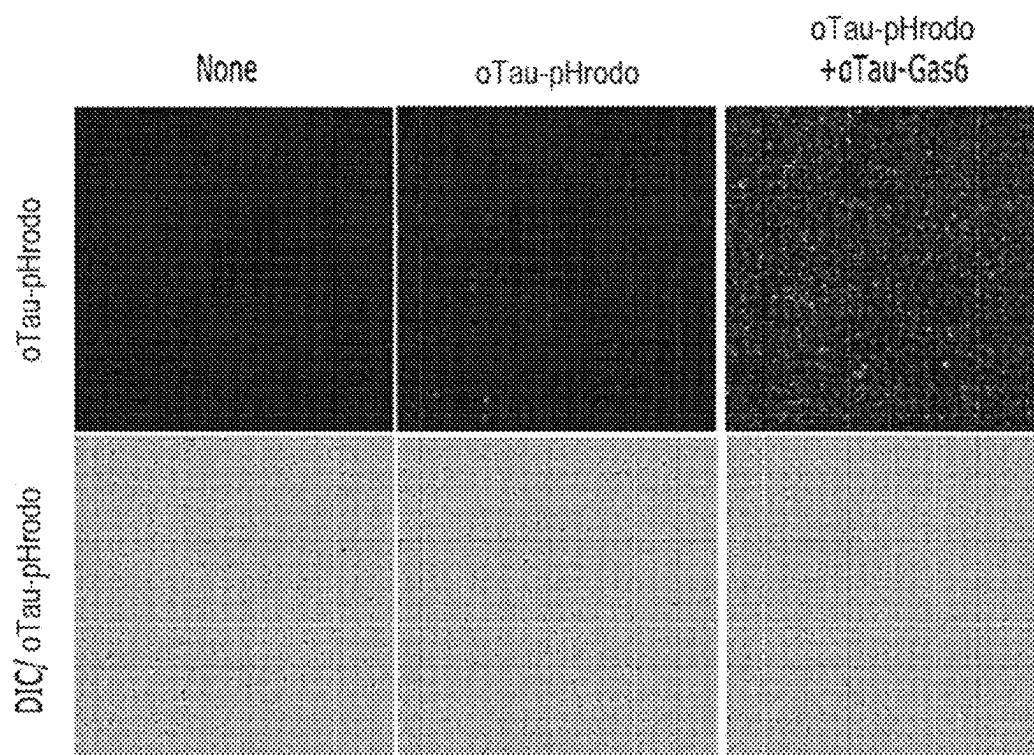
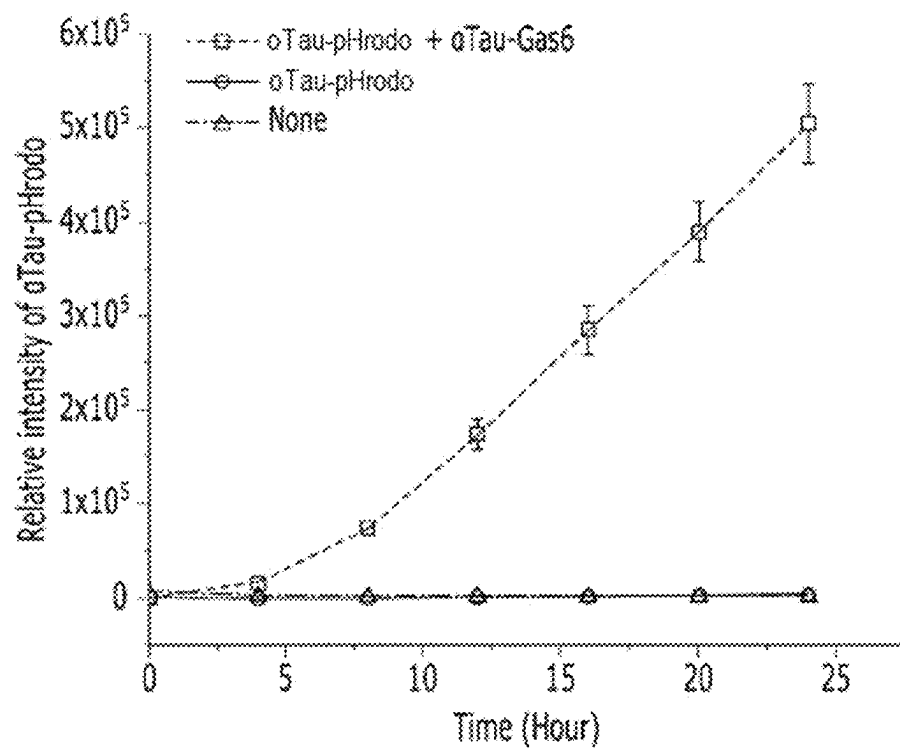


FIG. 28

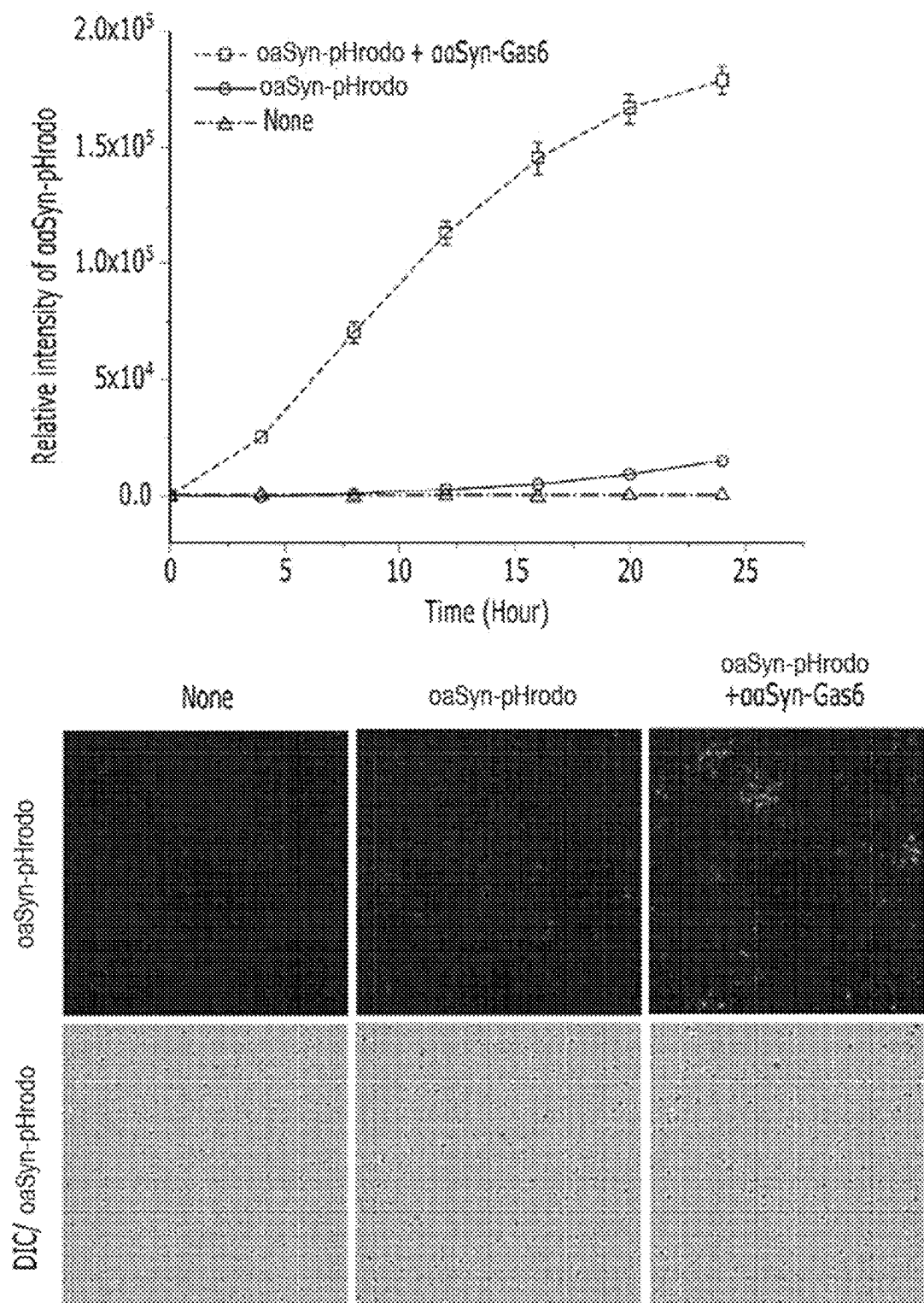


FIG. 29

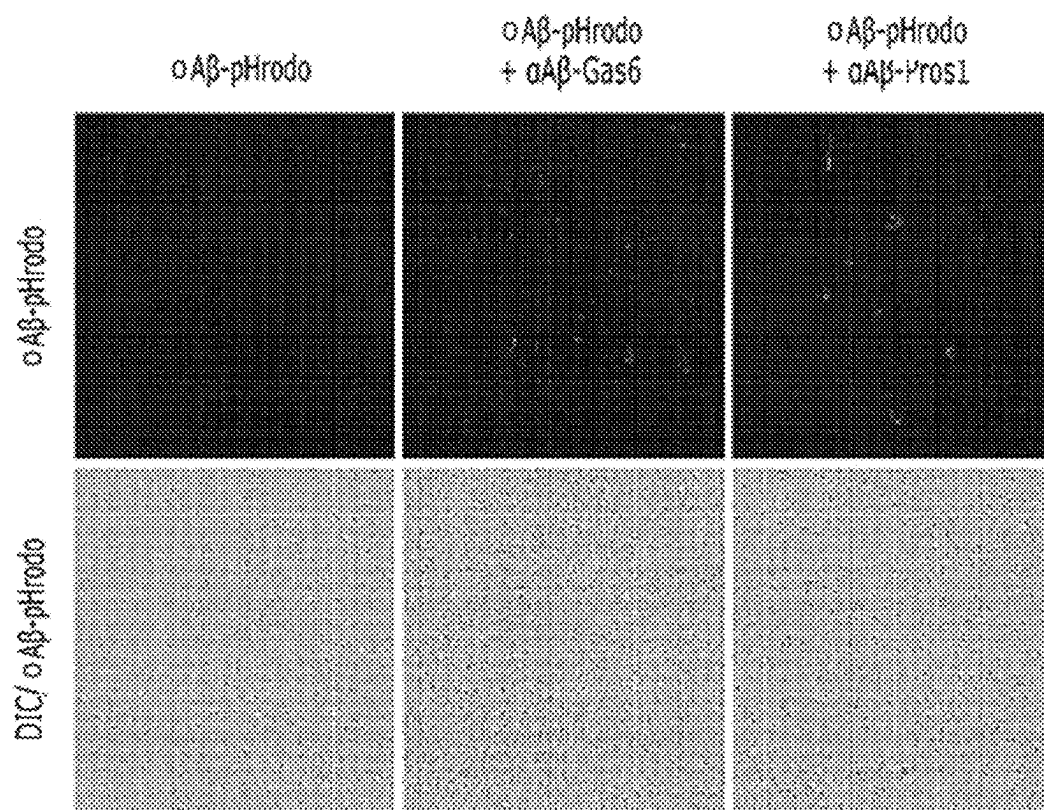
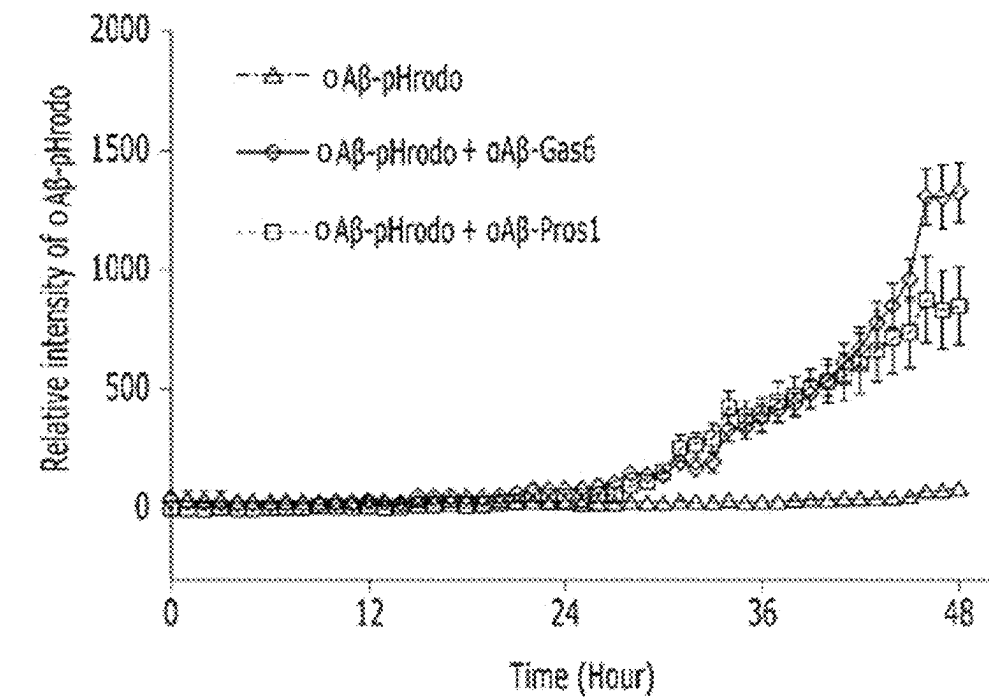


FIG. 30

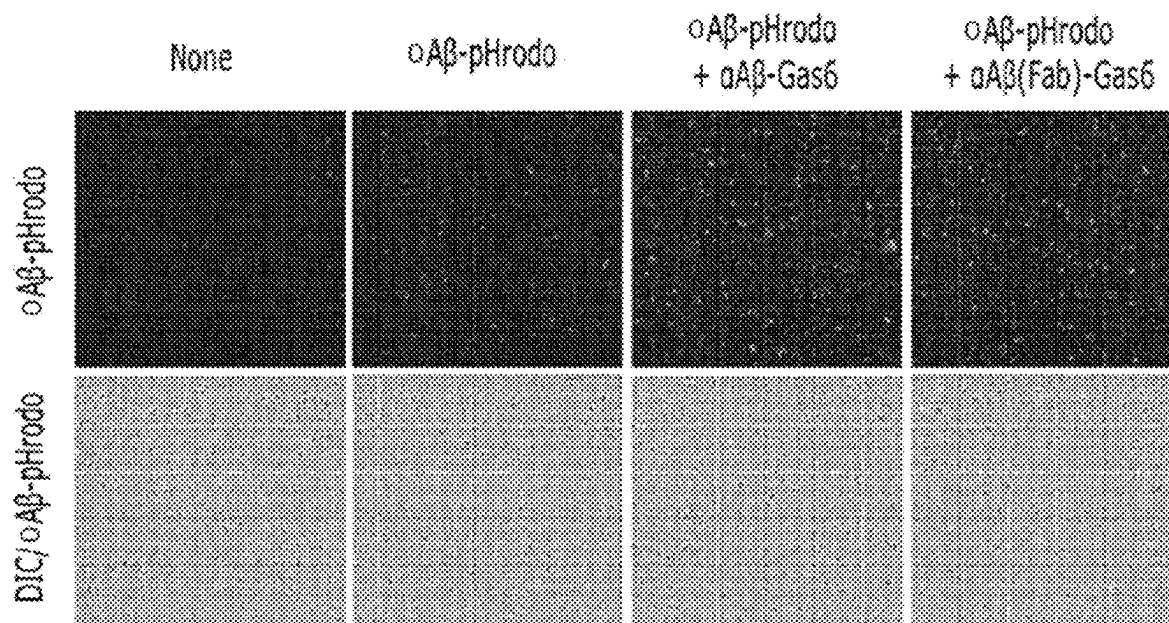
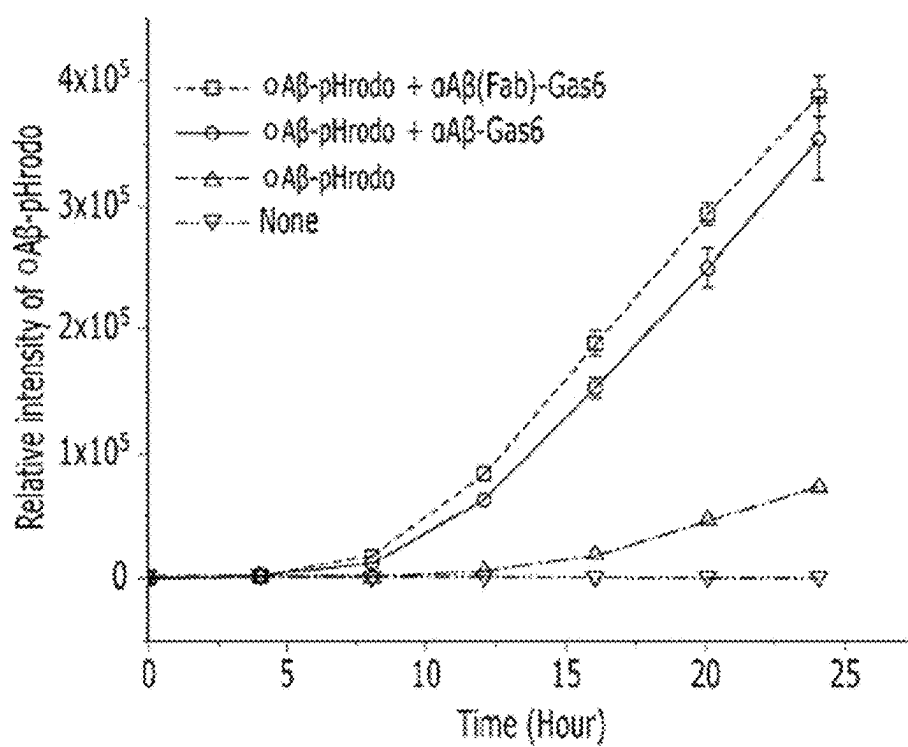
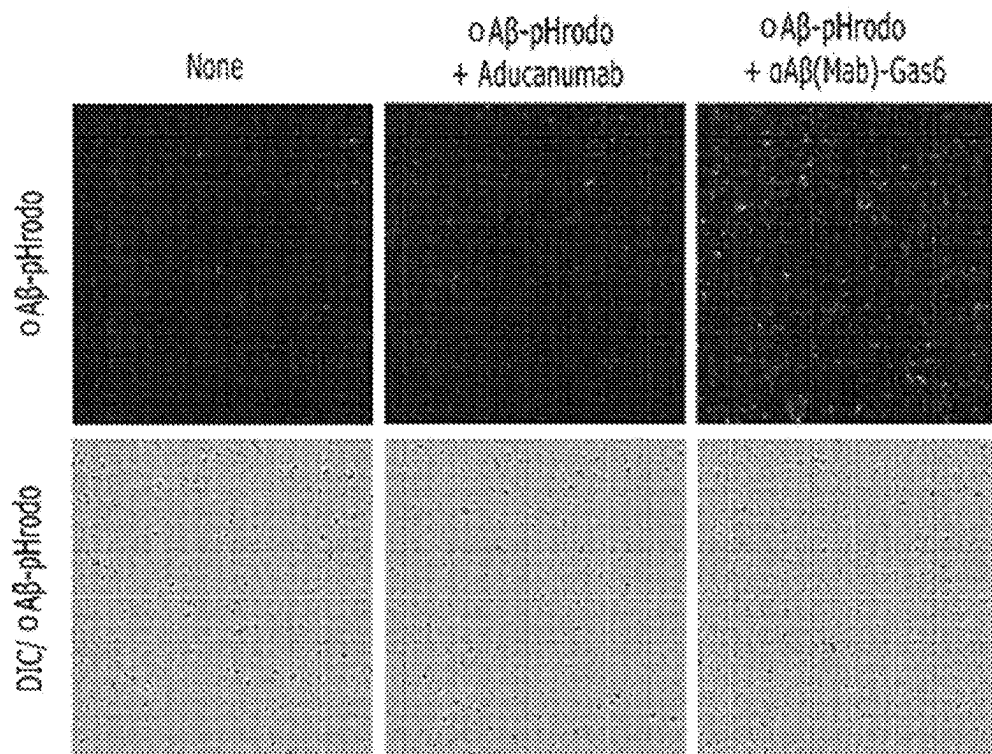
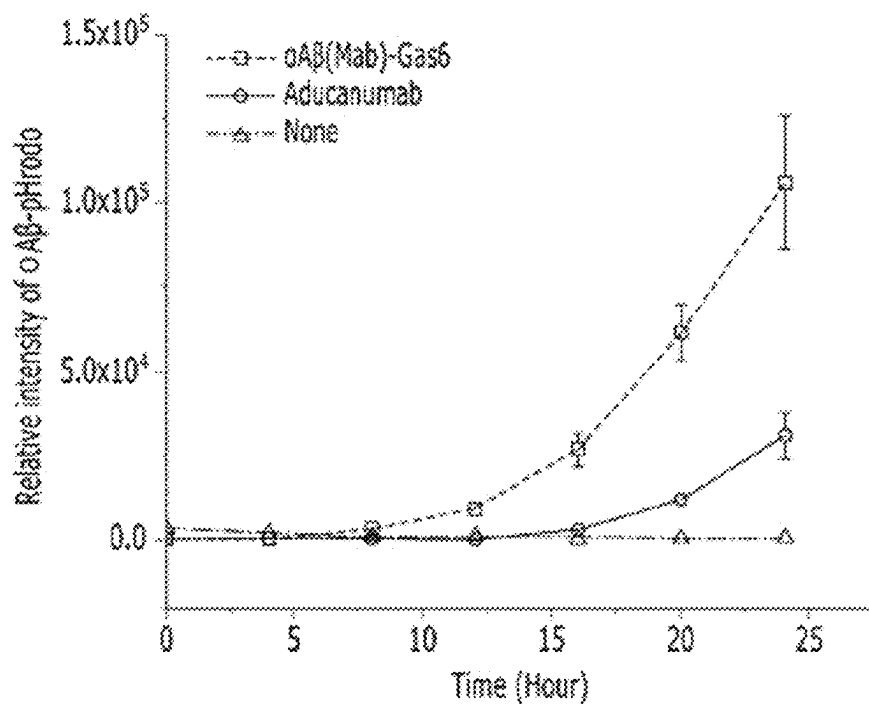


FIG. 31



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**TAM RECEPTOR-BINDING FUSION
MOLECULE HAVING
NON-INFLAMMATORY PHAGOCYTOSIS
INDUCING ACTIVITY**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation-in-part of and claims the benefit of PCT Application No. PCT/KR2022/001671 filed Jan. 28, 2022, which claims priority based on Korean Patent Application No. 10-2021-0013056 filed Jan. 29, 2021, of which the entire contents are incorporated by reference herein.

**INCORPORATION BY REFERENCE OF
SEQUENCE LISTING**

The content of the electronically submitted sequence listing, file name: Q289305_SEQ_LIST_ST26.xml; size: 268,800 bytes; and date of creation: Jul. 16, 2023, filed herewith, is incorporated herein by reference in its entirety.

FILED

The present disclosure relates to fusion molecules that are capable of inducing phagocytosis without inducing inflammatory reaction, their uses, and manufacturing method. The fusion molecules are useful for prevention or treatment of diseases that are caused by or characterized by abnormal accumulation of substances in the body, such as proteopathy. The present disclosure also relates to nucleic acid molecules encoding the fusion molecules. The present disclosure further relates to methods of suppressing abnormal accumulation of substances, promoting clearance of aggregates of substances, and/or treating disorders or diseases that are caused by or characterized by abnormal accumulation of substances, without inducing inflammatory reaction.

BACKGROUND

Numerous degenerative diseases are characterized by aberrant folding, polymerization and accumulation of proteins. These proteopathies include various types of amyloidosis.

Amyloidosis is a disease in which abnormal proteins called amyloid accumulate in tissues. Amyloid is a protein aggregate that has a diameter of 7-13 nm and a beta-sheet structure and exhibits a fibrous morphology when viewed under a microscope, and it is characterized by being stained with Thioflavin T (ThioT) and Congo red. Amyloid is not normally found in the body, and to date, 36 proteins have been identified as being amyloidogenic (Picken, *Acta Haematol.* (2020), 143:322-334). Representative examples of amyloidosis diseases include neurological diseases such as Alzheimer's disease, Parkinson's disease, Huntington's disease and prion disease. In addition, there are a number of amyloid diseases having various aspects depending on amyloid-causing proteins and affected organs.

Alzheimer's disease is the biggest cause of dementia and is a fatal disease accompanied by learning and memory impairment. 130 million people are expected to suffer from Alzheimer's disease by 2050 worldwide, and 1 in 9 people among the population above 65 years old have already been diagnosed with Alzheimer's disease.

A hallmark of Alzheimer's disease is that beta-amyloid (A β) protein caused by abnormal degradation of amyloid

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precursor protein (APP), deposits and accumulates around the brain cell membrane. Another hallmark is abnormal hyperphosphorylation of microtubule-associated tau protein.

It has recently been reported that beta-amyloid oligomers and fibrils cause synaptic dysfunction and cytotoxicity through various pathways, and create a vicious cycle that adversely affects nerve cells through functional changes in astrocytes and microglia, which are responsible for immunity in the brain.

Therapeutic drugs for Alzheimer's disease approved by the FDA to date inhibit acetylcholine degradation or inhibit the activity of NMDA receptors, and, thus, provide temporary relief of symptoms, but do not treat the underlying cause or the disease itself. Therefore, there is a need for the development of new treatments for treating Alzheimer's disease and other diseases characterized by aberrant accumulation or deposit of beta-amyloid.

For a therapeutic treatment of Alzheimer's disease, drug development has been conducted for decades with a focus on inhibiting formation of and eliminating beta-amyloid. Unfortunately, however, most of the therapeutic drugs for Alzheimer's disease developed to inhibit formation of and eliminate beta-amyloid failed during the clinical stage due to ineffectiveness or insufficient efficacy. For example, in the case of BACE (β -site amyloid precursor protein cleaving enzyme) inhibitors for reducing beta-amyloid, strategies that prevent additional beta-amyloid production are largely ineffective, because in Alzheimer's patients with cognitive decline, beta-amyloid plaques have already accumulated and neuronal cell death is taking place.

Since the recent studies reporting that monoclonal antibodies which specifically bind to beta-amyloid oligomers and fibrils induced beta-amyloid clearance and restored cognitive function in Alzheimer's disease patients, a strategy to treat Alzheimer's disease through anti-beta-amyloid antibodies has emerged as a new hope.

The mechanisms of action of beta-amyloid monoclonal antibodies proposed to date include inhibition of aggregation of beta-amyloid oligomers and fibrils by binding thereto, or the induction of microglial phagocytosis of beta-amyloid through Fc receptors that recognize the monoclonal antibodies.

However, despite the advances in the development of therapeutic drugs for Alzheimer's disease, current immunotherapy using anti-beta-amyloid monoclonal antibodies shows amyloid-related imaging abnormalities (ARIA) accompanied by severe edema in 55% of patients treated with the antibodies, and for this reason, about 35% of the ARIA patients were dropped from clinical trials. The ARIA phenomenon is known to be due to synaptotoxicity and cytotoxicity caused by inflammatory responses that are inevitably activated when anti-beta-amyloid monoclonal antibodies stimulate Fc receptors of microglia cells.

Since synapses and neurons in the brain respond sensitively to inflammatory cytokines, treatment using anti-beta-amyloid monoclonal antibodies has an inherent problem in that it inevitably causes damages to neurons and synapses, even if it clears beta-amyloid to some extent. In addition to monoclonal antibodies, companies such as Alector and Denali presented strategies to improve the microglia's ability to clear beta-amyloid by activating targets such as TREM2 that regulate the immunological mechanism of microglia, and these strategies have received a lot of attention. However, even in these strategies, when microglia are excessively activated, synaptic damage due to an increase in overall phagocytotic capacity is expected.

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Therefore, an important task in the treatment of Alzheimer's disease is to develop therapeutic modalities to selectively clear only beta-amyloid oligomers and fibrils without causing inflammatory responses and synaptic damage, and these drugs are expected to make a significant contribution to the treatment of Alzheimer's disease.

Furthermore, there is a need to selectively clear only a substance of which abnormal accumulation causes disorders or accumulates thereof as a target, for example, abnormally accumulated proteins causing proteopathy, without causing inflammatory responses and consequent additional tissue damage reported in conventional experimental drugs. The present disclosure meets this need by providing therapeutic modalities for selectively clearing abnormally accumulated proteins that cause or characterize certain disease.

SUMMARY

The present disclosure relates to fusion molecules having phagocytosis-inducing activity without inducing inflammatory responses. One aspect of the present disclosure provides a fusion molecule having phagocytosis-inducing activity, the fusion molecule comprising: a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor; and a second region that specifically binds to a target substance to be cleared or decreased, and the fusion molecule does not induce inflammatory responses. In embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses.

In some embodiments, the TAM receptor may be any one selected from the group consisting of Tyro3, Axl, MerTK, or a combination thereof, which are capable of inducing phagocytosis by binding to a laminin G-like domain (or LG domain) of a phagocytic cell including, but not limited to, macrophages or microglial cells. In embodiments, the TAM receptor may be Axl receptor.

In embodiments, the first region may comprise Gas6, ProS1, Tubby, Tulp1, Gal3, or an active fragment thereof, which each is capable of specifically binding to a TAM receptor. The first region may be selected from Gas6, ProS1, or an active fragment thereof, which each is capable of specifically binding to a TAM receptor. In embodiments, the first region may comprise or consist essentially of Gas6 or an active fragment thereof that is capable of binding to TAM receptor. In embodiments, the first region comprising or consisting essentially of Gas6 or an active fragment thereof is capable of binding to Axl receptor.

In certain embodiments, the first region may comprise a laminin G-like domain of Gas6 or ProS1, or an active fragment thereof, which contains a laminin G-like domain as a phagocytosis-related bridging molecule which is abundantly expressed in various tissue, and thus is able to induce phagocytosis through a TAM receptor. In embodiments, the laminin G-like domain may comprise an LG1 domain, an LG2 domain, or a combination thereof, and may preferably include both an LG1 domain and an LG2 domain, which are able to induce phagocytosis by binding to the TAM receptor.

Exemplary embodiments are directed to a binding molecule or fusion molecule comprising a first region capable of binding to a TAM receptor and a second region capable of specifically binding to a target substance, said target substance being a substance of which aberrant accumulation in a living tissue is characteristic of or associated with a disease, wherein the first region and the second region are coupled to each other directly or via a linker,

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wherein the first region comprises

- (a) a TAM receptor ligand;
- (b) an anti-Axl antibody or an antigen-binding fragment thereof;
- (c) an anti-Tyro3 antibody or an antigen-binding fragment thereof; or
- (d) an anti-MerTK antibody or an antigen-binding fragment thereof, with proviso that when the first region comprises an anti-MerTK antibody or an antigen-binding fragment thereof, the molecule is not a bispecific antibody; or
- (e) combinations thereof.

According to some embodiments, the binding molecule may further comprise a scaffold bound to the first region, to the second, or to both of the first region and the second region at different positions.

In embodiments, the first region is a TAM receptor ligand and the TAM receptor ligand comprises a sequence selected from the group consisting of SEQ ID NOS: 1-113 or a sequence having at least 85% of sequence identity thereto.

In still some embodiments, the first region is capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises one or more sequences selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 5, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30, SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 63, SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67, SEQ ID NO: 68, SEQ ID NO: 69, SEQ ID NO: 70, SEQ ID NO: 71, SEQ ID NO: 72, SEQ ID NO: 73, SEQ ID NO: 74, SEQ ID NO: 75, SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, SEQ ID NO: 80, SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, and SEQ ID NO: 87, or a sequence having at least 85% of sequence identity thereto.

In still some embodiments, the first region is capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises the sequence of SEQ ID NO: 1 or a sequence having at least 85% of sequence identity thereto, and/or the sequence of SEQ ID NO: 2 or a sequence having at least 85% of sequence identity thereto.

In still another embodiment, the first region capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises the sequence of SEQ ID NO: 5 or a sequence having at least 85% of sequence identity thereto.

In still some embodiments, the first region is capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises one or more sequences selected from the group consisting of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 6, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 41, SEQ ID NO: 42, SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47, SEQ ID NO: 48, SEQ ID NO: 49, SEQ ID NO: 50, SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54, SEQ ID NO: 55, SEQ ID NO: 56, SEQ ID NO: 57, SEQ ID NO: 58, SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61, SEQ ID NO: 62, SEQ ID NO: 88, SEQ ID NO: 89, SEQ ID NO: 90, SEQ ID NO: 91, SEQ ID NO: 92, SEQ ID NO: 93, SEQ ID NO: 94, SEQ ID NO: 95, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 98, SEQ ID NO: 99, SEQ ID NO:

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100, SEQ ID NO: 101, SEQ ID NO: 102, SEQ ID NO: 103, SEQ ID NO: 104, SEQ ID NO: 105, SEQ ID NO: 106, SEQ ID NO: 107, SEQ ID NO: 108, SEQ ID NO: 109, SEQ ID NO: 110, SEQ ID NO: 111, SEQ ID NO: 112, and SEQ ID NO: 113, or a sequence having at least 85% of sequence identity thereto.

In still some embodiments, the first region is capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises the sequence of SEQ ID NO: 3 or a sequence having at least 85% of sequence identity thereto, and/or the sequence of SEQ ID NO: 4 or a sequence having at least 85% of sequence identity thereto.

In still another embodiment, the first region capable of binding to an Axl receptor the first region capable of binding to an Axl receptor comprises the sequence of SEQ ID NO: 6 or a sequence having at least 85% of sequence identity thereto.

In embodiments, the fusion molecule (or binding molecule) may comprise the first region comprising the sequence of SEQ ID NO: 1, SEQ ID NO: 2, or a combination thereof, or a sequence having at least 85% sequence identity thereto. In an embodiment, the combination of the sequence of SEQ ID NO: 1 and SEQ ID NO: 2 may comprise the sequence of SEQ ID NO: 5 or a sequence having at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% to SEQ ID NO: 5.

In embodiments, the fusion molecule (or binding molecule) may comprise the first region comprising the sequence of SEQ ID NO: 3, SEQ ID NO: 4, or a combination thereof, or a sequence having at least 85% sequence identity thereto. In an embodiment, the combination of the sequence of SEQ ID NO: 3 and SEQ ID NO: 4 may comprise the sequence of SEQ ID NO: 6 or a sequence having at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% to SEQ ID NO: 6.

In embodiments, the fusion molecule (or binding molecule) may comprise the sequence of amino acid residues 31-871 of SEQ ID NO: 136, amino acid residues 31-687 of SEQ ID NO: 138, amino acid residues 31-697 of SEQ ID NO: 140, amino acid residues 31-684 of SEQ ID NO: 150, amino acid residues 31-676 of SEQ ID NO: 152, amino acid residues 25-673 of SEQ ID NO: 154, amino acid residues 22-662 of SEQ ID NO: 156, or amino acid residues 22-885 of SEQ ID NO: 158, amino acid residues 22-908 of SEQ ID NO: 162, amino acid residues 22-919 of SEQ ID NO: 163, amino acid residues 22-919 of SEQ ID NO: 165, or a sequence having at least 90% sequence identity thereto (wherein a different linker can be used in place the linkers as shown in Tables 3 and 5-10.

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In embodiments, the first region comprising a laminin G-like domain of Gas6 or ProS1, or an active fragment thereof, does not comprise a Gla domain. Without being bound to a particular theory, it is expected that the lacking of Gla domain in the first region may make the fusion molecule not be able to recognize phosphatidylserine (PS) of TAM receptor, while the second region is able to induce phagocytosis by recognizing a target substance.

In some embodiments, the first region comprising a laminin G-like domain of Gas6 or ProS1, or an active fragment thereof, does not comprise a Gla domain and does not comprise an EGF domain. The lacking of EGF domain in the first region provides an advantage in the manufacturing process of the fusion molecule to increase the yield by suppressing an aggregation of the fusion molecule during the purification step. In some embodiments, the fusion molecule (or binding molecule) may form a homomultimer or a heteromultimer, or form a linear multimer as a single chain.

According to embodiments, the target substance that is to be cleared or decreased and to which the second region specifically binds, may be a substance that accumulates in living tissue, causing a disease. For example, it may be a substance accumulated in an affected (i.e., diseased) tissue of a patient or circulating in blood of a patient. The substance may be protein. That is, the disease may be, but not limited thereto, proteopathy. In certain embodiments, the target substance may be amyloid. That is, the proteopathy may be amyloidosis. The target substance may be one or more of the amyloid substances listed in Table 1 below or APOE or apoptosis-associated spec-like protein containing a caspase activating recruitment domain (ASC-speck), of which abnormal accumulation or deposit is associated with or characteristic of a disease, and in this case, the disease may be a disease in which each abnormally accumulated substance is detected. For example, the proteopathy may be Alzheimer's disease, Parkinson's disease, Huntington's disease, and Prion disease, and in this case, target substances may be β -amyloid, tau, α -synuclein, huntingtin, and prion proteins, respectively, which are identified as proteins of which abnormal accumulation causes the diseases. Aberrant accumulation of APOE is associated with Alzheimer's disease, cerebral amyloid angiopathy, and/or cardiovascular disease. Aberrant accumulation of apoptosis-associated spec-like protein containing a caspase activating recruitment domain (ASC-speck) is associated with Alzheimer's Disease, Parkinson's Disease, Huntington's disease, Multiple System Atrophy, Amyotrophic Lateral Sclerosis, Spinocerebellar ataxia, Frontotemporal Dementia, Frontotemporal Lobar Degeneration, Mild Cognitive Impairment, Parkinson-plus syndromes, Pick disease, Progressive isolated aphasia, Grey-matter degeneration [Alpers], Subacute necrotizing encephalopathy, and Lewy body dementia.

TABLE 1

Target Substance	Abbreviation	Diseases associated with or characterized by aberrant accumulation of target substance
β -Amyloid	A β	Alzheimer's disease, Hereditary cerebral haemorrhage with amyloidosis, etc.
Amyloid precursor protein-derived β -amyloid	A β	Alzheimer's disease, Hereditary cerebral haemorrhage with amyloidosis, etc.
α -Synuclein	α -Syn	Parkinson's disease, Parkinson's dementia, dementia with Lewy bodies, multiple system atrophy, etc.

TABLE 1-continued

Target Substance	Abbreviation	Diseases associated with or characterized by aberrant accumulation of target substance
Prp ^{Sc}	PrP	Transmissible spongiform encephalopathy (fatal familial insomnia, Gerstmann-Straussler-Scheinker disease, Creutzfeldt-Jacob disease, new variant Creutzfeldt-Jacob disease, etc.), etc.
Microtubule-associated protein tau	Tau	Tauopathies (Pick's disease, progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia with parkinsonism linked to chromosome 17, argyrophilic grain disease, etc.), Alzheimer's disease, Parkinson's disease, etc.
Huntingtin exon 1	HTT exon 1	Huntington's disease, etc.
TAR DNA-binding protein 43	TDP43	Frontotemporal dementia, amyotrophic lateral sclerosis (ALS), etc.
Superoxide dismutase 1	SOD1	Amyotrophic lateral sclerosis (ALS), etc.
ABri peptide	bri	Familial British dementia
ADan peptide	dan	Familial Danish dementia
Immunoglobulin light-chain fragment	AL	Light-chain amyloidosis
Immunoglobulin heavy-chain fragment	AH	Heavy-chain amyloidosis
N-terminal fragment of serum amyloid A protein	AA	AA amyloidosis
Transthyretin	ATTR	Senile systemic amyloidosis, familial amyloid polyneuropathy, familial amyloid cardiomyopathy, leptomeningeal amyloidosis
β -2 microglobulin	A β 2M	Dialysis-related amyloidosis, hereditary visceral amyloidosis
N-terminal fragment of apolipoprotein AI	AApoAI	ApoAI amyloidosis
C-terminally extended apolipoprotein AII	AApoAII	ApoAII amyloidosis
N-terminal fragment of apolipoprotein AIV	AApoAIV	ApoAIV amyloidosis
apolipoprotein C-II	AApoCII	ApoCII amyloidosis
apolipoprotein C-III	AApoCIII	ApoCIII amyloidosis
Gelsolin fragment	AGel	Familial amyloidosis, Finnish type, Hereditary gelsolin amyloidosis
Lysozyme	ALys	Hereditary non-neuropathic systemic amyloidosis
Fibrinogen alpha chain fragment	AFib	Fibrinogen amyloidosis
N-terminally truncated cystatin C	ACys	Hereditary cerebral hemorrhage with amyloidosis, Icelandic type
Amylin, IAPP	IAPP	Diabetes mellitus type 2, insulinoma
Calcitonin	Cal	Medullary carcinoma of the thyroid
Atrial natriuretic factor	AANF	Cardiac arrhythmias, isolated atrial amyloidosis
Prolactin	PRL	Pituitary prolactinoma
Insulin	AIIns	Localized amyloidosis at insulin injection sites
Lactadherin or medin	AMed	Aortic medial amyloidosis
Lactotransferrin or lactoferrin	LTF	Gelatinous drop-like corneal dystrophy
Odontogenic ameloblast-associated protein	ODAM	Calcifying epithelial odontogenic tumors
pulmonary surfactant-associated protein C	SPC	Pulmonary alveolar proteinosis
Leukocyte cell-derived chemotaxin-2	ALECT2	Renal LECT2 amyloidosis
Galectin-7	Agal7	Lichen amyloidosis, macular amyloidosis
Corneodesmosin	Cor	Hypotrichosis simplex of the scalp
C-terminal fragment of TGFBI (or keratoepithelin)	Ker	Lattice corneal dystrophy; type I, 3A or Avellino
SGI (Semenogelin-1)	ASem1	Seminal vesicle amyloidosis
S100 protein (A8 or A9)	(no abbreviation)	Prostate cancer
Enfuvirtide	AEnf	Injection-localized amyloidosis
Apolipoprotein E	APOE	Alzheimer's disease, cerebral amyloid angiopathy, cardiovascular disease

TABLE 1-continued

Target Substance	Abbreviation	Diseases associated with or characterized by aberrant accumulation of target substance
Apoptosis-associated Spec-like protein containing a Caspase Activating Recruitment Domain	ASC	Alzheimer's Disease, Parkinson's Disease, Huntington's disease, Multiple System Atrophy, Amyotrophic Lateral Sclerosis, Spinocerebellar ataxia, Frontotemporal Dementia, Frontotemporal Lobar Degeneration, Mild Cognitive Impairment, Parkinson-plus syndromes, Pick disease, Progressive isolated aphasia, Grey-matter degeneration [Alpers], Subacute necrotizing encephalopathy, and Lewy body dementia

In embodiments, the present disclosure is directed to a nucleic acid or polynucleotide encoding the fusion proteins described above.

In embodiments, the present disclosure is directed to a vector containing the nucleic acid or polynucleotide.

Embodiments are directed to a host cell containing the vector.

Another aspect of the present disclosure provides a method of producing a therapeutic fusion molecule for treatment of a disease or disorder associated with or characterized by aberrant accumulation of substance in a subject, comprising expressing the fusion molecule by culturing a host cell under a condition for expressing the fusion molecule.

In embodiments, the present disclosure is directed to a method of reducing or enhancing a reduction of aberrant deposit of substance that causes or characterizes certain disorder or diseases in a subject, which method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject; and a second region that specifically binds to the substance. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fcγ receptor).

In embodiments, the present disclosure is directed to a method of removing or clearing or enhancing clearance of aberrant deposit of substance that causes or characterizes certain disorder or diseases in a subject, which method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject; and a second region that specifically binds to the substance. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fcγ receptor).

In embodiments, the present disclosure is directed to a method of suppressing formation of aberrant accumulations of substance in a subject. The method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject and a second region that specifically binds to the substance. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fcγ receptor).

In embodiments, the present disclosure is directed to a method of treating or preventing a disorder or disease in a subject, wherein the disorder or disease is characterized by or caused by aberrant accumulation of substance. The method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject; and a second region that specifically binds to the substance. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fcγ receptor).

In embodiments, the present disclosure is directed to a method of delaying development of a symptom associated with a disease that is characterized by, associated with, or caused by aberrant accumulation of substance, in a subject. The method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, a vector comprising the polynucleotide, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject; and a second region that specifically binds to the substance. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated

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inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fc γ receptor).

In embodiments, the present disclosure provides a method of reducing a substance of which aberrant accumulation is associated with or characteristic of a disease or disorder, in a subject. The method comprises administering to the subject an effective amount of a fusion molecule or a polynucleotide encoding the fusion molecule, wherein the fusion molecule comprises a first region that is capable of binding to a TAM (Tyro3, Axl and MerTK) receptor on surface of a cell in the subject, and a second region that specifically binds to the substance. The substance may be soluble, oligomeric, or aggregated form. In some embodiments, the toxic effects of accumulated substance are inhibited and/or reduced. Thus, the method of the disclosure can be used to treat any disease in which accumulation of a substance is present or suspected. In non-limiting embodiments, the substance and the disease or disorder may be one or more of those listed in Table 1. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fc γ receptor).

In the above methods, according to embodiments thereof, the aberrant deposits of substance are in the brain (brain tissue) of the subject. In some embodiments, the aberrant deposits of substance are in the cerebral vasculature. In some embodiments, the aberrant accumulation of substance is in the circulatory system. In some embodiments, the aberrant accumulation of substance is in various tissues such as heart, kidney, liver, and the like.

In embodiments, the present disclosure is directed to a pharmaceutical composition comprising an effective amount of any of the above-disclosed fusion molecule or polynucleotides encoding the fusion molecule, and a pharmaceutical acceptable excipient. In non-limiting embodiments, the fusion molecule does not have an effector function and does not induce Fc-mediated inflammatory responses. For example, the fusion molecule does not comprise a moiety to bind to an Fc receptor, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fc γ receptor).

In embodiments, the present disclosure is directed to kits comprising an effective amount of any of the above-disclosed fusion molecule or polynucleotides encoding the fusion molecule. The kits are generally in suitable packaging and provided with appropriate instructions, are useful for any of the methods described herein.

These and other aspects, objects, features and advantages of the example embodiments will become apparent to those having ordinary skill in the art upon consideration of the following detailed description of illustrated example embodiments.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1A schematically illustrates the structure of TAM receptors and TAM ligands. In the structure of TAM receptors: the N-terminal starts with 2 Ig-like domains, followed by 2 fibronectin type 3 domains, followed by a single-pass transmembrane domain and a protein tyrosine kinase at the C-terminal. In the structure of the TAM ligands protein S (Pros1) and Gas6, the N-terminal contains a GLA domain,

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followed by a thrombin-sensitive region (TSR), followed by 4 EGF-like domains, followed by a C-terminal (SHBG-like domain, consisting of 2 LG repeats. In FIG. 1A, EGF stands for epidermal growth factor, Ig stands for immunoglobulin, LG stands for laminin G, and SHBG stands for sex hormone-binding globulin.

FIG. 1B schematically shows beta-amyloid- and FITC-binding phagocytosis inducing fusion molecule comprising Gas6 as a non-limiting example of first region of the fusion molecule.

FIG. 1C through FIG. 1M schematically show non-limiting exemplary embodiments of the structure of the fusion molecules.

FIG. 2 shows the results of Western blot analysis of a non-limiting exemplary chimeric phagocytosis inducer comprising a FLAG Tag, produced according to Preparation Example 1.

FIG. 3 schematically shows an action of a non-limiting exemplary chimeric phagocytosis inducer, produced according to Preparation Example 1, on TAM receptor.

FIG. 4 shows the evaluation results for selective beta-amyloid clearing ability of α A β -Gas6.

FIG. 5 shows the evaluation results for beta-amyloid clearing ability of α A β -Gas6 in the HMC3 cell line by beta-amyloid engulfment assay in vitro.

FIG. 6 shows the evaluation results for beta-amyloid clearing ability of α A β -Gas6 in the HMC3 cell line by beta-amyloid engulfment assay in vitro.

FIG. 7 shows results indicating that the beta-amyloid clearing ability of α A β -Gas6 is associated with or dependent on Gas6 binding to Axl among TAM receptors.

FIG. 8 shows results indicating that the beta-amyloid clearing ability of α A β -Gas6 is associated with or dependent on Gas6 binding to Axl among TAM receptors.

FIG. 9 shows results indicating that the beta-amyloid clearing ability of α A β -Gas6 is associated with or dependent on Gas6 binding to Axl among TAM receptors.

FIG. 10 shows the results of comparative analysis of the activation of inflammatory response signaling by α A β -Gas6 and aducanumab using THP-Axl cells.

FIG. 11 shows the results of comparative analysis of the levels of pro-inflammatory cytokine secretion by α A β -Gas6 and aducanumab using THP-Axl cells.

FIG. 12 shows the evaluation results for anti-inflammatory activity of α A β -Gas6.

FIG. 13 shows results indicating that the beta-amyloid clearing ability of microglia was significantly increased by α A β -Gas6.

FIG. 14 shows results indicating that the beta-amyloid clearing ability of astrocytes was significantly increased by α A β -Gas6.

FIG. 15 shows results indicating that the transcriptional levels of pro-inflammatory cytokines in astrocytes were changed by α A β -Gas6 and aducanumab.

FIG. 16 shows results indicating that the transcriptional levels of pro-inflammatory cytokines in BV2 were changed by α A β -Gas6 and aducanumab.

FIG. 17 shows the evaluation results for beta-amyloid plaque clearing ability of α A β -Gas6 through administration of α A β -Gas6 protein in 5xFAD Alzheimer's disease model mice.

FIG. 18 shows the evaluation results for beta-amyloid plaque clearing ability of α A β -Gas6 through administration of α A β -Gas6 virus in 5xFAD Alzheimer's disease model mice.

FIG. 19 shows results indicating that beta-amyloid contained in lysosomes were increased by microglia-mediated

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clearance in 5xFAD Alzheimer's disease model mice upon administration of α A β -Gas6 protein.

FIG. 20 shows results indicating that beta-amyloid contained in lysosomes were increased by astrocyte-mediated clearance in 5xFAD Alzheimer's disease model mice upon administration of α A β -Gas6 protein.

FIG. 21 shows results indicating that beta-amyloid contained in lysosomes were increased by microglia-mediated clearance in 5xFAD Alzheimer's disease model mice upon administration of α A β -Gas6 virus.

FIG. 22 shows results indicating that beta-amyloid contained in lysosomes were increased by astrocyte-mediated clearance in 5xFAD Alzheimer's disease model mice upon administration of α A β -Gas6 virus.

FIG. 23 shows results indicating that microglia-mediated synapse engulfment that abnormally increased in 5xFAD Alzheimer's disease model mice due to the side effect of aducanumab was significantly restored upon administration of α A β -Gas6 virus.

FIG. 24 shows results indicating that microglia-mediated synapse engulfment that abnormally increased in 5xFAD Alzheimer's disease model mice due to the side effect of aducanumab was significantly restored upon administration of α A β -Gas6 virus.

FIG. 25 shows an experimental protocol for evaluating cognitive and memory abilities in 5xFAD Alzheimer's disease model mice upon administration of α A β -Gas6 virus.

FIG. 26 shows results indicating that cognitive and memory abilities in 5xFAD Alzheimer's disease model mice were more restored upon administration of α A β -Gas6 virus than administration of aducanumab.

FIG. 27 shows the evaluation results for tau clearing ability of α Tau-Gas6 in the HMC3 cell line by in vitro tau engulfment assay.

FIG. 28 shows the evaluation results for alpha-synuclein clearing ability of α Tau-Gas6 in the HMC3 cell line by in vitro tau engulfment assay.

FIG. 29 shows the evaluation results for beta-amyloid clearing ability of α A β -ProS1 in primary-cultured astrocytes by in vitro tau engulfment assay.

FIG. 30 shows the evaluation results for beta-amyloid clearing ability of α A β (Fab)-Gas6 in the HMC3 cell line by in vitro tau engulfment assay.

FIG. 31 shows the evaluation results for beta-amyloid clearing ability of α A β (Fab)-Gas6 in the HMC3 cell line by in vitro tau engulfment assay.

DETAILED DESCRIPTION

Methods and compositions are provided for reducing or suppressing formation of or clearing a target substance of which accumulation is associated with or characteristic of a disorder or disease via a phagocytosis, preventing or treating an individual having a disease or disorder characterized by an aberrant accumulation of a substance, improving symptoms of a disease or disorder characterized by an aberrant accumulation of a substance, and/or a target substance of which accumulation is associated with or characteristic of a disorder or disease via a phagocytosis.

Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Each smaller range between any stated value or intervening value in a stated range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller

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ranges may independently be included or excluded in the range, and each range where either, neither or both limits are included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

Definitions

As used herein, the singular forms "a," "an," and "the," refer to both the singular as well as plural, unless the context clearly indicates otherwise. Therefore, for example, reference to "a cell" includes a plurality of such cells and reference to "the peptide" includes reference to one or more peptides and equivalents thereof, e.g. polypeptides, known to those skilled in the art, and so forth.

As used herein, the terms "about" and "consisting essentially of" refers to a value or composition that is within an acceptable error range for the particular value or composition as determined by one of ordinary skill in the art, which will depend in part on how the value or composition is measured or determined, i.e., the limitations of the measurement system. For example, "about" or "consisting essentially of" can mean within 1 or more than 1 standard deviation per the practice in the art. Alternatively, "about" or "consisting essentially of" can mean a range of up to 10% (i.e., $\pm 10\%$). For example, "about 5 mg" can include any number between 4.5 mg and 5.5 mg (for 10%), between 4.75 mg and 6.25 mg (for 5%), between 4.8 mg and 6.2 mg (for 4%), between 4.85 mg and 6.15 mg (for 3%), between 4.9 mg and 6.1 mg (for 2%), or between 4.95 mg and 6.05 mg (for 1%). Furthermore, particularly with respect to biological systems or processes, the terms can mean up to an order of magnitude or up to 5-fold of a value. When particular values or compositions are provided in the application and claims, unless otherwise stated, the meaning of "about" or "consisting essentially of" should be assumed to be within an acceptable error range for that particular value or composition.

As used herein, "administration" or "administering" refers to the introduction of a composition into a subject by a chosen route. For example, if the chosen route is intravenous, the composition is administered by introducing the composition into a vein of the subject. In some examples, the peptides and antibodies disclosed herein are administered to a subject.

As used herein, "amino acid" refers to naturally occurring and synthetic amino acids, as well as amino acid analogs and amino acid mimetics that function in a manner similar to the naturally occurring amino acids. Naturally occurring amino acids are those encoded by the genetic code, as well as those amino acids that are later modified, e.g., hydroxyproline, gamma-carboxyglutamate, and O-phosphoserine. Amino acid analogs refers to compounds that have the same basic chemical structure as a naturally occurring amino acid, i.e., an α -carbon that is bound to a hydrogen, a carboxyl group, an amino group, and an R group, e.g., homoserine, norleucine, methionine sulfoxide, methionine methyl sulfonium. Such analogs have modified R groups (e.g., norleucine) or modified peptide backbones, but retain the same basic chemical structure as a naturally occurring amino acid. Amino acid mimetics refers to chemical compounds that have a structure that is different from the general chemical structure of an amino acid, but that functions in a manner similar to a naturally occurring amino acid.

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As used herein, "polypeptide," "oligopeptide," "peptide" and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. The terms also apply to amino acid polymers in which one or more amino acid residue is an artificial chemical mimetic of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers and non-naturally occurring amino acid polymer. The terms also encompass an amino acid polymer that has been modified naturally or by intervention; for example, disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation or modification, such as conjugation with a labeling component. Also included within the definition are, for example, polypeptides containing one or more analogs of an amino acid (including, for example, unnatural amino acids, etc.), as well as other modifications known in the art. It is understood that, because the polypeptides of this invention are based upon an antibody, the polypeptides can occur as single chains or associated chains.

As used herein, "polynucleotide," or "nucleic acid," as used interchangeably herein, refer to polymers of nucleotides of any length, and include DNA and RNA. The nucleotides can be deoxyribonucleotides, ribonucleotides, modified nucleotides or bases, and/or their analogs, or any substrate that can be incorporated into a polymer by DNA or RNA polymerase. A polynucleotide may comprise modified nucleotides, such as methylated nucleotides and their analogs. If present, modification to the nucleotide structure may be imparted before or after assembly of the polymer. The sequence of nucleotides may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component. Other types of modifications include, for example, "caps", substitution of one or more of the naturally occurring nucleotides with an analog, internucleotide modifications such as, for example, those with uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.) and with charged linkages (e.g., phosphorothioates, phosphorodithioates, etc.), those containing pendant moieties, such as, for example, proteins (e.g., nucleases, toxins, antibodies, signal peptides, poly-L-lysine, etc.), those with intercalators (e.g., acridine, psoralen, etc.), those containing chelators (e.g., metals, radioactive metals, boron, oxidative metals, etc.), those containing alkylators, those with modified linkages (e.g., alpha anomeric nucleic acids, etc.), as well as unmodified forms of the polynucleotide(s). Further, any of the hydroxyl groups ordinarily present in the sugars may be replaced, for example, by phosphonate groups, phosphate groups, protected by standard protecting groups, or activated to prepare additional linkages to additional nucleotides, or may be conjugated to solid supports. The 5 and 3 terminal OH can be phosphorylated or substituted with amines or organic capping group moieties of from 1 to 20 carbon atoms. Other hydroxyls may also be derivatized to standard protecting groups. Polynucleotides can also contain analogous forms of ribose or deoxyribose sugars that are generally known in the art, including, for example, 2'-O-methyl-, 2'-O-allyl-, 2'-fluoro- or 2'-azido-ribose, carbocyclic sugar analogs, a-anomeric sugars, epimeric sugars such as arabinose, xyloses or lyxoses, pyranose sugars, furanose sugars, sedoheptulose, acyclic analogs and abasic nucleoside analogs such as methyl riboside. One or more phosphodiester linkages may be replaced by alternative linking groups. These alternative linking groups include, but are not limited to, embodiments wherein phosphate is replaced by P(O)S ("thiolate"), P(S)S ("dithiolate"), "(O)NR₂" ("amidate"), P(O)R, P(O)OR', CO or CH₂ ("for-

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macetyl"), in which each R or R' is independently H or substituted or unsubstituted alkyl (1-20 C) optionally containing an ether (—O—) linkage, aryl, alkenyl, cycloalkyl, cycloalkenyl or araldyl. Not all linkages in a polynucleotide need be identical. The preceding description applies to all polynucleotides referred to herein, including RNA and DNA.

As used herein, "recipient", "individual", "subject", "host", and "patient", are used interchangeably herein and refer to any mammalian subject for whom diagnosis, treatment, or therapy is desired, particularly humans. "Mammal" for purposes of treatment refers to any animal classified as a mammal, including humans, domestic and farm animals, and zoo, sports, or pet animals, such as dogs, horses, cats, cows, sheep, goats, pigs, etc. In embodiments, the mammal is human.

As used herein, "antibody" refers to single chain, two-chain, and multi-chain proteins and glycoproteins belonging to the classes of polyclonal, monoclonal, chimeric and hetero immunoglobulins (monoclonal antibodies being preferred); it also includes synthetic and genetically engineered variants of these immunoglobulins.

As used herein, "specific binding," "specifically binds," and the like, refer to non-covalent or covalent preferential binding to a molecule relative to other molecules or moieties in a solution or reaction mixture (e.g., an antibody specifically binds to a particular polypeptide or epitope relative to other available polypeptides/epitopes). In some embodiments, the affinity of one molecule for another molecule to which it specifically binds is characterized by a KD (dissociation constant) of 10⁻⁵ M or less (e.g., 10⁻⁶ M or less, 10⁻⁷ M or less, 10⁻⁸ M or less, 10⁻⁹ M or less, 10⁻¹⁰ M or less, 10⁻¹¹ M or less, 10⁻¹² M or less, 10⁻¹³ M or less, 10⁻¹⁴ M or less, 10⁻¹⁵ M or less, or 10⁻¹⁶ M or less). "Affinity" refers to the strength of binding, increased binding affinity being correlated with a lower KD. As used herein, the "binding" and "specific binding" of the first region to TAM receptor and the second region to a target substance do not require modulating, changing, affecting, or modifying activity of the bound TAM receptor or the target substance.

As used herein, "variable" refers to the fact that certain portions of the variable domains differ extensively in sequence among antibodies and are used in the binding and specificity of each particular antibody for its particular antigen. However, the variability is not evenly distributed throughout the variable domains of antibodies. It is concentrated in three segments called complementarity-determining regions (CDRs) or hypervariable regions both in the light-chain and the heavy-chain variable domains. The more highly conserved portions of variable domains are called the framework (FR). The variable domains of native heavy and light chains each comprise four FR regions, largely adopting a b-sheet configuration, connected by three CDRs, which form loops connecting, and in some cases forming part of, the b-sheet structure. The CDRs in each chain are held together in close proximity by the FR regions and, with the CDRs from the other chain, contribute to the formation of the antigen-binding site of antibodies (see Kabat et al., Sequences of Proteins of Immunological Interest, Fifth Edition, National Institute of Health, Bethesda, Md. (1991)). The constant domains are not involved directly in binding an antibody to an antigen, but exhibit various effector functions, such as participation of the antibody in antibody-dependent cellular toxicity.

"Fv" is the minimum antibody fragment, which contains a complete antigen-recognition and -binding site. In a two-chain Fv species, this region consists of a dimer of one

heavy- and one light-chain variable domain in tight, non-covalent association. In a single-chain Fv species (scFv), one heavy- and one light-chain variable domain can be covalently linked by a flexible peptide linker such that the light and heavy chains can associate in a “dimeric” structure analogous to that in a two-chain Fv species. It is in this configuration that the three CDRs of each variable domain interact to define an antigen-binding site on the surface of the VH-VL dimer. Collectively, the six CDRs confer antigen-binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three CDRs specific for an antigen) has the ability to recognize and bind antigen, although at a lower affinity than the entire binding site.

The term “complementarity determining region” or “CDR,” as used herein, refers to the sequences of amino acids within antibody variable regions which confer antigen specificity and binding affinity. For example, in general, there are three CDRs in each heavy chain variable region (e.g., HCDR1, HCDR2, and HCDR3) and three CDRs in each light chain variable region (LCDR1, LCDR2, and LCDR3). The precise amino acid sequence boundaries of a given CDR can be determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), “Sequences of Proteins of Immunological Interest,” 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD (“Kabat” numbering scheme), Al-Lazikani et al., (1997) JMB 273,927-948 (“Chothia” numbering scheme), or a combination thereof. Under the Kabat numbering scheme, in some embodiments, the CDR amino acid residues in the heavy chain variable domain (VH) are numbered 31-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3); and the CDR amino acid residues in the light chain variable domain (VL) are numbered 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3). Under the Chothia numbering scheme, in some embodiments, the CDR amino acids in the VH are numbered 26-32 (HCDR1), 52-56 (HCDR2), and 95-102 (HCDR3); and the CDR amino acid residues in the VL are numbered 26-32 (LCDR1), 50-52 (LCDR2), and 91-96 (LCDR3). In a combined Kabat and Chothia numbering scheme, in some embodiments, the CDRs correspond to the amino acid residues that are part of a Kabat CDR, a Chothia CDR, or both. For instance, in some embodiments, the CDRs correspond to amino acid residues 26-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3) in a VH, e.g., a mammalian VH, e.g., a human VH; and amino acid residues 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3) in a VL, e.g., a mammalian VL, e.g., a human VL.

The “Fab fragment” also contains the constant domain of the light chain and the first constant domain (CH1) of the heavy chain. Fab fragments differ from Fab' fragments by the addition of a few residues at the carboxy terminus of the heavy chain CH1 domain including one or more cysteines from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. F(ab')₂ antibody fragments originally were produced as pairs of Fab fragments which have hinge cysteines between them. Other chemical couplings of antibody fragments are also known.

As used herein, the term “antibody fragment” or “antigen-binding fragment” or “active fragment” is defined as a portion of an intact antibody comprising the antigen binding site or variable region of the intact antibody, wherein the portion is free of the constant heavy chain domains (i.e., CH2, CH3, and CH4, depending on antibody isotype) of the Fc region of the intact antibody. Examples of antibody fragments include Fab, Fab', Fab'-SH, F(ab')₂, and Fv frag-

ments; diabodies; any antibody fragment that is a polypeptide having a primary structure consisting of one uninterrupted sequence of contiguous amino acid residues (referred to herein as a “single-chain antibody fragment” or “single chain polypeptide”), including without limitation (1) single-chain Fv (scFv) molecules, (2) single chain polypeptides containing only one light chain variable domain, or a fragment thereof that contains the three CDRs of the light chain variable domain, without an associated heavy chain moiety, (3) single chain polypeptides containing only one heavy chain variable region, or a fragment thereof containing the three CDRs of the heavy chain variable region, without an associated light chain moiety, (4) nanobodies comprising single Ig domains from non-human species or other specific single-domain binding modules; and multispecific or multivalent structures formed from antibody fragments. In an antibody fragment comprising one or more heavy chains, the heavy chain(s) can contain any constant domain sequence (e.g. CH1 in the IgG isotype) found in a non-Fc region of an intact antibody, and/or can contain any hinge region sequence found in an intact antibody, and/or can contain a leucine zipper sequence fused to or situated in the hinge region sequence or the constant domain sequence of the heavy chain(s), and (5) an isolated complementarity determining region (CDR).

The terms “phagocytic cells” and “phagocytes” are used interchangeably herein to refer to a cell that is capable of phagocytosis. There are four main categories of phagocytes: macrophages, mononuclear cells (histiocytes and monocytes), polymorphonuclear leukocytes (neutrophils), and dendritic cells.

As used herein “chimeric” refers to a molecule that includes sequences derived from two different molecules.

The term “Fc region” is used to define a C-terminal region of an immunoglobulin heavy chain. The “Fc region” may be a native sequence Fc region or a variant Fc region. Although the boundaries of the Fc region of an immunoglobulin heavy chain might vary, the human IgG heavy chain Fc region is usually defined to stretch from an amino acid residue at position Cys226, or from Pro230, to the carboxyl-terminus thereof. The numbering of the residues in the Fc region is that of the EU index as in Kabat. Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md., 1991. The Fc region of an immunoglobulin generally comprises two constant domains, CH2 and CH3, and when the Fc region is employed as a scaffold according to embodiments of the present disclosure, the Fc region may comprise CH2, CH3, or combinations thereof. The Fc region as a scaffold or in a heavy chain of an antibody may contain mutations. For example, heavy chain constant region or Fc region may contain substitutions selected from T250Q/M428L; M252Y/S254T/T256E+H433K/N434F; E233P/L234V/L235A/G236A+A327G/A330S/P331S; E333A; S239D/A330L/I332E; P257I/Q311; K326W/E333S; S239D/I332E/G236A; N297A; L234A/L235A; N297A+M252Y/S254T/T256E; K322A and K444A, wherein the numbering is according to the EU numbering (Edelman, G. M. et al., Proc. Natl. Acad. USA, 63, 78-85 (1969); www.imgt.org/IMGTScientificChart/Numbering/Hu_IgHGnber.html#refs).

As used herein, “Fc receptor” and “FcR” describe a receptor that binds to the Fc region of an antibody. The preferred FcR is a native sequence human FcR. Moreover, a preferred FcR is one which binds an IgG antibody (a gamma receptor) and includes receptors of the FcγRI, FcγRII, and FcγRIII subclasses, including allelic variants and alternatively spliced forms of these receptors. FcγRII receptors

include FcγRIIA (an “activating receptor”) and FcγRIM (an “inhibiting receptor”), which have similar amino acid sequences that differ primarily in the cytoplasmic domains thereof.

A “native sequence Fc region” or “wild-type Fc region” comprises an amino acid sequence identical to the amino acid sequence of an Fc region found in nature. A “variant Fc region” comprises an amino acid sequence which differs from that of a native sequence Fc region by virtue of at least one amino acid modification, yet retains at least one effector function of the native sequence Fc region. Preferably, the variant Fc region has at least one amino acid substitution compared to a native sequence Fc region or to the Fc region of a parent polypeptide, e.g. from about one to about ten amino acid substitutions, and preferably from about one to about five amino acid substitutions in a native sequence Fc region or in the Fc region of the parent polypeptide. The variant Fc region herein will preferably possess at least about 80% sequence identity with a native sequence Fc region and/or with an Fc region of a parent polypeptide, and most preferably at least about 90% sequence identity therewith, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99% sequence identity therewith.

A polynucleotide or polypeptide having a certain percent “sequence identity” to another polynucleotide or polypeptide, means that, when aligned, that percentage of bases or amino acids are the same, and in the same relative position, when comparing the two sequences. Sequence similarity can be determined in a number of different manners. To determine sequence identity, sequences can be aligned using the methods and computer programs, including BLAST, available over the world wide web (www.ncbi.nlm.nih.gov/BLAST). See, e.g., Altschul et al. (1990), *J. Mol. Biol.* 215:403-10. Another alignment algorithm is FASTA, available in the Genetics Computing Group (GCG) package. See also at ebi.ac.uk/Tools/sss/fasta/. Other techniques for alignment are described in *Methods in Enzymology*, vol. 266: Computer Methods for Macromolecular Sequence Analysis (1996), ed. Doolittle, Academic Press, Inc., a division of Harcourt Brace & Co., San Diego, Calif., USA. Of particular interest are alignment programs that permit gaps in the sequence. The Smith-Waterman is one type of algorithm that permits gaps in sequence alignments. See *Meth. Mol. Biol.* 70:173-187 (1997). Also, the GAP program using the Needleman and Wunsch alignment method can be utilized to align sequences. See *J. Mol. Biol.* 48:443-453 (1970), doi: 10.1016/0022-2836 (70) 90057-4.

As used herein, an “effective dosage” or “effective amount” drug, compound, or pharmaceutical composition is an amount sufficient to effect beneficial or desired results. For prophylactic use, beneficial or desired results include results such as eliminating or reducing the risk, lessening the severity, or delaying the outset of the disease, including biochemical, histological and/or behavioral symptoms of the disease, its complications and intermediate pathological phenotypes presenting during development of the disease. For therapeutic use, beneficial or desired results include clinical results such as inhibiting, suppressing or reducing the formation of substance accumulation (non-limiting example may include amyloid plaques), reducing, removing, clearing amyloid plaques, improving cognition, reversing or slowing cognitive decline, sequestering or increasing soluble substance circulating in biological fluids, decreasing one or more symptoms resulting from the disease (biochemical, histological and/or behavioral), including its complica-

tions and intermediate pathological phenotypes presenting during development of the disease, increasing the quality of life of those suffering from the disease, decreasing the dose of other medications required to treat the disease, enhancing effect of another medication, delaying the progression of the disease, and/or prolonging survival of patients. An effective dosage can be administered in one or more administrations. For purposes of this invention, an effective dosage of drug, compound, or pharmaceutical composition is an amount sufficient to accomplish prophylactic or therapeutic treatment either directly or indirectly. As is understood in the clinical context, an effective dosage of a drug, compound, or pharmaceutical composition may or may not be achieved in conjunction with another drug, compound, or pharmaceutical composition. Thus, an “effective dosage” may be considered in the context of administering one or more therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable result may be or is achieved.

As used herein, “treatment” or “treating” is an approach for obtaining beneficial or desired results including clinical results. For purposes of this invention, beneficial or desired clinical results include, but are not limited to, one or more of the following: inhibiting, suppressing or reducing the formation of deposit of substance, reducing, removing, or clearing amyloid deposits, improving cognition, reversing or slowing cognitive decline, sequestering soluble substance circulating in biological fluids, reducing a substance (including soluble, oligomeric and deposited) in a tissue, inhibiting, slowing and/or reducing accumulation of substance in the tissue, inhibiting, slowing and/or reducing toxic effects of a substance peptide in a tissue, decreasing symptoms resulting from the disease, increasing the quality of life of those suffering from the disease, decreasing the dose of other medications required to treat the disease, delaying the progression of the disease, and/or prolonging survival of patients. The tissue may include brain of an individual.

The term “development” of a disease means the onset and/or progression of the disease within an individual. A disease development can be detectable using standard clinical techniques as described herein. However, development also refers to disease progression that may be initially undetectable. For purposes of this invention, progression refers to the biological course of the disease state, in this case, as determined by a standard neurological examination, patient interview, or may be determined by more specialized testing. A variety of these diagnostic tests include, but not limited to, neuroimaging, detecting alterations of levels of specific proteins in the serum or cerebrospinal fluid (e.g., amyloid peptides and Tau), computerized tomography (CT), and magnetic resonance imaging (MRI). “Development” includes occurrence, recurrence, and onset. As used herein “onset” or “occurrence” of a disease includes initial onset and/or recurrence.

As used herein, “delaying” development of a disease means to defer, hinder, slow, retard, stabilize, and/or postpone development of the disease. This delay can be of varying lengths of time, depending on the history of the disease and/or individual being treated. As is evident to one skilled in the art, a sufficient or significant delay can, in effect, encompass prevention, in that the individual does not develop the disease. For example, a method that delays development of a disease is a method that reduces probability of disease development in a given time frame and/or reduces extent of the disease in a given time frame, when compared to not using the method. Such comparisons are

typically based on clinical studies, using a statistically significant number of subjects.

As used herein, “vector” means a construct, which is capable of delivering, and preferably expressing, one or more gene(s) or sequence(s) of interest in a host cell. Examples of vectors include, but are not limited to, viral vectors, naked DNA or RNA expression vectors, plasmid, cosmid or phage vectors, DNA or RNA expression vectors associated with cationic condensing agents, DNA or RNA expression vectors encapsulated in liposomes, and certain eukaryotic cells, such as producer cells.

A “host cell” includes an individual cell or cell culture that can be or has been a recipient for vector(s) for incorporation of polynucleotide inserts. Host cells include progeny of a single host cell, and the progeny may not necessarily be completely identical (in morphology or in genomic DNA complement) to the original parent cell due to natural, accidental, or deliberate mutation. A host cell includes cells transfected in vivo with a polynucleotide(s) of this invention.

As used herein, “expression control sequence” means a nucleic acid sequence that directs transcription of a nucleic acid. An expression control sequence can be a promoter, such as a constitutive or an inducible promoter, or an enhancer. The expression control sequence is operably linked to the nucleic acid sequence to be transcribed.

As used herein, “pharmaceutically acceptable carrier” includes any material which, when combined with an active ingredient, allows the ingredient to retain biological activity and is non-reactive with the subject’s immune system. Examples include, but are not limited to, any of the standard pharmaceutical carriers such as a phosphate buffered saline solution, water, emulsions such as oil/water emulsion, and various types of wetting agents. Preferred diluents for aerosol or parenteral administration are phosphate buffered saline or normal (0.9%) saline. Compositions comprising such carriers are formulated by well-known conventional methods.

TAM Receptor

TAM receptors (Tyro3, Axl, and Mer) belong to a family of receptor tyrosine kinases that have important effects on homeostasis and inflammation. Also, they affect cell proliferation, survival, adhesion, and migration. TAM receptors comprise 2 immunoglobulin-like and 2 fibronectin type III repeats in their extracellular domains in tandem. This is connected to a single-pass transmembrane domain and a cytoplasmic protein tyrosine kinase. Left figure of FIG. 1A.

TAM receptors enhance phagocytosis of apoptotic cells, also known as efferocytosis.

The Axl protein contains 894 amino acids with a glycine-rich loop (Gly543-Gly548), a catalytic loop (His670-Asn677), and a DFG motif (Asp690-Phe691-Gly692). Although the molecular weight of the full-length Axl is 104 kDa, post-translational modifications of the extracellular domains give rise to two modified forms with molecular weights 120 and 140 kDa. Potential N-linked glycosylation sites include Asn43, Asn157, Asn198, Asn339, Asn345, and Asn401. In various embodiments of the present disclosure, the term “Axl” or “Axl receptor” or “Axl protein” includes the full-length Axl of 104 kDa, post-translational modified Axl, and glycosylated Axl. In some embodiments, the human Axl polypeptide corresponds to Genbank accession no. NP_068713 or NP_068713.2 (isoform 1 precursor), SEQ ID NO: 114, or UniProt accession no. P30530.4, Q8N5L2, or Q9UD27, or their mature forms. Amino acid residues

1-32 of SEQ ID NO: 114 is reported as signal sequence, and regions ranging amino acid residues 26 to 92 of SEQ ID NO: 114 is reported as interaction with Gas6. In one embodiment, the nucleic acid encoding the human Axl polypeptide corresponds to Genbank accession no. NM_021913, version no. NM_021913.5. Murine Axl refers to the Axl member of the murine TAM family of receptor tyrosine kinases. In some embodiments, the murine Axl polypeptide corresponds to Genbank accession no. AAH46618, version no. AAH46618.1, SEQ ID NO: 115. In one embodiment, the nucleic acid encoding the murine Axl polypeptide corresponds to Genbank accession no. BC046618, version no. BC046618.1. Various natural variants and mutations, and posttranslational variants and mutants of Axl as well as orthologues of Axl have been reported. For example, human Axl proteins under accession nos. NP_001265528.1 (626 amino acid residues), NP_001690.2 (885 amino acid residues, isoform 2 precursor), EAW57022 (885 amino acid residues, isoform CRA_a), EAW57023.1 (894 amino acid residues, isoform CRA_b), AAH32229.1 (894 amino acid residues), AAH32229.1 (885 amino acid residues), and the like are considered as “Axl” or “Axl receptor” or “Axl protein” according to the embodiments of the present disclosure.

The cells expressing the TAM receptor(s) may be at least one type of professional phagocytes, at least one type of non-professional phagocytes, or a combination thereof. Here, the professional phagocytes refer to cells whose main role is to remove dead cells and accumulated debris through phagocytosis, and examples thereof include macrophages, neutrophils, dendritic cells, and mast cells. Macrophages usually stay in each tissue that can become a path of infection, and in many cases, they are called different names for tissues, including, for example, adipose tissue macrophages, bone marrow or blood monocytes, hepatic Kupffer cells, lymph node sinus histiocytes, alveolar macrophages, connective tissue histiocytes or giant cells, microglia of the central nervous system, placental Hofbauer cells, renal intraglomerular mesangial cells, bone osteoclasts, epithelioid cells of granulomas, red pulp macrophages of the spleen, peritoneal macrophage of the peritoneal cavity, LysoMac of Peyer’s patch, and the like. On the other hand, the non-professional phagocytes refer to cells that mainly perform functions specific to the tissue in which the phagocytes reside, but can perform phagocytosis when necessary, and examples thereof epithelial cells, endothelial cells, fibroblasts, mesenchymal cells, some tissue-specific cells, for example, astrocytes or oligodendrocyte of the central nervous system, retinal Muller glia, hepatocytes, muscular satellite cells, testicular Sertoli cells, etc., and some lymphocytes such as natural killer cells, large granular lymphocytes, eosinophils, basophils, B cells, etc. The fusion molecule according to the present disclosure is able to induce phagocytosis in phagocytes specific to a tissue in which a target substance to be cleared accumulates. For example, when abnormal proteins accumulated in the brain are to be cleared, the phagocytosis may be induced in astrocytes, microglia, oligodendrocytes, or combinations thereof. It may be induced, for example, by topically administering the fusion molecule according to the present disclosure to this tissue or by manipulating cells in the tissue to express and secrete the fusion molecule.

First Region Comprising a Sequence Capable of Binding to TAM Receptor

TAM receptors can be activated via their ligands, growth arrest specific 6 protein (Gas6) and Protein S (Pros1), which are members of the family of vitamin K-dependent proteins.

In exemplary embodiments, the first region that is capable of binding to TAM receptors may comprise, consist of, or consist essentially of one or more TAM ligands.

A TAM ligand, protein S contains an amino terminal γ carboxyglutamic acid (GLA) domain, followed by a thrombin-sensitive loop region and 4 epidermal growth factor-like domains ending with the carboxy-terminal (C-terminal), consisting of 2 laminin G repeats that together comprise the sex hormone-binding globulin domain (right figure of FIG. 1A). The C-terminal region is sufficient for TAM receptor binding and phosphorylation. Gas6 is a 75-kDa vitamin K-dependent protein and has high structural homology (~42%) with protein S and the modular composition is the same as shown in FIG. 1A.

In addition to Gas6 (SEQ ID NO: 7) and ProS1 (SEQ ID NO: 34), tubby (accession nos. P50607, U54644.1, AAB53494.1, U82467.1, AAB53699.1, CH471064.2, EAW68634.1, BC075031.2, AAH75031.1, BC075032.2, AAH75032.1, NP_003311.2, NP_813977.1, 1S31_A), tubby-like protein 1 (Tulp1) (accession nos. AAB53700.1, AAH32714.1, AAH65261.1, NP_001276324.1, AAB97966.1, EAX03840.1, EAX03839.1, BAJ84064.1, BAJ84063.1, AKU84911.1, NP_813977.1, NP_003311.2), and galectin-3 (Gal3) (accession nos. NP_002297, NP_002297.1) are reported as TAM receptor ligands. Tubby and Gal-3 specifically bind to Mer, whereas Tulp1 can activate all 3 of the TAM receptors.

Gas6, one of the ligands for TAM receptors, is reported to show the highest affinity for Axl compared to Tyro3 or Mer. Human Gas6 contains 678 amino acids (SEQ ID NO: 7), with gamma-carboxyglutamic acid (Gla) domains, four epidermal growth factor (EGF)-like domains, and two laminin G-like (LG) domains (FIG. 1A, right figure). Various isoforms of GAS6 are reported. For example, S6L, G8R, G8V, R14H, L18Q isoforms have been reported and these isoforms are included in the present disclosure.

In embodiments, the first region that is capable of binding to TAM receptor may comprise, consist of, or consist essentially of Gas6 protein or an active fragment thereof. The term "active fragment" as used herein denotes a fragment that is capable of binding to TAM receptor, in particular, Axl receptor. For example, an active fragment of Gas6 protein may comprise, consist of, or consist essentially of the sequence of SEQ ID NO: 1, 2, 5, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, or 87. For example, an active fragment of ProS protein may comprise, consist of, or consist essentially of the sequence of SEQ ID NO: 3, 4, 6, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, or 113. The present disclosure encompass the sequences having sequence identity of at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% to the sequence of any one of SEQ ID NOS. Sequences of SEQ ID NOS: 8-23 show sequence identity of at least 85% to SEQ ID NO: 1 (LG-1 domain of Gas6). Sequences of SEQ ID NOS: 24-33 show sequence identity of at least 85% to SEQ ID NO: 2 (LG-2 domain of Gas6). Sequences of SEQ ID NOS: 35-45 show sequence identity of at least 85% to SEQ ID NO: 3 (LG-1 domain of ProS). Sequences of SEQ ID NOS: 46-62 show sequence identity of at least 85% to SEQ ID NO: 4 (LG-2

domain of ProS). Sequences of SEQ ID NOS: 63-87 show sequence identity of at least 85% to SEQ ID NO: 5 (LG domains of Gas6). Sequences of SEQ ID NOS: 88-113 show sequence identity of at least 84% to SEQ ID NO: 6 (LG domains of ProS).

In other embodiments, the first region may comprise, consist, or consist essentially of variable region or CDRs of an anti-Axl antibody or a full-length anti-Axl antibody of which the effector function, in particular, Fc receptor-binding function is abolished or removed. The antibody or antigen-binding fragments may bind to extracellular domain of Axl, for example expressed on surface of phagocytic cells and induce internalization and phagocytosis without involving inflammatory reaction, in particular Fc-mediated inflammatory reaction. Non-limiting examples of anti-Axl antibody may include those described in, for example, WO2017200493A1, WO2015193430A1, WO2011159980A1, WO2016097370A1, WO2012175691A1, WO2015193428A1, WO2010131733A1, WO2017220695A1, WO2010130751A1, WO2016166302A1, WO2017009258A1, WO2016005593A1, and the like, all of which the contents are incorporated by reference herein in their entireties. According to embodiments of the present disclosure, whole antibody, variable region, CDRs, or scFv, F(ab), or F(ab') of those anti-Axl antibodies may be employed as the first region of the fusion molecule. In certain embodiments, anti-Axl antibody may be anti-Axl agonistic antibodies or antigen-binding fragments thereof. The antibody or an antigen-binding fragment thereof may be selected from among, for example, i) immunoglobulins such as IgG1, IgG2, IgG3 and IgG4; ii) native antibody fragments such as Fv, Fab, Fab', F(ab')₂, VHH, VNAR, etc.; and iii) engineered antibodies such as scFv, dsFv, ds-scFv, (scFv)₂, diabody, triabody, tetrabody, pentabody, etc. The antibody or antigen-binding fragment thereof may be, for example, a Mab, Fab, or single-chain variable fragment (scFv) based on an antibody that specifically binds to a corresponding target substance, or six complementarity-determining regions (CDRs) derived from the antibody.

In other embodiments, the first region may comprise, consist, or consist essentially of variable region or CDRs of an anti-MerTK (Mer Tyrosine Kinase) antibody or a full-length anti-MerTK antibody of which the effector function, in particular, Fc receptor-binding function is abolished or removed. The antibody or antigen-binding fragments may bind to extracellular domain of MerTK, for example expressed on surface of phagocytic cells and induce internalization and phagocytosis without involving inflammatory reaction, in particular Fc-mediated inflammatory reaction. Non-limiting examples of anti-MerTK antibody may include those described in, for example, WO2016106221A1, WO2020076799A1, WO2020176497A1, and the like, all of which the contents are incorporated by reference herein in their entireties. According to embodiments of the present disclosure, whole antibody, variable region, CDRs, or scFv, F(ab), or F(ab') of those anti-MerTK antibodies may be employed as the first region of the fusion molecule. The antibody or an antigen-binding fragment thereof may be selected from among, for example, i) immunoglobulins such as IgG1, IgG2, IgG3 and IgG4; ii) native antibody fragments such as Fv, Fab, Fab', F(ab')₂, VHH, VNAR, etc.; and iii) engineered antibodies such as scFv, dsFv, ds-scFv, (scFv)₂, diabody, triabody, tetrabody, pentabody, etc. The antibody or antigen-binding fragment thereof may be, for example, a Mab, Fab, or single-chain variable fragment (scFv) based on an antibody

that specifically binds to a corresponding target substance, or six complementarity-determining regions (CDRs) derived from the antibody.

In other embodiments, the first region may comprise, consist, or consist essentially of variable region or CDRs of an anti-Tyro3 antibody or a full-length anti-Tyro3 antibody of which the effector function, in particular, Fc receptor-binding function is abolished or removed. The antibody or antigen-binding fragments may bind to extracellular domain of Tyro3, for example expressed on surface of phagocytic cells and induce internalization and phagocytosis without involving inflammatory reaction, in particular Fc-mediated inflammatory reaction. Non-limiting examples of anti-Tyro3 antibody may include those described in, for example, WO2016166348A1, and the like, all of which the contents are incorporated by reference herein in their entireties. According to embodiments of the present disclosure, whole antibody, variable region, CDRs, or scFv, F(ab), or F(ab') of those anti-Tyro3 antibodies may be employed as the first region of the fusion molecule. The antibody or an antigen-binding fragment thereof may be selected from among, for example, i) immunoglobulins such as IgG1, IgG2, IgG3 and IgG4; ii) native antibody fragments such as Fv, Fab, Fab', F(ab')₂, VHH, VNAR, etc.; and iii) engineered antibodies such as scFv, dsFv, ds-scFv, (scFv)₂, diabody, triabody, tetrabody, pentabody, etc. The antibody or antigen-binding fragment thereof may be, for example, a Mab, Fab, or single-chain variable fragment (scFv) based on an antibody that specifically binds to a corresponding target substance, or six complementarity-determining regions (CDRs) derived from the antibody.

The peptide comprising the sequence of any one of SEQ ID Nos above includes not only the amino acid sequence of the peptide but also an amino acid sequence variant thereof. The term "sequence variant" refers to a protein having a sequence in which one or more amino acid residues differ from the amino acid sequence. As long as the activity of the fusion molecule is maintained, any truncation, deletion, insertion, substitution, or a combination thereof in the final structure of the protein is possible. One example of the sequence variant is a form in which amino acid residues at sites not essential for activity are truncated or deleted, or amino acid residues at sites important for autoinhibition are substituted. In some cases, it may also be modified by phosphorylation, glycosylation, methylation, farnesylation, or the like. These sequence variations and modifications are more preferable when the function and/or stability (thermal stability, pH stability, structural stability, etc.) and/or solubility of the protein are increased by mutation in the amino acid sequence.

The method for mutagenesis of the amino acid sequence is based on a method of producing a nucleic acid molecule comprising a nucleotide sequence corresponding to the amino acid sequence to be mutated by mutating a nucleotide sequence encoding the protein, and a method for obtaining the gene encoding the protein may be performed in vivo or in vitro using any mutagenesis technique well known in the art, for example, site-directed mutagenesis (Hutchinson et al., *J. Biol. Chem.*, 253:6551, 1978; Zoller and Smith, *DNA*, 3:479-488, 1984; Oliphant et al., *Gene*, 44:177, 1986; Hutchinson et al., *Proc. Natl. Acad. Sci. U.S.A.*, 83:710, 1986), TAB linker (Pharmacia), PCR technique (Higuchi, 1989, "Using PCR to Engineer DNA" in *PCR Technology: Principles and Applications for DNA Amplification*, H. Erlich, ed., Stockton Press, Chapter 6, pp. 61-70), or the like.

Amino acid sequence insertions include amino- and/or carboxyl-terminal fusions ranging in length from one resi-

due to polypeptides containing a hundred or more residues, as well as intrasequence insertions of single or multiple amino acid residues. Examples of terminal insertions include an antibody with an N-terminal methionyl residue or the antibody fused to an epitope tag. Other insertional variants of the antibody molecule include the fusion to the N- or C-terminus of the antibody of an enzyme or a polypeptide which increases the serum half-life of the antibody.

Examples of modified polypeptides include polypeptides with conservative substitutions of amino acid residues, one or more deletions or additions of amino acids which do not significantly deleteriously change the functional activity, or use of chemical analogs.

Substitution variants have at least one amino acid residue in the antibody molecule removed and a different residue inserted in its place. The sites of greatest interest for substitutional mutagenesis include the hypervariable regions, but FR alterations are also contemplated. Conservative substitutions are shown in Table 2 under the heading of "conservative substitutions". If such substitutions result in a change in biological activity, then more substantial changes, denominated "exemplary substitutions" in Table 2, or as further described below in reference to amino acid classes, may be introduced and the products screened.

TABLE 2

Amino acid substitutions		
Original Residue	Conservative Substitutions	Exemplary Substitutions
Ala (A)	Val	Val; Leu; Ile
Arg (R)	Lys	Lys; Gln; Asn
Asn (N)	Gln	Gln; His; Asp, Lys; Arg
Asp (D)	Glu	Glu; Asn
Cys (C)	Ser	Ser; Ala
Gln (Q)	Asn	Asn; Glu
Glu (E)	Asp	Asp; Gln
Gly (G)	Ala	Ala
His (H)	Arg	Asn; Gln; Lys; Arg
Ile (I)	Leu	Leu; Val; Met; Ala; Phe; Norleucine
Leu (L)	Ile	Norleucine; Ile; Val; Met; Ala; Phe
Lys (K)	Arg	Arg; Gln; Asn
Met (M)	Leu	Leu; Phe; Ile
Phe (F)	Tyr	Leu; Val; Ile; Ala; Tyr
Pro (P)	Ala	Ala
Ser (S)	Thr	Thr
Thr (T)	Ser	Ser
Trp (W)	Tyr	Tyr; Phe
Tyr (Y)	Phe	Trp; Phe; Thr; Ser
Val (V)	Leu	Ile; Leu; Met; Phe; Ala; Norleucine

Substantial modifications in the biological properties of the antibody are accomplished by selecting substitutions that differ significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. Naturally occurring residues are divided into groups based on common side-chain properties:

Non-polar: Norleucine, Met, Ala, Val, Leu, Ile;
 Polar without charge: Cys, Ser, Thr, Asn, Gln;
 Acidic (negatively charged): Asp, Glu;
 Basic (positively charged): Lys, Arg;
 Residues that influence chain orientation: Gly, Pro; and
 Aromatic: Trp, Tyr, Phe, His.

Non-conservative substitutions are made by exchanging a member of one of these classes for another class. Any

cysteine residue not involved in maintaining the proper conformation of the antibody also may be substituted, generally with serine, to improve the oxidative stability of the molecule and prevent aberrant cross-linking. Conversely, cysteine bond(s) may be added to the antibody to improve its stability, particularly where the antibody is an antibody fragment such as an Fv fragment.

Amino acid modifications can range from changing or modifying one or more amino acids to complete redesign of a region, such as the variable region. Changes in the variable region can alter binding affinity and/or specificity. In some embodiments, no more than one to five conservative amino acid substitutions are made within a CDR domain. In other embodiments, no more than one to three conservative amino acid substitutions are made within a CDR domain. In still other embodiments, the CDR domain is CDR H3 and/or CDR L3.

Target Substances and Diseases Characterized by Aberrant Accumulation or Deposit of Target Substances

The target substance may be a substance that accumulates in living tissue, causing a disease. For example, it may be a substance accumulated in an affected (i.e., diseased) tissue of a patient. The substance accumulated in a disease may be protein. That is, the disease may be proteopathy, without being limited thereto. For example, the target substance may be amyloid. That is, the proteopathy may be amyloidosis. The target substance may be selected from abnormally accumulated substances listed in Table 1 above, and in this case, the disease may be a disease in which each abnormally accumulated substance is detected. In some embodiments, the target substance, of which aberrant accumulation is associated with proteopathy and which is desired to be cleared or reduced or suppressed, may be APOE or apoptosis-associated spec-like protein containing a caspase activating recruitment domain (ASC-speck). For example, the proteopathy may be selected from Alzheimer's disease, Parkinson's disease, Huntington's disease, and Prion disease, and in this case, target substances may be abnormally accumulated proteins that cause the diseases. That is, the target substances may be β -amyloid, tau, α -synuclein, huntingtin, and prion proteins, respectively.

Aberrant accumulation of APOE is associated with Alzheimer's disease, cerebral amyloid angiopathy, and/or cardiovascular disease. Aberrant accumulation of apoptosis-associated spec-like protein containing a caspase activating recruitment domain (ASC-speck) is associated with Alzheimer's Disease, Parkinson's Disease, Huntington's disease, Multiple System Atrophy, Amyotrophic Lateral Sclerosis, Spinocerebellar ataxia, Frontotemporal Dementia, Frontotemporal Lobar Degeneration, Mild Cognitive Impairment, Parkinson-plus syndromes, Pick disease, Progressive isolated aphasia, Grey-matter degeneration [Alpers], Subacute necrotizing encephalopathy, and Lewy body dementia.

Second Region of Fusion Molecule

The second region that specifically binds to the target substance may be selected from among an antibody, an antigen-binding fragment thereof, an antibody-like protein, a peptide, an aptamer, and a soluble receptor, and is not particularly limited as long as it specifically binds to the target substance.

Here, the antibody or an antigen-binding fragment thereof may be selected from among, for example, i) immunoglobulins such as IgG1, IgG2, IgG3 and IgG4; ii) native antibody fragments such as Fv, Fab, Fab', F(ab')₂, VHH, VNAR, etc.; and iii) engineered antibodies such as scFv, dsFv, ds-scFv, (scFv)₂, diabody, triabody, tetrabody, pentabody, etc. The antibody or antigen-binding fragment thereof may be, for example, a Mab, Fab, or single-chain variable fragment (scFv) based on an antibody that specifically binds to a corresponding target substance, or six complementarity-determining regions (CDRs) derived from the antibody. That is, the protein or antigen-binding fragment thereof that specifically binds to the target substance comprises a portion necessary for an activity that specifically binds to the target substance, and the type or range thereof is not particularly limited as long as the protein or antigen-binding fragment thereof is linked to the first region and does not cause an inflammatory response and synaptic damage. For example, the target substance may be beta-amyloid, and in this case, the protein or antigen-binding fragment thereof that specifically binds to the target substance may comprise aducanumab or a single-chain variable fragment thereof. The second region comprise a Mab, Fab, or single-chain variable fragment based on based on six complementarity determining regions (CDRs) derived from commercially available antibodies such as aducanumab, semorinemab, and cinnanemab.

In non-limiting exemplary embodiments, the sequence information of the target substance can be obtained from public database and the target substance-binding sequences can be obtained from the publications or public database. By way of exemplary embodiments, some sequences from the public database are exemplified below. One skilled art should understand that sequences of target substance or a second region capable of binding to the target substance are not limited to the specific sequences exemplified below, but encompass isomers, orthologues, variants, and mutants. For example, when Amyloid precursor protein-derived β -amyloid is a target substance, the target subject may have the sequence available under GenBank accession no. AAB29908.1 or a fragment thereof (e.g., beta-amyloid (29-40) or a Chain A (Accession No. 1BJC_A)), and the second region that is capable of binding to the target substance can have a light chain variable region of SEQ ID NO: 161 and a heavy chain variable region of SEQ ID NO: 162. When the target substance is α -Synuclein, it can have the sequence available under UniProtKB/Swiss-Prot: P37840.1 or a fragment thereof, and the second region capable of binding to the target substance may have light chain variable region of SEQ ID NO: 163 and a heavy chain variable region of SEQ ID NO: 164. When the target substance is Microtubule-associated protein tau, it can have the sequence available under UniProtKB/Swiss-Prot: P10636.5 or a fragment thereof, and the second region capable of binding to the target substance may have light chain variable region of SEQ ID NO: 165 and a heavy chain variable region of SEQ ID NO: 166. When the target substance is PrP^{Sc}, it can have the sequence available under GenBank Accession No. NP_001073592.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in US 2021/0070870 A1. When the target substance is Huntingtin exon 1, it can have the sequence available under GenBank Accession No. NP_001375421.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in US 2022/0332808 A1. When the target substance is TAR DNA-binding protein 43

(TDP43), it can have the sequence available under UniProtKB/Swiss-Prot: Q13148.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 9,587,014 B2. When the target substance is superoxide dismutase 1 (SOD1), it can have the sequence available under GenBank: CAG46542.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 9,283,271 B2. When the target substance is an immunoglobulin light-chain fragment, it can have the sequence available under PDB: 6Z1O_A or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 8,268,973 B2. When the target substance is an N-terminal fragment of serum amyloid A protein, it can have the sequence available under GenBank: AAB24060.1, GenBank: AAA85338.1, NCBI NP_001372595.1, or NCBI Reference Sequence: NP_110381.2 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 8,268,973 B2. When the target substance is a transthyretin, it can have the sequence available under UniProtKB/Swiss-Prot: P02766.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 11,267,877 B2. When the target substance is an amylin, IAPP (AIAPP), it can have the sequence available under UniProtKB/Swiss-Prot: P10997 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in U.S. Pat. No. 10,882,902. When the target substance is an APOE, it can have the sequence available under UniProtKB/Swiss-Prot: P02649.1 or a fragment thereof, and the second region capable of binding to the target substance may be the sequences disclosed in US 2022/0411485A. When the target substance is an Apoptosis-associated Spec-like protein containing a Caspase Activating Recruitment Domain (ASC-speck), it can have the sequence available under UniProtKB/Swiss-Prot: Q9ULZ3.2 or a fragment thereof (e.g., US 2021/0079075A), and the second region capable of binding to the target substance may be the sequences disclosed in US No. 10,961,3-6 B22021/0079075A.

The antibody or antigen-binding fragment thereof may not comprise an Fc region, and preferably may comprise an Fc region variant that does not bind to an Fc receptor (particularly an Fcγ receptor). This Fc region variant may serve to improve properties such as purification. Fc variants with a reduced affinity to the human FcγRIIIA and/or FcγRIIA and/or FcγRI compared to a IgG Fc region by way of amino acid substitution are disclosed for example, WO2012130831 and U.S. Pat. No. 8,753,628, of which entire content is incorporated by reference herein. Fc regions may be aglycosylated or deglycosylated.

The antibody-like protein refers to a protein scaffold capable of specifically binding to a target substance, like an antibody. Antibody-like proteins may be designed to have a size of about 2 to 20 kDa, which is smaller than antibodies (about 150 kDa on average), and thus target a binding site that antibodies cannot reach. It is known that antibody-like proteins are more stable at high temperatures than antibodies and are much easier to synthesize using non-mammalian cells such as viruses and yeast or synthesize chemically, compared to antibodies.

As used herein, the term “aptamer” refers to a single-stranded DNA (ssDNA) or RNA having high specificity and affinity for a specific substance. Aptamers have a very high affinity for specific substances, are stable, may be synthe-

sized in a relatively simple way, may be modified in various ways to increase the binding affinity thereof, and can target cells, proteins, and even small organic substances. Thus, the aptamers are characterized by having very high specificity and stability compared to antibodies that have already been developed. In addition, the aptamer may be produced through a known SELEX (Systematic Evolution of Ligands by Exponential enrichment) method. As this aptamer, an aptamer that specifically binds to, for example, beta-amyloid, tau, or alpha-synuclein, may be produced through a known SELEX (Systematic Evolution of Ligands by Exponential enrichment) method and then linked to the first region, thereby producing the fusion molecule according to the present invention.

The aptamer of the present disclosure is not limited as long as it is able to specifically bind to beta-amyloid, tau, or alpha-synuclein, and bases that are used for the aptamer may be selected from among A, G, C, U, and deoxy forms thereof, unless otherwise specified.

In addition, the aptamer may be modified by linkage of at least one, selected from the group consisting of polyethylene glycol (PEG), inverted deoxythymidine (idT), locked nucleic acid (LNA), 2'-methoxy nucleoside, 2'-amino nucleoside, 2'-F-nucleoside, amine linker, thiol linker, and cholesterol, at the 5'-end region, intermediate region, 3'-end region, or both ends thereof in order to increase the stability thereof. Inverted deoxythymidine (idT) is a molecule that is generally used to prevent nuclease degradation of an aptamer having weak nuclease resistance. In the case of a nucleic acid unit, the 3'-OH of the previous nucleotide is attached to the 5'-OH of the next nucleotide to form a chain, but in the case of idT, the 3'-OH of the previous nucleotide is attached to the 3'-OH of the next unit so that 5'-OH, not 3'-OH, is exposed. Thus, idT is a molecule that has the effect of inhibiting degradation by 3' exonuclease, a type of nuclease. In exemplary embodiments, aptamers against beta-amyloid include, but are not limited to, those reported in Yan Zheng, *Advances in aptamers against Aβ and applications in Aβ detection and regulation for Alzheimer's disease*, *Theranostics*, 2022; 12(5): 2095-2114, of which the content is incorporated by reference herein in its entirety.

The soluble receptor of the present disclosure comprises a domain having an activity capable of binding to a target substance, that is, an endogenous ligand, wherein the domain may be one derived from an endogenous membrane receptor or an intracellular receptor, or a derivative thereof. In this case, the soluble receptor comprised in the second region of the fusion molecule of the present disclosure may preferably be one in which regions having activities other than binding to a target substance have been removed from the endogenous receptor. Exemplary soluble receptors that bind to beta-amyloid are reported by John E. Donahue et al., *RAGE, LRP-1, and amyloid-beta protein in Alzheimer's disease*, *Acta Neuropathol* (2006) 112:405-415, of which the content is incorporated by reference herein in its entirety.

In the embodiments, the peptide as a second region, means an entity other than the antibody or an active fragment thereof, antibody-like protein or soluble receptor among polypeptides having amino acids as monomers capable of binding specifically to a target substance. Various peptides that bind to beta amyloid are reported in, for example, Alexander L. Schwarzman, *Selection of peptides binding to the amyloid b-protein reveals potential inhibitors of amyloid formation*, *Amyloid*, December 2005; 12(4): 199-209, of which the content is incorporated by reference herein in its entirety.

Since the fusion molecule according to the present disclosure induces phagocytosis through interaction with the TAM receptor(s), the phagocytosis may be induced in cells expressing the TAM receptor(s). Phagocytosis generally means ingestion of cells or particles of 0.5 μm or more in size, and includes a process of tethering, engulfing, and degrading the cells or particles. In this case, phagocytosis forms a phagosome that surrounds the internalized cell or particle, and includes degradation within the phagolysosome by fusion of the phagosome and the lysosome. In phagocytosis, the process of cell death by apoptosis or necrosis is also referred to as efferocytosis.

Fusion Molecule or Binding Molecule

The induction of phagocytosis by the fusion molecule according to the present disclosure may not involve an inflammatory response. This enables clearance of the target substance without inducing an inflammatory response and tissue damage caused by an inflammatory response to be suppressed so that tissue dysfunction caused by accumulation of the target substance can be treated more safely than conventional techniques.

According to the embodiments, the fusion molecule or the binding molecule does not contain the target substance or a fragment thereof, to which the first region binds.

The first region and the second regions, described above, are coupled to each other directly or via a linker, to form a fusion molecule. Embodiments of the fusion molecule according to the present disclosure encompass monomers of a polypeptide comprising the first region and the second region in a single chain as well as multimers composed of two or more polypeptide chains. Multimers encompass various forms of multimers such as homo-multimers and hetero-multimers. Therefore, according to the embodiments, the fusion molecule may have a monovalent first region (i.e., having one binding site to a single TAM receptor) or a multivalent first region(s) (i.e., having multiple first regions or having one single first region capable of binding to two or more TAM receptors). Embodiments of the fusion molecule according to the present disclosure may have a monovalent second region (i.e., having one binding site to a single target substance) or a multivalent second region(s) (i.e., having multiple second regions each binding different target substances or having one single second region capable of binding to two or more target substances). Embodiments of the fusion molecule according to the present disclosure may have a monovalent first region and a monovalent second region; a monovalent first region and a multivalent second region; or a multivalent first region and a multivalent second region.

Schematic depiction of non-limiting exemplary embodiments of the fusion molecules are shown in FIGS. 1C through 1M.

FIG. 1C depicts fusion molecules comprising a first region and a second region coupled to each other. The first region of FIG. 1C may be any one of the first region described above, which is capable of binding to a TAM receptor. The first region may be a single TAM ligand or an antibody or antigen-binding fragment thereof. The first region may be of a plurality of same or different TAM binders. Similarly, the second region of FIG. 1C may be any one of the second region described above, which is capable of binding to a target substance. The second region may be a single target substance binder. The second region may be of a plurality of same or different target substance binders. A fusion molecule may comprise a scFv, Fab, nanobody, or

antibody as a first region and a scFv, Fab, nanobody, or antibody as a second region, a TAM ligand or a fragment thereof as a first region and a scFv, Fab, nanobody, or antibody as a second region, a scFv, Fab, nanobody, or antibody as a first region and a ligand or receptor or an aptamer as a second region, and the like. For example, a scFv, Fab, or nanobody as a first region or a second region may be a part of an antibody as a second region or a first region, forming a bispecific or multispecific antibody as a fusion molecule. A single or tandem of a scFv, Fab, or nanobody as a first or a second region may be linked to a whole antibody as a second or a first region. When the fusion molecule is a form of bispecific antibody, the first region is not an anti-MerTK antibody or a fragment thereof. The fusion molecule may be of a monomeric structure, a tandem structure where the first region, the second region, or the entire fusion molecule are repeated, or a multimeric structure containing two or more peptide chains. In case of multimeric fusion molecules, individual peptide chain may have a same sequence (homomultimer) or different sequences (heteromultimer). For heteromultimeric fusion molecules, the first region and the second region may present in all of the plural peptide chains forming the multimeric structure, or in different peptide chains, respectively, or a part of the plural peptide chains has both the first and the second region and the other(s) of the plural peptide chains has only either of the first region or the second region. While not shown in FIG. 1C, the fusion molecule may comprise one or more linker to couple the first region and the second region.

FIGS. 1D-1M depict fusion molecules comprising a first region and a second region as well as a scaffold. The first region, second region, and scaffold are as those described in this disclosure. As depicted in FIGS. 1D-1M, the scaffold may be bond to the first region and/or second region at different positions. The first region of FIGS. 1D-1M may be any one of the first region described above, which is capable of binding to a TAM receptor. The first region may be a single TAM ligand or an antibody or antigen-binding fragment thereof. The first region may be of a plurality of same or different TAM binders. Similarly, the second region of FIGS. 1C-1M may be any one of the second region described above, which is capable of binding to a target substance. The second region may be a single target substance binder. The second region may be of a plurality of same or different target substance binders. A fusion molecule may comprise a scFv, Fab, nanobody, or antibody as a first region and a scFv, Fab, nanobody, or antibody as a second region, a TAM ligand or a fragment thereof as a first region and a scFv, Fab, nanobody, or antibody as a second region, a scFv, Fab, nanobody, or antibody as a first region and a ligand or receptor or an aptamer as a second region, and the like. For example, a scFv, Fab, or nanobody as a first region or a second region may be a part of an antibody as a second region or a first region, forming a bispecific or multispecific antibody as a fusion molecule. A single or tandem of a scFv, Fab, or nanobody as a first or a second region may be linked to a whole antibody as a second or a first region. When the fusion molecule is a form of bispecific antibody, the first region is not an anti-MerTK antibody or a fragment thereof. The fusion molecule may be of a monomeric structure, a tandem structure, or a multimeric structure containing two or more peptide chains. In case of multimeric fusion molecules, individual peptide chains may have a same sequence (homomultimer) or different sequences (heteromultimer). In case of heteromultimeric fusion molecules, the first region and the second region may present in all of the plural peptide

chains forming the multimeric structure, or in different peptide chains, respectively, or a part of the plural peptide chains has both the first and the second region and the other(s) of the plural peptide chains has only either of the first region or the second region. While not shown in FIGS. 1D-1M, the fusion molecule may comprise one or more linker to couple the first region, the second region, and/or the scaffold.

The fusion molecule may further comprise a tag. When such a label is added to the fusion molecule, it may be used to check the purification, expression, action or mechanism of action of the fusion molecule.

Examples of the tag include, but are not limited to, His-tag, T7-tag, S-tag, FLAG-tag, Strep-tag, thioredoxin (Trx)-tag, His-patch thioredoxin-tag, lacZ (L-galactosidase)-tag, chloramphenicol acetyltransferase-tag, trpE-tag, avidin/streptavidin/Strep-tag, T7gene10-tag, staphylococcal protein A-tag, streptococcal protein G-tag, glutathione-S-transferase (GST)-tag, dihydrofolate reductase (DHFR)-tag, cellulose binding domains (CBDs)-tag, maltose binding protein (MBP)-tag, galactose-binding protein-tag, calmodulin binding protein (CBP)-tag, hemagglutinin influenza virus (HA)-tag, HSV-tag, B-(VP7 protein region of bluetongue virus)-tag, polycysteine-tag, polyphenylalanine-tag, (Ala-Trp-Trp-Pro)_n-tag, polyaspartic acid-tag, c-myc-tag, lac repressor-tag, and the like. The tag may be located at the N-terminus, C-terminus or internally of the target protein.

The fusion molecule may further comprise a signal peptide or leader sequence at the N-terminus. It is known that a signal peptide is a short peptide present at the N-terminus at the initial stage of protein synthesis toward the secretory pathway, and directs the intracellular localization of the corresponding protein, membrane topology (in the case of a membrane protein), and the like. The signal peptide may be cleaved during expression and extracellular secretion of the fusion molecule.

The above-mentioned first region, second region, tag, signal peptide, or regions having minimal functionality (e.g., LG1 and LG2 regions or scFv heavy chain variable region and light chain variable region) included in the fusion molecule may be linked together directly or by a linker comprising a short oligopeptide or polypeptide. In general, the linker may comprise 2 to 500 amino acid residues. The length or type of the linker is not particularly limited as long as the linker can link the above-described regions together so as to have the intended activity, thereby forming the fusion molecule. An example of the linker may be the commonly used oligopeptide linker (GGGGS)_n (SEQ ID NO: 116), that is, a linker in which one or more Gly-Gly-Gly-Gly-Ser (SEQ ID NO: 117) units are repeated. Other examples of the linker include, but are not limited to, (GSSGGS)_n (SEQ ID NO: 118), KESGVSSEQLAQFRSLD (SEQ ID NO: 119), EGKSSGSGSESKST (SEQ ID NO: 120), GSAGSAAGSGEF (SEQ ID NO: 121), (EAAAK)_n (SEQ ID NO: 122), CRRRRRREAEAC (SEQ ID NO: 123), A(EAAAK)₄ALEA(EAAAK)₄A (SEQ ID NO: 124), GGGGGGGG (SEQ ID NO: 125), GGGGGG (SEQ ID NO: 126), AEAAAKEAAAAKA (SEQ ID NO: 127), PAPAP (SEQ ID NO: 128), (Ala-Pro)_n, VSQTSKL-TRAETVFPDV (SEQ ID NO: 129), PLGLWA (SEQ ID NO: 130), TRHRQPRGWE (SEQ ID NO: 131), AGNRVRRSVG (SEQ ID NO: 132), RRRRRRRR (SEQ ID NO: 133), GFLG (SEQ ID NO: 134), and GSSGGSGSSGGSGGGDEADGSRGSQKAGVDE (SEQ ID NO: 135). Other suitable linkers comprise the sequences described in WO2012/088461A, of which the content is incorporated by reference herein in its entirety.

The fusion molecule according to embodiments of the present disclosure may further comprise a scaffold bound to the first region, to the second, or to both of the first region and the second region at different positions of the scaffold, as schematically depicted in FIGS. 1D-1D-1M. The scaffold refers to a protein or peptide, when incorporated into a protein or peptide of interest, improves properties of the total protein such as PK, enhances stability and/or in vivo half-life, or enhances productivity of the protein. In some embodiments, blood-brain barrier (BBB) permeable scaffold may optimize BBB permeability and/or optimizing distribution of the protein of interest (e.g., the fusion molecules according to the present disclosure) in the brain.

The scaffold may include, but not limited thereto, a single chain Fc region with reduced or abolished Fc receptor binding affinity, a multimer Fc region with reduced or abolished Fc receptor binding affinity, an antibody without variable region, or an Fc-hinge region with reduced or abolished Fc receptor binding affinity. In some embodiments, the scaffold may include albumin. The first region may be linked or fused to one position of the scaffold and the second region may be linked or fused to another position of the scaffold. The link or fusion between the first/the second region and the scaffold may be a direct bond or via a linker described above. It is known that heavy chain constant region or Fc region may contain mutations selected among T250Q/M428L; M252Y/S254T/T256E+H433K/N434F; E233P/L234V/L235A/G236A+A327G/A330S/P331S; E333A; S239D/A330L/I332E; P257I/Q311; K326W/E333S; S239D/I332E/G236A; N297A; L234A/L235A; N297A+M252Y/S254T/T256E; K322A and K444A, wherein the numbering is according to the EU numbering (Edelman, G. M. et al., Proc. Natl. Acad. USA, 63, 78-85 (1969); imgt.org/IMGTSscientificChart/Numbering/HuIGHGnber.html#refs).

The fusion molecules according to aspects of the present disclosure may have a structure as schematically shown in non-limiting exemplary illustrations of, for example, FIG. 1B and FIGS. 1C through 1D-1M.

Structures of non-limiting exemplary fusion molecules (including signal sequences, optional linkers, and tags) are illustrated in Tables 3 and 5-13 and SEQ ID NOs: 136, 138, 140, 142, 150, 152, 154, 156, 158, 162-169. Non-limiting exemplary fusion molecules may comprise, consist of, or consist essentially of the fusion molecule (or binding molecule) of the sequence of amino acid residues 31-871 of SEQ ID NO: 136, amino acid residues 31-687 of SEQ ID NO: 138, amino acid residues 31-697 of SEQ ID NO: 140, amino acid residues 31-684 of SEQ ID NO: 150, amino acid residues 31-676 of SEQ ID NO: 152, amino acid residues 25-673 of SEQ ID NO: 154, amino acid residues 22-662 of SEQ ID NO: 156, or amino acid residues 22-885 of SEQ ID NO: 158, or a sequence having at least 90% sequence identity thereto, wherein a different linker can be used in place the linkers as shown in Tables 3 and 5-Another aspect of the present disclosure provides a nucleic acid molecule encoding the fusion molecule, and an expression vector containing the same.

As described above, the nucleic acid molecule sequence encoding the fusion molecule may be mutated by substitution, deletion, insertion, or a combination thereof, of one or more nucleotide residues, as long as it encodes a protein having an activity equivalent thereto. The nucleic acid molecule sequence encoding the fusion molecule may be isolated from nature or artificially produced through synthesis or genetic recombination. The nucleic acid molecule

sequence encoding the fusion molecule is operatively linked to an expression vector capable of expressing the same.

The term "expression vector" is a vector capable of expressing a protein or RNA of interest by introducing a nucleic acid sequence encoding a gene of interest into a suitable host cell, and refers to a gene construct containing essential regulatory elements operably linked to express the gene insert. Such expression vectors include all vectors such as plasmid vectors, cosmid vectors, bacteriophage vectors, and viral vectors.

A suitable expression vector has expression control elements such as a promoter, a start codon, a stop codon, a polyadenylation signal and an enhancer. The start codon and the stop codon are generally considered to be part of a nucleic acid sequence encoding a protein, and the sequence encoding the protein is designed to be in frame so as to be operable in the vector. The promoter may be constitutive or inducible. In addition, a conventional expression vector contains a selectable marker. Operational linkage with the expression vector can be performed using genetic recombination techniques well known in the art, and site-specific DNA cleavage and ligation can be performed using enzymes generally known in the art.

The expression vector may preferably be configured to express the fusion molecule in a host cell for isolation and purification of the fusion molecule or such that the vector may be introduced into a cell in vivo and the corresponding cell may express and secrete the fusion molecule. For the purpose of introducing into cells in vivo, the vector may preferably be a non-integrating vector, that is, a vector that is not integrated into the genome of a host cell.

Still another aspect of the present disclosure provides a cell expressing the fusion molecule.

The cells may be transformed to contain the nucleic acid molecule or an expression vector containing the same, and the "transformation" may be performed using suitable standard techniques selected depending on the host cell as known in the art, including any method of introducing the nucleic acid molecule into an organism, cell, tissue, or organ. These methods include, but are not limited to, electroporation, protoplast fusion, calcium phosphate (CaPO_4) precipitation, calcium chloride (CaCl_2) precipitation, agitation using silicon carbide fibers, agrobacterium-mediated transformation, PEG-, dextran sulfate-, lipofectamine-, and desiccation/inhibition-mediated transformation methods.

Examples of the host cells include, but are not limited to, prokaryotic host cells such as *Escherichia coli*, *Bacillus subtilis*, *Streptomyces*, *Pseudomonas* (e.g., *Pseudomonas putida*), *Proteus mirabilis*, or *Staphylococcus* (e.g., *Staphylococcus carnosus*). Other examples of the host cell include fungal cells such as *Aspergillus*, yeast cells, including *Pichia pastoris*, *Saccharomyces cerevisiae*, *Schizosaccharomyces*, and *Neurospora crassa*, lower eukaryotic cells, or cells derived from higher eukaryotes including insect cells, plant cells, or mammalian cells.

After the fusion molecule is expressed in the cells, it may be isolated and purified using conventional biochemical isolation techniques, such as treatment with a protein precipitating agent (salting out method), centrifugation, sonication, ultrafiltration, dialysis, or various chromatography such as molecular sieve chromatography (gel filtration), adsorption chromatography, ion exchange chromatography, and affinity chromatography, which are generally used in combination in order to isolate proteins with high purity (Sambrook et al., Molecular Cloning: A laboratory Manual, 2nd Ed., Cold Spring Harbor Laboratory Press(1989); Deus-

cher, M., Guide to Protein Purification Methods Enzymology, Vol. 182. Academic Press. Inc., San Diego, CA (1990)).

Pharmaceutical Composition

Yet another aspect of the present disclosure provides a pharmaceutical composition for preventing or treating a disease caused by accumulation of the target substance in living tissue, the pharmaceutical composition containing the fusion molecule or the expression vector. Here, the composition may be administered topically to a site where the substance that causes the disease, that is, the target substance, accumulates.

A further aspect of the present disclosure provides the use of the fusion molecule for manufacture of a medicament for preventing or treating proteopathy.

The fusion molecule, which is an active ingredient in the pharmaceutical composition, is contained in a "pharmaceutically effective amount".

The pharmaceutical composition may be administered orally or parenterally, preferably parenterally. More preferably, it may be administered topically to a tissue in which the target substance to be cleared accumulates.

As used herein, the term "parenteral administration" includes subcutaneous injection, intravenous, intramuscular, intrasternal injection or infusion techniques.

When the pharmaceutical composition is prepared as an injectable formulation, it may be prepared as the injectable formulation a conventional method known in the art. The injectable formulation may be in a form dispersed in a sterile medium so that it may be administered directly to a patient or may be in a form that may be administered after being dispersed in distilled water for injection at an appropriate concentration.

When the pharmaceutical composition is formulated for oral administration, it may contain one or more carriers selected from among diluents, lubricants, binders, disintegrants, sweeteners, stabilizers, and preservatives, and may contain one or more additives selected from among flavorings, vitamins, and antioxidants.

Techniques necessary for formulation of the pharmaceutical composition, and pharmaceutically acceptable carriers, additives, etc. are widely known to those skilled in the art (see, for example, the Handbook of Pharmaceutical Excipients, 4th edition, Rowe et al., Eds., American Pharmaceutical Association (2003); Remington: The Science and Practice of Pharmacy, 20th edition, Gennaro, Ed., Lippincott Williams & Wilkins (2000); Remington's Pharmaceutical Sciences (19th ed., 1995)).

The appropriate dosage of the pharmaceutical composition may vary depending on factors such as formulation method, administration mode, patient's age, weight, sex, medical condition, diet, administration time, administration route, excretion rate, and response sensitivity. The dosage of the pharmaceutical composition of the present disclosure is 0.0001 to 1,000 $\mu\text{g/kg}$ body weight for an adult.

Advantageous Effects

The present disclosure relates to a fusion molecule having phagocytosis-inducing activity, which can solve the problem of tissue damage caused by activation of an inflammatory response, which occurs in the prior art. Accordingly, the fusion molecule is able to effectively clear abnormally accumulated substances such as beta-amyloid, tau, alpha-synuclein, huntingtin or prion protein, and thus may be used to prevent or treat diseases caused by these abnormally

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accumulated substances, for example, Alzheimer's disease, Parkinson's disease, Huntington's disease, or prion disease. The fusion molecule may be administered to a patient in the form of a purified fusion molecule or a gene therapy vector capable of expressing and secreting the fusion molecule when introduced into a cell.

However, it should be understood that effects of the present disclosure are not limited to the above effects, and include all effects that may be inferred from the configuration of the invention described in the detailed description or claims.

EXAMPLES

Hereinafter, the present disclosure will be described in more detail with reference to examples and experimental examples. However, the following examples and experimental examples are illustrative only, and the scope of the invention is not limited thereto.

Preparation Example 1. Preparation of Gas6-Based Fusion Molecule Having Beta-Amyloid Clearance Activity (I): Beta-Amyloid Binding Region in the Form of scFv

To prepare a beta-amyloid (A β)-specific chimeric phagocytosis inducer based on Gas6 protein, the Gla domain,

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which recognizes PS (phosphatidylserine) in apoptotic cells, was first removed, and a single-chain variable fragment (scFv) of aducanumab, an amyloid-specific antibody, was introduced at that position [α A β -Gas6(E)].

In addition, for the efficiency of protein production, the EGF repeat domain present in the internal residues of the Gas6 protein was also removed and an scFv of aducanumab was introduced at that position, thereby preparing α A β -Gas6 (FIG. 1B).

In addition, as controls for verifying beta-amyloid-specific binding of the scFv of aducanumab, α FITC-Gas6(E) and α FITC-Gas6, each introduced with an E2 scFv that selectively recognizes FITC, instead of the scFv of aducanumab, were prepared.

Table 3 below shows amino acid sequences related to the preparation of the fusion molecules, and Table 4 below shows nucleotide sequences related to the preparation of the fusion molecules (the underlined sequences are flag tags). In Table 3, information for sequences constituting the final binding molecules are included within parenthesis. The full length sequences contain, from the N-terminal to the C-terminal direction, as an example, signal sequence (SS), first region, linker (when applicable), second region, and flag or his tags, which are linked consecutively. The sequence identifiers are intended for the full length sequences.

TABLE 3

1. α A β -Gas6(E) (FLAG tag, Gla delete, G-/-) SEQ ID NO: 136
MAPSLSPGPAALRRAPQLLLLLLAECALA (SS)
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASLQSGV
PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKEIKRGGGSGGG
GSGGGGSEVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHVRQAPGKGLEWV
AVIWFDTGTTKYYTDSVKGRFTISRDNSKNTLYLQMNTLRAEDTAVYYCARDRGIGA
RRGPYYMDVWGKTTTVSS (Adu-scFv)
GGGGSGGGGS (Linker)
CINKYGSPTYKNSGFATCVQNLDPDQCTPNPCDRKGTQACQDLGMNFFCLCKAGWG
GRLCDKDVNECSQENGCLQICHNKPGSFHCSCHSGFELSSDGRTCQDIDECADSEA
CGEARCKNLPGSYSCLCDEGFAYSSQEKACRDVDECLQGRCEQVCVNSPGSYTCHC
DGRGGLKLSQDMDTCEIILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRL
VAEFDRTFDPEGILLFAGGHQDSTWIVLALRAGRLQLRLYNGVGRVTSSEGVINHH
GMWQTSVEELARNLVIKVNDAVMKIAVAGDLFQPERGLYHLNLTVGGIPTFHEKD
LVQPINPRLDGCMRSWNWLNGETTTIQTETVKVNTMRQCFSVTERGSFYPGSGFAFY
SLDYMRTPLDVGTSTWEVEVVAHIRPAADTGVLFWALWAPDLRAVPLSVALVDYH
STKKLKKQLVVLAVEHTALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSE
VSAALQERLAVLERHLRSPVLTFFAGGLPDVPVTSAPVTAIFYRGCMTELVNRRLLD
LDEAAAYKHSIDTAHSCPPVEPAAA (Gas6-Gla deleted)
QGSRADYKDHGDYKDHIDYKDDDDK* (FLAG)
2. α FITC-Gas6(E) (FLAG tag, Gla delete, G-/-) SEQ ID NO: 137
MAPSLSPGPAALRRAPQLLLLLLAECALA (SS)
QVQLVESGNNLVQPGGSLRLSCAASGFTFGSFSMSWVRQAPGGGLEWVAGLSARSS
LTHYADSVKGRFTISRDNANKSVYLMNSLRVEDTAVYYCARRSYDSSGYWGHFY
SYMDVWGQGLTVTVSGGGSGGGSGGGSSVLTQPSSVSAAPGQKVTISCSGSTS
NIQNNYVSWYQQHPGKAPKLMIVDSKRPSGVPDRFSGSKSGNSALDLSGLQSEDE
ADYYCAAWDDSLSEFLFGTGTCLTVLG (α FITC-scFv)
GGGGSGGGGS (Linker)
CINKYGSPTYKNSGFATCVQNLDPDQCTPNPCDRKGTQACQDLGMNFFCLCKAGWG
GRLCDKDVNECSQENGCLQICHNKPGSFHCSCHSGFELSSDGRTCQDIDECADSEA
CGEARCKNLPGSYSCLCDEGFAYSSQEKACRDVDECLQGRCEQVCVNSPGSYTCHC
DGRGGLKLSQDMDTCEIILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRL
VAEFDRTFDPEGILLFAGGHQDSTWIVLALRAGRLQLRLYNGVGRVTSSEGVINHH
GMWQTSVEELARNLVIKVNDAVMKIAVAGDLFQPERGLYHLNLTVGGIPTFHEKD
LVQPINPRLDGCMRSWNWLNGETTTIQTETVKVNTMRQCFSVTERGSFYPGSGFAFY
SLDYMRTPLDVGTSTWEVEVVAHIRPAADTGVLFWALWAPDLRAVPLSVALVDYH
STKKLKKQLVVLAVEHTALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSE
VSAALQERLAVLERHLRSPVLTFFAGGLPDVPVTSAPVTAIFYRGCMTELVNRRLLD
LDEAAAYKHSIDTAHSCPPVEPAAA (Gas6-Gla deleted)
QGSRADYKDHGDYKDHIDYKDDDDK* (FLAG)
3. α A β -Gas6 (FLAG tag, Gla EGF delete, GE-/-) SEQ ID NO: 138
MAPSLSPGPAALRRAPQLLLLLLAECALA (SS)
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASLQSGV
PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKEIKRGGGSGGG

TABLE 3-continued

GSGGGGSEVQLVESGGGVQPGRSRLRLSCAASGFAPSSYGMHWVRQAPGKGLEWV
 AVIWFDTGKKYYTDSVKGRFTISRDN SKNTLYLQMNTRLAEDTAVYYCARDRGIGA
 RRGPIYMDVWGKGTITVTVSS (Adu-scFv)
 GGGSGGGGS (Linker)
 DILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLELQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVI
 KVN RDAVMKIAVAGDLFQPERGLYHLNLT VGGIPFHEKDLVQPINPRLDGCMRSWN
 WLN GEDTTIQETVKV NTRMQCF SVTERGSFYPGSGFAPYSLDYMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHT
 ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTA FYRGCMTEVNRRLLDLDEAAYKHSDITAHSCPPV
 EPAAA (Gas6-Gla EGF deleted)
 QGSRADYKDHGDYKDHIDIDYKDDDDK* (FLAG)

4. α FITC-Gas6 (FLAG tag, Gla EGF delete, GE/-) SEQ ID NO: 139
 MAPSLSPGPAALRRAPQLLLLLLAAECALA(SS)
 QQVLVESGGNVLQPGGSLRLSCAASGFTFGSFSMSWVRQAPGGGLEWVAGLSARSS
 LTHYADSVKGRFTISRDN AKNSVYLQMN SLRVEDTAVYYCARRSYDSSGYWGHFY
 SYMDVWGGQTLTVTS GGGSGGGSGGGSGGGSSVLTQPSSVSAAPGQKVTISCSGSTS
 NIGNNVSWYQQHPGKAPKLMYDVSKRPSGVPDRFSGSKSGNSASLDISGLQSEDE
 ADYYCAAWDDSLSEFLFTGTGKLTVLG (aFITC-scFv)
 GGGSGGGGS (Linker)
 DILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLELQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVI
 KVN RDAVMKIAVAGDLFQPERGLYHLNLT VGGIPFHEKDLVQPINPRLDGCMRSWN
 WLN GEDTTIQETVKV NTRMQCF SVTERGSFYPGSGFAPYSLDYMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHT
 ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTA FYRGCMTEVNRRLLDLDEAAYKHSDITAHSCPPV
 EPAAA (Gas6-Gla EGF deleted)
 QGSRADYKDHGDYKDHIDIDYKDDDDK* (FLAG)

5. $\alpha\beta$ -Gas6 (HA tag, Gla EGF delete, GE/-) SEQ ID NO: 140
 MAPSLSPGPAALRRAPQLLLLLLAAECALA(SS)
 DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLSQGV
 PSRFSGSGSGTDFTLTITSSLPEDFATYYCQQSYSTPLTFGGGTKVEIKRGGSGSGG
 GSGGGGSEVQLVESGGGVQPGRSRLRLSCAASGFAPSSYGMHWVRQAPGKGLEWV
 AVIWFDTGKKYYTDSVKGRFTISRDN SKNTLYLQMNTRLAEDTAVYYCARDRGIGA
 RRGPIYMDVWGKGTITVTVSS (Adu-scFv)
 GGGSGGGGS (Linker)
 DILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLELQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVI
 KVN RDAVMKIAVAGDLFQPERGLYHLNLT VGGIPFHEKDLVQPINPRLDGCMRSWN
 WLN GEDTTIQETVKV NTRMQCF SVTERGSFYPGSGFAPYSLDYMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHT
 ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTA FYRGCMTEVNRRLLDLDEAAYKHSDITAHSCPPV
 EPAAA (Gas6-Gla EGF deleted)
 GSGSGSGSGSYYPYDVPDYA* (HA)

6. Lentiviral Aducanumab IgG_IRES ZsGreen deleted SEQ ID NO: 141
 MGWSCIIILFLVATATG (SS)
 DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLSQGV
 PSRFSGSGSGTDFTLTITSSLPEDFATYYCQQSYSTPLTFGGGTKVEIKRRTVAAPSV
 FIFPPSDEQLKSGTASVCLLNNFYPREAKVQWVDNALQSGNSQESVTEQDSKDST
 YLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (Adu-Light chain)
 RKRKSGEGRGSLLTCGDVEENPGP (T2A)
 MGWSCIIILFLVATATG (SS)
 EVQLVESGGGVQPGRSRLRLSCAASGFAPSSYGMHWVRQAPGKGLEWVAVIWFDG
 TKKYYTDSVKGRFTISR
 DNSKNTLYLQMNTRLAEDTAVYYCARDRGIGARRGPIYMDVWGKGTITVTVSSA
 STKGPSVFP LAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVL
 QSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSSDKTHTSPPCPA
 PELLGGPSVFLFPPPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN
 AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKCKVSNKALPAPIEKTIKAKG
 QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPV
 LDSGGSFFLYSKLTVDKSRWQQGNV FSCVMHEALHNHYTQKSLSLSPGK* (Adu-
 Heavy chain)

7. Endogenous full sequence human Gas6 protein SEQ ID NO: 142
 MAPSLSPGPAALRRAPQLLLLLLAAECALALLLPAREATQFLRPRQRRAPQVFEEAK
 QGHLERECVEELCSREEAREVFENDPETDYFYPRYLDCINKYGSPTYKNSGFATCVQ
 NLPDQCTPNPCDRKGTQACQDLMGNFFCLCKAGWGGRLCDKDVNECSQENGCGCL
 QICHNKP GSFHCSCHSGFELS DGRTCQDIDECADSEACGEARCKNLPGSYSCLCDE
 GFAYSSQEKACRDVDECLQGRCEQVCVNSPGSYTCHCDGRGGLKLSQDMDCEDIL
 GPFPFSVAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFAGG
 HQDSTWIVLALRAGRLELQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVIK
 NRDAVMKIAVAGDLFQPERGLYHLNLT VGGIPFHEKDLVQPINPRLDGCMRSWNW
 LNGEDTTIQETVKV NTRMQCF SVTERGSFYPGSGFAPYSLDYMRTPLDVGTESTWE
 VEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHTA

TABLE 3-continued

LALMEIKVCDGQEHVVTVSLRDGEATLEVDGTRGQSEVSAAQLQERLAVLERHLRS
PVLTFAGGLPDVPVTSAPVTAIFYRGCMTELVNRRLLDLDEAAYKHS DITAHSCPPVE
PAAA (full-length human Gas6)
QGSRADYKDHGDYKDHDIDYKDDDDK* (FLAG)

TABLE 4

1. $\alpha\beta$ -Gas6(E) (FLAG tag, Gla delete, G-/-) SEQ ID NO: 143
ATGGCCCCCTTCGCTCTCGCCCCGGCCCCGCCCTGCGCCGCGCGCCGAGCTGC
TGCTGCTGCTGCTGGCCGCGGAGTGC CGCTTGCC (SS)
GACATTAGAGTACTCAATCTCTAGCTCTCTGAGCGCTCCGTTGGAGATAGAG
TCACTATTACCTGCAGAGCCAGCCAATCCATCAGCTCTTATCTAAATTGGTACCA
ACAGAAAGCCCGGCAAGCGCCAAAGCTGCTCATCTACGCTGCAAGCTCCTTACA
GAGCGGAGTACCCAGCAGATTCTCAGGCAGTGGCAGTGGGACTGACTTCACATT
GACGATTAGCTCTCTGCAGCCTGAAGACTTTGCCACATACTATTGTCAGCAGAGC
TATAGCACCCCGCTGACGTTTGGAGGCGGAACCTAAGGTGGAAATCAAGAGAGG
AGGCGGGGCTCCGCGGGGGTGGCTCGGGGGAGGAGGCTCAGAGGTTGAGC
TTGTGCGAGTCTGGGGGGGAGTCGTTGAGCCAGGTAGAAGCCTCAGACTGAGCT
GTGCGCAAGTGGGTTTGCTTTTTCATCTTACGGTATGCACTGGGTGAGCAGGC
TCCTGGCAAGGACTCGAGTGGGTGCTGTAATATGGTTCGATGGTACAAAGAA
ATACTATACCGATAGTGTGAAAGGAAGATTACCATTTACGAGACAACAGTAA
AAATACCTTGTACCTTCAGATGAACACCTGAGAGCAGAAGACAGCCGTGTA
CTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCTATTATATGGA
TGTGTGGGGAAGGGAACAACAGTACTGTGAGCTCT (Adu-scFv)
GGCGGGGCGGCGAGCGGCGGCTGGCAGC (Linker)
TGCATCAACAAGTATGGGTCTCCGTACACCAAAAACCTCAGGCTTCGCCACCTGC
GTGCAAAACCTGCCTGACCAGTGCACGCCCAACCCCTGCGATAGGAAGGGGACC
CAAGCCTGCCAGGACCTCATGGGCAACTTCTTCTGCCTGTGTAAGCTGGCTGGG
GGGCGGGCTCTGCGACAAGATGTCAACGAATGCAGCCAGGAGAACGGGGG
TGCCCTCAGATCTGCCACAACAAGCCGGTAGCTTCCACTGTTCTGCCACAGCG
GCTTCGAGCTCTCCTCTGATGGCAGGACCTGCCAAGACATAGACAGTGCAGCAG
ACTCGGAGGCTCTGCGGGAGGCGCGCTGCAAGAACCTGCCCGGCTCCTACTCCT
GCCTCTGTGACGAGGGCTTTGCGTACAGCTCCAGGAGAAGGCTTGCCGAGATG
TGGACGAGTGTCTGAGGGCGCTGTGAGCAGGTCTGCGTGAACCTCCCGAGGGA
GCTACACCTGCCACTGTGACGGGCTGGGGGCTCAAGCTGTCCAGGACATGG
ACACCTGTGAGGACATCTTGCCTGCGTGCCTTTCAGCGTGGCCAAGAGTGTGA
AGTCTCTGTACTTGGGCGGATGTTTCACTGGGACCCCGTGTCCGACTGCGCTT
CAAGAGGCTGCAGCCCAACAGGCTGGTAGCTGAGTTGACTTCCGACCTTTGA
CCCCGAGGGCATCCTCCTTTTGCCGGAGGCCACAGGACAGCACCTGGATCGT
GCTGGCCCTGAGAGCCGCGCGCTGGAGCTGCAGCTGCGCTACAAACGGTGTCCG
CCGTGTACCAAGCAGCGGCCCGGTCTCATCAACCATGGCATGTGGCAGACAATCTC
TGTTGAGGAGCTGGCGCGGAATCTGGTCTCAAGGTCAACAGGGATGCTGTCTCAT
GAAATCGCGGTGGCGGGGACTTGTTCACACCGGAGCGAGGACTGTATCATCT
GAACCTGACCTGGGAGGTATTCCCTTCCATGAGAAGGACCTCGTGCAGCCTAT
AAACCTCGTCTGGATGGCTGCATGAGGAGCTGGAACCTGGCTGAACGGAGAAGA
CACCACCATCCAGGAAACGGTGAAAGTGAACACGAGGATGCAGTCTTCTCGGT
GACGAGAGAGGCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCCTGGA
CTACATGCGGACCCCTCTGGAGCTCGGACTGAATCAACCTGGGAAGTAGAAGT
CGTGCTCATACCCCGCAGCGCAGACAGGCGTGTGTTTGCCTCTGGG
CCCCGACCTCCGTGCGTGCCTCTCTCTGTGGCACTGGTAGACTATCACTCCACG
AAGAACTCAAGAAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCCTTGGCC
CTAATGGAGATCAAGGTCTGCGACGGCCAAGAGCACGTGGTCAACGCTCTCGCTG
AGGACGCTGAGGCCACCTGGAGGTGGACGGCACAGGGGCCAGAGCGAGGT
GAGCGCCGCGCAGCTGCAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGGA
GCCCCGTGCTACCTTTGCTGGCGGCTGCCAGATGTGCCGTGACTTCAGCGCC
AGTCAACGCGTTCTACCGCGGCTGCATGACACTGGAGGTCAACCGGAGGCTGCT
GGACCTGGACGAGGCGGCGTACAAGCACAGCGACATCACGGCCCACTCCTGCC
CCCCGTGGAGCCCGCGCAGCC (Gas6-Gla deleted)
caagGATCCCGGCTGACTACAAAGACCATGACGGTGATTATAAAGATCATGACA
TCGACTACAAGGATGACGATGACAAGTga (FLAG)

2. α FITC-Gas6(E) (FLAG tag, Gla delete, G-/-) SEQ ID NO: 144
ATGGCCCCCTTCGCTCTCGCCCCGGCCCCGCCCTGCGCCGCGCGCCGAGCTGC
TGCTGCTGCTGCTGGCCGCGGAGTGC CGCTTGCC (SS)
CAGGTTACAGTGGTTGAGAGCGGAGGCAATCTGGTTACGCCCCTGGTAGTCTG
CGTCTGTCTTGTGCGGCTCAGGGTTCACTTTCGGTAGTTTTCATGAGCTGGG
TCCGTGAGGCACAGGCGGTGGGCTGGAATGGGTGGCAGGCTGTCTGCACGTA
GCTCCTTGACCCATATGCAGATAGTGTAAAGGGCGGTTCACAATTTTACGCG
ACAACGCTAAGAATAGCTCTACCTGCAATGAACCTCCTGCGGGTCGAGGATA
CCGCACTGTATTACTGCGCTCGCCGTTCTTATGACTCTAGTGGATACTGGGGCCA
TTTTTATAGCTACATGGATGTGTGGGACAGGGCACTCTGGTGACCGTTTCCGGA
GGCGGTGGGTCTGGAGGCGGTGGGAGTGGAGCGGTGGGTCAAGCGTTCTGACC
CAGCGCTCTCTGTGAGCGCGCGCCAGGCCAGAAAGTGACAATTTCTGTCTG
GAAGTACTTCAACATCGGCAACAATTATGTTTCTGTGATCAGCAGCACCCGG

TABLE 4-continued

GCAAAGCGCCCAAGCTGATGATTATGATGTGTCTAAACGTCCAAGTGGTGTTC
TGACCGGTTACGCGTTTCCAAGTCTGGGAATAGTGCCTCACTGGACATCTCAGG
CTGTCAAAGCGAAGATGAGGCGGACTATTACTGCGCAGCTTGGGATGACAGCCT
GTCCGAATTTCTGTTTCGGCACCGGACAAAGCTGACCGTGCTGGGC (aFITC-scFv)
GGCGGGGGCGGAGCGGCGGCGGTGGCAGC (Linker)
TGCATCAACAAGTATGGGTCTCCGTACACCAAAAACTCAGGCTTCGCCACCTGC
GTGCAAAACCTGCCTGACCAAGTGACGCGCCCAACCCCTGCGATAGGAAGGGGACC
CAAGCCTGCCAGGACCTCATGGGCAACTTCTTCTGCCTGTGTAAAGCTGGCTGGG
GGGGCCGGCTCTGCGACAAAGATGTCAACGAATGCAGCCAGGAGAACGGGGGC
TGCCCTCAGATCTGCCACAACAAGCGGGTAGCTTCCACTGTTCTGCCACAGCG
GCTTCGAGCTCTCCTCTGATGGCAGGACCTGCCAAGACATAGACGAGTGGCGAG
ACTCGGAGGCTGCGGGGAGGCGCGCTGCAAGAACCTGCCCGGCTCCTACTCCT
GCCTCTGTGACGAGGGCTTTGCGTACAGCTCCAGGAGAAGGCTTGCCGAGATG
TGGACGAGTGTCTGACGGGCGCTGTGAGCAGGTCTGCGTGAACCTCCCGAGGA
GCTACACCTGCCACTGTGACGGGCGTGGGGGCTCAAGCTGTCCAGGACATGG
ACACCTGTGAGGACATCTTGCCGTGCGTGCCCTTCAAGCTGGCCAGAGGTGTGA
AGTCCTTGTAACCTGGGCGGATGTTCAAGTGGGACCCCGTGATCCGACTGCGCTT
CAAGAGGCTGCAGCCACCAGGCTGGTAGCTGAGTTGACTTCCGGACCTTTGA
CCCCGAGGACATCCTCCTTTGCGGAGGCGCCAGGACAGCAGCTGGATCGT
GCTGCGCTTGAGAGCGGCGCGCTGGAGCTGCAGCTGCGCTACACCGGTGTGCG
CCGTGTCAACAGCAGCGGCGCGGTGCATCAACCATGGCATGTGGCAGACAATCTC
TGTTGAGGAGCTGGCGCGGAATCTGGTGCATCAAGGTCAACAGGGATGCTGTCT
GAAATCGCGGTGGCGGGGACTTGTTCACCGGAGCGAGGACTGTATCATCT
GAACCTGACCGTGGGAGGTATTCCTTCCATGAGAAGGACCTCGTGACGCTAT
AAACCTCGTCTGGATGGCTGCATGAGGAGCTGGAACCTGGCTGAACGGAGAAGA
CACCACCATTCAGGAAACGGTGAAGTGAACACGAGGATGCAGTCTTCTCGGT
GACGAGAGAGGCTCTTCTACCCCGGAGCGGCTTCGCCCTTCTACAGCCTGGA
CTACATGCGGACCCCTCTGGACGTGCGGACTGAATCAACCTGGGAAGTGAAGT
CGTGCTCATACCCGCCAGCGCAGACACAGGCGTGCTGTTTGCCTCTCGGGC
CCCCGACCTCCGTGCGGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACG
AAGAACTCAAGAAGCAGCTGGTGGTCTGGCCGTGGAGCATACGCGCTTGCC
CTAATGGAGATCAAGGTGTGCGACGGCCAGAGCACGTGGTCAACGCTCTCGCTG
AGGGACCGTGAGGCCACCTGGAGGTGGACGGCACAGGGGCCAGAGCGAGGT
GAGCGCGCGCAGCTGCAGGAGAGGCTGGCGGTGCTCGAGAGGCACCTGCGGA
GCCCCGTGCTCACCTTTGCTGGCGGCTGCGCAGATGTCCCGGTGACTTCAGCGCC
AGTCACCGCGTCTACCGCGGTGCATGACACTGGAGGTCAACCGGAGGCTGCT
GGACCTGGAGCGGCGGCGGTACAAGCACAGCGACATCACGGCCCATCTGCCCC
CCCCGTGGAGCCCGCGCAGCC (Gas6-Gla deleted)
caagGATCCCGGGCTGACTACAAAGACCATGACGGTGATTATAAGATCATGACA
TCGACTACAAGGATGACGATGACAAGtga (FLAG)

3. $\alpha\beta$ -Gas6 (FLAG tag, Gla EGF delete, GE-/-) SEQ ID NO: 145
ATGGCCCCCTTCGCTCTCGCCCCGGGCGCGCCCTGCGCGCGCGCGCGAGCTGC
TGCTGCTGCTGCTGGCCGCGGAGTGCGCGCTTGCC (SS)
GACATTCAAGTACTCAATCTCCTAGCTCTCTGAGCGCTCCGTTGGAGATAGAG
TCACTATTACCTGCAGAGCCAGCCAAATCCATCAGCTCTTATCTAAATTGGTACCA
ACAGAAAGCCCGCAAGCGCCAAAGCTGCTCATCTACGCTGCAAGCTCCTTACA
GAGCGGAGTACCCAGCAGATTCTCAGGCACTGGCAGTGGGACTGACTTCACTT
GACGATTAGCTCTCTGAGCCTGAAGACTTTGCCACATACTATTGTGACGAGAGC
TATAGCACCCGCTGACGTTTGGAGGCGGAACTAAGGTGGAATCAAGAGAGG
AGGCGGGGGCTCCGCGGGGGTGGCTCGGGGGGAGGAGGCTCAGAGGTTGAGC
TTGTGAGTCTGGGGGGGGAGTCTTACGCCAGGTAGAAGCCTCAGACTGAGCT
GTGCGCAAGTGGGTTTGCTTTTTCATCTTACGGTATGCACTGGGTGAGACAGGC
TCCTGGCAAGGACTCGAGTGGGTGCTGTAATATGGTTCGATGGTACAAAGAA
ATACTATACCGATAGTGTGAAGGAAGATTCAACATTTACAGAGACAACAGTAA
AAATACCTTTGATCTTCAAGTGAACACCTGAGAGCAGAAGACAGCCGTGTA
CTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCTATTATATGGA
TGTGTGGGGGAAGGGAACAACAGTGAAGTGTGAGCTCT (Adu-scFv)
GGCGGGGGCGGAGCGGCGGCGGTGGCAGC (Linker)
GACATCTTGCCGTGCGTGCCCTTCAAGCTGGCCAAAGAGTGTGAAGTCTTGTACC
TGGGCGGGATGTTCAAGTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
AGCCACCAAGGCTGGTAGCTGAGTTTGAAGTCCCGGACCTTGAACCCGAGGGCA
TCCTCCTCTTTGCGGAGGCGCCAGGAGCAGCAGCTGGATCGTGTGGCCCTGAG
AGCGGGCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCGGTGCACAG
CAGCGGCGCGGTGCATCAACCATGGCATGTGGCAGACAATCTCTGTGAGGAGCT
GGCGCGGAATCTGGTGCATCAAGGTCAACAGGGATGCTGTGATGAAAATCGCGGT
GGCGGGGAGCTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTGAGCGT
GGGAGGTATTCCTTCCATGAGAAGGACCTCGTGAGCCTATAAACCCCTCGTCTG
GATGGCTGCATGAGGAGCTGGAACCTGGCTGAACGGAGAAGACACCAACATCCA
GGAAACGGTGAAGTGAACACGAGGATGCAGTGTCTTCTCGGTGACGGAGAG
GCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCTGGACTACATGCGGAG
CCCTCTGAGCTCGGGACTGAATCAACCTGGGAAGTAGAAGTCTGGCTCATCAT
CGCCCGGAGCGCAGACACAGGCGTGTGTTTGGCTCTGGGCCCCCGACCTCCGT
GCCGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
AAGCAGCTGGTGGTCTGGCGTGGAGCATACGGCTTGGCCCTAATGGAGATC
AAGGTCTGCGACGGCCAAGAGCACGTGGTCAACGCTCTGCTGAGGAGCGGTGAG
GCCACCTGGAGGTGACGGCACAGGGGCGAGAGCGAGGTGAGCGCGCGCA
GCTGAGGAGAGGCTGGCGGTGCTGAGAGGACCTGCGGAGCCCCGTGCTCAC
CTTGTGCGCGCTGCGCAGATGTGCGGTGACTTCAGCGCCAGTCAACCGGCTT
TACCGCGGTGCATGACACTGGAGTCAACCGGAGGCTGCTGGACCTGGACGAG

TABLE 4-continued

CGGGCGTACAAGCACAGCGACATCACGGCCCACTCTGCCCCCGTGGAGCCC
 GCCGCAGCC (Gas6-Gla EGF deleted)
 caaGGATCCCGGGCTGACTACAAAGACCATGACGGTGATTATAAAGATCATGACA
 TCGACTACAAGGATGACGATGACAAGTGA (FLAG)

4. α FITC-Gas6 (FLAG tag, Gla EGF delete, GE-/-) SEQ ID NO: 146

ATGGCCCCCTTCGCTCTCGCCCCGGGCGCGCCCTGCGCCGCGCGCCGAGCTGC
 TGCTGCTGCTGCTGGCCGCGAGTGCGCGCTTGCC (SS)
 CAGGTTACAGTGGTTGAGAGCGGAGGCAATCTGGTTACAGCCCGTGGTAGTCTG
 CGTCTGTCTTGTGCGGCGTCAGGGTTCACTTTCGGTAGTTTTTCAATGAGCTGGG
 TCCGTGAGGCACCGCGGTGGGCTGGAATGGGTGGCAGGTCTGTCTGCACGTA
 GCTCCCTGACCCACTATGCAGATAGTGTAAAGGGCGGTTCACAATTTACGCG
 ACAACGCTAAGAATAGCGTCTACCTGCAATGAACTCCCTGCGGGTCGAGGATA
 CCGCAGTGATTACTGCGCTCGCCGTTCTATGACTCTAGTGGATACTGGGGCCA
 TTTTATAGCTACATGGATGTGTGGGACAGGGCACTCTGGTGACCGTTTCCGGG
 GGCGGTGGGTCTGGAGGCGGTGGGAGTGGAGGCGGTGGGTCAAGCGTTCTGACC
 CAGCGCTCTCTGTGAGCGCCGCGCAGGCCAGAAAGTGACAATTTCTGTCTGTG
 GAAGTACTTCAAACATCGGCAACAATTATGTTTCTGGTATCAGCAGCACCCGG
 GCAAAGCGCCCAAGCTGATGATTATGATGTGTCTAAACGTCCTCAAGTGGTGTCC
 TGACCGGTTACGCGGTTCCAACTCTGGGAATAGTGCCTCACTGGACATCTCAGG
 CCTGCAAAAGCAGATGAGGCGGACTATTACTGCGCAGCTTGGGATGACAGCCT
 GTCCGAATTTCTGTTTCGCGCACCGGACAAAGCTGACCGTGCTGGGC (aFITC-scFv)
 GGCGGGGGCGGCGAGCGGCGCGGTGGCAGC (Linker)
 GACATCTTGCCGTGCGTGCCCTTCAGCGTGGCCAAGAGTGTGAAGTCCTTGATCC
 TGGGCGGATGTTTCACTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
 AGCCACCAAGGCTGGTAGCTGAGTTTGAATTCGCGACCTTTGACCCCGAGGGCA
 TCCTCCTCTTTGCGGAGGCGCACAGGACAGCACCTGGATCGTGTGGCCCTGAG
 AGCCGGCCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCCGTGTACACG
 CAGCGGCCCGGTCACTCAACCATGGCATGTGGCAGACAATCTCTGTGAGGAGCT
 GGCGCGGAATCTGGTCACTAAGGTCAACAGGGATGCTGTGATGAAAATCGCGGT
 GGCGGGGAGCTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTCACCGT
 GGGAGGTATTCCTTCCATGAGAAGGACCTCGTGAGCTATAAACCTCGTCTG
 GATGGCTGTATGAGGAGCTGGAATGGCTGAACGAGAGAAGACACCCATCCA
 GGAAACGGTGAAAGTGAACACGAGGATGCAGTGCTTCTCGGTGACGGAGAGAG
 GCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCTGGACTACATGCGGAC
 CCCTCTGAGCTCGGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT
 CCGCCGAGCCGAGACACAGGCGTGCTGTTTGGCTCTGGGCCCCCGACCTCCGT
 GCCGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
 AAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCCTTGCCCTAATGGAGATC
 AAGGTCTGCGACGGCCAAAGAGCACGTGGTCAACCGTCTCGCTGAGGAGCGGTGAG
 GCCACCTTGAGGTGGACGGCACAGGGGCGAGAGCGAGGTGAGCGCCGCGCA
 GCTGCAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGGAGCCCCGTGCTCAC
 CTTTGGCGGCGCTGCGCAGATGTGCGGTGACTTCAGCGCCAGTCAACCGGCTT
 TACCGCGGTGCTGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
 GCGGCGTACAAGCACAGCGACATCACGGCCCACTCTGCCCCCGTGGAGCCC
 GCCGCAGCC (Gas6-Gla EGF deleted)
 caaGGATCCCGGGCTGACTACAAAGACCATGACGGTGATTATAAAGATCATGACA
 TCGACTACAAGGATGACGATGACAAGTga (FLAG)

5. α β -Gas6 HA tag (Gla EGF delete, GE-/-) SEQ ID NO: 147

ATGGCCCCCTTCGCTCTCGCCCCGGGCGCGCCCTGCGCCGCGCGCCGAGCTGC
 TGCTGCTGCTGCTGGCCGCGAGTGCGCGCTTGCC (SS)
 GACATTCAGATGACTCAATCTCTAGCTCTCTGAGCGCTCCGTTGGAGATAGAG
 TCATATTACCTGCAGAGCCAGCCAATCCATCAGCTCTTATCTAAATGGTACCA
 ACAGAAAGCCCGCAAGCGCCAAAGCTGCTCATCTACGCTGCAAGCTCTTACA
 GAGCGAGGTACCCAGCAGATTCTCAGGCACTGGCAGTGGGACTGACTTCACATT
 GACGATTAGCTCTCTGCAGCTGAAGACTTTGCCACATACTATTGTGACGAGAGC
 TATAGCACCCGCTGACGTTTGGAGGCGGAATAAGGTGGAAATCAAGAGAGG
 AGGCGGGGGCTCCGGCGGGGGTGGCTCGGGGGGAGGAGGCTCAGAGGTTACAG
 TTGTGAGTCTGGGGGGGAGTCGTTACGCGAGGTAGAAGCTCAGACTGAGCT
 GTGCCGCAAGTGGGTTGCTTTTTCATCTTACGGTATGCATGGGTGAGACAGGC
 TCCTGGCAAAGGACTCGAGTGGGTGCTGTAATATGGTTCGATGGTACAAAGAA
 ATACTATACCGATAGTGTGAAGGAAGATTACCATTTACGAGACAACAGTAA
 AAATACCTTGTACCTTCAGATGAACACCTGAGAGCAGAAGACACAGCGGTGTA
 CTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCTATTATATGA
 TGTGTGGGGGAAGGGAACAACAGTGAAGTGTGAGCTCT (Adu-scFv)
 GGCGGGGGCGGACGCGCGGCGGTGGCAGC (Linker)
 GACATCTTGCCGTGCGTGCCCTTCAGCGTGGCCAAGAGTGTGAAGTCCTTGATCC
 TGGGCGGATGTTTCACTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
 AGCCACCAAGGCTGGTAGCTGAGTTTGAATTCGCGACCTTTGACCCCGAGGGCA
 TCCTCCTCTTTGCGGAGGCGCACAGGACAGCACCTGGATCGTGTGGCCCTGAG
 AGCCGGCCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCCGTGTACACG
 CAGCGGCCGGTCACTCAACCATGGCATGTGGCAGACAATCTCTGTTGAGGAGCT
 GGCGCGGAATCTGGTCACTAAGGTCAACAGGGATGCTGTGATGAAAATCGCGGT
 GGCGGGGAGCTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTGACCGT
 GGGAGGTATTCCTTCCATGAGAAGGACCTCGTGAGCCTATAAACCTCGTCTG
 GATGGCTGTATGAGGAGCTGGAATGGCTGAACGAGAGAAGACACCCATCCA
 GGAAACGGTGAAAGTGAACACGAGGATGCAGTGCTTCTCGGTGACGGAGAGAG
 GCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCTGGACTACATGCGGAC
 CCCTCTGAGCTGCGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT

TABLE 4-continued

CCGCCAGCCGACAGACACAGGCGTGCTGTTTGGCTCTGGGCCCCGACCTCCGT
 GCGGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
 AAGCAGCTGGTGGTCTGGCCCTGGAGCATACGGCCTTGGCCCTAATGGAGATC
 AAGGTCTGCGACGGCCAAAGAGCACGTGGTCACCGTCTCGCTGAGGGACGGTGAG
 GCCACCTTGAGGTGGACGGCACCAGGGGCCAGAGCGAGGTGAGCGCCGCGCA
 GCTGCAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGGAGCCCCGTGCTCAC
 CTTTGTGCGCGCCTGCCAGATGTGCCGGTGACTTCAGCGCCAGTCACCGCTTC
 TACCGCGGTGCTGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
 GCGGCGTACAAGCAGCGACATCAACGGCCCACTCTGCCCCCGTGGAGCCC
 GCCGAGCC (Gas6-Gla EGF deleted)
 GGCAGCGGCAGCGGAGCGGACGCGGACGTCAGCTACCCATACGATGTTCC
 AGATTACGCTTGA (HA)

6. Lentiviral Aducanumab IgG IRES Zsgreen deleted SEQ ID NO: 148

GGATCCATGGGCTGGTCTGCATCATCTGTTCTGGTGGCCACCGCCACCGGC
 (SS)
 GACATTCAGATGACTCAATCTCTAGCTCTCTGAGCGCTCCGTTGGAGATAGAG
 TCATTATTACCTGCAGAGCCAGCCAATCCATCAGCTCTTATCTAAATTGGTACCA
 ACAGAAGCCCGCAAAGCGCCAAAGCTGCTCATCTACGCTGCAAGCTCCTTACA
 GAGCGAGTACCACAGCAGATTCTCAGGCACTGGCAGTGGGACTGACTTCACATT
 GACGATTAGCTCTCTGCAGCCTGAAGACTTTGCCACATACTATTGTCAGCAGAGC
 TATAGCACCCCGCTGACGTTTGGAGGCGGAACCTAAGGTGGAATCAAGAGAAA
 ACGAAGTGTGGTGCACCATCTGTCTTCTCTCCGCCATCTGATGAGCAGTTG
 AAATCTGGAATGCTCTGTGCTGTGCTGCTGAATAACTTCTATCCAGAGAGG
 CCAAGTACAGTGGAGGTGGATAACGCCCTTCAATCGGGTAACTCCAGGAGA
 GTGTACAGAGCAGGACAGCAAGGACAGCACCACAGCTCAGCAGCAGCTG
 ACGCTGAGCAAAGCAGACTACGAGAAACACAAAGTCTACGCTGCGAAGTCAC
 CCATCAGGCGCTGTCTCGCCGCTCACAAGAGCTTCAACAGGGGAGAGTGT
 (Adu-Light chain)
 CGCAGAAACGCGGAAGCGGAGAGGGCAGAGGAAGTCTTCTAACATGCGGTGA
 CGTGAGGAGAATCCCGGCCCT (T2A)
 ATGGGCTGGTCTCTGCATCATCTGTCTCTGGTGGCCACCGCCACCGGC (SS)
 GAGGTTGAGCTTGTGAGTCTGGGGGGGAGTCGTTACGCGAGGTAGAAGCCTC
 AGACTGAGCTGTGCCGCAAGTGGTGTGCTTTTCTATCTTACGGTATGCACTGGG
 TGAGACAGGCTCTGGCAGGACTCGAGTGGGTGCTGTAATATGGTTCGATG
 GTACAAAGAAATACTATACCGATAGTGTGAAGGAAGATTACCATTTTCAGAG
 ACAACAGTAAAAATACTTGTACCTTCAGATGAACACCTGAGAGCAGAAGACA
 CAGCGGTGTACTACTGCGCCAGAGATAGAGGTATCGGAGCAAGCGTGGTCCCT
 ATTATATGGATGTGTGGGGGAAGGAACAACAGTACTGTGAGCTCTGCCTCCA
 CCAAGGGCCATCGGTCTTCCCCCTGGCACCTCTCTCCAAGAGCACCTCTGGGG
 CACAGCGGCCCTGGGCTGCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
 GTCGTGGAACCTCAGGCGCCCTGACCAGCGCGTGACACCTTCCCGGTGTCCT
 ACAGTCTCAGGACTCTACTCCTCAGCAGCGTGGTACTGTGCTCTAGCAGC
 TTGGGACCCAGACCTACATCTGCAACGTGAATCACAAGCCAGCAACACCAAG
 GTGGACAAGAAAGTTGAGCCCAATCTAGCGACAAAACCTCACACAAGCCACC
 GTGCCAGCACCTGAACTCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAA
 CCAAGGACACCTCATGATCTCCCGGACCTGAGGTACATGCGTGGTGGTG
 GACGTGAGCCACGAAGACCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTG
 GAGGTGCAATAATGCCAAGACAAAGCCGCGGAGGAGCAGTACAACAGCACGTA
 CCGTGTGGTCAAGCTCTCACCCTCTGCACAGGACTGGCTGAATGGCAAGGA
 GTACAAGTGCAAGGTCTCAACAAAGCCCTCCAGCCCCATCAGAAAACCAT
 CTCGAGGACCAAGGGCAGCCCCGAGAACCAAGGTGTACACCTGCCCCCATC
 CCGGATGAGCTGACCAAGAACAGGTGAGCTGACCTGCTGCTGCTCAAGGCTT
 CTATCCAGCGACATCGCCGTGGAGTGGAGAGCAATGGGCGCCGAGAGACA
 ACTACAAGACACCGCTCCGCTGCTGGACTCCGACGGCTCTTCTCTCTACAG
 CAAGCTCACCCTGGACAGAGCAGGTGGCAGCAGGGGAACGCTTCTCATGCTC
 CGTGATGCATGAGGCTCTGCACAACCACTACACGAGAGGCTCTCTCTGTC
 CCGGGTAAAtga (Adu-Heavy chain)

7. Endogenous full sequence human Gas6 protein SEQ ID NO: 149

ATGGCCCCCTTCGCTCTCGCCCGGGCCCGCCCTGCGCCGCGCGCCGAGCTGC
 TGCTGCTGCTGCTGCGCCGCGAGTGCGCGCTTGCCGCGCTGTTGCGCGCGCGG
 AGGCCACGCACTTCTGCGGCCAGGCAGCCGCGCCCTTTCAGTCTTCGAGG
 AGGCCAAGCAGGGCCACCTGGAGAGGGAGTGCCTGGAGGAGCTGTGACGCGC
 GAGGAGGCGCGGAGGTGTTGAGAACGACCCGAGACGGATTATTTTACCCA
 AGATACTTAGACTGCATCAACAAGTATGGGTCTCCGTACACCAAAAACCTAGGC
 TTCGCCACTGCTGTCAAAACCTGCTGACCAAGTGCACGCCCAACCCCTGCGAT
 AGGAAGGGGACCAAGCTGCTGACGACCTCATGGGCAACTTCTTCTGCTGTGT
 AAAGCTGGCTGGGGGGCCGGCTCTGCGACAAGATGTCAACGAATGCAGCCA
 GGAGAACGGGGCTGCTCCAGATCTGCCACAACAGCCGGTAGCTTCCACTG
 TTCTTGCCACAGCGCTTCTGAGCTCTCTCTGATGGCAGGACCTGCCAAGACATA
 GACGAGTGCAGACTCGAGGCTGCGGGGAGGCGCTGCAAGAACCTGCC
 CGGCTCTACTCTGCTGCTGTGACGAGGCTTTGCGTACAGCTCCAGGAGAAG
 GCTTGCCGAGATGTGACGAGTGTCTGACGGGCCGTGTGAGCAGGTCTGCGTG
 AACTCCCCAGGAGCTACACTGCCACTGTGACGGGCGTGGGGCCCTCAAGCTG
 TCCAGGACATGGACACTGTGAGGACATCTTGGCGTGCCTGCTTCAAGCTG
 GCCAAGAGTGTGAAGTCTTGTACCTGGGCCGATGTTAGTGGACCCCGTG
 ATCCGACTGCGCTTCAAGAGGTGCGAGCCACAGGCTGGTAGCTGAGTTTGAC
 TTCCGGACCTTGACCCCGAGGACATCTCTCTTTGCGGAGGCCACAGGACA
 GCACCTGGATCGTGTGCGCTGAGAGCCGCGCTGGAGCTGCAGTGCCT

TABLE 4-continued

ACAACGGTGTGCGCCGTGTACCAGCAGCGGCCCGGTCAACCATGGCATGT
GGCAGACAATCTCTGTTGAGGAGCTGGCGCGGAATCTGGTCATCAAGGTCAACA
GGGATGCTGTGATGAAATCGCGGTGGCCGGGGAATGTTCCAAACCGGAGCGAG
GACTGTATCATCTGAACCTGACCGTGGGAGGTATTCCTTCCATGAGAAGGACCT
CGTGACGCTATAAACCTCTGCTGGATGGTGCATGAGGAGCTGGAAGTGGCT
GAACGGAGAAGACACCCATCCAGGAAACGGTGAAAGTGAACACGAGGATGC
AGTGCTTCTCGGTGACGAGAGAGGCTCTTCTACCCCGGAGCGGCTTCGCTT
CTACAGCTTGGACTACATGCGGACCCCTCTGGACGTCGGGACTGAATCAACCTG
GGAAGTAGAAGTCGTGGCTCACATCCGCCAGCCGAGACACAGCGCTGCTGTT
TGCGCTCTGGGCCCCGACCTCCGTGCCGTGCTCTCTCTGTGGCACTGGTAGAC
TATCACTCTCACGAAGAACTCAAGAAGCAGCTGGTGGTCTTGGCGTGGAGCAT
ACGGCCTTGGCCCTAATGGAGATCAAGGTCTGCGACGGCCAGAGCAGTGGTC
ACCGTCTCGCTGAGGACGGTGAGGCCACCCCTGGAGGTGGACGGCAGCAGGGG
CCAGAGCAGGTGAGCGCCGCGCAGCTGAGGAGAGGCTGGCCGTGCTCGAGA
GGCACCTGCGGAGCCCCGTGCTACCTTTGCTGGCGGCTGCCAGATGTGCCGGT
GACTTCAGCGCCAGTACCGCGTCTTACCGCGGCTGCATGACACTGGAGGTCAA
CCGGAGGCTGTGGACCTGGACGAGCGCGGTACAGCACAGCGACATCACGG
CCCACTCTGCCCCCGGTGGAGCCCGCCGAGCC (full-length human Gas6)
caaggGATCCGGGCTGACTACAAAGACCATGACGGTGATTATAAAGATCATGACA
TCGACTACAAGGATGACGATGACAAGTGA (FLAG)

20

Preparation Example 2. Gas6-Based Fusion Molecule Targeting Tau

To prepare a tau-specific chimeric phagocytosis inducer
based on Gas6 protein, the Gla domain and the EGF repeat

domain were first removed, and a single-chain variable
fragment (scFv) of semorinemab, a tau-specific antibody
fragment; scFv), was introduced at that position (α Tau-
Gas6). Table 5 below shows the amino acid sequence and
nucleotide sequence of the chimeric phagocytosis inducer.

TABLE 5

1. α Tau-Gas6 (Tau-VL-G4Sx3-VH-LG-HA-T2A-EGFP, amino acid
sequence) SEQ ID NO: 150
MAPSLSPGPAALRRAPQLLLLLLAAECALA (SS)
DDVLTQTPLSLPVTGQPASISCRSSQSI VHSNGNTYLEWYLQKPGQSPQLLIYKVS
NRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQGSVPWTFGGGTKEIKGGG
GSGGGSGGGGSEVQLVESGGGLVQPGGSLRLSCAASGLIFRSGMSWVRQAPGKG
LEWVATINSGGTYTYYPDSVKGRFTISRDNKNTLYLQMNSLRRAEDTAVYYCANSYS
GAMDYWGQGLTVTVSS (aTau-scFv)
GGGSGGGGS (Linker)
DILPCVPFSVAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFAG
GHQDSTWIVLALRAGRLQLRYNGVGRVTS SGPVINHG MWQTSVEELARNLVIK
VNRDAVMKIAVAGDLFQPERGLYHLNLTVGGLPFHEKDLVQPINRDLGCMRSWNW
LNGEDTTIQETVKVNTRMQCFSVTERGSFYPSGFAFYSLDYMRTPLDVGTETSTWEV
EVVAHIRPAADTGVLFWALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHTALAL
MEIKVCDGQEHVVTVSLRDGEATLEVDGTRGQSEVSAALQERLAVLERHLRSPVL
TFAGGLPDVPVTSAPVTAIFYRGCMTELVNRRLLDLDEAAYKHSITAHSCPPVEPAA
A (Gas6-Gla EGF deleted)
GSGSGSGSGSGSYYPDVPDYA (HA)
EGRGSLTTCGDVEENPGP (T2A)
VSKGEELFTGVVPIVLVELDGDVNGHKFSVSGEGEGDATYGLTLKFICTTGKLPVWP
PTLVTTLYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAEVK
FEGDTLVNRIELKGIDFKEDGNILGHKLEYINYNHNHYIMADKQKNGIKVNFIRHN
IEDGSVQLADHYQNTPIGDGPVLLPDNHYLSTQSAISKDPNEKRDMVLLFEVTA
AGITLGMDELYK* (EGFP)

2. α Tau-Gas6 (Tau-VL-G4Sx3-VH-LG-HA-T2A-EGFP, nucleotide
sequence) SEQ ID NO: 151
ATGCCCCCTCGCTCTCGCCCGGCGCCCGCTGCGCCGCGCGCCGCGAGCTGC
TGCTGCTGCTGCTGGCCGCGAGTGCGCGCTTGCC (SS)
GACGATGTATTAACACAACTCCCTATCATTTGCCGGTGACCCCGGGCCAAACCA
GCTTCGATCAGCTGCCGTAGCTCTCAGAGCATCGTGACAGCAACGGTAATACC
TACCTGGAATGGTATTTGCAAAAACCGGTCAATCCCCGAGTTGCTGATTATATA
AAGTTTCGAATCGTTTACGCGGTGTTCCGGATCGTTTACGCGGCTCTGGTCCGG
CACCGATTTTACGCTGAAGATCAGTCGCGTGAAGACGGAGGACGTGGGTGCTCTA
CTACTGCTTTACGGGTAGTTTGGTGCCGTGGACCTTTGGTCAGGCTACTAAGGTG
GAAATTAAGGTGTTGGGGATCAGGTGGCGCGCGCAGCGCGGTGGCGGGAG
CGAGGTACAACAGTTGAATCAGGTGGAGGTTGGTTACGCCAGGTGGTTTCGCT
CGCTCAGTTGTGCGGCAAGCGGTTTGATCTTTCGAGCTATGGTATGAGCTGG
GTTCTGTCAGGCGCCGGCAAGGCTCTGAGTGGGTGGCGACCATTAACCTCTGGC
GGCAGGTACACCTACTATCCGACTCCGTGAAAGGCGGTTTACCATTCTCCCGCG
ACAATAGCAAAAACACCTGTATTTGAGATGAACCTCGTCCGCGCAGAGGACA
CCGCTGTGTACTACTGCGCAATTCCTACAGCGGTGCTATGGATTATTGGGGTCA
GGGCACATTTGGTGACTGTAAGCAGC (aTau-scFv)
GGCGGGGCGGACGCGCGCGGTGGCAGC (Linker)
GACATCTTGCCGTGCGTGCCCTTCAGCGTGGCCAGAGTGTGAAGTCTTGTAC
TGGGCCGATGTTTCAGTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
AGCCACACAGCTGGTAGCTGAGTTGACTTCGGACCTTTGACCCGAGGGCA
TCTCTCTCTTTGCCGAGGCCACAGGACAGCACCTGGATCGTCTGGCCCTGAG

TABLE 5-continued

AGCCGGCCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCCGTGTCACCAG
CAGCGGCCCGGTTCATCAACCATGGCATGTGGCAGACAATCTCTGTGAGGAGCT
GGCGCGGAATCTGGTCATCAAGGTCAACAGGGATGCTGTCATGAAAAATCGCGGT
GGCCGGGGACTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTGACCGT
GGGAGGTATTCCTTCATGAGAAGGACCTCGTGCAGCCTATAAACCTCGTCTG
GATGGCTGCATGAGGAGCTGGAACCTGGCTGAACGAGAGACACCCACATCCA
GGAAACGGTGAAAGTGAACACGAGGATGCAGTGCTTCTCGGTGACGGAGAGAG
GCTCTTTCTACCCCGGAGCGGCTTCGCCTTCTACAGCCTGGACTACATGCGGAC
CCCTCTGGACGTCGGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT
CCGCCAGCCGAGACACAGCGTGCTGTTTGGCTCTGGGCCCCGACCTCCGT
GCCGTGCCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
AAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCCTTGGCCCTAATGGAGATC
AAGGTCTGCGACGGCCAAGAGCACGTGGTCACCGTCTCGCTGAGGACGGTGAG
GCCACCTCGAGGTGGACGGCACCAGGGGCCAGAGCGAGGTGAGCGCCGCGCA
GCTGCAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGGAGCCCCGTGCTCAC
CTTTGCTGGCGGCTGCCAGATGTGCCGCTGACTTCAGCGCCAGTCACCGCTTC
TACCGCGGTGCTGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
GCGGCGTACAAGCAGCGACATCAACGGCCACTCTGCCCCCGTGGAGCCC
GCCGAGCC (Gas6-Gla EGF deleted)
GGCAGCGGAGCGGAGCGGCAGCGGCAGCGCAGCTacccatacgatgttcca
gattacgct (HA)
GAGGGCAGAGGAAGTCTGCTAACATGCGGTGACGTCGAGGAGAATCTTGGCCC
A (T2A)
GTGAGCAAGGGCGAGGAGCTGTTACCGGGGTGGTGGCCATCCTGGTCGAGCTG
GACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGA
TGCCACCTACGGCAAGCTGACCTGAAGTTCATCTGCACACCGCAAGCTGCC
CGTGCCCTGGCCCCCCTCGTGACCACTGACCTACGGCGTGCACTGCTTCAGC
CGCTACCCCGACCATGAAGCAGCAGCACTTCTCAAGTCCGCGATGCCCGAA
GGCTACGTCCAGGAGCGCACCATCTCTTCAAGGACGACGGCAACTACAAGACC
CGCGCCGAGGTGAAGTTCGAGGGCGACACCTGGTGAACCGCATCGAGCTGAAG
GGCATCGACTTCAAGGAGGACGGCAACATCTTGGGGCACAAGCTGGAGTACAA
CTACAACAGCCACAACGTCTATATCATGGCCGACAGCAGAAAGACGCGATCAA
GGTGAACCTCAAGATCCGCCACAACATCGAGGACGGCAGCGTGCACTGCGCGA
CCACTACAGCAGAACACCCCATCGGCGACGGCCCCGTGCTGCTGCGCGACAA
CCACTACCTGAGCAGCCAGTCCGCCCTGAGCAAGACCCCAACGAGAAGCGCGA
TCACATGCTCTGCTGGAGTTCGTGACCGCCCGGGATCACTCTCGGCATGGAC
GAGCTGTACAAGTaa (EGFP)

Preparation Example 3. Gas6-Based Fusion Molecule Targeting Alpha-Synuclein

To prepare an alpha-synuclein-specific chimera phagocy-
tosis inducer based on Gas6 protein, the Gla domain and the

EGF repeat domain were first removed, and a single-chain
variable region (scFv) of cinpanemab, an alpha-synuclein-
specific antibody, was introduced at that position (α Syn-
Gas6). Table 6 below shows the amino acid sequence and
nucleotide sequence of the chimeric phagocytosis inducer.

TABLE 6

1. α Syn-Gas6 (Cinpanemab (aSyn)_VL-G4Sx3-VH-LG-HA-T2A-EGFP,
amino acid sequence) SEQ ID NO: 152
MAPSLSPGPAALRRAPQLLLLLLAAECALA (SS)
SYELTQPPSVSVSPGQTARITCSGEALPMQFAHWYQQRPGKAPVIVYKDSERPSGV
PERFSGSSSGTTATLTITGVQAEDEADYYCQSPDSTNTYEVFGGGTKLTVLGGGGSG
GGSGGGGSEVQLVESGGGLVEPGGSLRLSCAVSGGFDPEKAWMSWVRQAPGQGLQ
WVARIKSTADGGTTSYAAPVEGRFIISRDDSRNMLYLQMNLSKTEDTAVYYCTSAH
WGQGLTVTVSS (aaSyn-scFv)
GGSGSGGGGS (Linker)
DILPCVPFVSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDRTFDPEGILLFA
GGHQDSTWIVLALRAGRLQLRLYNGVGRVTSSGPVINHGMMQTIISVEELARNLVI
KVNDRDAVMKIAVAGDLFQPERGLYHLNLTGGIPFHEKDLVQPINPRLDGCMRSWN
WLNGETTIQETVKVNTRMQCFVTERGSFYPGSGFAFYSLDYMRTPLDVGTESTW
EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLVAEHT
ALALMEIKVCDGQEHVVTVSLRDGEATLEVDGTRGQSEVSAQAQLQERLAVLERHLR
SPVLTFAGGLPDVPVTSAPVTAFYRGCMTELVNRRLLDLDEAAKYHSDITAHSCPPV
EPAAA (Gas6-Gla EGF deleted)
GSGSGSGSGSYPDVDPDYA (HA)
EGRGSLTTCGDVEENPGP (T2A)
VSKGEELFTGVVPIVLDELGDVNGHKFSVSGEGEGDATYKGLTLKFICTTGKLPVPW
PTLVTTLTLYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAIVK
FEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHN
IEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALS KDPNEKRDMHMLLEFVTA
AGITLGMDLEYK*

2. α Syn-Gas6 (Cinpanemab (aSyn)_VL-G4Sx3-VH-LG-HA-T2A-EGFP,
nucleotide sequence) SEQ ID NO: 153
ATGCCCCCTCGCTCTCGCCGGGCCCGCCCTGCGCCGCGCGCCGAGCTGC
TGCTGCTGCTGCTGGCCGGGAGTGCGCGCTTGCC (SS)
TCCTATGAGCTGACTCAGCCACCTCGGTGTCACTGTCCCCAGGACAGACGGCC
AGGATCACCTGCTCTGGAGAAGCATTGCCAATGCAATTGCTCATTGGTACCAAC

TABLE 6-continued

AGAGGCCAGGCAAGCCCCAGTGATAGTGGTGTACAAAGACAGTGAGAGACCG
 TCAGTGTCCCTGAGCGATTCTCTGGCTCCAGCTCAGGGACAACAGCCACGTTG
 ACCATCACTGGAGTCCAGGCAGAAGATGAGGCTGACTATTACTGCCAGTCGCCA
 GACAGCACTAACACTTATGAAGTCTTCGGCGGAGGGACCAAGCTGACCGTCCTA
 GGTGGTGGGGGATCAGGTGGCGGCGGCAGCGCGGTGGCGGGAGCGAGGTGCA
 GCTGGTGGAGTCTGGGGGAGGTCTGGTCGAGCCGGGGGGGTCCCTAAGACTCTC
 CTGTGCAGTCTCCGGATTTCGATTTCGAAAAAGCCTGGATGAGTTGGGTCCGCCA
 GGCTCCAGGGCAGGGGTACAGTGGGTGGCCGTATCAAGAGCACAGCTGATGG
 TGGGACAACAAGCTACGCCGCCCGTGGAAAGGCAAGTTTCATCATCTCAAGAGA
 TGATTTCAGAAAACATGCTTTATCTGCAAAATGAACAGTCTGAAAACCTGAAGACAC
 AGCCGTCTATTATTGTACATCAGCCACTGGGGCCAGGGAACCTGGTACCGTC
 TCCTCG (aaSyn-scFv)
 GGCGGGGGCGGCAGCGCGGCGGTGGCAGC (Linker)
 GACATCTTGCCTGCGTGCCTTCAGCGTGGCCAAGAGTGTGAAGTCCTTGTACC
 TGGGCGGATGTTCACTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
 AGCCCAACAGGCTGGTAGCTGAGTTGACTTCCGGACCTTTGACCCCGAGGGCA
 TCCTCCTCTTTGCCGAGGCCACAGGACAGCACCTGGATCGTGTGCGCTGAG
 AGCGCGCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCGTGTCAACAG
 CAGCGGCCCGGTCAACAACATGGCATGTGGCAGACAATCTCTGTGAGGAGCT
 GGCGCGGAATCTGGTCATCAAGGTCAACAGGGATGCTGTCATGAAAATCGCGGT
 GGCGGGGACTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTGACCGT
 GGGAGGTATTCCTTCCATGAGAAGGACCTCGTGCAGCCTATAAACCCCTCGTCTG
 GATGGCTGATGAGGAGCTGGAACCTGGCTGAACGAGAGAAGACACCACTCCA
 GGAACCGGTGAAAGTGAACACGAGGATGCAGTGTCTCGGTGACGGAGAGAG
 GCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCCTGGACTACATCGCGAC
 CCCTCTGGACGTGGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT
 CCGCCAGCCCGCAGACACAGGCGTGTCTTTGCGCTCTGGGCCCCCGACCTCCGT
 GCCGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
 AAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCTTGGCCCTAATGGAGATC
 AAGGTCTGCGACGGCCAAGAGCACGTGGTCAACCGTCTCGCTGAGGGACGGTGAG
 GCCACCTGGAGGTGGACGGCACAGGGGCCAGAGCGAGGTGAGCGCCGCGCA
 GCTGCAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGGAGCCCCGTGCTCAC
 CTTTGTGGCGGCCGTGCAGATGTGCCGGTGACTTCAGCGCCAGTCACCGCGTTC
 TACCGCGGCTGCATGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
 GCGGCGTACAAGCACAGCGACATCAGCGCCACTCTGCCCCCGTGGAGCCC
 GCCGCAGCC (Gas6-Gla EGF deleted)
 GGCAGCGGCAGCGCAGCGCAGCGCAGCGCAGCTACCCATACGATGTTCC
 AGATTACGCT (HA)
 GAGGGCAGAGGAAGTCTGCTAACATGCGGTGACGTGAGGAGAATCCTGGCCCC
 A (T2A)
 GTGAGCAAGGGCGAGGAGCTGTTCAACGGGGTGGTGCCCATCTGGTCGAGCTG
 GACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGA
 TGCCACCTACGGCAAGCTGACCTGAAGTTTATCTGCACCAACGGCAAGCTGCC
 CGTGCCCTGGCCCCACCTCGTGACCACTGACCTACGGCGTGCAGTGCTTCAGC
 CGCTACCCCGACCATGAAGCAGCACGACTTCTCAAGTCCGCCATGCCCGAA
 CTACAACAGCCACAACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAA
 GGTGAACCTCAAGATCCGCCACAACATCGAGGACGGCAGCTGCAGCTGAGC
 CCACTACCAGCAGAACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAA
 CCACTACCTGAGCACCCAGTCCGCCCTGAGCAAGACCCCAACGAGAAGCGCGA
 TCACATGTTCTGCTGAGTTCGTGACCGCCCGGGATCACTCTCGCATGGAC
 GAGCTGTACAAGTAA (EGFP)

Preparation Example 4. ProS1-Based Fusion Molecule Targeting Beta-Amyloid

To prepare a beta-amyloid ($A\beta$)-specific chimeric phago-
cytosis inducer based on ProS1 protein, the Gla domain and

the EGF repeat domain were first removed, and a single-
chain variable fragment (scFv) of aducanumab, a beta-
amyloid-specific antibody, was introduced at that position
($\alpha A\beta$ -ProS1). Table 7 below shows the amino acid sequence
and nucleotide sequence of the chimeric phagocytosis
inducer.

TABLE 7

1. $\alpha A\beta$ -ProS1 ($\alpha A\beta$ -ProS1 (GE)-FLAG, amino acid sequence)
 SEQ ID NO: 154
 MRVLGGRCGALLACLLLVLPVSEA (SS)
 DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGV
 PSRFSGSGSGTDFLTITSLQPEDFATYYCQQSYSTPLTFGGGKVEIKRGGGSGGG
 GSGGGSEVQLVESGGGVVQPGRLRLSCAASGFAPSSYGMHWVRQAPGKLEWV
 AVIWFDTGKKYTDTSVKGRFTISRDNKNTLYLQMNTLRADTAVYYCARDRGIGA
 RRPYYMDVWGKTTTVTVSS (Adu-scFv)
 GGGSGGGGS (Linker)
 VVSVCLPLNLDTKYELLYLAQFAGVVLYLKFRLPFISRFSAEFDFRTYDSEGVILYA
 ESIDHSAWLLIALRGGKIEVQLKNEHTSKITGGDVINNGLWNMVSVELEHSISIKIA
 KEAVMDINKGPLFKPENGLLETKVYFAGFPRKVESELIKPINRLDGCIRSWNLK

TABLE 7-continued

QGASGIKEIIQEKQNKHCLVTVEKGSYYPGSGIAQPHIDYNNVSSAEGWHVNVTLNI
 RPSTGTGVMALVSGNNTVPFAVSLVDSTSEKSDILLSEVENTVIYRIQALSLCSDQQ
 SHLEFRVNRNNLELSTPLKIETISHEDLQRQLAVLDKAMKAKVATYLGGLPDVPFSA
 TPVNAFYNGCMEVNINGQLDLDEAISKHNDIRAHSCPSVWKTKNS (ProS1 (GE-))
 QGSRADYKDHGDYKDHIDYKDDDK* (FLAG)

2. $\alpha\text{A}\beta$ -ProS1 ($\alpha\text{A}\beta$ -ProS1 (GE-)-FLAG, nucleotide sequence)
 SEQ ID NO: 155

ATGAGGGTCCTGGGTGGGCGCTGCGGGGCGCTGTGCGGTGTCTCTCTCTAGTG
 CTTCCCGTCTCAGAGGCA (SS)
 GACATTACAGTACTCAATCTCTAGCTCTCTGAGCGCTCCGTTGGAGATAGAG
 TCACATTACCTGCAGAGCCAGCCAAATCCATCAGCTCTTATCTAAATTGGTACCA
 ACAGAAAGCCCGCAAGCGCCAAAGCTGCTCATCTACGCTGCAAGCTCCTTACA
 GAGCGGAGTACCCAGCAGATTCTCAGGCACTGGCAGTGGGACTGACTTCACATT
 GACGATTAGCTCTCTGCAGCCTGAAGACTTTGCCACATACTATTGTGACGAGAGC
 TATAGCACCCCGCTGACGTTTGGAGGCGGAATAAGGTGGAATCAAGAGAGG
 AGGCGGGGGCTCCGCGGGGGTGGCTCGGGGGGAGGAGCTCAGAGGTTTCAGC
 TTGTGCGAGTCTGGGGGGGAGTCGTTACGCGAGGTAGAAAGCTCAGACTGAGCT
 GTGCCGCAAGTGGGTTTGTCTTTTCATCTTACGGTATGCACTGGGTGAGACAGGC
 TCCTGGCAAGGACTCGAGTGGGTGCTGTAATATGGTTCGATGGTACAAAGAA
 ATACTATACCGATAGTGTGAAGGAAGATTACCATTTTCAGGAGACACAGTAA
 AAATACCTTGTACCTTCAGATGAACACCTGAGAGCAGAAGACACAGCCGTGTA
 CTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCTATTATATGGA
 TGTGTGGGGGAAGGGAACAACAGTACTGTGAGCTCT (Adu-scFv)
 GGCGGGGGCGGCGAGCGGCGGCGGTGGCAGC (Linker)
 GTTGTTCAGTGTGCTTCCCTTGAACCTTGACACAAAGTATGAATTACTTTACTT
 GGCGGAGCAGTTTGCAGGGGTGTTTTATATTTAAATTTTCGTTTGCAGAAATC
 AGCAGATTTTTCAGCAGAATTTGATTTCGGACATATGATTGAGAAGGCGTGATA
 CTGTACGCAAGATCTATCGATCACTCAGCGTGGCTCCTGATTGCCTTCGTGGT
 GAAAGATTGAAGTTTTCAGCTTAAGAATGAACATACATCCAAATCACAAGTGGAG
 GTGATGTTTATTAATAATGGTCTATGGAATATGGTGTCTGTGGAAGAATTAGAAC
 ATAGTATTAGCATTAAATAGCTAAAGAAGCTGTGATGGATATAAATAAACCTG
 GACCCCTTTTAAAGCCGAAATGGATTGCTGGAAACCAAGTATACCTTTGCAG
 GATTCCCTCGAAAGTGAAGTGAACCTCATTAAACCGATTAAACCTCGTCTAG
 ATGGATGTATACGAAGCTGGAATTTGATGAAGCAAGGAGCTTCTGGAATAAAGG
 AAATTATTCAAGAAAAACAAATAAGCATTGCCTGGTTACTGTGGAGAAGGGCT
 CCTACTATCTGGTTCTGGAATTGCTCAATTTACATAGATATAATAATGTATC
 CAGTGTGAGGGTTGGCATGTAAATGTGACCTTGAATATTCGTCCATCCACGGGC
 ACTGGTGTATGCTTGCCTTGGTTTCTGGTAACAACACAGTGCCCTTTGCTGTGTC
 CTTGGTGGACTCCACCTCTGAAAAATCACAGGATATTCTGTTATCTGTTGAAAT
 ACTGTAATATATCGGATACAGGCCCTAAGTCTATGTTCGATCAACAATCTCATC
 TGGAAATTTAGAGTCAACAGAAACAATCTGGAGTTGTGACACCACTTAAAAATAG
 AAACCATCTCCCATGAAGACCTTCAAAGACAACTTGCCGTCTTGACAAAGCAA
 TGAAAGCAAAAGTGCCACATACCTGGGTGGCCTTCCAGATGTTCCATTTCAGTG
 CCACACCACTGAATGCCCTTTTATAATGGCTGCATGGAAGTGAATATTAATGGTGT
 ACAGTTGGATCTGGATGAAGCAATTTCTAAACATAATGATATTAGAGCTCACTCA
 TGTCCATCAGTTTGGAAAAAGACAAAGAATTCT (ProS1 (GE-))
 CAAGGATCCCGGGCTGACTACAAAGACCATGACGGTGATTATAAGATCATGAC
 ATCGACTACAAGGATGACGATGACAAGTga (FLAG)

Preparation Example 5. Gas6-Based Fusion
 Molecules Targeting Beta-Amyloid (II):
 Beta-Amyloid-Binding Regions in the Form of Fab
 or Mab

To prepare gas6 protein-based beta-amyloid ($\text{A}\beta$)-specific
 chimera phagocytosis inducer, the Gla domain, which rec-

ognizes PS (phosphatidylserine) in apoptotic cells, was first
 removed, and an antigen-binding fragment (Fab) or mono-
 clonal antibody (Mab) of the beta-amyloid-specific antibody
 aducanumab was introduced at that position ($\alpha\text{A}\beta$ [Fab]-
 Gas6, and $\alpha\text{A}\beta$ [Mab]-Gas6). Tables 8-10 below show the
 amino acid sequences and nucleotide sequences of the two
 chimeric phagocytosis inducers.

TABLE 8

Second Region Light Chain (SEQ ID NO: 161) (the light
 chain which is capable of specifically binding to beta-
 amyloid can form a dimer with the peptide of sequence of
 SEQ ID NO: 156 or SEQ ID NO: 158 to form Fab Adu-Gas6
 or Mab Adu-Gas6, respectively):
 MGWSCIILFLVATATG (SS)
 DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGV
 PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKVEIKRRTVAAPSV
 FIFPPSPDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDST
 YLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC* (Adu Light Chain)

1. $\alpha\text{A}\beta$ [Fab]-Gas6 (Aducanumab VH-CH1 (Fab)-Gas6-FLAG, amino
 acid sequence) SEQ ID NO: 156
 METDTLLLVLLLWVPGSTGD (SS)
 EVQLVESGGGVVQGRSLRLSCAASGFAFSSYGMHWVRQAPGKGLEWVAIVFDG
 TKKYYTDSVKGRFTISRDNKNTLYLQMNTLRAEDTAVYYCARDRTGARRGPYY

TABLE 8-continued

MDVWVGKGTVTTVSSASTKGPSVFPPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN
SGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVE
PKSCDKTH (Adu-VH-CH1)
GGGSGGGGS (Linker)
DILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDRTFDPEGILLFA
GGHQDSTWIVLALRAGRLEQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVI
KVNRRDAVMKIAVAGDLFQPERGLYHLNLTVGGI PFHEKDLVQPINPRLDGCMRSWN
WLNGEDTTIQETVKVNTRMQCF SVTERGSFYPGSGFAFYSLDYMRTPLDVGTESTW
EVEVVAHIRPAADTVGLFALWAPDLRAVPLSVLVDYHSTKKLKKQLVVLAVEHT
ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQLQERLAVLERHLR
SPVLTFAGGLPDVPVTSAPVTA FYRGCMTEVNRRLLDLDEAAYKHS DITAHSCPPV
EPAAA (Gas6-Gla EGF deleted)
DYKDHGDYKDHIDYKDDDDK* (FLAG)

2 α A β [Fab]-Gas6 (Aducanumab (Fab)-Gas6-FLAG, nucleotide
sequence) SEQ ID NO: 157
ATGGAGACAGACACTCTCTGCTATGGGTACTGCTGCTCTGGGTTCCAGGTTCCA
CTGGTGAC (SS)
GAGGTT CAGCTTGT CGAGTCTGGGGGGGAGTCGTT CAGCCAGGTAGAAAGCCTC
AGACTGAGCTGTGCGCGCAAGTGGGTTTGCTTTTTCATCTTACGTTATGCATGGG
TGAGACAGGCTCCTGGCAAAGGACTCGAGTGGGTGCTGTAATATGGTTTCGATG
GTACAAAGAAATACTATAACCGATAGTGTGAAGGAAGATT CACCATTTACGAG
ACAACAGTAAAAATACCTTGTA CTT CAGATGAACACCTTGAGAGCAGAAGACA
CAGCCGTGTACTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCT
ATTATATGGATGTGTGGGGGAAGGGAAACAACAGTGA CTGTGAGCTCTGCCTCCA
CCAAGGGCCCATCGGTCTTCCCCCTGGCACCTCTCCAAGAGCACCTCTGGGGG
CACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGCGTGCACACCTTCCCGGCTGTCTT
ACAGTCTCTCAGGACTCTACTCCTCAGCAGCGTGGTGACTGTGCCCTCTAGCAGC
TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCAGCAACACCAAG
GTGGACAAGAAAGTTGAGGCCAAATCTGTGACAAAACCTCAC (Adu-VH-CH1)
GGCGGAGGTGGAAGCGGAGGCGGTGGAAGC (Linker)
GACATCTTGCCGTGCGTGCCCTTCAGCGTGGCCAAGAGTGTGAAGTCCTTGTTACC
TGGGCGGGATGTT CAGTGGGACCCCGTGATCCGACTGCGCTTCAAGAGGCTGC
AGCCACCAAGGCTGGTAGCTGAGTTTGACTTCCGGACCTTTGACCCCGAGGGCA
TCCTCTCTTTTGCCGGAGGCCACCAAGGACAGCACCTGGATCGTGCTGGCCCTGAG
AGCCGGCCGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCCGTGTCAACGAG
CAGCGGCCCGGT CATCAACCATGGCATGTGGCAGACAATCTCTGTTGAGGAGCT
GGCGCGGAATCTGGTCAATCAAGGTCAACAGGGATGCTGTGATGAAAATCGCGGT
GGCCGGGGGACTTGTTC AACCGGAGCGAGGACTGTATCATCTGAACCTCACCGT
GGGAGGTATTCCTTCCATGAGAAGGACCTCGTG CAGCCTATAAACCTCTGCTG
GATGGCTGTATGAGGAGCTGGAAC TGGCTGAACGAGAGAAGACACCAACATCCA
GGAAACGGTGAAAGTGAACACGAGGATGCAGTGCTTCTCGGTGACGGAGAGAG
GCTCTTTCTACCCCGGAGCGGCTTCGCTTCTACAGCTTGACTACATGCGGAC
CCCTCTGGACGTCGGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT
CCGCCAGCCGAGACACAGGCGTGCTGTTTGCCTCTGGGCCCCGACCTCCGT
GCCGTGCCCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
AAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCTTGGCCCTAATGGAGATC
AAGGCTCTGCGACGGCCAAGAGCACGTGGTCAACGCTCTCGCTGAGGAGCGGTGAG
GCCACCTTGAGGTGGACGGCACCAAGGGCCAGAGCGAGGTGAGCGCCGCGCA
GCTG CAGGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGAGCCCCGTGCTCAC
CTTTGCCGCGGCCCTGCCAGATGTGCCGTGACTTCAGCGCCAGTCAACCGCTTC
TACCGCGGCTGCATGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
GCGCGGTACAAGCACAGCGACATCACGGCCACTCTGCCCCCGGTGGAGCCC
GCCGCAGCC (Gas6-Gla EGF deleted)
GACTACAAAGACCATGACGGTGATTATAAGATCATGACATCGACTACAAGGAT
GACGATGACAAGTga (FLAG)

TABLE 9

1. α A β [Mab]-Gas6 (Aducanumab heavy chain (Mab)-Gas6-FLAG,
amino acid sequence) SEQ ID NO: 158
METDTLLLVLLLVPGSTGD (SS)
EVQLVESGGGVVQPGRSLRLSCAASGFAFSSYGMHWVRQAPGKGLEWVAWIWFDG
TKKYITDSVKGRFTISRDNKNTLYLQMNTLRAEDTAVYYCARDRGIGARRGPYY
MDVWVGKGTVTTVSSASTKGPSVFPPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN
SGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVE
PKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTVVVDVSHEDPEVK
FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLT VLVHQDWLNQKEYCKKVSNAKAL
PAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSGSEFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLS
LSPGK (Adu-Heavy chain)
GGGSGGGGS (Linker)
DILPCVPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDRTFDPEGILLFA
GGHQDSTWIVLALRAGRLEQLRYNGVGRVTS SGPVINHGMMQTI SVEELARNLVI
KVNRRDAVMKIAVAGDLFQPERGLYHLNLTVGGI PFHEKDLVQPINPRLDGCMRSWN
WLNGEDTTIQETVKVNTRMQCF SVTERGSFYPGSGFAFYSLDYMRTPLDVGTESTW

TABLE 9-continued

EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHT
 ALALMEIKVCDGQEHVTVVSLRDGEATLEVDGTRGQSEVSAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTAFYRGCMTEVNRRLDLDDEAAYKHSITAHSCPPV
 EPAAA (Gas6-Gla EGF deleted)
 DYKDHGDYKDHIDYKDDDDK* (FLAG)

2. α A β [Mab]-Gas6 (Aducanumab (Mab)-Gas6-FLAG, nucleotide
 sequence) SEQ ID NO: 159
 ATGGAGACAGACACTCCTGCTATGGGTACTGCTGCTCTGGGTTCAGGTTCCA
 CTGGTGAC (SS)
 GAGGTTGAGTCTGTCGAGTCTGGGGGGGAGTCTGTTAGCCAGGTAGAACCTC
 AGACTGAGCTGTGCCGCAAGTGGGTTTGCTTTTTCATCTTACGGTATGCACTGGG
 TGAGACAGGCTCCTGGCAAAGGACTCGAGTGGGTGCTGTAATATGGTTGATG
 GTACAAAGAAATACTATACCGATAGTGTGAAAGGAAGATTACCATTTACGAG
 ACAACAGTAAAAATACCTTGTACCTTCAGATGAACACCTTGAGAGCAGAAGACA
 CAGCCGTGTACTACTGCGCCAGAGATAGAGGTATCGGAGCAAGGCGTGGTCCCT
 ATTATATGGATGTGTGGGGGAAGGGAACAACAGTGACTGTGAGCTCTGCCTCCA
 CCAAGGGCCCATCGGTCTTCCCCCTGGCACCTCTCTCCAAGAGCACCTCTGGGGG
 CACAGCGGCCCTGGGTGCTGGTCAAGGACTACTTCCCGAACCGGTGACGGT
 GTCGTGGAACCTCAGCGCCCTGACCAGCGCGTGACACCTTCCCGGCTGTCTT
 ACAGTCTCTCAGGACTTACTTCCCTCAGCAGCGTGGTGACTGTGCCCTCTAGCAG
 TTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCAGCAACACCAAG
 GTGGACAAGAAAGTTGAGGCCAAATCTGTGACAAACTCACACATGCCACCG
 TGCCAGCACCTGAACCTCTGGGGGACCGTCAGTCTTCTCTTCCCCCAAAAC
 CCAAGGACACCTCATGATCTCCCGACCCCTGAGGTACATGCGTGGTGGTGG
 ACGTGAGCCACGAAGACCTGAGGTCAAGTTCAACTGGTACGTGGACGCGGTGG
 AGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTAC
 CGTGTGGTCAGCGTCTCACCCTCTGCACAGGACTGGTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAGCCCTCCAGCCCCATCGAGAAACCATC
 TCCAAGCCAAAGGGCAGCCCGAGAACCACAGGTGTACACCTGCCCCATCC
 CGGGATGAGCTGACCAAGAACCAGGTGAGCTGACCTGCCTGGTCAAAGGCTTC
 TATCCACGCGACATCGCGTGGAGTGGGAGAGCAATGGGCAGCCGAGAAACAA
 CTACAAGACCAAGCCCTCCCGTGTGGACTCCGACGGCTCCTTCTCTCTACAGC
 AAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGCTTCTCATGCTCC
 GTGATGCATGAGGCTCTGCACAACCACTACACGCAAGAGCCTCTCCCTGTCC
 CCGGTAAA (Adu-Heavy chain)
 GGCGGAGGTGGAAGCGGAGGCGGTGGAAGC (Linker)
 GACATCTTGCCGTGCGTGCCCTTCAGCGTGGCCAAGAGTGTGAAGTCCTTGTAAC
 TGGGCGGATGTTTCACTGAGGAGCCCGGTGATCCGACTGCGCTTCAAGAGGCTGC
 AGCCACCAAGGCTGGTAGCTGAGTTTGAATTCGGACCTTTGACCCGAGGGCA
 TCTCCTCTTTTGGCGAGGCCACAGGACGACCTGGATCGTGCTGGCCCTGAG
 AGCCGCGGCTGGAGCTGCAGCTGCGCTACAACGGTGTGCGCCGTGTACCCAG
 CAGCGGCCCGTCAACCATGGCATGTGGCAGACAATCTCTGTGAGGAGCT
 GGCGCGGAATCTGGTCATCAAGGTCAACAGGATGCTGTGATGAAAAATCGCGGT
 GGCCGGGGACTTGTTCACCGGAGCGAGGACTGTATCATCTGAACCTCACCGT
 GGGAGGTATTCCTTCCATGAGAAGGACCTCGTGACGCTATAAACCTCGTCTG
 GATGGCTGTATGAGGAGCTGGAACCTGGCTGAACGGAAGACACCACTCCA
 GGAAACGGTGAAAGTGAACACGAGGATGCAGTGCTTCTCGGTGACGGAGAGAG
 GCTCTTTCTACCCCGGAGCGGCTTCGCCTTCTACAGCCTGGACTACATGCGGAC
 CCCTCTGGACGTGGGACTGAATCAACCTGGGAAGTAGAAGTCGTGGCTCACAT
 CCGCCAGCCGCGAGACACAGGCGTGTCTTTGCGCTCTGGGCCCCGACCTCCGT
 GCCGTGCTCTCTCTGTGGCACTGGTAGACTATCACTCCACGAAGAACTCAAG
 AAGCAGCTGGTGGTCTGGCCGTGGAGCATACGGCTTGGCCCTAATGGAGATC
 AAGGTCTGCGACGGCCAAGAGCACGTGGTCACCGTCTCGCTGAGGGACGGTGAG
 GCCACCTCGGAGGTGGACGGCACCAGGGCCAGAGCGAGGTGAGCGCCGCGCA
 GGTGACGAGAGGCTGGCCGTGCTCGAGAGGCACCTGCGAGCCCCGTGCTCAC
 CTTTGCCGCGGCTGCCAGATGTGCCGTGACTTCAGCGCCAGTCACCGCGTTC
 TACCGCGCTGCATGACACTGGAGGTCAACCGGAGGCTGCTGGACCTGGACGAG
 GCGGCGTACAAGCACAGCGACATCACGGCCACTCTGCCCCCGTGGAGCCC
 GCCGCAGCC (Gas6-Gla EGF deleted)
 GACTACAAGACCATGACGGTGATTATAAGATCATGACATCGACTACAAGGAT
 GACGATGACAAG (FLAG)
 TGA

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Preparation Examples 6-8

As non-limiting exemplary embodiment of the binding molecule containing a scaffold protein between the first region and the second region, wherein the Gas6 and anti-amyloid antibody scFv (in this example, aducanumab scFv) are employed as the first region and the second region, respectively, a single chain Fc region with reduced or abolished Fc receptor binding affinity are manufactured. The sequences employed in the construction are shown in Table 10.

As another non-limiting exemplary embodiment of the binding molecule containing a scaffold protein between the first region and the second region, wherein the Gas6 and

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anti-amyloid antibody scFv (in this example, aducanumab scFv) are employed as the first region and the second region, respectively, a heterodimeric binding molecule is manufactured. The first polypeptide of the heterodimeric binding molecule comprises anti-amyloid antibody scFv, Fc region (DD), and Gas6, and the second polypeptide of the heterodimeric binding molecule comprises anti-amyloid antibody scFv region and Fc region (KK). Still another non-limiting exemplary embodiment of the binding molecule containing a scaffold protein between the first region and the second region, a homodimer comprising two polypeptides which each comprise anti-amyloid antibody scFv (as second region), Fc region (scaffold), and Gas6 is manufactured. The peptide sequences are shown in Table 10

TABLE 10

Preparation Example 6: Single polypeptide fusion molecule: anti-amyloid antibody-MFc-Gas6-His (SEQ ID NO: 162)
 METDTLLWVLLWVPGSTGD (SS)
 DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGV
 PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKVEIKGGGSGGGG
 SGGGSEVQLVESGGGVVQPGSRSLRLSCAASGFAFSSYGMHWRQAPGKGLEWVA
 VIWFDGTKKYYTDSVKGRFTISRDNKNTLYLQMNTLRAEDTAVVYCARDRGIGAR
 RGPYYMDVWGKTTVTSS (anti-amyloid antibody-scFv)
 GGGGSGGGGS (linker)
 APEFLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNA
 KTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQPR
 EPQVYTFPPEQEEMTKNQVSLRCLVKGFYPSDIAVEWESNGQPENN
 YKTTKPVLDSDGSPRLESRLTVDKSRWQEGNVECSVMHEACSWHLCKSLSLSLGK
 (Monomeric Fc with reduced or abolished Fc gamma receptor
 binding affinity)
 GGGGSGGGSGGGGS (linker)
 DILPCVPFSVAKSVKSLYLGRMESGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLQLRYNGVGRVTSSGSPVINHGMMWQTIISVEELARNLVI
 KVNDAVMKIAVAGDLFQPERGLYHLNLTVGGLPFHEKDLVQPINPRLDGCMRSWN
 WLNGETTTIQETVKVNTRMQCFVS TERGSFYPGSGFAFYSLDMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLVAEHT
 ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTAFYRGCMTELVNRRLLDLDEAAYKHS DITAHSCPPV
 EPAAA (Gas6)
 HHHHHH (His)

Preparation Example 7: Heterodimeric fusion molecule comprising monovalent first region and monovalent second region: First polypeptide comprising anti-amyloid antibody-Fc(DD)-Gas6-His (SEQ ID NO: 163)
 METDTLLWVLLWVPGSTGD (SS)
 DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGV
 PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKVEIKGGGSGGGG
 SGGGSEVQLVESGGGVVQPGSRSLRLSCAASGFAFSSYGMHWRQAPGKGLEWVA
 VIWFDGTKKYYTDSVKGRFTISRDNKNTLYLQMNTLRAEDTAVVYCARDRGIGAR
 RGPYYMDVWGKTTVTSS (anti-amyloid antibody-scFv)
 GGGGSGGGGS (linker)
 DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNW
 YVDGVEVHNAKTKPREEQYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPI
 EKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPEN
 NYDTTPVLDSDGSFFLYSDDLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSP
 GK (Fc(DD))
 GGGGSGGGSGGGGS (linker)
 DILPCVPFSVAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLQLRYNGVGRVTSSGSPVINHGMMWQTIISVEELARNLVI
 KVNDAVMKIAVAGDLFQPERGLYHLNLTVGGLPFHEKDLVQPINPRLDGCMRSWN
 WLNGETTTIQETVKVNTRMQCFVS TERGSFYPGSGFAFYSLDMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFALWAPDLRAVPLSVALVDYHSTKKLKKQLVVLVAEHT
 ALALMEIKVCDGQEHVTVSLRDGEATLEVDGTRGQSEVSAQAQLQERLAVLERHLR
 SPVLTFAGGLPDVPVTSAPVTAFYRGCMTELVNRRLLDLDEAAYKHS DITAHSCPPV
 EPAAA (Gas6)
 HHHHHH (His)
 Second polypeptide comprising anti-amyloid antibody-Fc(KK)-Gas6-His (SEQ ID NO: 164)
 METDTLLWVLLWVPGSTGD (SS)
 DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGV
 PSRFGSGSGTDFTLTISLQPEDFATYYCQQSYSTPLTFGGGTKVEIKGGGSGGGG
 SGGGSEVQLVESGGGVVQPGSRSLRLSCAASGFAFSSYGMHWRQAPGKGLEWVA

TABLE 10-continued

VIWFDGTKKYYTDSVKGRFTISRDN SKNTLYLQMN TLRAEDTAVYYCARDRGIGAR
 RGPYYMDVWGKTTTVTVSS (anti-amyloid antibody-scFv)
 GGGGSGGGGS (linker)
 DKHTCPCPCAPPELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNW
 YVDGVEVHNAKTKPREEQYASTYRVSVLTVLHQDWLNGKEYCKVSNKALPAPI
 EKTISKAKGQPREPQVYTLPPSRKEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPEN
 NYKTTTPPVLSKSGSFFLYSKLTVDKSRWQQGNVFC SVMHEALHNNHYTQKSLSLSP
 GK (Fc (KK))

Preparation Example 8: Single polypeptide fusion molecule
 comprising monovalent first region, scaffold, and monovalent
 second region: Anti-amyloid antibody-Fc-Gas6-His
 (SEQ ID NO: 165)
 METDTLLLVLLWVPGSTGD (SS)
 DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNNWYQQKPKAPKLLIYAASSLQSGV
 PRRFSGSGSGTDFTLTITSLQPEDFATYYCQSYSTPLTFGGGTKEIKGGGSGGGG
 SGGGSEVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHWVRQAPGKGLEWVA
 VIWFDGTKKYYTDSVKGRFTISRDN SKNTLYLQMN TLRAEDTAVYYCARDRGIGAR
 RGPYYMDVWGKTTTVTVSS (anti-amyloid antibody-scFv)
 GGGGSGGGGS (linker)
 DKHTCPCPCAPPELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNW
 YVDGVEVHNAKTKPREEQYASTYRVSVLTVLHQDWLNGKEYCKVSNKALPAPI
 EKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPEN
 NYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFC SVMHEALHNNHYTQKSLSLSP
 GK (Fc region with reduced or abolished Fc gamma receptor
 binding affinity)
 GGGGSGGGSGGGGS (linker)
 DILPCVFPFSAKSVKSLYLGRMFSGTPVIRLRFKRLQPTRLVAEFDFTFDPEGILLFA
 GGHQDSTWIVLALRAGRLQLRYNGVGRVTS SGPVINHGMMQTSVEELARNLVI
 KVNRAVAMKIAVAGDLFQPERGLYHLNLTVG GIPFHEKDLVQPINRDLGCMRSWN
 WLNEDDTTQETVKVNTRMQCFSVTERGSFYPGSGFAFYSLDYMRTPLDVGTESTW
 EVEVVAHIRPAADTGVLFAWAPDLRAVPLSVALVDYHSTKKLKKQLVVLAVEHT
 ALALMEIKVCDGQEHVVTVSLRDGEATLEVDGTRGQSEVSAALQQLERLAVLERHLR
 SPVLTFAAGGLPDVPVTSAPVTA FYRGCMTEVNRRLLDLDEAAYKHSDI TAHS CPPV
 EPAAA (Gas6)
 HHHHHH (His)

Preparation Example 9

As non-limiting exemplary embodiment of the binding
 molecule containing a scaffold protein between the first
 region and the second region, a bispecific antibody, wherein
 a scFv of anti-Axl antibody and anti-amyloid antibody are
 employed as the first region and the second region, respec-
 tively, is manufactured. See, FIG. 1J. A heavy chain of the
 bispecific antibody has the following sequence of Table 11
 and the light chain of anti-amyloid antibody light chain has
 the sequence of SEQ ID NO: 161. The Fc region of the
 heavy chain contains NA mutation to reduced or abolish Fc
 gamma receptor binding affinity.

TABLE 11

Bispecific Antibody fusion molecule: Anti-amyloid
 Ab-anti-Axl ScFv (first polypeptide)
 (SEQ ID NO: 166)
 METDTLLLVLLWVPGSTGD (SS)
 EVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHWVRQAPGKGLEWVA
 VIWFDGTKKYYTDSVKGRFTISRDN SKNTLYLQMN TLRAEDTAVYYCAR
 DRIGARRGPYYMDVWGKTTTVTVSSASTKGPSVFPLAPSSKSTSGGTA
 ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVP
 SSSLTGTQTYI CNVNHKPSNTKVDKKEPKSCDKHTCPCPCAPPELLGGP
 SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
 KTKPREEQYASTYRVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTI
 SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNG
 QPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFC SVMHEALHN
 HYTQKSLSP
 LSPGK (anti-amyloid-antibody heavy chain)
 GGGGSGGGGS (Linker)
 EVKLVESGGDLVKGKSLKLSAASGFTFSSYGMWSVRQTPDKRLEWVA
 TISSGGSYTYPD SVKGRFTISRDN AKNTLYLQMSLSKSEDTAMYYCAR
 HPIYYTYDDTMDYWGQTSVTVSSGGGSGGGSGGGGSDIVLTQSPA

TABLE 11-continued

MAASPGEKVTMTCSASSSVSSGNFHWYQQKPGTSPKLIWYRTSNLASGV
 PARFSGSGSGTSYSLTITSSMEADAATYYCQQWSGYPWTFGGGTKEIK
 (anti-Axl scFv)

Preparation Example 10

As non-limiting exemplary embodiment of the binding
 molecule containing a scaffold protein between the first
 region and the second region, a homodimeric bispecific
 antibody, wherein a scFv of anti-Axl antibody and a scFv
 region of an anti-amyloid antibody are employed as the first
 region and the second region, respectively, is manufactured.
 See FIG. 11. The bispecific antibody comprises a first
 polypeptide and a second polypeptide, which are identical to
 each other and each comprises the sequence of SEQ ID NO:
 167. The structure of the first/second polypeptide is illus-
 trated in Table 12, and the Fc region scaffold contains N—A
 mutation to reduced or abolish Fc gamma receptor binding
 affinity.

TABLE 12

Bispecific homodimeric comprising two polypeptides:
 anti-amyloid Ab-anti-Axl ScFv (SEQ ID NO: 167)
 METDTLLLVLLWVPGSTGD (SS)
 EVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHWVRQAPGKGLEWVA
 WFDGTKKYYTDSVKGRFTISRDN SKNTLYLQMN TLRAEDTAVYYCARDGI
 GARRGPYYMDVWGKTTTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
 KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQT
 YI CNVNHKPSNTKVDKKEPKSCDKHTCPCPCAPPELLGGPSVFLFPPKPK
 DTLMSRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYAST
 YRVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYT

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TABLE 12-continued

LPSPRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
 GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
 (anti-amyloid-antibody heavy chain)
 GGGSGGGGS (Linker)
 EVKLVESGGDLVKGPSGLKLSCAASGFTFSSYGMWSVRQTPDKRLEWVATI
 SSGGSYTYYPDSVKGFRFTISRDNKNTLYLQMSSSLKSEDTAMYYCARHPIY
 YTYDDTMDYWGQGTSTVTVSSGGGGSGGGSGGGSDIVLTQSPAIMAASPG
 EKVMTTCSASSVSSGNFHWYQQKPGTSPKLIWYRTSNLASGVPARFSGSG
 SGTSYSLTISSEAEADAATYYCQQWSGYPWTFGGGKLEIK (anti-Ax1
 scFv)

Preparation Example 11

As another non-limiting exemplary embodiment of the
 binding molecule containing a scaffold protein between the
 first region and the second region, a heterodimeric bispecific
 antibody, wherein a scFv of anti-Ax1 antibody and an
 anti-amyloid antibody are employed as the first region and
 the second region, respectively, is manufactured. The first
 polypeptide of a heavy chain of the bispecific antibody has
 the following sequence of SEQ ID NO: 168 of Table 13
 below, the second polypeptide of a heavy chain of the
 bispecific antibody comprises the sequence of SEQ ID NO:
 169 of Table 13 below, and the light chain of anti-amyloid
 antibody has the sequence of SEQ ID NO: 161. The Fc
 region contains NA mutation to reduce or abolish Fc gamma
 receptor binding affinity and the polypeptides of the Fc
 region form a hetero dimer (DD-KK).

TABLE 13

Bispecific heterodimeric comprising bivalent second
 region and bivalent first region: first polypeptide
 of heavy chain of anti-amyloid Ab-anti-Ax1 ScFv
 (with Fc region DD) (SEQ ID NO: 168)
 METDTLLLVLLWVPGSTGD (SS)
 EVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHWVRQAPGKLEWVAVI
 WFDGTYKYYTDSVKGFRFTISRDNKNTLYLQMNTLRAEDTAVYYCARDRI
 GARRGPYYMDVWGKGTITVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
 KDYFPEPVTWSNNGALTSKVHTFPFAVLQSSGLYSLSSVTVTPSSSLGTQT
 YICNVNHPKSTNTKVDKKEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPPKPK
 DTLMTISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNATKPREEQYAST
 YRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
 LPSPRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYDTTPPVLDSD
 GSFFLYSCLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
 (anti-amyloid antibody heavy chain (DD))
 GGGSGGGGS (Linker)
 EVKLVESGGDLVKGPSGLKLSCAASGFTFSSYGMWSVRQTPDKRLEWVATI
 SSGGSYTYYPDSVKGFRFTISRDNKNTLYLQMSSSLKSEDTAMYYCARHPIY
 YTYDDTMDYWGQGTSTVTVSSGGGGSGGGSGGGSDIVLTQSPAIMAASPG
 EKVMTTCSASSVSSGNFHWYQQKPGTSPKLIWYRTSNLASGVPARFSGSG
 SGTSYSLTISSEAEADAATYYCQQWSGYPWTFGGGKLEIK (anti-Ax1
 scFv)
 Second polypeptide of heavy chain of anti-amyloid
 Ab (with Fc region KK) (SEQ ID NO: 169)
 METDTLLLVLLWVPGSTGD (SS)
 EVQLVESGGGVQPGSRSLRLSCAASGFAFSSYGMHWVRQAPGKLEWVAVI
 WFDGTYKYYTDSVKGFRFTISRDNKNTLYLQMNTLRAEDTAVYYCARDRI
 GARRGPYYMDVWGKGTITVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
 KDYFPEPVTWSNNGALTSKVHTFPFAVLQSSGLYSLSSVTVTPSSSLGTQT
 YICNVNHPKSTNTKVDKKEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPPKPK
 DTLMTISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNATKPREEQYAST
 YRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
 LPSPRKELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
 GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
 (anti-amyloid antibody heavy chain (KK))

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Experimental Example 1. Gas6-Based Fusion
 Molecule Targeting Beta-Amyloid (I):
 Beta-Amyloid Binding Domain in scFv Form

1-1. Analysis of Expression of Fusion Molecule in
 Transfected Cells

After plasmid transfection into HEK293 cells, the expres-
 sion of the fusion molecule containing the Flag tag accord-
 ing to Preparation Example 1 analyzed by Western blot
 analysis using the Flag tag, and the results are shown in FIG.
 2.

1-2. Analysis of Beta-Amyloid Specific Binding
 Affinity of Prepared Fusion Molecules

To verify whether each of α A β -Gas6(E), α A β -Gas6,
 α FITC-Gas6(E), and α FITC-Gas6 can selectively recognize
 beta-amyloid and FITC, the culture broth secreted from the
 HEK293 transfected with each plasmid was collected and
 subjected to an experiment using beta-amyloid oligomer and
 FITC-conjugated beads. The results showed that α A β -Gas6
 (E) and α A β -Gas6 recognized only beta-amyloid oligomer
 beads, and α FITC-Gas6 (E) and α FITC-Gas6 recognized
 only FITC beads, thereby inducing phagocytosis, as shown
 in FIG. 4.

Although α A β -Gas6 (E) and α A β -Gas6 were shown to
 exhibit similar activities, it was found that α A β -Gas6
 obtained by additionally removing the EGF domain of Gas6
 could be obtained in high yield without aggregation in the
 protein purification process. Thus, α A β -Gas6 was used in
 subsequent experiments.

1-3. Analysis of Mechanism of Action of Prepared
 Fusion Molecule

(1) Analysis Using Cell Line

An in vitro A β engulfment assay was developed, in which
 beta-amyloid oligomers are conjugated with a pH indicator
 and hence can emit red fluorescence in intracellular lyso-
 somes when they are uptaken by phagocytosis.

As a result of performing the in vitro A β engulfment assay
 with HMC3 cells, a human microglial cell line expressing
 TAM receptors, it was shown that beta-amyloid oligomers
 were selectively cleared by α A β -Gas6 (FIGS. 5 and 6).

In particular, in an experiment where cells were treated
 additionally with an antibody that interferes with the func-
 tion of TAM receptors, it was confirmed that α A β -Gas6
 cleared beta-amyloid oligomers mainly through Ax1 among
 Tyro3, MerTK, and Ax1 (FIGS. 7 to 9). In fact, when Ax1
 was removed from HMC3 cells, the activity of α A β -Gas6
 significantly decreased. In addition, THP-1, which is a
 human monocyte cell line that does not express TAM
 receptors, did not show an increase in beta-amyloid clear-
 ance by α A β -Gas6, while THP-Ax1 cells overexpressing
 Ax1 exhibited a significantly increased ability to clear beta-
 amyloid fibrils in a manner dependent on α A β -Gas6.

Next, since THP-Ax1 cells express both Ax1 and Fc
 receptors, the degree of inflammatory response induced
 upon beta-amyloid uptake by each of α A β -Gas6 and adu-
 canumab was analyzed in those cells. To this end, the NF-kB
 reporter was first expressed in THP-Ax1 cells, and each of a
 control, α A β -Gas6 and aducanumab was added to the cells
 together with beta-amyloid oligomers. As a result, it was
 confirmed that, when aducanumab was added, the expres-
 sion of the NF-kB reporter significantly increased, but when
 α A β -Gas6 was added, the NF-kB reporter was expressed at

or below the control level (FIG. 10). In addition, as a result of measuring the secreted protein levels of IL-1b, IL-6 and TNF, which are the three most representative inflammatory cytokines, it was shown that, when THP-Axl cells were treated with aducanumab, the protein levels of the inflammatory cytokines in the treated cells significantly increased compared to those in the control group (FIG. 11). In contrast, importantly, the levels of these inflammatory cytokines in the cells treated with α A β -Gas6 did not increase compared to those in the control group. This is a key result that, as our hypothesis suggests, the α A β -Gas6 fusion phagocytosis-inducing protein does not induce an inflammatory response when phagocytosing a target substance through a TAM receptor, which is similar to recognition and efferocytosis of naturally apoptotic cells.

In addition, unlike aducanumab, α A β -Gas6 increased the expression of Twist1/2 gene, which is known as a mechanism of suppressing inflammatory responses (FIG. 12).

(2) Analysis Using Astrocytes and Microglia

To examine whether astrocytes and microglia, which are cells expressing TAM receptors in the brain, can clear beta-amyloid through α A β -Gas6, primary astrocytes and microglia obtained from mouse brains were separately purified and then cultured. Then, each of purified α A β -Gas6 and aducanumab was added to the cells together with beta-amyloid fibrils, and the degree of clearance of beta-amyloid fibrils was measured in real time.

The results showed that α A β -Gas6 increased the beta-amyloid clearing ability of microglia in a concentration-dependent manner, which is similar to the results obtained in HMC3 which is a cell line expressing Axl (FIG. 13). Importantly, it was shown that, when aducanumab was added, the beta-amyloid clearing ability of astrocytes did not change at all, but when α A β -Gas6 was added, the beta-amyloid clearing ability of astrocytes significantly increased in a concentration-dependent manner (FIG. 14). This suggests that α A β -Gas6 significantly enhances the beta-amyloid clearing ability of astrocytes, which was previously insignificant, because astrocytes do not express Fc receptors but express TAM receptors.

Each of α A β -Gas6 and aducanumab was added to astrocytes and the microglia cell line BV2 together with beta-amyloid fibrils to increase beta-amyloid uptake, and then the mRNA levels of TNF, IL-1a and IL-1b in each cell line were measured to determine the degree of inflammatory responses (FIGS. 15 and 16). As a result, similar to the results obtained in the cell lines, it was shown that, when the cells were treated with aducanumab, the levels of transcripts and proteins of the above inflammatory cytokines in the astrocytes and BV2 cells significantly increased compared to those in the control group, but when the cells were treated with α A β -Gas6, the levels of these inflammatory cytokines in the cells did not increase compared to those in the control group.

As described above, it has been found that the use of the α A β -Gas6 fusion phagocytosis inducer may be a groundbreaking method of effectively clearing beta-amyloid plaques accumulated in the patient's brain, through astrocytes and microglia without causing an inflammatory response, which is a serious side effect of existing monoclonal antibody therapeutics. This could be a very encouraging result that can significantly improve current treatment strategies.

1-4. Evaluation of In Vivo Efficacy

(1) Efficacy According to Introduction of Fusion Molecule or Expression Vector Containing the Same

5x FAD mice were used as Alzheimer's disease model mice. Since 5x FAD mice simultaneously express 5 genes with mutations, the onset at which beta-amyloid plaques are generated in the mice is early, and pathological symptoms caused by beta-amyloid plaques can be studied from 3 to 4 months of age regardless of aging.

To verify the effect of α A β -Gas6 in vivo through the 5x FAD model, α A β -Gas6 was delivered to the brain in two different ways. Through previous studies, it is known that aducanumab is not delivered well to the brain by intravascular injection or intraperitoneal injection even in Alzheimer's disease model mice. Thus, to accurately compare and analyze the effects of α A β -Gas6 with aducanumab, 1) direct cannulation was performed in the mouse brain, and each of purified α A β -Gas6 and aducanumab was injected once a day into the ventricle of the brain for 3 weeks, and 2) each of α A β -Gas6 and aducanumab was made in lentiviral form to be expressed in the hippocampus of the mouse through stereotaxic injection. Importantly, it was found that the number of beta-amyloid plaques significantly decreased both when the purified α A β -Gas6 protein was added and when the gene was expressed in lentiviral form (FIGS. 17 and 18).

In addition, by quantifying the levels of beta-amyloid in lysosomes of microglia and astrocytes after α A β -Gas6 was delivered to the brain in the form of protein or virus, it was shown that the ability to clear beta-amyloid significantly increased in both types of cells (FIGS. 19 to 22).

This suggests that, since TAM receptors are expressed in both microglia and astrocytes, microglia and astrocytes can recognize and clear beta-amyloid when α A β -Gas6 is introduced therein, which is similar to the results of the in vitro studies.

(2) Comparison of Effects of Antibody Therapeutics and Fusion Molecule of the Present Invention

It is known that, in Alzheimer's disease, synapses are indiscriminately removed by microglia, resulting in a decrease in the number of synapses. Surprisingly, this phenomenon was aggravated when aducanumab was delivered to Alzheimer's model mice, but when α A β -Gas6 was expressed in a viral form, abnormal removal of synapses by microglia was restored to a normal level (FIGS. 23 and 24).

In addition, as in the results from a cognitive and memory test for remembering the shape or location of a new object in Alzheimer's model mice according to the protocol shown in FIG. 25, it was confirmed that the expression of α A β -Gas6 exhibited significantly superior cognitive and memory recovery effects compared to aducanumab (FIG. 26).

In addition, to verify whether the chimeric phagocytic protein of the present disclosure is effective in clearing various target substances, phagocytosis-inducing proteins specific for tau and alpha-synuclein (α Syn) in addition to beta-amyloid were prepared as described in Preparation Examples 2 and 3, and the target substance clearing effects thereof were tested following protocols in Experimental Examples 2 and 3.

1-5. Assay of Fusion Molecules of Preparation Examples 6-11

By following the procedure of Example 1-1 through 1-4, the properties and in vivo efficacy of clearing amyloids are evaluated for the fusion molecules of Preparation Examples 6-11.

Experimental Example 2. Gas6-Based Fusion Molecule Targeting Tau

An in vitro tau engulfment assay was developed, in which tau oligomers are conjugated with a pH indicator and hence can emit red fluorescence in intracellular lysosomes when they are uptaken by phagocytosis. HMC3 cells, a human microglial cell line expressing TAM receptors, were treated with a culture medium expressing the phagocytosis-inducing protein [α Tau-Gas6] according to Preparation Example 2, and in vitro tau engulfment assay was performed. As the result shown in FIG. 27, it was confirmed that tau oligomers were selectively cleared by α Tau-Gas6.

Experimental Example 3. Gas6-Based Fusion Molecule Targeting Alpha-Synuclein

An in vitro α Syn engulfment assay was developed, in which alpha-synuclein (α Syn) oligomers are conjugated with a pH indicator and hence can emit red fluorescence in intracellular lysosomes when they are uptaken by phagocytosis. HMC3 cells, a human microglial cell line expressing TAM receptors, were treated with a culture medium expressing the phagocytosis-inducing protein [α Syn-Gas6] according to Preparation Example 3, and in vitro tau engulfment assay was performed. As the result shown in FIG. 27, it was confirmed that α Syn oligomers were selectively cleared by α Syn-Gas6.

Experimental Example 4. ProS1-Based Fusion Molecule Targeting Beta-Amyloid

Next, to verify whether the chimeric phagocytosis-inducing protein prepared using a ligand for TAM receptor other than Gas6 is also effective, α A β -ProS1 was prepared as described in Preparation Example 4 using the ProS1 ligand, and the efficacy thereof was evaluated. To this end, primary-cultured mouse astrocytes expressing TAM receptors were treated with a culture medium expressing α A β -ProS1, and

the in vitro A β engulfment assay used in Experimental Example 1-3 was performed. As the result shown in FIG. 29, it was confirmed that beta-amyloid oligomers were selectively cleared by α A β -ProS1.

Experimental Example 5. Gas6-Based Fusion Molecule Targeting Beta-Amyloid (II): Beta-Amyloid Binding Regions in the Forms of Fab and Mab

Next, to verify whether various target-binding regions other than scFv may be used as target protein-binding domains in the preparation of chimeric phagocytosis-inducing proteins, phagocytosis-inducing proteins were prepared according to Preparation Example 5 using an antigen-binding fragment (Fab) or a complete-form monoclonal antibody (Mab) instead of an scFv and were subjected to an experiment (α A β [Fab]-Gas6 and α A β [Mab]-Gas6). To this end, HMC3 cells, a human microglial cell line expressing TAM receptors, were treated with a culture medium expressing each of α A β [Fab]-Gas6 and α A β [Mab]-Gas6, and the in vitro A β engulfment assay used in Experimental Example 1-3 was performed. As the results shown in FIGS. 30 and 31, it was confirmed that beta-amyloid oligomers were selectively cleared by each of α A β [Fab]-Gas6 and α A β [Mab]-Gas6.

The scope of the present disclosure is defined by the appended claims, and all changes or modifications derived from the meaning and scope of the claims and equivalents thereto should be construed as being included in the scope of the present invention.

The fusion molecules having phagocytosis-inducing activity according to the embodiment of the present disclosure can solve the problem of tissue damage caused by activation of an inflammatory response, which occurs in the prior art. Accordingly, the fusion molecules could effectively clear abnormally accumulated substances such as beta-amyloid, tau, alpha-synuclein, huntingtin or prion protein, and thus may be used to prevent or treat diseases caused by these abnormally accumulated substances, for example, Alzheimer's disease, Parkinson's disease, Huntington's disease, or prion disease. Therefore, it may be used in the therapeutics industry for treatment of the above diseases.

All publications, patent applications, patents, and other references mentioned herein are expressly incorporated herein by reference in their entireties.

SEQUENCE LISTING

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Sequence total quantity: 169
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source               1..173
                     mol_type = protein
                     organism = synthetic construct

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GRMFSGTPVI RLFKRLQPT RLVAEFDERT FDPEGILLFA GGHQDSTWIV LALRAGRLLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAGAG DLFQPERGLY 120
HLNLTVGIGP FHEKDLVQPI NPRLDGCMRS WNWLNGETT IQETVKVNTR MQC 173

SEQ ID NO: 2          moltype = AA length = 194
FEATURE              Location/Qualifiers
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source               1..194
                     mol_type = protein
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                                organism = synthetic construct
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VALVDYHSTK KLKKQLVVLVA VEHTALALME IKVCDGQEHV VTVSLRDGEA TLEVDTGTRGQ 120
SEVSAALQLE RLAVLERHLR SPVLTFAAGL PDVPVTSAPV TAFYRGCMTL EVNRRLLDL 180
EAAYKHSdit AHSC 194

SEQ ID NO: 3                moltype = AA length = 177
FEATURE                     Location/Qualifiers
REGION                     1..177
                           note = TAM(Ax1)-binding sequence
source                     1..177
                           mol_type = protein
                           organism = synthetic construct

SEQUENCE: 3
LLYLAEQFAG VVLYLKFRLLP EISRFSAEFD FRTYDSEGLV LYAESIDHSA WLLIALRGKK 60
IEVQLKNEHT SKITTGGDVI NNLGNMVS EELHSISIK IAKEAVMDIN KPGPLFKPEN 120
GLEETKVYFA GFPRKVESEL IKPINRLDG CIRSWNLKQ GASGIKEIIQ EKQNKHC 177

SEQ ID NO: 4                moltype = AA length = 183
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source                     1..183
                           mol_type = protein
                           organism = synthetic construct

SEQUENCE: 4
YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSDILLS VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 5                moltype = AA length = 400
FEATURE                     Location/Qualifiers
REGION                     1..400
                           note = TAM(Ax1)-binding sequence
source                     1..400
                           mol_type = protein
                           organism = synthetic construct

SEQUENCE: 5
DILPCVPFVS AKSVKSLYL RMFSGTPVIR LRFKRLQPTR LVAEFDRTFD DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNRD 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKVNTRM QCFSVTERGS FYPGSGFAPY SLDYMRTPLD VGTESTWEVE VVAHIRPAAD 240
TGVLFWALWAP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HTALALMEIK VCDGQEHVVT 300
VSLRDGEATL EVDGTRGQSE VSAAQLQERL AVLERHLRSP VLTFAAGLDP VPVTSAPVTA 360
FYRGCMTELV NRRLLDLDEA AYKHSditAH SCPPVEPAAA 400

SEQ ID NO: 6                moltype = AA length = 393
FEATURE                     Location/Qualifiers
REGION                     1..393
                           note = TAM(Ax1)-binding sequence
source                     1..393
                           mol_type = protein
                           organism = synthetic construct

SEQUENCE: 6
VVSVCPLPNL DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA LRGGKIEVQL KNEHTSKITT GGDVINNLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL FKPENGLLET KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK HCLVTVEKGS YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV PFAVSLVDST SEKSDILLS VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN 300
LELSTPLKIE TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE AISKHNDIRA HSCPSVWKKT KNS 393

SEQ ID NO: 7                moltype = AA length = 678
FEATURE                     Location/Qualifiers
REGION                     1..678
                           note = GAS6 protein
source                     1..678
                           mol_type = protein
                           organism = Homo sapiens

SEQUENCE: 7
MAPSLSPGPA ALRRAPQLLL LLLAAECALA ALLPAREATQ FLRPRQRRAF QVFEEAKQGH 60
LERECVEELC SREEAREVFE NDPETDYFYP RYLDCKINKY SPYTKNSGFA TCQNLDPDQC 120
TPNPCDRKGT QACQDLGMNF FCLCKAGWGG RLCDKDVNEC SQENGGCLQI CHNKPGSFHC 180
SCHSGFELSS DGRTCQDIDE CADSEACGEA RCKNLPGSYS CLCDEGPAYS SQEKACRDVD 240
ECLQGRCEQV CVNSPGSYTC HCDGRGGLKL SQMDTDCEDI LPCVPFVSAK SVKSLYLGRM 300
FSGTPVIRLR FKRLQPTRLV AEFDFRTFD EGIILFAGGH QDSTWIVLAL RAGRLQLRL 360
YNGVGRVTSS GPVINHGMWQ TISVEELARN LVIKVNDAV MKIAVAGDLF QPERGLYHLN 420

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LTVGGIPFHE KDLVQPINPR LDGCMRSWNW LNGEDTTIQE TVKVNTRMQC FSVTERGSFY 480
PGSGFAFYSL DYMRTPLDVG TESTWEVEVV AHIRPAADTG VLFALWAPDL RAVPLSVALV 540
DYHSTKCLKK QLVVLAVEHT ALALMEIKVC DGQEHVVTVS LRDGEATLEV DGTRGQSEVS 600
AAQLQERLAV LERHLRSPVL TFAGGLPDVP VTSAPVTAFY RGCMTLEVNR RLLDLDEAAY 660
KHSDITAHSC PPVEPAAA 678

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SEQ ID NO: 8      moltype = AA length = 173
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REGION           1..173
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                 organism = synthetic construct

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SEQUENCE: 8
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV LALRAGRLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAVAG DLFQPERGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCMRS WNWLNEDTT IQETVKANTK MQC 173

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SEQ ID NO: 9      moltype = AA length = 173
FEATURE          Location/Qualifiers
REGION           1..173
                 note = TAM(Ax1)-binding sequence
source           1..173
                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 9
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV LALRAGRLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAVAG DLFQPERGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCMRS WNWLNEDTT IQETVKANTK MQC 173

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SEQ ID NO: 10     moltype = AA length = 172
FEATURE          Location/Qualifiers
REGION           1..172
                 note = TAM(Ax1)-binding sequence
source           1..172
                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 10
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV LALRAGRLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAVAG DLFQPERGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCMRS WNWLNEDTT IQETVKANTR MQ 172

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SEQ ID NO: 11     moltype = AA length = 173
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REGION           1..173
                 note = TAM(Ax1)-binding sequence
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                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 11
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGVLFFA GGHQDSTWIV LALRAGRLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ERNLVIKVN R DAVMKIAVAG DLFQLDRGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCLRS WNWLNEDTT IQETVKVNTK MQC 173

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SEQ ID NO: 12     moltype = AA length = 173
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REGION           1..173
                 note = TAM(Ax1)-binding sequence
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                 organism = synthetic construct

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SEQUENCE: 12
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV LALRAGRLEL 60
QLRYNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAVAG DLFQPERGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCMRS WNWLNEDTT IQETVKANAK MQC 173

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SEQ ID NO: 13     moltype = AA length = 173
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                 note = TAM(Ax1)-binding sequence
source           1..173
                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 13
GRMFSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV LALRAGRLEL 60
QLRYSGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKIAVAG DLFQPERGLY 120
HLNLTVGGINP FHEKDLVQPI NPLRDGCMRS WNWLNEDTT IHEAVKVNAR MQC 173

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SEQ ID NO: 14     moltype = AA length = 171

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	mol_type = protein	
	organism = synthetic construct	
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QLRYHGVSRV TSSGPVINHG MWQTISVEEL DRNLVVKVNR DAVMKIAVAG DLFQLDRGLY		120
HLNLTVGGIN PKERDLVQPI NPLDGCVRN WNWLNGETT IQETVKANPK M		171
SEQ ID NO: 15		
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	mol_type = protein	
	organism = synthetic construct	
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QLRYGGVGRV TSSGPVINHG TWQTISVEEL ERNVVVKVNR DAVMKIAVAG DLFQDRGLY		120
HLNLTVGGIN PKEKDLVQPI NPLDGCVRN WNWLNGETT IQETVRANAR MQC		173
SEQ ID NO: 16		
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source	1..173	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 16		
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QLRYGGVGRV TSSGPVINHG TWQTISVEEL DRNLVIKVNK DAVMKIAVAG DLFQDRGLY		120
HLNLTVGGIN PKEKDLVQPI NPLDGCVRN WNWLNGETT IQETVKANSK MQC		173
SEQ ID NO: 17		
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	organism = synthetic construct	
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QLRYGGVGRV TSSGPVINHG TWQTISVEEL ERNVVVKVNR DAVMKIAVAG DLFQDRGLY		120
HLNLTVGGIN PKEKDLVQPI NPLDGCVRN WNWLNGETT IQETVRANAR MQC		173
SEQ ID NO: 18		
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	mol_type = protein	
	organism = synthetic construct	
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QLRYGGVGRV TSSGPVINHG MWQTISVEEL ERNLVIKVNK DAVMKIAVAG DLFQDRGLY		120
HLNLTVGGIN PKEKDLVQPI NPLDGCVRN WNWLNGETT IQETVKVNVK MQC		173
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	mol_type = protein	
	organism = synthetic construct	
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QLHYNGVGRV TSSGPVINHG AWQTISVEEM ARSLVIKVNK DAVMKIAVAG DLFQPERGMF		120
HLNLTVGGIN PHEKDLVQPI NPLDGCVRN WNWLNGETT IQETVKVNTK MQC		173
SEQ ID NO: 20		
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SEQUENCE: 20		

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QLRYQGVRV	TSSGPVINHG	MWQTISVEEL	ERNLVIKVNK	DAVMKIAVAG	DLPQLDRGLY	120
HLNLTVGIP	FKEKDLVQPM	NPRLDGCMS	WNWLNGETT	IQETVKVNVK	MQC	173
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	organism = synthetic construct					
SEQUENCE: 21						
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TSSGPVINHG	MWQTISVEEL	ERNLVIKVNK	DAVMKIAVAG	DLPQLDRGLY	HLNLTVGIP	120
FKEKDLVQPM	NPRLDGCMS	WNWLNGETT	IQETVKVNVK	MQC		163
SEQ ID NO: 22	moltype = AA length = 170					
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source	1..170					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 22						
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YNGVGRVTSS	GPVINHGAWQ	TISVEEMARS	LVIKVNRDAV	MKIAVAGDLF	QPERGMFHLN	120
LTVGGIPHE	KDLVQPINPR	LDGCIRSWNW	MNGEDTTIQE	TVKVNTKMQC		170
SEQ ID NO: 23	moltype = AA length = 171					
FEATURE	Location/Qualifiers					
REGION	1..171					
	note = TAM(Ax1)-binding sequence					
source	1..171					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 23						
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RYQGVGRVTS	SGPVINHGMM	QTISVEELER	NLVIKVNKDA	VMKIAVAGDL	FQLDRGLYHL	120
NLTVGGIPFK	EKDLVQPMNP	RLDGCMSWN	WLNGEDTTIQ	ETVKVNVKMQ	C	171
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	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 24						
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VALVDYHSTK	KLKKQLVLA	VEHVALALME	IKVCDGQEHM	VTISLRGEA	TLEVDGTRGQ	120
SEVSAQLQE	RLAVLERHLR	SPVLTFAAGL	PDVPVTSAPV	TAFYRGCMTL	EVNRRLLDLD	180
EAYKHSDIT	AHSC					194
SEQ ID NO: 25	moltype = AA length = 194					
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source	1..194					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 25						
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VALVDYHSTK	KLKKQLVLA	VEHVALALME	IKVCDGQEHV	VTVSLRDGEA	TLEVDGTRGQ	120
SEVSATQLQE	RLAVLERHLR	SPVLTFAAGL	PDVPVTSAPV	TAFYRGCMTL	EVNRRLLDLD	180
EAYKHSDIT	AHSC					194
SEQ ID NO: 26	moltype = AA length = 189					
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	note = TAM(Ax1)-binding sequence					
source	1..189					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 26						
GSFYPGSGFA	FYSLDYMRT	LDIGESTWE	IEVVAHIRPA	ADTGVLFALW	VPDLRAVPLS	60
VALVDYHSTK	KLKKQLVLA	VEHVALALME	IKVCDGQEHV	VTVSLRDGEA	TLEVDGTRGQ	120
SEVSATQLQE	RLAVLERHLR	SPVLTFAAGL	PDVPVTSAPV	TAFYRGCMTL	EVNRRLLDLD	180
EAYKHSDI						189

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SEQ ID NO: 27      moltype = AA  length = 194
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source            1..194
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 27
GSFYPGSGFA FYSLDYMRTPLDVGTESAWIEVVAHIRPAADTGVLFALWVPDLRAVPLS 60
VALVDYHSTK KLKKQLVVLA VEHVALALMEIKVCDGQEHV VTVSLRDGEA TLEVDGTRGQ 120
SEVSAAQLQE RLAVLERHLR SPVLTFAGGL PDVPVTSAPV TAFYRGCMTEVNRRLLDLD 180
EAAYKHSDIT AHSC 194

SEQ ID NO: 28      moltype = AA  length = 194
FEATURE           Location/Qualifiers
REGION            1..194
                  note = TAM(Axl)-binding sequence
source            1..194
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 28
GSFYPGSSFA FYSLDYVRTPLDVGTESTWEIDVVAHIRPAADTGVLFALWVPDLRAVPLS 60
VALVDYHSTK KLKKQLVVLA VEHVALALMEIKVCDGQEHV VTVSLRDGEA TLEVDGTRGQ 120
SEVSATQLQE RLAVLERHLR SPVLTFAGGL PDVPVTSAPV TAFYRGCMTEVNRRLLDLD 180
EAAYKHGDIT AHSC 194

SEQ ID NO: 29      moltype = AA  length = 194
FEATURE           Location/Qualifiers
REGION            1..194
                  note = TAM(Axl)-binding sequence
source            1..194
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 29
GSFYPGSGFA FYSLDYMRTPLDIRTESTWEIEVVAHIRPAADTGVLFALWVPDLRAVPLS 60
VALVDYHSTK KLKKQLVVLA VEHVALALMEIKVCDGQEHM VTISLREGEA TLEVDGTRGQ 120
SEVSAAQLQE RLAVLEKHLQ SPVLTFAGGL PDVPVTSAPV TAFYRGCMTEVNRRLLDLD 180
EAAYKHSDIT AHSC 194

SEQ ID NO: 30      moltype = AA  length = 190
FEATURE           Location/Qualifiers
REGION            1..190
                  note = TAM(Axl)-binding sequence
source            1..190
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 30
GSFYPGSGFA FYSLDYMRTPLDIRTESTWEIEVVAHIRPAADTGVLFALWVPDLRAVPLS 60
VALVDYHSTK KLKKQLVVLA VEHVALALMEIKVCDGQEHM VTISLREGEA TLEVDGTRGQ 120
SEVSAAQLQE RLAVLEKHLQ SSVLTFAGGL PDVPVTSAPV TAFYRGCMTEVNRRLLDLD 180
EAAYKHSDIT 190

SEQ ID NO: 31      moltype = AA  length = 191
FEATURE           Location/Qualifiers
REGION            1..191
                  note = TAM(Axl)-binding sequence
source            1..191
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 31
FYPGGFCLPS LDYMRTPLDI GTESTWEIEVVAHIRPAADTGVLFALWVPDLRAVPLSVAL 60
VDYHSTKKLK KQLVVLAVEH VALALMEIKVCDGQEHMTI SLREGEATLEVDGTRGQSEV 120
SAAQLQERLA VLEKHLRSPV LTFAGGLPDV PVTAPVTAF YRGCMTEVNRRLLDLEAA 180
YKHS DITAHSC 191

SEQ ID NO: 32      moltype = AA  length = 195
FEATURE           Location/Qualifiers
REGION            1..195
                  note = TAM(Axl)-binding sequence
source            1..195
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 32
GRDRPASHGA QPVSPQAVRT PLDIGTESTW EIEVVAHIRP AADTGVLFALWVPDLRAVPL 60
SVALVDYHST KKLKKQLVVLA AVEHVALALMEIKVCDGQEH MVTVSLREGEATLEVDGTRG 120
QSEVSAAQLQ ERLAVLEKHL QSPVLTFAGG LPDVPVTSAP VTAFYRGCMTEVNRRLLDLD 180
DEAAYKHSDI TAHSC 195

SEQ ID NO: 33      moltype = AA  length = 194

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FEATURE Location/Qualifiers
 REGION 1..194
 note = TAM(Ax1)-binding sequence
 source 1..194
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 33
 GSFYPGNGFA FYSLNYM RTP LDVGTESTWE IEVVAHIRPA ADTGVLFWALW AADLRAVPLS 60
 VALVDYHSTK KLKKQLVVLA VERVALALME IKVCDGQEHV VTVSLREGEA TLAVDGTGRGQ 120
 SEVSAAQLQE RLATLHRHLQ SPVLTFAGGL PDVPVTSAPV TAFYRGCM TL EVNRRLLDLD 180
 EAAKKGHDIT SHSC 194

SEQ ID NO: 34 moltype = AA length = 676
 FEATURE Location/Qualifiers
 REGION 1..676
 note = Pro S protein
 source 1..676
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 34
 MRVLGGRGCA PLACLLLVLP VSEANLLSKQ QASQVLVRKR RANSLEETK QGNLERECIE 60
 ELCNKEEARE VFENDPETDY FYPKYLVLCLR SPQTGLFTAA RQSTNAYPDL RSCVNAIPDQ 120
 CSPLPCNEDG YMSCKDGKAS FTCTCKPGWQ GEKCEFDINE CKDPSNINGG CSQICDNTPG 180
 SYHSCCKNGF VMLSNKKDCK DVDECSLKPS ICGTAVCKNI PGDFECECPE GYRYNLKSKS 240
 CEDIDECSEN MCAQLCVNYP GGYTCYCDGK KGFKLAQDQK SCEVSVVCLP LNLDTKYELL 300
 YLAEQFAGVV LYLKFRLEPI SRFSAEFDFR TYDSEGVILY AESIDHSAWL LIALRGKKIE 360
 VQLKNEHTSK ITTGGDVINN GLWNMVSVEE LEHSISIKIA KEAVMDINKP GPLFKPENGL 420
 LETKVYFAGF PRKVESELK PINPRLDGCI RSWNLMKQGA SGIKEIIQEK QNKHCLVTVE 480
 KGSYYPGSGI AQPHIDYNNV SSAEGWHVNV TLNIRPSTGT GVMLALVSGN NTVPPAVSLV 540
 DSTSEKSDI LLSVENTVIY RIQALSLCSD QQSHLEFRVN RNNLELSTPL KIETISHEDL 600
 QRQLAVLDKA MKAKVATYLG GLPDVPFSAT PVNAFYNGCM EVNINGVQLD LDEAISKHND 660
 IRAHSCPSVW KKTKNS 676

SEQ ID NO: 35 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 35
 LLYLAEQFAG VVLYLKFRLEP EISRFSAEFD FRTYDSEGVI LYAESIDHSA WFLIALRGKK 60
 IEIQLKNEHT SKITTGGDII NNGLWNMVSV EELEHSISIK IAKEAVMDIN KPGPLFKPEN 120
 GLEETKVYFA GFPRKVESEL IKPINPRLDG CIRSWNLMKQ GASGIKEIIQ EKQNKHC 177

SEQ ID NO: 36 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 36
 LLYLAEQFAG VVLYLKFRLEP EISRFSAEFD FRTYDSQGI LYAESIDHSA WLLIALRGKK 60
 IEVQLKNEHT SKITTGGDII NNGLWNMVSV EELEHSISIK IAKEAVMDIN KPGPLFKPEN 120
 GLEETKVYFA GFPRKVESEL IKPINPRLDG CIRSWNLMKQ GASGIKEIIQ EKQNKHC 177

SEQ ID NO: 37 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 37
 LLYLAEQFAG VVLYLKFRLEP EISRFSAEFD FRTYDSQGI LYAESIDHSA WFLIALRGKK 60
 IEIQLKNEHT SKITTGGDII NNGLWNMVSV EELEHSISIK IAKEAVMDIN KPGPLFKPEN 120
 GLEETKVYFA GFPRKVESEL IKPINPRLDG CIRSWNLMKQ GASGIKEIIQ EKQNKHC 177

SEQ ID NO: 38 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 38
 LLYLAEQFAG VVLYLKFRLEP EISRFSAEFD FRTYDSQGI LYAESIDHSA WLLIALRGKK 60
 IEVQLKNEHT SKITTGGDII NNGLWNMVSV EELEHSISIK IAKEAVMDIN KPGPLFKPEN 120

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GLETKVYFA GFPRKVESEL IKPINRLDG CIRSWNLMKQ GASGIKEIIQ EKQNKHC 177

 SEQ ID NO: 39 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 39
 LLYLAEQFAG VVLYLKFRP EISRFSAEFD FRTYDSQGI LYAESIDHSA WLLIALRGGK 60
 IEVQLKNEHT SKITTGGGII NNGLWNMVS EEEHSSISIK IAKEAVMDIN KGPPLFKPEN 120
 GLETKVYFA GFPRKVESEL IKPINRLDG CIRSWNLMKQ GASGIKEIIQ EKQNKHC 177

 SEQ ID NO: 40 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 40
 LLYLAEQFAG VVLYLKFRP EISRFTAED FRTYDSEGVV LYAESIDHSA WILIAVRDGK 60
 FEVQLKNEQT SKITTGGGII NNGVWHTVS EEEHSSVSLK IAKEAVMNIN KGPPLFKPEH 120
 GFLETKVYFA GFPRKVESQF IKPINRLDG CIRGWNLMKQ GASGVKEIIQ EKQNKHC 177

 SEQ ID NO: 41 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 41
 LLYLAEQFVG VVLYLKFRP EISRFSAEFD FRTYDSEGI LYAESLDHSA WFLVALRDGK 60
 IEIQPKNEHT TKITTGGKVI SNGLWNMVS EEEHSSISVK IAKEAVMNIN KGPSLFNPTN 120
 GFLETKVYFA GFPRKVENAL IKPINRLDG CIRGWNLMKQ GASGVKEIIQ EKQNKHC 177

 SEQ ID NO: 42 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 42
 LLYLAEQFAG VVLYLKFRP EISRFTAED FRTYDSEGVV LYAESIDHSA WILIAVRDGK 60
 FEVQLKNEQT SKITTGGGII NNGVWHTVS EEEHSSVSLK IAKEAVMNIN KGPPLFKPEH 120
 GFLETKVYFA GFPRKVESQF IKPINRLDG CIRGWNLMKQ GASGVKEIIQ EKQNKHC 177

 SEQ ID NO: 43 moltype = AA length = 177
 FEATURE Location/Qualifiers
 REGION 1..177
 note = TAM(Ax1)-binding sequence
 source 1..177
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 43
 LLYLAEQFAG VVLYLKFRP EISRFTAED FRTYDSEGVV LYAESLDHSA WILIAVRDGK 60
 FEVQLKNEQT SRITTGGGVV NNGVWHTVS EEEHSSVSLK IAKEAVMNIN KGPPLFKPEH 120
 GFLETKVYFA GFPRQVESQF IKPINRLDG CIRGWNLMKQ GASGVKEIIQ EKQNKHC 177

 SEQ ID NO: 44 moltype = AA length = 182
 FEATURE Location/Qualifiers
 REGION 1..182
 note = TAM(Ax1)-binding sequence
 source 1..182
 mol_type = protein
 organism = synthetic construct

 VARIANT 42
 note = X can be any amino acid

 SEQUENCE: 44
 LLYLAEQISG VVLYLKFRP EISRFSAEFH FWTYDSEGMI LXAESVNHSA WLLIALRGGK 60
 IEVQLKNEHT SKITTEGDVI NNGLWNELST SQVSVEELEH SISIKIAKEA VMDIDKGPL 120
 FKPENGLLET KVFAGFPQK VESELIKPIN PCLDGCIRGW NLMKQASGI KEIIQEKQNK 180
 HC 182

 SEQ ID NO: 45 moltype = AA length = 177
 FEATURE Location/Qualifiers

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REGION          1..177
                note = TAM(Axl)-binding sequence
source          1..177
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 45
LLYLAEQFVG VVLYLKFRLLP EITRFSAEFD FRTYDSEGVI LYAESSDHSA WFLIALRDGK 60
IEIQFKNEHT TKITTGGRVI NDGLWNMVSV EELEQSISVK IAKEAVMNIN KPESLFPKPTN 120
GFLETKVYFA GLPRKVENAL IKPINRLDG CIRGWNLMNQ GASGVKEIIQ EKQNKHC 177

SEQ ID NO: 46    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(Axl)-binding sequence
source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 46
YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSDILL S VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAILDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 47    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(Axl)-binding sequence
source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 47
YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSDILL S VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 48    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(axl)-binding sequence
source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 48
YYPGSGIAQF HIDYNNVSSA EGWHINVTLN IRPSMTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSDILL S VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 49    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(axl)-binding sequence
source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 49
YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSDILL S VENTVIYRIQ APSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 50    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(axl)-binding sequence
source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 50
YYPGSGIAQF HIDYNNVSSA EGWHINVTLN IRPSMTGTGVM LALVSGNSTV PFAVSLVDST 60
SEKSDILL S VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC 183

SEQ ID NO: 51    moltype = AA length = 183
FEATURE         Location/Qualifiers
REGION          1..183
                note = TAM(Axl)-binding sequence

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source          1..183
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 51
YYPGSGIAEF HIDYNNGSNA EGVHINVTLN IRPSMTGTVM LALVSSNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYRIQ ALSLCNQRS HLEFRVNRNN LELLTPLKIE TISQEELQTQ 120
LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN VNGVELDLDE AISKHNDIRA 180
HSC                                                    183

SEQ ID NO: 52      moltype = AA length = 183
FEATURE           Location/Qualifiers
REGION            1..183
                  note = TAM(Ax1)-binding sequence
source            1..183
                  mol_type = protein
                  organism = synthetic construct
VARIANT           80
                  note = X can be any amino acid

SEQUENCE: 52
YYPGSGIAQF HVDYNNVSSA EGVHINVTLN IHPSMTGTVM LALVSGNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN LELLTPLKIE TISHEDLQRQ 120
LAILDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC                                                    183

SEQ ID NO: 53      moltype = AA length = 183
FEATURE           Location/Qualifiers
REGION            1..183
                  note = TAM(Ax)-binding sequence
source            1..183
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 53
YYPGSGIAQF HIDYNNGSNA EGVHINVTLN IRPSMTGTVM LALVSSNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYRIQ ALSLCNQRS HLEFRVNRNN LELLTPLKIE TISHEELQRQ 120
LAILDKAMTG KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC                                                    183

SEQ ID NO: 54      moltype = AA length = 183
FEATURE           Location/Qualifiers
REGION            1..183
                  note = TAM(Ax1)-binding sequence
source            1..183
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 54
YYPGSGIAQF HIDYNNGSNA EGVHINVTLN IRPSMTGTVM LALVSGNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYRIQ ALSLCNQRS HLEFRVNRNN LELLTPLKIE TISDEELRRQ 120
LAILDKAMTG KVATYLGGLP DVPFSATPVN AFYNGCMEVN INDVQLDLDE AISKHNDIRA 180
HSC                                                    183

SEQ ID NO: 55      moltype = AA length = 183
FEATURE           Location/Qualifiers
REGION            1..183
                  note = TAM(Ax1)-binding sequence
source            1..183
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 55
YYPGSGIAEF HIDYNNGSNA EGVHINVTLN IRPSMTGTVM LALVSSNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYRIQ ALSLCNQRS HLEFRANRNN LELLTPLKIE TISQEELQTQ 120
LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN INGVQLDLDE AISKHNDIRA 180
HSC                                                    183

SEQ ID NO: 56      moltype = AA length = 183
FEATURE           Location/Qualifiers
REGION            1..183
                  note = TAM(Ax1)-binding sequence
source            1..183
                  mol_type = protein
                  organism = synthetic construct
VARIANT           80
                  note = X can be any amino acid

SEQUENCE: 56
YYPGSGIAQF RIDYNNVSSV EGVHINVTLN IHPSMTGTVM LALVSGNNTV PFAVSLVDST 60
SEKSDIVLS VENTVIYLIQ ALSLCSDQQS HLEFIVNRNN LELLTPLKIE TISHEDLQRQ 120
LAILDKAMKA KVATYLGGLP DVPFSATPVN ALYKGCMEVN INGVQLDLDE AISKHNDIIA 180
HSC                                                    183

SEQ ID NO: 57      moltype = AA length = 183

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FEATURE	Location/Qualifiers	
REGION	1..183	
	note = TAM(Ax1)-binding sequence	
source	1..183	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 57		
YYPGSGIAKF	IVDYNVSSA	EGWYVNVSLN IRPSKGTGVM LALVSHNNTV PFAVSLVDST 60
SEKLDQDILLS	VEKTVIYRIQ	ALSLCSDQQF HLEFKVNRHN LEVSTPLKME TISHEDLQKQ 120
LAILDKAMQG	EVVTYLGGLP	DVPFSATPVN AFYNGCMEVN INGVLLDLDE AISKHNDIRA 180
HSC		183
SEQ ID NO: 58		
	moltype = AA length = 183	
FEATURE	Location/Qualifiers	
REGION	1..183	
	note = TAM(Ax1)-binding sequence	
source	1..183	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 58		
YYPGSGIAKF	IIDYNNASNA	EGWYVNVSLN IRPSTGTGVM LALVSRNNTV PFAVSLVDST 60
SEKLDQDILLS	VEKTVVCRIQ	ALSLCSDQQS HLEFKVNRHN LEVLTPLKME TISHEDLQKQ 120
LAILDKAMQG	DVVTYLGGLP	DVPFSATPVN AFYNGCMEVN INGVLLDLDE AISKHNDIRA 180
HSC		183
SEQ ID NO: 59		
	moltype = AA length = 183	
FEATURE	Location/Qualifiers	
REGION	1..183	
	note = TAM(Ax1)-binding sequence	
source	1..183	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 59		
YYPGSGIAKF	IIDYNNVSSA	EGWHVNVSLN IRPSMTGTGVM LALVSRNNTV PFAVSLVDST 60
SEKLDQDILLS	VEKTVIYRIE	ALSLCSDQQS HLELKVNRHS LEVSTPLKME TVSHEDIQKE 120
LAILDKAMQG	EVVTYLGGLP	DVSFSATPAN AFYNGCMEVN MNGVLLDLDE AISKHNDIRA 180
HSC		183
SEQ ID NO: 60		
	moltype = AA length = 176	
FEATURE	Location/Qualifiers	
REGION	1..176	
	note = TAM(Ax1)-binding sequence	
source	1..176	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 60		
SGIAQPHIDY	NNVSSAEGWH	VNVTLNIRPS TGTGVMLALV SGNNTVPFAV SLVDSTSEKS 60
QDILLSVENT	VIYRIQALSL	CSDQQSHLEF RVNRNNLELS TPLKIETISH EDLQRQLAVL 120
DKAMKAKVAT	YLGGLPDVPF	SATPVNAFYN GCMEVNINGV QLDLDEAISK HNDIRA 176
SEQ ID NO: 61		
	moltype = AA length = 173	
FEATURE	Location/Qualifiers	
REGION	1..173	
	note = TAM(Ax1)-binding sequence	
source	1..173	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 61		
HIDYNNVSSA	EGWHVNVTLN	IRPSTGTGVM LALVSGNNTV PFAVSLVDST SEKSQDILLS 60
VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN LELSTPLKIE TISHEDLQRQ LAVLDKAMKA 120
KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN INGVQLDLDE AISKHNDIRA HSC 173
SEQ ID NO: 62		
	moltype = AA length = 173	
FEATURE	Location/Qualifiers	
REGION	1..173	
	note = TAM(Ax1)-binding sequence	
source	1..173	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 62		
YYPGSGIAQF	HIDYNNVSSA	EGWHVNVTLN IRPSTGTGVM LALVSGNNTV PFAVSLVDST 60
SEKSQDILLS	VENTVIYRIQ	ALSLCSDQQS HLEFRVNRNN LELSTPLKIE TISHEDLQRQ 120
LAVLDKAMKA	KVATYLGGLP	DVPFSATPVN AFYNGCMEVN INGVQLDLDE AIS 173
SEQ ID NO: 63		
	moltype = AA length = 395	
FEATURE	Location/Qualifiers	
REGION	1..395	
	note = TAM(Ax1)-binding sequence	
source	1..395	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 63
VPFSVAKSVK  SLYLGRMFSG  TPVIRLRFKR  LQPTRLVAEF  DFRTFDPEGI  LLFAGGHQDS  60
TWIVLALRAG  RLEQLQRYNG  VGRVTSSGPV  INHGMWQTIS  VEELARNLVI  KVNRAVMKI  120
AVAGDLFQPE  RGLYHLNLTV  GGIPFHEKDL  VQPINPRLDG  CMRSWNWNLG  EDTTIQETVK  180
VNTRMQCFVS  TERGSFYPGS  GFAFYSLDYM  RTPLDVGTES  TWEVEVVAHI  RPAADTGVLF  240
ALWAPDLRAV  PLSVALVDYH  STKKLKKQLV  VLAVEHTALA  LMEIKVCDGQ  EHVVTVSLRD  300
GEATLEVDGT  RGQSEVSAAQ  LQERLAVLER  HLRSPVLTFA  GGLPDVPVTS  APVTAFYRGC  360
MTLEVNRRLL  DLDEAAYKHS  DITAHSCPPV  EPAAA  395

SEQ ID NO: 64      moltype = AA  length = 390
FEATURE            Location/Qualifiers
REGION             1..390
                   note = TAM(Ax1)-binding sequence
source             1..390
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 64
DILPCVPFVS  AKSVKSLYL  RMFSGTPVIR  LRFKRLQPTR  LVAEFDFRTE  DPEGILLFAG  60
GHQDSTWIVL  ALRAGRLELQ  LRYNGVGRVT  SSGPVINHGM  WQTISVEELA  RNLVIKVNRD  120
AVMKIAVAGD  LFQPERGLYH  LNLTVGGIPF  HEKDLVQPIN  PRLDGCMRSW  NWLNGEDTTI  180
QETVKVNTRM  QCFSVTERGS  FYPGSGFAFY  SLDYMRTPLD  VGTESTWEVE  VVAHIRPAAD  240
TGVLFALWAP  DLRAVPLSVA  LVDYHSTKKL  KKQLVVLAVE  HTALALMEIK  VCDGQEHVVT  300
VSLRDGEATL  EVDGTRGQSE  VSAAQLQERL  AVLERHLRSP  VLTFAAGGLPD  VPVTSAPVTA  360
FYRGCMTLEV  NRRLDLDEA  AYKHS DITAH  390

SEQ ID NO: 65      moltype = AA  length = 400
FEATURE            Location/Qualifiers
REGION             1..400
                   note = TAM(Ax1)-binding sequence
source             1..400
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 65
DILPCVPFVS  AKSVKSLYL  RMFSGTPVIR  LRFKRLQPTR  LVAEFDFRTE  DPEGILLFAG  60
GHQDSTWIVL  ALRAGRLELQ  LRYNGVGRVT  SSGPVINHGM  WQTISVEELA  RNLVIKVNKD  120
AVMKIAVAGD  LFQPERGLYH  LNLTVGGIPF  HEKDLVQPIN  PRLDGCMRSW  NWLNGEDTTI  180
QETVKANTRM  QCFSVTERGS  FYPGSGFAFY  SLDYMRTPLD  IGTESTWEIE  VVAHIRPAAD  240
TGVLFALWAP  DLRAVPLSVA  LVDYHSTKKL  KKQLVVLAVE  HVALALMEIK  VCDGQEHVVT  300
VSLRDGEATL  EVDGTRGQSE  VSATQLQERL  AVLERHLRSP  VLTFAAGGLPD  VPVTSAPVTA  360
FYRGCMTLEV  NRRLDLDEA  AYKHS DITAH  SCPPVEPAAA  400

SEQ ID NO: 66      moltype = AA  length = 400
FEATURE            Location/Qualifiers
REGION             1..400
                   note = TAM(Ax1)-binding sequence
source             1..400
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 66
DILPCVPFVS  AKSVKSLYL  RMFSGTPVIR  LRFKRLQPTR  LVAEFDFRTE  DPEGILLFAG  60
GHQDSTWIVL  ALRAGRLELQ  LRYNGVGRVT  SSGPVINHGM  WQTISVEELA  RNLVIKVNRD  120
AVMKIAVAGD  LFQPERGLYH  LNLTVGGIPF  HEKDLVQPIN  PRLDGCMRSW  NWLNGEDTTI  180
QETVKVNTRM  QCFSVTERGS  FYPGSGFAFY  SLDYMRTPLD  IGTESTWEIE  VVAHIRPAAD  240
TGVLFALWAP  DLRAVPLSVA  LVDYHSTKKL  KKQLVVLAVE  HVALALMEIK  VCDGQEHVVT  300
VSLRDGEATL  EVDGTRGQSE  VSAAQLQERL  AVLERHLRSP  VLTFAAGGLPD  VPVTSAPVTA  360
FYRGCMTLEV  NRRLDLDEA  AYKHS DITAH  SCPPVEPTTA  400

SEQ ID NO: 67      moltype = AA  length = 380
FEATURE            Location/Qualifiers
REGION             1..380
                   note = TAM(Ax1)-binding sequence
source             1..380
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 67
RMFSGTPVIR  LRFKRLQPTR  LVAEFDFRTE  DPEGILLFAG  GHQDSTWIVL  ALRAGRLELQ  60
LRYNGVGRVT  SSGPVINHGM  WQTISVEELA  RNLVIKVNRD  AVMKIAVAGD  LFQPERGLYH  120
LNLTVGGIPF  HEKDLVQPIN  PRLDGCMRSW  NWLNGEDTTI  QETVKVNTRM  QCFSVTERGS  180
FYPGSGFAFY  SLDYMRTPLD  VGTESTWEVE  VVAHIRPAAD  TGVLFALWAP  DLRAVPLSVA  240
LVDYHSTKKL  KKQLVVLAVE  HTALALMEIK  VCDGQEHVVT  VSLRDGEATL  EVDGTRGQSE  300
VSAAQLQERL  AVLERHLRSP  VLTFAAGGLPD  VPVTSAPVTA  FYRGCMTLEV  NRRLDLDEA  360
AYKHS DITAH  SCPPVEPAAA  380

SEQ ID NO: 68      moltype = AA  length = 400
FEATURE            Location/Qualifiers
REGION             1..400
                   note = TAM(Ax1)-binding sequence

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source                1..400
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 68
DILPCVPFVS AKSVKSLYLGM RMFSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKVNTRM QCFSVTERGS FYPGSGFAPY SLDYMRTPLD VGTESAWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHVVT 300
VSLRDGEATL EVDGTRGQSE VSAAQLQERL AVLERHLRSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLLDLDEA AYKHS DITAH SCPPVEPAAA 400

SEQ ID NO: 69         moltype = AA length = 400
FEATURE              Location/Qualifiers
REGION               1..400
                      note = TAM(Ax1)-binding sequence
source               1..400
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 69
DILPCVPFVS AKSVKSLYLGM RMFSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKANTRM QCFSVTERGS FYPGSGFAPY SLDYMRTPLD IGTESTWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHVVT 300
VSLRDSEATL EVDGTRGQSE VSATQLQERL AVLERHLRSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLLDLDEA AYKHS DITAH SCPPVEPAAA 400

SEQ ID NO: 70         moltype = AA length = 400
FEATURE              Location/Qualifiers
REGION               1..400
                      note = TAM(Ax1)-binding sequence
source               1..400
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 70
DILPCVPFVS AKSVKSLYLGM RMFSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKVNTRM QCFSVTERGS FYPGSGFAPY SLDYMRTPLD IGTESTWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHVVT 300
VSLRDGEATL EVDGTRGQSE VSAAQLQERL AVLERHLRSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLLDLDEA AYKHS DITAH SCPPVEPTTA 400

SEQ ID NO: 71         moltype = AA length = 395
FEATURE              Location/Qualifiers
REGION               1..395
                      note = TAM(Ax1)-binding sequence
source               1..395
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 71
LPCVPFVS AKSVKSLYLGRM FSGTPVIRL FRKRLQPTRLV AEFDFRTFDP EGILLFAGGH 60
QDSTWIVL ALRAGRLELQLR YNGVGRVTSS GPVINHGMMQ TISVEELARN LVIKVNDRDAV 120
MKIAVAGDLF QPERGLYHLN LTVGGIPFHE KDLVQPINPR LDGCMRSWNW LNGEDTTIQE 180
TVKVNTRMQC FSVTERGSFY PGSGFAPYSL DYMRTPLDVG TESTWEIEV VVAHIRPAADTG 240
VLPALWAPDL RAVPLSVALV DYHSTKKLKK QLVLVAVEHT ALALMEIKVC DGQEHVVTVS 300
LRDGEATLEV DGTGTRGQSEV AAQLQERLAV LERHLRSPVL TFAGGLPDVP VTSAPVTA 360
RGCMTELVNR RLLDLDEAAY KHS DITAHSC PPVEP 395

SEQ ID NO: 72         moltype = AA length = 400
FEATURE              Location/Qualifiers
REGION               1..400
                      note = TAM(Ax1)-binding sequence
source               1..400
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 72
DILPCVPFVS AKSVKSLYLGM RMFSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKANTKM QCFSVTERGS FYPGSGFAPY SLDYMRTPLD IGTESTWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHVMT 300
ISLREGATL EVDGTRGQSE VSAAQLQERL AVLEKHLRSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLLDLDEA AYKHS DITAH SCPPVEPAAA 400

SEQ ID NO: 73         moltype = AA length = 400
FEATURE              Location/Qualifiers
REGION               1..400

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source          note = TAM(Axl)-binding sequence
                1..400
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 73
DILPCVPFMS AKSVKSLYLGM RMSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSSWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKANTKM QCFSVTERGS FYPGSGFAFY SLDYMRPLD IGTESTWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHMVT 300
ISLREGEATL EVDGTRGQSE VSAAQLQERL AVLEKHLRSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLDLDEA AYKHS DITAH S C P P V E P A A A 400

SEQ ID NO: 74      moltype = AA length = 400
FEATURE           Location/Qualifiers
REGION            1..400
                  note = TAM(Axl)-binding sequence
source            1..400
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 74
DILPCVPFMS AKSVKSLYLGM RMSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
AVMKIAVAGD LFQPERGLYH LNLTVGGIPF HEKDLVQPIN PRLDGCMRSW NWLNGEDTTI 180
QETVKANPKM QCFSVTERGS FYPGSGFAFY SLDYMRPLD IGTESTWEIE VVAHIRPAAD 240
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHMVT 300
ISLREGEATL EVDGTRGQSE VSATQLQERL AVLEKHLRSP VLTFAGGLPD VPLTSAPVTA 360
FYRGCMTELV NRRLDLDEA AYKHS DITAH S C P P V E P T A A 400

SEQ ID NO: 75      moltype = AA length = 340
FEATURE           Location/Qualifiers
REGION            1..340
                  note = TAM(Axl)-binding sequence
source            1..340
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 75
DILPCVPFSL AKSVKSLYLGM RMSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGILLFAG 60
GHQDSTWIVL ALRAGRLELQ LRYNGVGRVT SSGPVINHGM WQTISVEELA RNLVIKVNDR 120
QETVKANAKM QCFSVTERGS FYPGSGFAFY SLDYMRPLD IRTESTWEIE VVAHIRPAAD 180
TGVLPALWVP DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE HVALALMEIK VCDGQEHMVT 240
ISLREGEATL EVDGTRGQSE VSAAQLQERL AVLEKHLQSP VLTFAGGLPD VSVTSAPVTA 300
FYRGCMTELV NRRLDLDEA AYKHS DITAH S C P P V E P A T A 340

SEQ ID NO: 76      moltype = AA length = 400
FEATURE           Location/Qualifiers
REGION            1..400
                  note = TAM(Axl)-binding sequence
source            1..400
                  mol_type = protein
                  organism = synthetic construct
VARIANT          100
                  note = X can be A or T

SEQUENCE: 76
DILPCVPFSV AKSVKSLYLGM RMSGTPVIR LRFKRLQPTR LVAEFDRTF DPEGVLLFAG 60
GHQDGTWVML ALRAGRLELQ LHYNGVGRVT SSGPVINHGX WQTISVEEMA RSLVIKVNDR 120
AVMKIAVAGD LFQPERGMFH LNLTVGGIPF HEKDLVQPIN PRLDGCIRSW NWMNGEDTTI 180
QETVKVNTKM QCFSVTERGS FYPGNGFAFY SLNYMRPLD VGTESTWEIE VVAHIRPAAD 240
TGVLPALWAA DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE RVALALMEIK VCDGQEHVVT 300
VSLREGEATL AVDGTGQSE VSAAQLQERL ATLERHLQSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLDLDEA AYKHGDITSH S C P T V E P A A A 400

SEQ ID NO: 77      moltype = AA length = 400
FEATURE           Location/Qualifiers
REGION            1..400
                  note = TAM(Axl)-binding sequence
source            1..400
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 77
DILPCVPFMS AKSVKSLYLGM RMSGTPVIR LRYRRLQPTR LVAEFDRTF DPEGVLLFAG 60
GHQDGTWIML ALRAGRLELQ LRYNGVGRIT SSGPVINHGT WQTISVEELA RSLVIKVNDR 120
AVMKIAVAGD LFQPERGMFH LNLTVGGIPF HEEDLVQPIN PRLDGCIRSW NWMNGEDTTI 180
QETVKVNSKM QCFSVTERGS FYPGNGFAFY SLNYMRPLD VGTESTWEIE VVAHIRPAAD 240
TGVLPALWAA DLRAVPLSVA LVDYHSTKKL KKQLVVLAVE RVALALMEIK VCDGQEHVVT 300
VSLREGEATL AVDGTGQSE VSAAQLQERL ATLERHLQSP VLTFAGGLPD VPVTSAPVTA 360
FYRGCMTELV NRRLDLDEA AYKHGDITSH S C P T V E P A A A 400

SEQ ID NO: 78      moltype = AA length = 399

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FEATURE Location/Qualifiers
 REGION 1..399
 note = TAM(Ax1)-binding sequence
 source 1..399
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78

DILPCVPFVS	AKSLKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDRTF	DPEGILFFAG	60
GHRDSTWIVL	ALRAGRLELQ	LLFNGVGRVT	SSGPVINHGM	WQTISVEELE	RNLVVKVNKD	120
AVMKIAVPGD	LFQLDRGLYH	LNLTVGGIPF	KEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKANAKM	QCFSLTEKGS	FFPGSGFAYF	SLGYVRTSLD	VGTTTWEIE	VEARIRPAAD	240
TGVLLALWAP	DHRAVPLSVA	LVDYHSTKKL	KKQLVVLAVE	SVALALMEIK	VCDGQEHVVS	300
VSVREDEATL	EVDGTKGQSE	VSAAQLQERL	AALGRHLRDP	VLTFIGGLPE	VPVTSAPVTA	360
FYRGCMTLEV	NRRPLDLDEA	SYKHSBITAH	SCPPVEPAA			399

SEQ ID NO: 79 moltype = AA length = 399

FEATURE Location/Qualifiers
 REGION 1..399
 note = TAM(Ax1)-binding sequence
 source 1..399
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 79

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDRTF	DPEGVLFAG	60
GRQDSTWIVL	ALRAGRLELQ	LRNGVGRVT	SSGPVINHGT	WQTISVEELE	RNLVVKVNKD	120
AVMKIAVAGD	LFQDRGLYH	LNLTVGGIPF	KEKELVQPIN	PRLDGCMRSW	NWLNSDITII	180
QETVKVNTKM	QCFSVTEKGS	FYPGTGFAYF	SLNYMRTSLD	TGTETTWEIK	VMARIRPATD	240
TGVLLALCAP	DHRTVPLSVA	LVDYHSTKKL	KKQLVVLAVE	SVVLALVEIK	ACDQGEHEVS	300
VSLKEGEATL	EVDGTRGWSE	ASATQLQERL	DTLRRHLHDP	VLTFGTGLPD	VPVTAAPVTA	360
FYRGCMTLEV	NRRLLDLDEA	AYKHSBITSH	SCPPVEPTA			399

SEQ ID NO: 80 moltype = AA length = 399

FEATURE Location/Qualifiers
 REGION 1..399
 note = TAM(Ax1)-binding sequence
 source 1..399
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 80

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDRTF	DPEGILLFAG	60
GHQDSTWIVL	ALRAGRLELQ	LRNGVGRVT	SSGPVINHGM	WQTISVEELA	RNLVVKVNKD	120
AVMKIAVAGD	LFQPERGLYH	LNLTVGGIPF	HEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKVNTKM	QCFSVTERGS	FYPGSGFAYF	SLDYMRTPLD	VGTESTWEVE	VVAHIRPAAD	240
TGVLPALWAP	DLRAVPLSVA	LVDYHSTKKL	KKQLVVLAVE	HTALALMEIK	VCDGQEHVVT	300
VSLRDGEATL	EVDGTRGQSE	VSAAQLQERL	AVLERHLRSP	VLTFAGGLPD	VPVTSAPVTA	360
FYRGCMTLEV	NRRLLDLDEA	AYKHSBITAH	SCPPVEPAA			399

SEQ ID NO: 81 moltype = AA length = 398

FEATURE Location/Qualifiers
 REGION 1..398
 note = TAM(Ax1)-binding sequence
 source 1..398
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 81

DILPCVPFVS	AKSMKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDRTF	DPEGVLFAG	60
GHQDSTWIVL	ALRAGRLELQ	LHYNGVGRVT	SSGPVINHGM	WQTISVEELE	RNVVVKVNKD	120
AVMKIAVAGD	LFQLDRGLYH	LNLTVGGIPF	KEKHLVQPIN	PRLDGCMRSW	SWLNGEDTTI	180
QETVKVNTKM	QCFSVTERGS	FYPGNGLAIY	SLNYSRTPPD	VGTTTWEVE	VVARIRPATD	240
TGVLLALVGD	GHTVPLSVAL	VDYHSTKKLK	KQLVVLAVEG	MALALMEIKV	CDGQEHVAV	300
SLKAGEASLE	VDGTRGQSEV	SAAQLQERLA	VLGRHLQGPL	LTFIGGLPDV	PVTSAPVTAF	360
YRGCMTLEVN	QKRLDLDEAT	YKHSBITSHS	CPPVEHAA			398

SEQ ID NO: 82 moltype = AA length = 397

FEATURE Location/Qualifiers
 REGION 1..397
 note = TAM(Ax1)-binding sequence
 source 1..397
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 82

DILPCVPFTM	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LLAEFDRTF	DPEGVLFAG	60
GRSDSTWIVL	GLRAGRLELQ	LRNGVGRIT	SSGPVINHGM	WQTISVEELE	RNLVVKVNKD	120
AVMKIAVAGE	LFQLERGLYH	LNLTVGGIPF	KEQDLVQPIN	PRLDGCMRSW	NWLNGEDSAI	180
QETVKANTKM	QCFSVTERGS	FFPGGAFAYF	SLNYTRTSLD	VGTTTWEVE	IVAWIRPATD	240
TGVLLALVGD	NHVVPLSVAL	VDYHSTKKLK	KQLVVLAVED	VALALMEIKV	CDSQEHVTV	300
SLRDGEATLE	VDGTRGQSEV	SAAQLQERLD	TLGHLQGTV	LTFVGGLEPEV	PVTSAPVTAF	360
YRGCMTLEVN	QKTLDLDEAS	YKHSBITSHS	CPPVEHA			397

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SEQ ID NO: 83 moltype = AA length = 398
 FEATURE Location/Qualifiers
 REGION 1..398
 note = TAM(Ax1)-binding sequence
 source 1..398
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 83

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDPRTF	DPEGILLFAG	60
GHQDSTWIVL	ALRAGRLELQ	LRYNVGRVT	SSGPVINHGM	WQTISVEELA	RNLVIKVNDR	120
AVMKIAVAGD	LFQPERGLYH	LNLTVGGIPF	HEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKVNTRM	QCFSVTERGS	FYPGSGFAPY	SLDYMRTPLD	VGTESTWEVE	VVAHIRPAAD	240
TGVLFPALWAP	DLRAVPLSVA	LVDYHSTKKL	KKQLVVLAVE	HTALALMEIK	VCDGQEHVVT	300
VSLRDGEATL	EVDGTRGQSE	VSAACLQERL	AVLERHLRSP	VLTFAGGLPD	VPVTSAPVTA	360
FYRGCMTLEV	NRRLDLDEA	AYKHS DITAH	SCPPVEPA			398

SEQ ID NO: 84 moltype = AA length = 398
 FEATURE Location/Qualifiers
 REGION 1..398
 note = TAM(Ax1)-binding sequence
 source 1..398
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 84

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDPRTF	DPEGVLFAG	60
GRQDSTWIVL	GLRAGRLELQ	LRYNVGRVT	SSGPVINHGM	WQTISVEELD	RNLVIKVNDR	120
AVMKIAVAGD	LFQDRLGLYH	LNLTVGGIPF	KEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKANSKM	QCFSVTERGS	FYPGTGFAPY	SLDYTRTSTA	VGTEATWEIE	VVAQIRPATD	240
TGVLALVAG	DHVVALSVAL	VDYHSTKKLK	KQLVVLAVEG	VTALALMEIKV	CDGQEHVAV	300
SVRKGEATLE	VDGTRGQSEV	TAAGLQESLA	VLGRHLQASV	LTFVGGLPDV	PVTSAPVTAF	360
YRGCMTLEVN	QQALDLDEAA	YKHS DITSHS	CPPVEQAG			398

SEQ ID NO: 85 moltype = AA length = 395
 FEATURE Location/Qualifiers
 REGION 1..395
 note = TAM(Ax1)-binding sequence
 source 1..395
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 85

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDPRTF	DPEGILLFAG	60
GHQDSTWIVL	ALRAGRLELQ	LRYNVGRVT	SSGPVINHGM	WQTISVEELA	RNLVIKVNDR	120
AVMKIAVAGD	LFQPERGLYH	LNLTVGGIPF	HEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKVNTRM	QCFSVTERGS	FYPGSGFAPY	SLDYMRTPLD	VGTESTWEVE	VVAHIRPAAD	240
TGVLFPALWAP	DLRAVPLSVA	LVDYHSTKKL	KKQLVVLAVE	HTALALMEIK	VCDGQEHVVT	300
VSLRDGEATL	EVDGTRGQSE	VSAACLQERL	AVLERHLRSP	VLTFAGGLPD	VPVTSAPVTA	360
FYRGCMTLEV	NRRLDLDEA	AYKHS DITAH	SCPPV			395

SEQ ID NO: 86 moltype = AA length = 394
 FEATURE Location/Qualifiers
 REGION 1..394
 note = TAM(Ax1)-binding sequence
 source 1..394
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 86

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDPRTF	DPEGVLFAG	60
GHQDSTWIVL	ALRAGRLELQ	LRYNGIGRVT	SSGSVINHGM	WQTISVEELE	RNLVIKINKD	120
AVMKIAVAGD	LFQDRLGLYH	LNLTVGGIPF	KDKDLVQPIN	PRLDGCMRSW	NWLNGEDSTI	180
QDTVKNVTM	QCFSVTEKGS	FYPGNFGAPF	SLNYVRTSLD	VGTEMTWEIE	VIARIRPATD	240
TGVLALVAK	NHTIPLSVAL	VDYHSTKKLK	KQLVILAIEN	VALSLMEIKV	CDGQEHVTV	300
SVKEGEATLV	VDGTRGQSEV	SPTQLQERLA	VLKRHLQDSV	HTFVGGVPDV	PVTSAPVTAF	360
YHGCMTLEVN	KKALDLDEAV	YKHS DITSHS	CPPI			394

SEQ ID NO: 87 moltype = AA length = 398
 FEATURE Location/Qualifiers
 REGION 1..398
 note = TAM(Ax1)-binding sequence
 source 1..398
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 87

DILPCVPFVS	AKSVKSLYL	RMFSGTPVIR	LRFKRLQPTR	LVAEFDPRTF	DPEGVLFAG	60
GHQDSAWIVL	GLRAGRLELQ	LRYGVGRVT	SSGPVINHGT	WQTISVEELD	RNLVIKVNDR	120
AVMKIAVAGD	LFQDRLGLYH	LNLTVGGIPF	KEKDLVQPIN	PRLDGCMRSW	NWLNGEDTTI	180
QETVKANSKM	QCFSVTERGS	FYPGTGFAPY	SLDYTRTSTA	VGTEAAWEIE	VVAWIRPATD	240
TGVLALVGG	DRVVALSVAL	VDYHSTKKLK	KQLVVLAVEG	VTALALMEIKV	CDGQEHVAV	300
SVQKEVTLE	VDGTRGQSEV	SAAGLQESLA	VLERHLQGSV	LTFVGGLPDV	PVTSAPVTAF	360
YRGCMTLEVN	QQTLDLDEAA	YKHS DITSHS	CPPVEQAS			398

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SEQ ID NO: 88      moltype = AA  length = 392
FEATURE           Location/Qualifiers
REGION            1..392
                  note = TAM(Ax1)-binding sequence
source            1..392
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 88
VSVCLPLNLD TKYELLYLAE QFAGVVLYLK FRLPEISRFS AEFDFRTYDS EGVILYAESI 60
DHSALLIAL RGGKIEVQLK NEHTSKITTG GDVINNGLWN MVSVEELEHS ISIKIAKEAV 120
MDINKPGPLF KPENGLLETK VYFAGFPRKV ESELIKPINP RLDGCIRSWN LMKQGASGIK 180
EIIQEKQNKH CLVTVEKGSY YPGSGIAQFH IDYNNVSSAE GWHVNVTLNI RPSTGTGVM 240
ALVSGNNTVP FAVSLVDSTS EKSQDILLSV ENTVIYRIQA LSLCSDQQSH LEFRVNRNRL 300
ELSTPLKIET ISHEDLQRQL AVLDKAMKAK VATYLGGLPD VPFSATPVNA FYNGCMEVNI 360
NGVQLDLDEA ISKHNDIRAH SCPSVWKKTK NS 392

SEQ ID NO: 89      moltype = AA  length = 388
FEATURE           Location/Qualifiers
REGION            1..388
                  note = TAM(Ax1)-binding sequence
source            1..388
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 89
LPLNLDTKYE LLYLAEQFAG VVLYLKFRLP EISRFSAEFD PRTYDSEGI LYAESIDHSA 60
WLLIALRGSK IEVQLKNEHT SKITTTGGDVI NNGLWNMVS EEEHESISIK IAKEAVMDIN 120
KPGPLFKPEN GLLETKVYFA GFPRKVESEL IKPINPRLDG CIRSWNLMKQ GASGIKEIIQ 180
EKQNKHCLVT VEKGSYYPGS GIAQFHIDYN NVSSAEGWHV NVTLNIRPST GTGVMALVS 240
GNNTVPFAVS LVDSTSEKSO DILLSVENTV IYRIQALSLC SDQQSHLEFR VNRNNLELST 300
PLKIETISHE DLQRQLAVLD KAMKAKVATY LGGLPDVPFS ATPVNAFYNG CMEVNINGVQ 360
LDLDEAISKH NDIRAHSCPS VWKKTNS 388

SEQ ID NO: 90      moltype = AA  length = 392
FEATURE           Location/Qualifiers
REGION            1..392
                  note = TAM(Ax1)-binding sequence
source            1..392
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 90
VSVCLPLNLD DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA LRGGKIEVQL KNEHTSKITT GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL FKPENGLLET KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK HCLVTVEKGS YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV PFAVSLVDST SEKSQDILLS VENTVIYRIQ ALSLCDQQS HLEFRVNRNN 300
LELSTPLKIE TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE AISKHNDIRA HSCPSVWKKTK KN 392

SEQ ID NO: 91      moltype = AA  length = 390
FEATURE           Location/Qualifiers
REGION            1..390
                  note = TAM(Ax1)-binding sequence
source            1..390
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 91
VSVCLPLNLD DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA LRGGKIEVQL KNEHTSKITT GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL FKPENGLLET KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK HCLVTVEKGS YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV PFAVSLVDST SEKSQDILLS VENTVIYRIQ ALSLCDQQS HLEFRVNRNN 300
LELSTPLKIE TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE AISKHNDIRA HSCPSVWKKTK 390

SEQ ID NO: 92      moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequence
source            1..393
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 92
VSVCLPLNLD DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA LRGGKIEVQL KNEHTSKITT GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL FKPENGLLET KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK HCLITVEKGS YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV PFAVSLVDST SEKSQDILLS VENTVIYRIQ ALSLCDQQS HLEFRVNRNN 300
LELSTPLKIE TISHEDLQRQ LAIDLKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360

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-continued

INGVQLDLDE AISKHNDIRA HSCPSVWKKT KNS 393

SEQ ID NO: 93 moltype = AA length = 393
 FEATURE Location/Qualifiers
 REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 93

VVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL	KFRLPEISRF	SAEFDPRTYD	SEGVILYAES	60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT	GGDVINNGWL	NMVSVEELEH	SISIKIAKEA	120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK	VESELIKPIN	PRLDGCIRSW	NLMKQGASGI	180
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAQF	HIDYNNVSSA	EGWHINVTLN	IRPSMTGTVM	240
LALVSGNNTV	PFAVSLVDST	SEKSDILLS	VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN	300
LELSTPLKIE	TISHEDLQRQ	LAVLDKAMKA	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVQLDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 94 moltype = AA length = 393
 FEATURE Location/Qualifiers
 REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 94

VVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL	KFRLPEISRF	SAEFDPRTYD	SEGVILYAES	60
IDHSAWLLIA	LRGGKIEIQL	KNEHTSKITT	GGDVINNGWL	NMVSVEELEH	SISIKIAKEA	120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK	VESELIKPIN	PRLDGCIRSW	NLMKQGASGI	180
KEIIQEKQNK	HCLITVEKGS	YYPGSGIAQF	HIDYNNVSSA	EGWHVNVTLN	IRPSTGTGVM	240
LALVSGNNTV	PFAVSLVDST	SEKSDILLS	VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN	300
LELSTPLKIE	TISHEDLQRQ	LAILDKAMKA	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVQLDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 95 moltype = AA length = 393
 FEATURE Location/Qualifiers
 REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 95

VVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL	KFRLPEISRF	SAEFDPRTYD	SEGVILYAES	60
IYHSAWLLIA	LRGGKIEVQL	KNEHTSKITT	GGDVINNGWL	NMVSVEELEH	SISIKIAKEA	120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK	VESELIKPIN	PRLDGCIRSW	NLMKQGASGI	180
KEIIQEKQNK	HCLITVEKGS	YYPGSGIAQF	HIDYNNVSSA	EGWHVNVTLN	IRPSTGTGVM	240
LALVSGNNTV	PFAVSLVDST	SEKSDILLS	VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN	300
LELSTPLKIE	TISHEDLQRQ	LAVLDKAMKA	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVQLDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 96 moltype = AA length = 393
 FEATURE Location/Qualifiers
 REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 96

VVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL	KFRLPEISRF	SAEFDPRTYD	SEGVILYAES	60
IDHSAWLLIA	LRGGKIEIQL	KNEHTSKITT	GGDVINNGWL	NMVSVEELEH	SISIKIAKEA	120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK	VESELIKPIN	PRLDGCIRSW	NLMKQGASGI	180
KEIIQEKQNK	HCLITVEKGS	YYPGSGIAQF	HIDYNNVSSA	EGWHVNVTLN	IRPSTGTGVM	240
LALVSGNNTV	PFAVSLVDST	SEKSDILLS	VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN	300
LELSTPLKIE	TTSHEDLQRQ	LAILDKAMKA	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVQLDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 97 moltype = AA length = 393
 FEATURE Location/Qualifiers
 REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 97

VVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL	KFRLPEISRF	SAEFDPRTYD	SEGVILYAES	60
IDHSAWLLIA	LRGGKIEIQL	KNEHTSKITT	GGDVINNGWL	NMVSVEELEH	SISIKIAKEA	120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK	VESELIKPIN	PRLDGCIRSW	NLMKQGASGI	180
KEIIQEKQNK	HCLITVEKGS	YYPGSGIAQF	HIDYNNVSSA	EGWHVNVTLN	IRPSTGTGVM	240
LALVSGNNTV	PFAVSLVDST	SEKSDILLS	VENTVIYRIQ	ALSLCSDQQS	HLEFRVNRNN	300

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LELSTPLKIE TISHEDLQRQ LAILDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE AISKHNDIRA HSCPSVWKKT KNS 393

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```

SEQ ID NO: 98      moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequence
source            1..393
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 98
VVSVCPLPLNL DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
VDHSAWLLIA  LRGGKIEVQL KNEHTSKITT  GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL  FKPENGLLET KVFYAGFPQK VESELIKPIN  PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK  HCLVTVEKGS  YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV  PFAVSLVDST SEKSDILL  VENTVIYRIQ APSLCSQQS HLEFRVNRNN 300
LELSTPLKIE  TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE  AISKHNDIRA HSCPSVWKKT KNS 393

```

```

SEQ ID NO: 99      moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequence
source            1..393
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 99
VVSVCPLPLNL DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA  LRGGKIEVQL KNEHTSKITT  GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL  FKPENGLLET KVFYAGFPQK VESELIKPIN  PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK  HCLVTVEKGS  YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSMGTGVM 240
LALVSGNSTV  PFAVSLVDST SEKSDILL  VENTVIYRIQ ALSLCSQQS HLEFRVNRNN 300
LELSTPLKIE  TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE  AISKHNDIRA HSCPSVWKKT KNS 393

```

```

SEQ ID NO: 100     moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequence
source            1..393
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 100
VVSVCPLPLNL NTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SEGVILYAES 60
IDHSAWLLIA  LRGGKIEVQL KNEHTSKITT  GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL  FKPENGLLET KVFYAGFPQK VESELIKPIN  PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK  HCLVTVEKGS  YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
LALVSGNNTV  PFAVSLVDST SEKSDILL  VENTVIYRIQ ALSLCSQQS HLEFRVNRNN 300
LELSTPLKIE  TISHEDLQRQ LAVLDKAMKA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE  AISKHNDIRA HSCPSVWKKT KNS 393

```

```

SEQ ID NO: 101     moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequences
source            1..393
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 101
AVSVCPLPLNL DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA  LRGGKIEVQL KNEHTSKITT  GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL  FKPENGLLET KVFYAGFPQK VESELIKPIN  PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK  HCLVTVEKGS  YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSMGTGVM 240
LALVSSNNTV  PFAVSLVDST SEKSDILL  VENTVIYRIQ ALSLCSQQS HLEFRVNRNN 300
LELLTPLKIE  TISHEDLQRQ LAILDKAMTA KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVQLDLDE  AISKHNDIRA HSCPSVWKKT KNS 393

```

```

SEQ ID NO: 102     moltype = AA  length = 393
FEATURE           Location/Qualifiers
REGION            1..393
                  note = TAM(Ax1)-binding sequence
source            1..393
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 102
AVSVCPLPLNL DTKYELLYLA EQFAGVVLYL KFRLPEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA  LRGGKIEVQL KNEHTSKITT  GGDVINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL  FKPENGLLET KVFYAGFPQK VESELIKPIN  PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK  HCLVTVEKGS  YYPGSGIAQF HIDYNGSSA EGWHVNVTLN IRPSMGTGVM 240

```


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LALVSSNNTV	PFAVSLVDST	SEKSQDIVLS	VENTVIYRIQ	ALSLCSNQRS	HLEFRVNRNN	300
LELLTPLKIE	TISHEELQRQ	LAILDKAMTG	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVELDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 103	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = tAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 103		
AVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL KFRLEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT GGDIINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAEF HIDYNNGSNA EGWHINVTLN IRPSMTGTVM 240
LALVSSNNTV	PFAVSLVDST	SEKSQDIVLS VENTVIYRIQ ALSLCSNQRS HLEFRVNRNN 300
LELLTPLKIE	TISHEELQTQ	LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVELDLDE	AISKHNDIRA	HSCPSVWKKT KNS 393

SEQ ID NO: 104	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 104		
AVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL KFRLEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT GGDIINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAEF HIDYNNGSNA EGWHINVTLN IRPSMTGTVM 240
LALVSSNNTV	PFAVSLVDST	SEKSQDIVLS VENTVIYRIQ ALSLCSNQRS HLEFRVNRNN 300
LELLTPLKIE	TISHEELQTQ	LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVELDLDE	AISKHNDIRA	HSCPSIWKKT KNS 393

SEQ ID NO: 105	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 105		
AVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL KFRLEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT GGDIINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAEF HIDYNNGSNA EGWHINVTLN IRPSMTGTVM 240
LALVSSNNTV	PFAVSLVDST	SEKSQDIVLS VENTVIYRIQ ALSLCSNQRS HLEFRANRNN 300
LELLTPLKIE	TISHEELQTQ	LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVELDLDE	AISKHNDIRA	HSCPSIWKKT KNS 393

SEQ ID NO: 106	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 106		
AVSVCLPLNL	DTKYELLYLA	EQFAGVVLYL KFRLEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT GGDIINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAEF HIDYNNGSNA EGWHINVTLN IRPSMDTGMV 240
LALVSSNSTV	PFAVSLVDST	SEKSQDIVLS VENTVIYRIQ ALSLCSNQRS HLEFRVNRNN 300
LELLTPLKIE	TISHEELQTQ	LAILDKAMKG KVATYLGGLP DVPFSATPVN AFYNGCMEVN 360
INGVELDLDE	AISKHNDIRA	HSCPSVWEKT KNS 393

SEQ ID NO: 107	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 107		
AVSVCLPLDL	DTKYELLYLA	EQFAGVVLYL KFRLEISRF SAEFDFRTYD SQGVILYAES 60
IDHSAWLLIA	LRGGKIEVQL	KNEHTSKITT GGAIINNGLW NMVSVEELEH SISIKIAKEA 120
VMDINKPGPL	FKPENGLLET	KVYFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180

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KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAEF	HIDYNNGSNA	EGWHINVTLN	IRPSMTGVM	240
LALVSSNNTV	PFAVSLVDST	SEKSQDIVLS	VENTVIYRIQ	ALSLCSNQRS	HLEFRVNRNN	300
LELLTPLKIE	TISQEEELQTQ	LAILDKAMKG	KVATYLGGLP	DVPFSATPVN	AFYNGCMEVN	360
INGVELDLDE	AISKHNDIRA	HSCPSVWKKT	KNS			393

SEQ ID NO: 108	moltype = AA	length = 398
FEATURE	Location/Qualifiers	
REGION	1..398	
	note = TAM(Ax1)-binding sequence	
source	1..398	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	57	
	note = X can be any amino acid	
VARIANT	285	
	note = X can be any amino acid	

SEQUENCE: 108		
VVSVCLPLNL	DTQYELLYLA	EQISGVVLYL
VNHSALLIA	LRGGKIEVQL	KNEHTSKTTT
IAKEAVMDID	KPGPLFKPEN	GLLETKVYFA
GASGIKEIIQ	EKQNKHCLVT	VEKGSYYPGF
GTGVMALVLS	GNNTVPFAVS	LVDSTSEKSQ
VNRNNLELLI	PLKIETISHE	DLQRQLAILD
CMEVNINGVQ	LDLDEAISKH	NDIRAHSCPS
		FWKKTKNS

SEQ ID NO: 109	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 109		
AVPVCPLPLNL	DTKSELLYLA	EQFAGVVLYL
IDHSAWILIA	VRDGKFEVQL	KNEQTSKITT
VMNINKLGPL	FKPEHGFLFET	KVYFAGFPRK
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAKF
LALVSHNNTV	PFAVSLVDST	SEKSQDILLS
LEVSTPLKME	TISHEDLQKQ	LAILDKAMQG
INGVLLDLDE	AISKHNDIRA	HSCPSVWKKT
		KSS

SEQ ID NO: 110	moltype = AA	length = 416
FEATURE	Location/Qualifiers	
REGION	1..416	
	note = TAM(Ax1)-binding sequence	
source	1..416	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	57	
	note = X can be any amino acid	
VARIANT	303	
	note = X can be any amino acid	

SEQUENCE: 110		
VVSVCLPLNL	DTQYELFYLA	EQFAGVVLYL
VNHSALLIA	LRGGKIEVQL	ENEHTSKITT
IFCVVSVVEE	LEHSISIKIA	KEAVMDINKP
PINPCLDGC	RGWNLMKQGA	SGIKEIIQEK
SSVEGWHINV	TLNIHPSMGT	GVMLALVSGN
LIXALSLCSD	QQSHLEFIVN	RNNLELLTPL
GLPDVPFSAT	PVNALYKGC	EVNINGVQLD
		LDEAISKHND
		IIAHSCPSFW
		KKTKNS

SEQ ID NO: 111	moltype = AA	length = 393
FEATURE	Location/Qualifiers	
REGION	1..393	
	note = TAM(Ax1)-binding sequence	
source	1..393	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 111		
AVPVCLLNL	DTKSELLYLA	EQFAGVVLYL
IDHSAWILIA	VRDGKFEVQL	KNEQTSKITT
VMNINKLGPL	FKPEHGFLFET	KVYFAGFPRK
KEIIQEKQNK	HCLVTVEKGS	YYPGSGIAKF
LALVSRNNTV	PFAVSLVDST	SEKLQDILLS
LEVSTPLKME	TISHEDLQKQ	LAILDKAMQG
INGVLLDLDE	AISKHNDIRA	HSCPSVWKKT
		KSS

SEQ ID NO: 112	moltype = AA	length = 393
FEATURE	Location/Qualifiers	

-continued

REGION 1..393
 note = TAM(Ax1)-binding sequence
 source 1..393
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 112
 AIPVCLPLNL DTKSELLYLA EQFAGVVLYL KFLRLPEISRF TAEFDFRTYD SEGVVLYAES 60
 LDHSAWILIA VRDGKFEVQL QNEQTSRITT GGGVVNNGVW HTVSVEELEH SVSLKIAKEA 120
 VMNINKLGPL FKPEHGPLET KVPFAGFPRQ VESQFIKPIN PRLDGCIRGW NLMKQGASGV 180
 KEIIQEKQNK HCLVTVEKGS YYPGSGIAKF IIDYNNVSSA EGWHVNVSLN IRPSMTGTVM 240
 LALVSRNNTV PFAVSLVDST SEKLQDILLS VEKTVIYRIE ALSLCSDQQS HLELKVNRHS 300
 LEVSTPLKME TVSHEDIQKE LAILDKAMQG EVVTYLGGLP DVFSFATPAN AFYNGCMEVN 360
 MNGVLLDLDE AISKHNDIRA HSCPSVWRKT KSS 393

SEQ ID NO: 113 moltype = AA length = 388
 FEATURE Location/Qualifiers
 REGION 1..388
 note = TAM(Ax1)-binding sequence
 source 1..388
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 113
 VVSVCLPLNL DTKYELLYLA EQFAGVVLYL KFLRLPEISRF SAEFDFRTYD SEGVILYAES 60
 IDHSAWLLIA LRGGKIEVQL KNEHTSKITT GGDVINNGLW NMVSVEELEH SISI KIAKEA 120
 VMDINKPGPL FKPENGPLET KVPFAGFPRK VESELIKPIN PRLDGCIRSW NLMKQGASGI 180
 KEIIQEKQNK HCLVTVEKGS YYPGSGIAQF HIDYNNVSSA EGWHVNVTLN IRPSTGTGVM 240
 LALVSGNNTV PFAVSLVDST SEKSDILLS VENTVIYRIQ ALSLCSDQQS HLEFRVNRNN 300
 LELSTPLKIE TISHEDLQRO LAVLDKAMKA KVATYLGGLP DVPPSATPVN AFYNGCMEVN 360
 INGVQLDLDE AISKHNDIRA HSCPSVWK 388

SEQ ID NO: 114 moltype = AA length = 894
 FEATURE Location/Qualifiers
 source 1..894
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 114
 MAWRCPRMGR VPLAWCLALC GWACMAPRGT QABESPFVGN PGNITGARGL TGTLCRLQV 60
 QGEPPEVHWL RDGQILELAD STQTQVPLGE DEQDDWIVVS QLRITSLQLS DTGQYQCLVF 120
 LGHQTFVSPQ GYVGLEGLPY FLEEPEDRTV AANTPFNLSC QAQGPPEPVD LLWLQDAVPL 180
 ATAPGHGPQR SLHVPGLNKT SSFSCAHPA KGVTSRTAT ITVLPQQPRN LHLVSRQPT 240
 LEVATPGLS GIYPLTHCTL QAVLSDGDMG IQAGEPDPE EPLTSQASVP PHQLRLGSLH 300
 PHTPYHIRVA CTSSQGPSSW THWLPVETPE GVPLGPPENI SATRNGSQAF VHWQEPAPL 360
 QGTLGLYRLA YQGQDTPEVL MDIGLRQEV LELQGDGVS NLTVCVAAYT AAGDGPWSLP 420
 VPLEAWRPQ AQPVHQLVKE PSTPAFSWPW WYVLLGAVVA AACVLILALF LVHRRKKETR 480
 YGEVFEPTVE RGEIVVRYRV RKSYSRRTTE ATNLNLGISE ELKEKLRDVM VDRHKVALGK 540
 TLGEGEFGAV MEGQLNQDDS ILKVAVKTMK IAICTRSELE DFLSEAVCMK EFDHPNVML 600
 IGVCFQGSER ESFPAPVVIL PFMKHGDLHS FLLYSRLGDQ PVYLPQTMLV KFMADIASGM 660
 EYLSKTRFIH RDLAARNMCL NENMSVCVAD FGLSKKIYNG DYYRQGRIAK MPVKWIAIES 720
 LADRVYTSKS DVWSFGVMTW EIATRGTQPY PGVENSEIYD YLRQGNRLKQ PADCLDGLYA 780
 LMSRCWELNP QDRPSFTELR EDLENTLKAL PPAQEPDEIL YVNMEDEGGY PEPGAAGGA 840
 DPPTQPDPIK SCSCLTAAEV HPAGRIVLCP STTPSPAQPA DRGSPAAPGQ EDGA 894

SEQ ID NO: 115 moltype = AA length = 888
 FEATURE Location/Qualifiers
 source 1..888
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 115
 MGRVPLAWWL ALCCWGCAAH KDTQTEAGSP FVGNPGNITG ARGLTGTLRC ELQVQGEPP 60
 VVWL RDGQIL ELADNTQTQV PLGEDWQDEW KVSQRLISA LQLSDAGEYQ CMVHLEGRTF 120
 VSQPGFVGL ELPYFLEEPE DKAVPANTPF NLSCQAQGGP EPVTLWLQD AVPLAPVTGH 180
 SSQHSLSQTPG LNKTSFSC EAHNAKGVTS RTATITVLPQ RPHHLHVSR QPTELEVAVT 240
 PGLSGIYPLT HCNLQAVLSD DGVGIWLGKS DPPEDPLTLQ VSVPPHQLRL EKLPHPTPYH 300
 IRISCSGSG PFWTHWLPV ETTGVPPLGP PENVSAMRNG SQVLVRWQEP RVPLQGTLLG 360
 YRLAYRGQDT PEVLMDIGLT REVTLERLGD RPVANLTVSV TAYTSAGDGP WSLFPVPLEPW 420
 RFGQGPPLHH LVSEPPPRAF SWPWWYVLLG ALVAAACVLI LALFLVHRRK KETRYGEVFE 480
 PTVERGELVV RYRVKRSYSR RTTEATLNSL GISEELKEKL RDMVDRHKV ALGKTLGEGE 540
 FGAVMEGQLN QDSILKVAV KTMKIAICTR SELEDLSEA VCMKEFDHPN VMRLIGVCFQ 600
 GSDREGFPEP VVILPFMKHG DLHSFLLYSR LGDQPVFLPT QMLVKFMADI ASGMEYLSK 660
 RFIHRDLAAR NCMLNENMSV CVADFGLSK IYNGDYRQG RIAKMPVKWI AIESLADRVY 720
 TSKSDVWSFG VTWMEIATRG QTPYPGVENS EIYDYLROQN RLKQPVDCLD GLYALMSRCW 780
 ELNPRDRPSF AELREDLENT LKALPPAQEP DEILYVNME GSHLEPRGA AGGADPPTQP 840
 DPKDSCSCLT AADVHSAGRY VLPSTAPGP TLSADRGCPA PPGQEDGA 888

SEQ ID NO: 116 moltype = AA length = 5
 FEATURE Location/Qualifiers
 source 1..5
 mol_type = protein
 organism = synthetic construct

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REGION	1..5	
	note = The entire sequence of amino acids 1-5 can be repeated one or more times	
SEQUENCE: 116		
GGGGS		5
SEQ ID NO: 117	moltype = length =	
SEQUENCE: 117		
000		
SEQ ID NO: 118	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
REGION	1..6	
	note = The entire sequence of amino acids 1-6 can be repeated one or more times	
SEQUENCE: 118		
GSSGGS		6
SEQ ID NO: 119	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 119		
KESGSVSSEQ LAQFRSLD		18
SEQ ID NO: 120	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 120		
EGKSSGSGSE SKST		14
SEQ ID NO: 121	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 121		
GSAGSAAGSG EF		12
SEQ ID NO: 122	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
REGION	1..5	
	note = The entire sequence of amino acids 1-5 can be repeated one or more times	
SEQUENCE: 122		
EAAAK		5
SEQ ID NO: 123	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 123		
CRRRRRREAE AC		12
SEQ ID NO: 124	moltype = AA length = 46	
FEATURE	Location/Qualifiers	
source	1..46	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 124		
AEAAAKEAAA KEAAAKEAAA KALEAEAAAK EAAAKEAAK EAAAKA		46
SEQ ID NO: 125	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 125		
GGGGGGGG		8

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SEQ ID NO: 126	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 126		
GGGGGG		6
SEQ ID NO: 127	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 127		
AEEAAKEAAA AKA		13
SEQ ID NO: 128	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 128		
PAPAP		5
SEQ ID NO: 129	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 129		
VSQTSKLTRA ETVFPDV		17
SEQ ID NO: 130	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 130		
PLGLWA		6
SEQ ID NO: 131	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 131		
TRHRQPRGWE		10
SEQ ID NO: 132	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 132		
AGNRVRRSVG		10
SEQ ID NO: 133	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 133		
RRRRRRRR		8
SEQ ID NO: 134	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 134		
GFLG		4
SEQ ID NO: 135	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
source	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 135		

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GSSGGSGSSG GSGGGDEADG SRGSQKAGVD E 31

SEQ ID NO: 136 moltype = AA length = 898
 FEATURE Location/Qualifiers
 source 1..898
 mol_type = protein
 organism = synthetic construct
 SIGNAL 1..30

SEQUENCE: 136
 MAPSLSPGPA ALRRAPQLLL LLLAAECALA DIQMTQSPSS LSASVGDRVIT ITCRASQSI 60
 SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS RFSGSGSGTD FTLTISSLQP EDFATYYCQQ 120
 SYSTPLTFGG GTKVEIKRGG GSGGGGSGGG GGSEVQLVES GGGVVQPGRS LRLSCAASGF 180
 AFSSYGMHWV RQAPGKGLEW VAVIWFDTGK KYTDSVKGR FTISRDN SKN TLYLQMNLT 240
 AEDTAVYYCA RDRGIGARRG PYYMDVWGKG TTVTVSSGGG GSGGGGSCIN KYGSPYTKNS 300
 GFATCVQNLQ DQCTPNPCDR KGTQACQDLM GNFFCLCKAG WGGRLCDKDV NECSQENGGC 360
 LQICHNKPGS FHCSCHSGFE LSSDGRTCQD IDECADSEAC GEARCKNLPG SYSCLCDEGF 420
 AYSSQEKACR DVDECLQGRG EQVCVNSPGS YTHCDGRGG LKLSQDMDTC EDILPCVPFS 480
 VAKSVKSLYL GRMFGSGTPVI RLRFKRLQPT RLVAEFDFT FDPEGILLFA GGHQDSTWIV 540
 LALRAGRLEL QLRVNGVGRV TSSGPVINHG MWQTISVEEL ARNLVIKVN R DAVMKI AVAG 600
 DLFQPERGLY HLNLTVG GIP FHEKDLVQPI NPLDGC MRS WNWLN GEDTT IQETVKVNTR 660
 MQCFSVTERG SFYPGSGFAP YSLDYMRTPL DVGTESTWEV EVVAHIRPAA DTGVLFPALWA 720
 PDLRAVPLSV ALVDYHSTKK KKLQLVVLAV EHTALALMEI KVC DGEHV TVSLRDGEAT 780
 LEVDGTRGQS EVSAAQLQER LAVLERHLRS PVLTFAGGLP DVPVTSAPVT AFYRGCMTLE 840
 VNRRLLDLE AAYKHS DITA HSCPPVEPAA AQGSRADYKD HDGDYKD HDI DYKDDDDK 898

SEQ ID NO: 137 moltype = AA length = 902
 FEATURE Location/Qualifiers
 source 1..902
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 137
 MAPSLSPGPA ALRRAPQLLL LLLAAECALA QVQLVESGGN LVQPGGSLRL SCAASGFTFG 60
 SFSMSWVRQA PGGGLEWVAG LSARSSLTHY ADSVKGRFTI SRDN AKNSVY LQMNSLRVED 120
 TAVYYCARRS YDSSGYWGHF YSYMDVWGQG TLVTVSGGGG SGGGGSGGGG SSVLTQPSSV 180
 SAAPGQKVTI SCSGSTSNIG NNYVSWYQQH PGKAPKLMY DVSKRPSGVP DRFSGSKSGN 240
 SASLDSGLQ SEDEADYYCA AWDDSLSEFL FGTGKLTVL GGGGGSGGGG SCINKYGSY 300
 TNSGFGATCV QNLDPQCTPN PCDRKGTQAC QDLMGNFFCL CKAGWGGRLC DKDVNECSQE 360
 NGGCLQICHN KPGSFHCSCH SGFELSSDGR TCQDIDECAD SEACGEARCK NLPGSYSCLC 420
 DEGFAYSSQE KACRDVDECL QGRCEQVCVN SPGSYTHCD GRGGLKLSQD MDTCEDILPC 480
 VPFSAKSVK SLYLGRMFSG TPVIRLRFKR LQPTRLVAEF DFTFDPEGI L LFAGGHQDS 540
 TWIVLALRAG RLELQLRYNG VGRVTSSGPV INHGMWQTIS VEELARNLVI KVN RDAVMKI 600
 AVAGDLFQPE RGLYHLNLTV GGI PFHEKDL VQPINRDLG CMRSWNWLN EDTTIQETVK 660
 VNTRMQCFSV TERGSFYPGS GFAFYSLDYM RTPLDVGTES TWEVEVVAHI RPAADTGVL 720
 ALWAPDLRAV PLSVALVDYH STKCLKKQLV VLAVEHTALA LMEIKVCDGQ EHVVTVSLRD 780
 GEATLEVDGT RGQSEVSAQA LQERLAVLER HLRSPLVTFA GGLPDVPVTS APVTAFYRG 840
 MTLVNNRLL DLEAAYKHS DITAHSCPPV EPAAAQGSRA DYKDHDGDYK DHDIDYKDD 900
 DK 902

SEQ ID NO: 138 moltype = AA length = 714
 FEATURE Location/Qualifiers
 source 1..714
 mol_type = protein
 organism = synthetic construct
 SIGNAL 1..30

SEQUENCE: 138
 MAPSLSPGPA ALRRAPQLLL LLLAAECALA DIQMTQSPSS LSASVGDRVIT ITCRASQSI 60
 SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS RFSGSGSGTD FTLTISSLQP EDFATYYCQQ 120
 SYSTPLTFGG GTKVEIKRGG GSGGGGSGGG GGSEVQLVES GGGVVQPGRS LRLSCAASGF 180
 AFSSYGMHWV RQAPGKGLEW VAVIWFDTGK KYTDSVKGR FTISRDN SKN TLYLQMNLT 240
 AEDTAVYYCA RDRGIGARRG PYYMDVWGKG TTVTVSSGGG GSGGGGSDIL PCVPFSAKS 300
 VKSLYLGRMF SGTPTVIRLF KRLQPTRLVA EFDFTFDPE GILLFAGGHQ DSTWIVLALR 360
 AGRLELQLRY NGVGRVTSSG PVINHG MWQT ISVEELARNL VIKVN RDAVM KIAGVAGDLFQ 420
 PERGLYHLNL TVGGIPFHEK DLVQPINRPL DGC MRSWNWL NGEDTTIQET VKVNTRMQCF 480
 SVTERGSFYP GSGFAFYSLD YMRTPLDVGT ESTWEVEVVA HIRPAADTG V L FALWAPDLR 540
 AVPLSVALVD YHSTKCLKKQ LVLVAEHTA LALMEIKVCD GQEHVVTVSL RDGEATLEVD 600
 GTRGQSEVSA AQLQERLAVL ERHLRSPVLT FAGGLPDVPV TSAPVTAFYR GCMTLEVNRR 660
 LLDLEAAYK HSDITAHSCP PVEPAAQGS RADYKDHDGD YKDHDIDYK DDDK 714

SEQ ID NO: 139 moltype = AA length = 718
 FEATURE Location/Qualifiers
 source 1..718
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 139
 MAPSLSPGPA ALRRAPQLLL LLLAAECALA QVQLVESGGN LVQPGGSLRL SCAASGFTFG 60
 SFSMSWVRQA PGGGLEWVAG LSARSSLTHY ADSVKGRFTI SRDN AKNSVY LQMNSLRVED 120
 TAVYYCARRS YDSSGYWGHF YSYMDVWGQG TLVTVSGGGG SGGGGSGGGG SSVLTQPSSV 180
 SAAPGQKVTI SCSGSTSNIG NNYVSWYQQH PGKAPKLMY DVSKRPSGVP DRFSGSKSGN 240
 SASLDSGLQ SEDEADYYCA AWDDSLSEFL FGTGKLTVL GGGGGSGGGG SDILPCVPFS 300

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VAKSVKSLYL	GRMFSGTPVI	RLRFKRLQPT	RLVAEFDFT	FDPEGILLFA	GGHQDSTWIV	360
LALRAGRLEL	QLRYNGVGRV	TSSGPVINHG	MWQTISVEEL	ARNLVIKVN	DAVMKIAVAG	420
DLFQPERGLY	HLNLTVGIP	FHEKDLVQPI	NPRLDGCMRS	WNWLNGETT	IQETVKVNT	480
MQCFSVTERG	SFYPGSGFAF	YSLDYMRTPL	DVGTESTWEV	EVVAHIRPAA	DTGVLFALWA	540
PDLRAVPLSV	ALVDYHSTKK	LKKQLVVLA	EHTALALMEI	KVCDGQEHV	TVSLRDGEAT	600
LEVDTGRGQS	EVSAQLQER	LAVLERHLRS	PVLTFAGGLP	DVPVTSAPVT	AFYRGCMTLE	660
VNRLLDLDE	AAYKHSITA	HSCPPVEPAA	AQGSRADYKD	HDGDYKDHI	DYKDDDDK	718

SEQ ID NO: 140 moltype = AA length = 708
 FEATURE Location/Qualifiers
 source 1..708
 mol_type = protein
 organism = synthetic construct
 SIGNAL 1..30

SEQUENCE: 140

MAPSLSPGPA	ALRRAPQLLL	LLLAEECALA	DIQMTQSPSS	LSASVGDRVT	ITCRASQSI	60
SYLNWYQQK	GKAPKLLIYA	ASSLQSGVPS	RFSGSGSGTD	FTLTISLQ	EDFATYYCQ	120
SYSTPLTFGG	GTKVEIKRGG	GGSGGGSGG	GGSEVQLVES	GGGVVQGRS	LRLSCAASGF	180
AFSSYGMHW	RQAPGKLEW	VAVIWFDTGK	KYTDVSVKGR	FTISRDN	SKN TLYLQMN	240
AEDTAVYYCA	RDRGIGARRG	PYYMDVWGKG	TTVTVSSGGG	GSGGGGSDIL	PCVPFVSAKS	300
VKSLLYGRMF	SGTPVIRLRF	KRLQPTRLVA	EFDFTFDPE	GILLFAGGHQ	DSTWIVLALR	360
AGRLLEQLRY	NGVGRVTSSG	PVINHGMMQT	ISVEELARNL	VIKVNDRDAV	KIAGAGDLFQ	420
PERGLYHLNL	TVGGIPFHEK	DLVQPINPRL	DGCMRSWNWL	NGEDTTI	QET VKVNTRM	480
SVTERGSFYD	GSQFAFYSLD	YMRTPLDVGT	ESTWEVEVVA	HIRPAADTGV	LFALWAPDLR	540
AVPLSVALVD	YHSTKKLKKQ	LVVLAVEHTA	LALMEIKVCD	GQEHVTVVSL	RDGEATLEVD	600
GTRGQSEVSA	AQLQERLAVL	ERHLRSPVLT	FAGGLPDVPV	TSAPVTAFYR	GCMTLEVNRR	660
LLDLDEAAYK	HSDITAHSCP	PVEPAAAGSG	SGSGSGSGSY	PYDVDPYA		708

SEQ ID NO: 141 moltype = AA length = 727
 FEATURE Location/Qualifiers
 source 1..727
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 141

MGWSCIIILFL	VATATGDIQM	TQSPSSLSAS	VGDRVTITCR	ASQSISSYLN	WYQQKPGKAP	60
KLLIYAASSL	QSGVPSRFSG	SGSGTDFTLT	ISSLQPEDFA	TYQCQSYST	PLTFGGGTKV	120
EIKRKRRTVAA	PSVFIFPPSD	EQLKSGTASV	VCLLNNFYPR	EAKVQWKVDN	ALQSGNSQES	180
VTEQDSKDS	YLSSTLTLS	KADYEKKHVV	ACEVTHQGLS	SPVTKSFNRG	ECRRKRGSGE	240
GRGSLTCTGD	VEENPGPMGW	SCIIILFLVAT	ATGEVQLVES	GGGVVQGRS	LRLSCAASGF	300
AFSSYGMHW	RQAPGKLEW	VAVIWFDTGK	KYTDVSVKGR	FTISRDN	SKN TLYLQMN	360
AEDTAVYYCA	RDRGIGARRG	PYYMDVWGKG	TTVTVSSAST	KGPSVFPLAP	SSKSTSGGTA	420
ALGCLVKDYF	PEPVTVSWNS	GALTSGVHTF	PAVLQSSGLY	SLSSVTVPS	SSLGTQTYIC	480
NVNHKPSNTK	VDKKVEPKSS	DKTHTSPPCP	APELLGGPSV	FLFPPKPKDT	LMISRTPEVT	540
CVVVDVSHED	PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVVSVLTVLH	QDWLNGKEYK	600
CKVSNKALPA	PIEKTISKAK	GQPREPQVYT	LPPSRDELTK	NQVSLTCLVK	GFYPSDIAVE	660
WESNGQPENN	YKTTTPVLDS	DGSFFLYSKL	TVDKSRWQQG	NVFSCSVMHE	ALHNYHTQKS	720
LSLSPGK						727

SEQ ID NO: 142 moltype = AA length = 705
 FEATURE Location/Qualifiers
 source 1..705
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 142

MAPSLSPGPA	ALRRAPQLLL	LLLAEECALA	ALLPAREATQ	FLRPRQRRAF	QVFEEAKQGH	60
LERECVEELC	SREEAREVFE	NDPETDYFYP	RYLDCINKYG	SPYTKNSGFA	TCVQNLPDQC	120
TPNPCRDKGT	QACQDLMGNF	FCLCKAGWGG	RLCDKDVNEC	SQENGGCLQI	CHNKPGSFHC	180
SCHSGFELSS	DGRTCDIDE	CADSEACGEA	RCKNLPGSYS	CLDEGFAYS	SQEKACRDVD	240
ECLQGRCEQV	CVNSPGSYTC	HCDGRGGLKL	SQDMTDCEDI	LPCVPFVSAK	SVKSLYLGRM	300
FSGTPVIRLR	FKRLQPTRLV	AEFDFTFDPE	EGILLFAGGH	QDSTWIVLAL	RAGRLELQLR	360
YNGVGRVTSS	GPVINHGMMQ	TISVEELARN	LVIKVNDRDAV	MKIAGAGDLF	QPERGLYHLN	420
LTVGGIPFHE	KDLVQPINPR	LDGCMRSWNW	LNGEDTTIQE	TVKVNTRMQC	FSVTERGSFY	480
PGSGFAFYSL	DYMRTPLDVG	TESTWEVEV	AHIRPAADTG	VLFWALWAPDL	RAVPLSVALV	540
DYHSTKKLKK	QLVVLAVEHT	ALALMEIKVC	DGQEHVTVVS	LRDGEATLEV	DGTRGQSEVS	600
AAQLQERLAV	LERHLRSPVL	TFAGGLPDVP	VTSAPVTAFY	RGCMTLEVN	RLLDLDEAAY	660
KHSDITAHSC	PPVEPAAAGQ	SRADYKDHDG	DYKDHIDYK	DDDDK		705

SEQ ID NO: 143 moltype = DNA length = 2697
 FEATURE Location/Qualifiers
 source 1..2697
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 143

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ctgctgctg	cgcgaggagtg	cgcgcttgcc	gacattcaga	tgactcaatc	tcctagctct	120
ctgagcgcct	cgttgagaga	tagagtcact	attacctgca	gagccagcca	atccatcagc	180
tcctatctaa	attggtacca	acagaagccc	ggcaaacgcg	caaagctgct	catctacgct	240
gcaagctcct	tacagagcgg	agtaaccagc	agattctcag	gcagtgccag	tgggactgac	300
ttcacattga	cgattagctc	tctgcagcct	gaagactttg	ccacatacta	ttgtcagcag	360

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agctatagca ccccgctgac gtttggagge ggaactaagg tggaaatcaa gagaggagge 420
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ggggggggag tcgttcagcc aggtagaagc ctacagactga gctgtgccgc aagtgggttt 540
gctttttcat cttacgggtat gcaactgggtg agacagggtc ctggcaaaagg actcgagtgg 600
gtcgctgtaa tatgggtcga tggtaaaaag aaatactata ccgatagtgt gaaagggaaga 660
ttcaccattt caccagacaa cagtaaaaat acctgtgacc ttccagatgaa caccctgaga 720
gcagaagaca cagccgtgta ctactgcgcc agagatagag gtatcggagc aaggcgtggt 780
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ctctcctctg atggcaggac ctgccaagac atagacgagt gcgcagactc ggagcctgc 1200
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gcgtacagct cccaggagaa ggcttgccga gatgtggacg agtgtctgca gggccgctgt 1320
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cgactgcgtc tcaagaggct gcagccacc aggtcggtag ctgagtttga cttccggacc 1560
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ctggccctga gagccggccg gctggagctg cagctgcgct acaacggtgt cggccgtgtc 1680
accagcagcg gcccggtcat caacctggc atgtggcaga caatctctgt tgaggagctg 1740
gcgcgggaatc tggatcatca ggtcaacagg gatgctgtca tgaaaatcgc ggtggccggg 1800
gacttgttcc aaccggagcg aggaactgtat catctgaacc tgaccgtggg aggtattccc 1860
ttccatgaga aggaacctcg gcagcctata aacctcgtc tggatggctg catgaggagc 1920
tggaaactggc tgaacggaga agacaccacc atccaggaaa cggtgaaagt gaacacgagg 1980
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tacagcctgg actacatgcg gacccctctg gacgtcggga ctgaatcaac ctgggaaagta 2100
gaagtctgtg ctacatcccg cccagccgca gacacaggcg tgctgtttgc gctctgggcc 2160
ccgacctcc gtgcgctgcc tctctctgtg gcactggtag actatcactc cacgaagaaa 2220
ctcaagaagc actggtgggt cctggccgtg gagcatacgg ccttgccctt aatggagatc 2280
aaggtctgcg acggccaaga gcacgtggtc accgtctcgc tgagggaagg tgaggccacc 2340
ctggagggtg acggccacagc gggccagagc gaggtgagcg ccgctcagct gcaggagagg 2400
ctggccgtgc tcgagaggca cctgcggagc cctgtgctca cctttgctgg cggcctgcca 2460
gatgtgccgg tgacttcagc gccagtcacc gcgttctacc gcggtcgtat gacactggag 2520
gtcaaccgga ggtctgtcgc cctggacgag gcggcgtaca agcacagcga catcacggcc 2580
cactcctgcc ccccgctgga gcccgccgca cccgggctga ctacaaagac 2640
catgacgggtg attataaaga tcatgacatc gactacaagg atgacgatga caagtga 2697

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SEQ ID NO: 144      moltype = DNA length = 2709
FEATURE             Location/Qualifiers
source              1..2709
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 144
atggccctt cgtctcgcc cgggcccgcc gccctgcgcc gcgcgccga gctgctgctg 60
ctgctgctgg ccgcggagtg ccgcttgcc caggttcagc tgggtgagag cggaggcaat 120
ctggttcagc ccggtggtag tctgctgtg cgtcagggtt cactttcggt 180
agtttttcaa tgagctgggt ccgtcaggca ccaggcgggt ggctggaatg ggtggcagg 240
ctgtctgcac gtactcctc gacccactat gcagatagtg ttaaaggggc gttcacaaat 300
tcacgcgaca acgctaagaa tagcgtctac ctgcaaatga actcctgcg ggtcaggat 360
accgcagtgt attactgcgc tcgcggttct tatgactcta gtggatactg gggccatttt 420
tatagctaca tggatgtgtg gggacagggc actctggtag ccgtttccgg aggcgggtgg 480
tctggaggcg gtgggagtg aggcgggtgg tcaagcgttc tgaccagacc gtctctgtc 540
agcgcgcgc caggccagaa agtgacaatt tctgtttctg gaagtacttc aaacatcgcc 600
aacaattatg tttcctggta tcagcagcac ccgggcaaaag cgcccaagct gatgatttat 660
gatgtgtcta aacgtccaag tgggttctc gaccgggttc gcggttccaa gtctgggaat 720
agtgcctcac tggacatctc aggcctgcaa agcgaagatg aggcggacta ttactgcga 780
gcttgggatg acagcctgtc cgaatttctg ttcggcaccg ggacaaagct gaccgtgctg 840
ggcggccggg gcgcgacggc cggcggtggc agctgcatac acaagtatgg gtctccgtac 900
acccaaaaact caggcttcgc cacctgcgtg caaaacctgc ctgaccagtg cagcccaac 960
ccctgcgata ggaaggggac ccaagcctgc caggacctca tgggcaactt cttctgcctg 1020
tgtaaagctg cctggggggg ccggctctgc gacaaagatg tcaacgaatg cagccaggag 1080
aacgggggct gcctccagat ctgccacaac aagccgggta gcttccactg ttctgccac 1140
agcggcttcg agctctctc tgatggcagg acctgccaa acatagacga gtgcgcagac 1200
tcggaggcct cgggggaggg gcgctgcaag aaacctgccc gctcctactc ctgcctctgt 1260
gacgagggtc ttgcgtacag ctcccaggag aaggcttgcc gagatgtgga cgagtgtctg 1320
cagggccgct gtgagcaggt ctgcgtgaac tcccaggga gctacacctg cactgtgac 1380
gggctgtggg gcctcaagct gtcccaggac atggacacct gtgaggacat cttgcgctg 1440
tgcccttca gcgtggccaa gagtgtgaag tcttgtacc tggcccgatg gttcagtggt 1500
accccgctga tccgactgcg cttcaagagg ctgcagccca ccaggctggt agctgagttt 1560
gacttcggga cctttgaccc cgagggcata cctctctttg ccggaggcca ccaggacagc 1620
acctggatcg tctggccct gagagccggc cggctggagc tgacgtgcg ctacaacggt 1680
gtcggccgtg tcaccagcag cggcccggtc atcaacctag gcatgtggca gacaatctct 1740
gttgaggagc tggcgggaa tctggtcatc aaggtcaaca gggatgctgt catgaaatc 1800
gcggtggccg gggacttgtt ccaaccggag cgaggactgt atcatctgaa cctgaccgtg 1860
ggaggtatct ccttccatga gaaggacctc gtgcagccta taaacctcgt tctggatggc 1920
tgcatgagga gctggaactg gctgaacgga gaagacacca ccatccagga aacggtgaa 1980

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gtgaacacga ggatgcagtg cttctcggtg acggagagag gctctttcta ccccgaggag 2040
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acctgggaag tagaagtcgt ggctcacatc cggccagccg cagacacagg cgtgctgttt 2160
gcgctctggg ccccgacact cctgtccgtg cctctctctg tggcactggg agactatcac 2220
tccacgaaga aactcaagaa gcagctggtg gtccctggcg tggagcatat ggccctggcc 2280
ctaattggaga tcaaggtctg cgacggccaa gagcacgtgg tcaccgtctc gctgaggggac 2340
ggtgaggcca ccttgagggt ggaacggcacc aggggccaga gcgaggtgag cgcgcgcag 2400
ctgcaggaga ggctggccgt gctcgagagg cacctgcgga gcccgtgct cacccttgct 2460
ggcggcctgc cagatgtgcc ggtgacttca gcgccagtc cgcgttcta ccgcggtgc 2520
atgacactgg aggtcaaccg gaggctgctg gacctggacg aggcggcgta caagcacagc 2580
gacatcacgg cccactctg ccccccgtg gagcccgccg cagcccaagg atccccggct 2640
gactacaaag accatgacgg tgattataaa gatcatgaca tcgactacaa ggatgacgat 2700
gacaagtga                                     2709

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SEQ ID NO: 145      moltype = DNA length = 2145
FEATURE
source              Location/Qualifiers
                    1..2145
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 145
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ctgctgctgg ccgcgaggag cgcgcttgcc gacattcaga tgactcaatc tcctagctct 120
ctgagcgcc cgttggaga tagagtcact attacctgca gagccagcca atccatcagc 180
tcttatctaa attggtacca acagaagccc ggcaaacgcg caaagctgct catctacgct 240
gcaagctcct tacagagcgg agtaccagc agattctcag gcagtggcag tgggactgac 300
ttcacattga cgattagctc ttgcagcct gaagactttg ccacatacta ttgtcagcag 360
agctatagca ccccgctgac gtttggaggc ggaactaagg tggaaatcaa gagaggaggc 420
gggggctccg cggggggtgg ctggggggga ggaggtcag aggttcagct tgtcgagctc 480
ggggggggag tcgttcagcc aggtagaagc ctcagactga gctgtgcgc aagtgggttt 540
gctttttcat cttacggtat gcaactgggtg agacaggctc ctggcaaaag actcgagtgg 600
gtcgtgtaaa tatggttcga tggtaaaaag aaataactata ccgatagtgt gaaaggaaga 660
ttcacccatt cagagacaaa cagtaaaaaa accttgtaac ttcagatgaa caccctgaga 720
gcagaagaca cagccgtgta ctactgcgcc agagatagag gtatcggagc aaggcggtgt 780
ccctattata tggatgtgtg ggggaaggga acaacagtga ctgtgagctc tggcgggggc 840
ggcagcggcg agccaccag cgacatcttg ccgtgcgtgc ccttcagcgt ggccaagagt 900
gtgaagtcct tgtacctggg ccggtatgtt agtgggaccc ccgtgatccg actgcgcttc 960
aagaggctgc agccaccag gctggtagct gactttgact tccggacctt tgaccccgag 1020
ggcctcctcc tctttgcggg agggccaccg gacagcacct ggatcgtgct ggccctgaga 1080
gcgggcccgc tggagctgca gctgcgctac aacggtgtcg ccctgtcac cagcagcggc 1140
ccggtcatca accatggcat gtggcagaca atctctgttg aggagctggc gcggaatctg 1200
gtcatcaagg tcaacaggga tctgtctatg aaaatcgagg tggccgggga cttgtccaa 1260
ccggagcgag gactgtatca tctgaacctg accgtgggag gtattccctt ccatgagaag 1320
gaactcgtgc agcctataaa cctcgtctg gatggctgca tgaggagctg gaactggctg 1380
aacggagaag acaccaccat ccaggaaaag gtgaaaagta acacgaggat gcagtgcttc 1440
tcggtgacgg agagaggctc ttctaccccc gggagcggct tcgcttcta cagcctggac 1500
tacatgcgga cccctctgga cgtcgggact gaatcaacct ggaagtaga agtcgtggct 1560
cacatccgcc cagcccgaga cacaggcgtg ctgtttgcgc tctgggcccc cgacctccgt 1620
gcgctgcctc tctctgtggc actggtagac tatcactcca cgaagaaact caagaagcag 1680
ctggtgtgct tggccgtgga gcatacggcc ttggccctaa tggagatcaa ggtctgcgac 1740
ggccaagagc acgtggtcac gctctcgtg agggacggtg agggccacct ggagggtggac 1800
ggcaccaggg gccagagcga ggtgagcgcc gcgcagctgc aggagaggct ggccgtgctc 1860
gagaggcacc ttcgggagccc cgtgctcacc ttgtctggcg gcctgcgaga tgtgaggtg 1920
acttcagcgc cagtcaccgc gttctaccgc ggctgcatga cactggagggt caaccggagg 1980
ctgctggacc tggacgaggg ggctgacaag cacagcgaca tcacggccca ctctgcccc 2040
cccggtgagc ccgcccagc caaggatcc cgggctgact acaagacca tgacgggtgat 2100
tataaagatc atgacatcga ctacaaggat gacgatgaca agtga                                     2145

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SEQ ID NO: 146      moltype = DNA length = 2157
FEATURE
source              Location/Qualifiers
                    1..2157
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 146
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ctgctgctgg ccgcgaggag cgcgcttgcc caggttcagc tgggtgagag cggaggcaat 120
ctggttcagc ccggtggtag tctgcgtctg tcttgtcgcg cgtcagggtt cactttcgtt 180
agtttttcaa tgagctgggt ccgtcaggca ccaggcgggt ggctggaatg ggtggcagggt 240
ctgtctgcac tagctccct gaccactat gcagatagtg ttaaggggcg gttcacaatt 300
tcacgcgaca acgctaagaa tagcgtctac ctgcaaatga actccctgcg ggtcgaggat 360
accgcagtgt attactgcgc tcgcccgttct tatgactcta gtggatactg gggccatttt 420
tatagctaca tggatgtgtg gggacagggc actctgggtg ccgtttcccg aggcggtggg 480
tctggaggcg gtgggagtg aggcggtggg tcaagcgttc tgacccagcc gtcctctgtc 540
agcgcgcgcg cagccagaaa agtgacaatt tctgttcttg gaagtacttc aaacatcgcc 600
aacaattatg tttcctggta tcaagcgcac ccgggcaaaag cgcaccaagct gatgatttat 660
gatgtgtcta aacgtccaag tgggtgttct gaccggttca gcggttccaa gtttgggaat 720
agtgccctac tggacatctc aggcctgcaa agcgaagatg aggcggacta ttactgcgca 780
gcttgggatg acagcctgtc cgaattttct ttcggcaccg ggacaaaagct gaccgtgctg 840
ggcgcggggg ccgcgagcgg cggcggtggc agcgacatct tgccgtgcgt gccctcagc 900
gtggccaaga gtgtgaagtc cttgtacctg ggccggatgt tcagtgggac ccccgatgat 960

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cgactgcgct tcaagagget gcagcccacc aggctggtag ctgagtttga ctteccggacc 1020
tttgaccccg agggcatcct cctctttggc ggaggccacc aggcagcac ctggatcgtg 1080
ctggccctga gagccgccc gctggagctg cagctgcgct acaacgggtg cggccgtgtc 1140
accagcagcg gcccggtcat caaccatggc atgtggcaga caatctctgt tggagagctg 1200
gcgcgggaatc tggtoatcaa ggtcaacagg gatgctgtca tgaaaatcgc ggtggccggg 1260
gacttgttcc aaccggagcg aggaactgtat catctgaacc tcaccgtggg aggtattccc 1320
ttccatgaga aggacctcgt gcagcctata aacctcgtc tggatggctg tatgaggagc 1380
tggaactggc tgaacggaga agacaccacc atccaggaaa cggtgaaagt gaacacgagg 1440
atgcagtgct tctcgggtgac ggagagaggg tctttctacc ccgggagcgg ctctcgcttc 1500
tacagcctgg actacatgcg gacctctctg gacgtcggga ctgaatcaac ctgggaagta 1560
gaagtctggc ctacatcccg ccagcccgca gacacaggcg tctgttttgc gctctgggcc 1620
cccgacctcc gtgcgctgac tctctctgtg gcaactgtag actatcactc caccgaagaa 1680
ctcaagaagc agctgggtgg cctggccgtg gacgatacgg ccttggccct aatggagatc 1740
aaggctctcg accgccaaga gcacgtgggt accgtctcgc tgagggaagg tgaggccacc 1800
ctggaggttg accgcccagg gggccagagg gaggtgagcg ccgcgcagct gcaggagagg 1860
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gatgtgccgg tgacttcagg gccagtcacc gcgtcttacc gcgctgcat gacactggag 1980
gtcaaccgga ggctgctgga cctggacgag gcggcgtaga agcacagcga catcacggcc 2040
cactcctgcc ccccggtgga gcccgccgca gcccaaggat cccgggctga ctacaaagac 2100
catgacggtg attataaaga tcatgacatc gactacaagg atgacgatga caagtga 2157

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SEQ ID NO: 147      moltype = DNA length = 2127
FEATURE            Location/Qualifiers
source              1..2127
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 147
atggccccctt cgtctcgcgc cgggccccgcc gccctgcgcc gcgcgcgcga gctgctgctg 60
ctgctgctgg ccgcggagtg ccgcgttgcc gacattcaga tgactcaatc tctagctctt 120
ctgagcgcct ccgttgagga tagagtcact attacctgca gagccagcca atccatcagc 180
tcttatctaa attggtacca acagaagccc ggcaaaagcg caaagctgct catctacgct 240
gcaaagctcct tacagagcgg agtaccacgc agattctcag gcagtggcag tgggactgac 300
tccacattga cgatttagctc tctgcagcct gaagactttg ccacatacta ttgtcagcag 360
agctatagca ccccgtgacg gtttggaggc ggaactaagg tggaaatcaa gagaggaggg 420
gggggctccg gcgggggttg ccggggggga ggaggtcag aggttcagct tgtcagctct 480
ggggggggag tcgttcagcc aggtagaagc ctcagactga gctgtgccgc aagtgggttt 540
gctttttcat cttacgggat gcactgggtg agacaggctc ctggcaaaagg actcagtggt 600
gtcgtgtaaa tatggttcga tgggtacaaa aaatactata ccgatagtg gaaaggaaga 660
tccaccattt cagcagacaa cagtaaaaat accctgtacc ttcagatgaa caccctgaga 720
gcagaagaca cagcgtgata ctactgcgcc agagatagag gtatcggagc aaggcgtggt 780
ccctattata tggatgtgtg ggggaaggga acaacagtga ctgtgagctc tggcgggggg 840
ggcagcggcg gcggtggcag cgacatcttg ccgtgcgtgc ccttcagcgt ggccaagagt 900
gtgaagtcct tgtacctggg ccggatgttc agtgggaccc ccgtgatccg actgcgcttc 960
aagaggctgc agcccaccag gctggtagct gagtttgact tccggacctt tgaccccgag 1020
ggcatcctcc tctttgcggg aggccaccag gacagcacct ggatcgtgct ggcctgaga 1080
gcggcccgcg tggagctgca gctgcgtac aacggtgtcg gccgtgtcac cagcagcggc 1140
ccggtcacga accatggcat gtggcagaca atctctgttg agggagctggc gcggaatctg 1200
gtcatcaagg tcaacaggga tctgtctcat aaaatcggcg tggccggggga cttgttccaa 1260
ccggagcgag gactgtatca tctgaacctg accgtgggag gtattccctt ccatgagaag 1320
gacctcgtgc agcctataaa cctcgtctg gatggctgca tgaggagctg gaactggctg 1380
aacggagaaag acaccaccat ccaggaaaag gtgaaaagta acacgaggat gcagtgttc 1440
tcggtgacgg agagaggctc ttctacccc gggagcggct tcgcttctca cagcctggac 1500
tacatgcgga cccctctgga cgtcgggact gaatcaacct gggaagtaga agtctgtggt 1560
cacatccgcc cagcccgaga cacaggcgtg ctgtttgcgc tctgggcccc cgacctcgt 1620
gcggtgcctc tctctgtggc atctgttaga tatcactcca cgaagaaact caagaagcag 1680
ctgggtggtcc tggccgtgga gcatacggcc ttggccctaa tggagatcaa ggtctcgcag 1740
ggccaagagc acgtgggtcac cgtctcgtg agggacggtg aggccacctt ggagggtgag 1800
ggcaccaggg gccagagcga ggtgagcgcc gcgcagctgc aggcagggct ggcctgtctc 1860
gagaggcacc tcgggagccc cgtgtcacc tttgtggcg gccctgccga tgtgcgggtg 1920
acttcagcgc cagtcaccgc gttctaccgc ggctgcatga cactggagggt caaccggagg 1980
ctgctggacc tggacgaggc ggcgataaag cacagcgaca tcacggcccc ctctgcccc 2040
cccgtaggag ccgcccagcg cggcagcggc agcggcagcg gcagcggcag cggcagctac 2100
ccatcacgatg ttccagatta cgcttga 2127

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SEQ ID NO: 148      moltype = DNA length = 2190
FEATURE            Location/Qualifiers
source              1..2190
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 148
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cagatgactc aatctcctag ctctctgagc gcctccgttg gagatagagt cactattacc 120
tgacagacca gccaatccat cagctcttat ctaaaattggt accaacagaa gccccgcaaa 180
gcgccaaaagc tggctactca cgctgcaagc tccttacaga gcggagtacc cagcagattc 240
tcaggcagtg gcagtgggac tgacttcaca ttgacgatta gctctctgca gcctgaagac 300
tttggcccat actattgtca gcagagctat agcacccccc tgacgttttg aggcggaaact 360
aagggtgaaa tcaagagaaa acgaactgtg gctgcaccat ctgtcttcat cttcccggca 420
tctgatgagc agttgaaatc ttgaaactgc tctgtcgtgt gccctgctga taactctat 480
ccagagaggg ccaaagtaca gtggaagggt gataacgccc tccaatcggg taactcccag 540

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gagagtgtca cagagcagga cagcaaggac agcacctaca gcctcagcag caccctgacg 600
ctgagcacaag cagactacga gaaacacaaa gtctacgcct gcgaagtcac ccatcagggc 660
ctgtcctcgc ccgtcacaaa gagcttcaac aggggagagt gtcgcagaaa acgcggaagc 720
ggagagggca gaggaagtct tctaaccatgc ggtgacgtgg aggagaatcc cgccctatg 780
ggctgggtcct gcatacctct gttcctgggt gccacgcgca ccggcgaggt tcagcttgtc 840
gagtcctgggg ggggagtcgt tcagccaggt agaagcctca gactgagctg tgcgcgaagt 900
gggtttgctt ttcatctta cggtatgcac tgggtgagac aggtcctgg caaaggactc 960
gagtggttcg ctgtaatatg gttcgatggt acaaagaaat actatacga tagtgtgaaa 1020
ggaagattca ccatttcacg agacaacagt aaaaatacct tgtaccttca gatgaacacc 1080
ctgagagcag aagacacagc cgtgtactac tgcccgagag atagaggatg cggagcaagg 1140
cgtggtccct attatatgga tgtgtggggg aagggaaaca cagtgactgt gagctctgcc 1200
tcacccaagg gcccatcggg cttccccctg gcacctcct ccaagagcac ctctgggggc 1260
acagcggccc tgggctgctt ggtcaaggac tacttccccg aaccggtgac ggtgtcgtgg 1320
aaactcaggcg cctcagcagc cggtcgtgac accttccccg ctgtcctaca gtctcagga 1380
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atctgcaacg tgaatcacia gcccgcaac accaagggtg acaagaaagt tgagccaaa 1500
tctagcgaca aaactcacac aagcccacgc tgccagcac ctgaaactct ggggggacgc 1560
tcagtcttcc tcttcccccc aaaacccaag gacacctca tgatctcccc gacctctgag 1620
gtcacatgcy tgggtgtgga cgtgagccac gaagaccctg aggtcaagtt caactggtac 1680
gtggacggcg tggaggtgca taatgccaag acaaagccgc gggaggagca gtacaacagc 1740
acgtaccgtg tggtcagcgt cctcacgcgt ctgcaccagg actggctgaa tggcaaggag 1800
tacaagtga aggtctccaa caaagccctc ccagcccca tcgagaaaac catctccaaa 1860
gccaaggggc agcccccaga accacaggtg tacacctgc ccccatcccc ggatgagctg 1920
accaagaacc aggtcagcct gacctgcctg gtcaaaggct tctatccag cgacatcgcc 1980
gtggagtggg agagcaatgg gcagccggag aacaactaca agaccacgcc tccgtgctg 2040
gactccgagc gctctcttct cctctacagc aagctcacgc tggacaagag caggtggcag 2100
caggggaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacacgcag 2160
aagagcctct cctgtcccc gggtaaatga 2190

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SEQ ID NO: 149      moltype = DNA length = 2118
FEATURE
source              Location/Qualifiers
                    1..2118
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 149
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ctgctgctgg ccgcggagtg cgcgcttgcc gcgctgttgc cggcgcgcga ggccacgcag 120
ttcctgcggc ccaggcagcg ccgcgccttt caggctctcg agggagccaa gcaggggcac 180
ctggagaggg agtgcggtgga ggaagctgtg agccgcgagg aggcgcggga ggtgttcgag 240
aacgaccccc agacggatta tttttacca agatacttag actgcatcaa caagtatggg 300
tctccgtaca ccaaaaactc aggtcttcgc acctgcgtgc aaaacctgcc tgaccagtgc 360
acgcccaccc cctgcgatag gaagggggacc caagcctgcc aggcacctcat gggcaacttc 420
ttctgcctgt gtaaaagctgg ctggggggggc cggctctgcy acaaagatgt caacgaatgc 480
agccagagga acggggggctg cctccagatc tgccacaaca agccgggtag cttccactgt 540
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gagtgctcgc agggccgctg tgagcaggtc tgcgtgaact ccccgaggag ctacacctgc 780
cactgtgacg ggcgtggggg cctcaagctg tcccaggaca tggacacctg tgaggacatc 840
ttgccgtgcy tgcccttcag cgtggccaag agtgtgaagt ccttgtacct gggccggatg 900
ttcagtgggg ccccctgatg ccgactgcgc ttcaagaggg tgcagcccac caggctggta 960
gctgagtttg acttccggag ctttgacccc gagggcatcc tctcttttgc cggaggccac 1020
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aagcacagcg acatcacggc ccaactcctgc ccccccggtg agccgcgcgc agcccaagg 2040
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SEQ ID NO: 150      moltype = AA length = 961
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source              Location/Qualifiers
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                    organism = synthetic construct
SIGNAL
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SEQUENCE: 150

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YYCFQGS LVP  WTFQGQTKVE  IKGGGSGSGG  GSGGGGSEVQ  LVESGGGLVQ  PGGSRLRLSCA  180
ASGLIFRSYG  MSWVRQAPGK  GLEWVATINS  GGTYYYPDS  VKGRFTISR  NSKNTLYLQM  240
NSLRAEDTAV  YYCANSYSYA  MDYWGQGTLV  TVSSGGGSGG  GGGSDILPCV  PFSVAKSVKS  300
LYLGRMFSGT  PVIRLRFKRL  QPTRLVAEFD  FRTFDPEGIL  LFAGGHQDST  WIVLALRAGR  360
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GLYHLNLTVG  GIPHEKDLV  QPINPRLDGC  MRSWNWLNGE  DTTIQETVKV  NTRMQCFSVT  480
ERGSFYPGSG  FAFYSLDYMR  TPLDVGTST  WEVEVVAHIR  PAADTGVLEA  LWAPDLRAVP  540
LSVALVDYHS  TKKLKQLV  LAVEHTALAL  MEIKVCDGQE  HVVTVSLRDG  EATLEVDGTR  600
GQSEVSAAQL  QERLAVLERH  LRSPVLTFFAG  GLPDVPVTS  PVTAFYRGCM  TLEVNRRLLD  660
LDEAAYKHS  D  ITAHS C PVE  PAAAGSGSGS  GSGSGSYPYD  VPDYAEGRGS  LLTCGDVEEN  720
PGPVSKGEEL  FTGVVPI  LVE  LDGDVNGHKF  SVS GEGEGDA  TYGKLT LKFI  CTTGKL PVPW  780
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SEQ ID NO: 151      moltype = DNA length = 2886
FEATURE             Location/Qualifiers
source               1..2886
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 152      moltype = AA length = 953
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                    mol_type = protein
                    organism = synthetic construct

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 DSTNTYEVFG GGTCLTVLGG GSGGGGGSGG GGSEVQLVES GGGLVEPGGS LRLSCAVSGF 180
 DFEKAWMSWV RQAPGQGLQW VARIKSTADG GTTSYAAPVE GRFII SRDDS RNMLYLQMNS 240
 LKTEDTAVVY CTSAHWGQGT LVTVSSGGGG SGGGGS DILP CVPFVSAKSV KSLYLGRMFS 300
 GTPVIRLRPK RLQPTRLVAE FDFRTFDPEG ILLFAGGHQD STWIVLALRA GRLELQLRYN 360
 GVGRVTSSGP VINHGMWQTI SVEELARNLV IKVNRDAVMK IAVAGDLFQP ERGLYHLNLT 420
 VGGIPFHEKD LVQPINRPLD GCMRSWNWLN GEDTTIQETV KVNTRMQCFS VTERGSFYPG 480
 SGFAFYSLDY MRTPLDVGTE STWEVEVVAH IRPAADTGVL FALWAPDLRA VPLSVALVDY 540
 HSTKKLKKQL VVLAVEHTAL ALMEIKVCDG QEHVVTVSLR DGEATLEV DG TRGQSEVSAA 600
 QLQERLAVLE RHLRSPVLTF AGGLPDVPVT SAPVTAFYRG CMTLEVNRRLL LDLDEAAYKH 660
 SDITAHSCPP VEPAAAGSGS GSGSGSGSYP YDVPDYAAGR GSLLTCGDVE ENPGPVSKGE 720
 ELFTGVVPTL VELDGVNGH KFSVSGEGEG DATYGKLTLL FICTTGKLPV PWPTLVTTLT 780
 YGVQCFSRYP DHMKQHDFFK SAMPEGYVQE RTIFFKDDGN YKTRAEVKFE GDTLVNRIEL 840
 KGIDFKEDGN ILGHKLEyny NSHNVIYIMAD KQKNGIKVNF KIRHNIEDGS VQLADHYQQN 900
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SEQ ID NO: 153 moltype = DNA length = 2862
 FEATURE Location/Qualifiers
 source 1..2862
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 154 moltype = AA length = 701
 FEATURE Location/Qualifiers
 source 1..701
 mol_type = protein
 organism = synthetic construct

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SIGNAL 1..24

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MHWVRQAPGK	GLEWVAVIWF	DGKKYYTDS	VKGRFTISR	NSKNTLYLQM	NTLRAEDTAV	240
YYCARDRGIG	ARRGPYYMDV	WGKGTTVTVS	SGGGGSGGGG	SVVSVCLPLN	LDTKYELLYL	300
AEQFAGVVLY	LKFRLLPEISR	FSAEFDPRTY	DSEGVILYAE	SIDHSAWLLI	ALRGGKIEVQ	360
LKNEHTSKIT	TGGDVINNL	WNMVSVEELE	HSISIKIAKE	AVMDINKPGP	LFPKENGLE	420
TKVYFAGFPR	KVESELIKPI	NPRLDGCIRS	WNLKQGASG	IKEIIQEKQN	KHCLVTVEKG	480
SYYPGSGIAQ	PHIDYNNVSS	AEGWHVNVTL	NIRPSTGTGV	MLALVSGNNT	VPFAVSLVDS	540
TSEKSDILL	SVENTVIYRI	QALSLCSDQQ	SHLEFRVNRN	NLELSTPLKI	ETISHEDLQR	600
QLAVLDKAMK	AKVATYLGGL	PDVPFSATPV	NAFYNGCMEV	NINGVQLDLD	EAISKHNDIR	660
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SEQ ID NO: 155 moltype = DNA length = 2106

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mol_type = other DNA

organism = synthetic construct

SEQUENCE: 155

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SEQ ID NO: 156 moltype = AA length = 684

FEATURE Location/Qualifiers

source 1..684

mol_type = protein

organism = synthetic construct

SIGNAL 1..21

SEQUENCE: 156

METDTLLLV	LLLWVPGSTG	DEVQLVESGG	GVVQGRSLR	LSCAASGFAP	SSYGMHWVRQ	60
APKGLEWVA	VIWFDGTKKY	YTPSVKGRFT	ISRDNKNL	YLQMNTRAE	DTAVYYCARD	120
RGIGARRGYP	YMDVWGKGT	VTSSASTKG	PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	180
PVTVSWNSGA	LTSVGHFPA	VLQSSGLYSL	SSVTVPSSS	LGTQTYICNV	NHKPSNTKVD	240
KKVEPKSCDK	THGGGSGGGG	GSDILPCVPF	SVAKSVKSLY	LGRMFGSTPV	IRLRFKRLQP	300
TRLVAEFDFR	TFDPEGLLLF	AGGHQDSTWI	VLALRAGRL	LQLRYNGVGR	VTSSGPVINH	360
GMWQTISVEE	LARNLVIKVN	RDAVMKIAVA	GDLFQPERGL	YHLNLTGGI	PFHEKDLVQP	420
INPRLDGCMR	SWNLNGEDT	TIQETVKVNT	RMQCFSVTER	GSFYPGSGFA	FYSLDYMRT	480
LDVGTSTWE	VEVVAHRPA	ADTVLFAFW	APDLRAVPLS	VALVDYHSTK	KLKQLVLLA	540
VEHTALALME	IKVCDGQEHV	VTVSLRDGEA	TLEVDGTRGQ	SEVSAAQLQE	RLAVLERHLR	600
SPVLTFAGGL	PDVPVTSAPV	TAFYRGCMTL	EVNRRLLDLD	EAAYKHSIT	AHSCPPVEPA	660
AADYKDHGDG	YKDHIDYKD	DDDK				684

SEQ ID NO: 157 moltype = DNA length = 2055

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FEATURE
source
Location/Qualifiers
1..2055
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 157
atggagacag acacactcct gctatgggta ctgctgctct gggttccagg ttccactggt 60
gacgaggttc agcttgtega gtctgggggg ggagtcgttc agccaggtag aagcctcaga 120
ctgagctgtg ccgcaagtgg gtttgctttt tcatcttacg gtatgcactg ggtgagacag 180
gctcctggca aaggactcga gtgggtcgct gtaatatggt tcgatggtag aaagaataac 240
tataccgata gtgtgaaagg aagattcacc atttcacgag acaacagtaa aaataccttg 300
taccttcaga tgaacacctt gagagcagaa gacacagccg tgtactactg cgccagagat 360
agaggtatcg gagcaaggcg tggtccttat tataatggatg tgtgggggaa gggacaacaaca 420
gtgactgtga gctctgcctc caccaagggc ccatcggtct tccccctggc accctcctcc 480
aagagcacct ctggggggcac agcggccttg ggctgcctgg tcaaggacta cttccccgaa 540
cgggtgacgg tgtcgtggaa ctacggcgcc ctgaccagcg gcgtgcacac cttccccggt 600
gtcctacagt cctcaggact ctactccctc agcagcgctg tgactgtgcc ctctagcagc 660
ttgggcaccc gagcctacat ctgcaacgtg aatcacaagc ccagcaacac caagggtggac 720
aagaaagtgg agcccaaatc ttgtgacaaa actcacggcg gaggtggaag cggaggcggt 780
ggaagcgaca tcttgccgtg cgtgcccttc agcgtggcca agagtgtgaa gtccttgtag 840
ctgggcccga ttgtcagttg gacccccttg atccgactgc gcttcaagag gctgcagccc 900
accaggctgg tagctgagtt tgacttcctg accttgtagc ccgaggcgat cctcctcttt 960
gccggaggcc accaggacag cacctggatc gtgctggccc tgagagccgg ccggtgggag 1020
ctgcagctgc gctacaacgg ttgctggcctg gtcaccagca gcggcccgtt catcaacct 1080
ggcatgtggc agacaatctc ttgtgaggag ctggcgcgga atctggtcat caaggtaaac 1140
agggatgctg tcatgaaaat gcggtggccc ggggacttgt tccaaccgga gcgaggactg 1200
tatcatctga acctcacctg gggaggtatt ccttccatg agaaggacct cgtgcagcct 1260
ataaaccttc gtctggatgg ctgtatgagg agctggaact ggctgaacgg agaagacacc 1320
accttcagg aacgggtgaa agtgaacacg aggatgcagt gcttctcggt gacggagaga 1380
ggctctttct accccgggag cggcttcgcc ttctacagcc tggactacat gcggaccctc 1440
ctggagctcg ggactgaatc aacctgggaa gtagaagtcg tggctcacat ccgcccagcc 1500
gcagacacag gcgtgtgttt tgcgtctctg gcccccgacc tccgtgccgt gcctctctct 1560
gtggcacttg tagactatca ctccacgaag aaactcaaga agcagctggg ggtcctggcc 1620
gtggagcata cggccttgcc cctaatggag atcaaggctc tgcagcgcca agagcacgtg 1680
gtcacctctc cctgtaggga cggtagggcc accctggagg tggacggcac caggggcccag 1740
agcgaggtga gcgcgcgcga gctgcaggag aggctggccg tgetcgagag gcacctggcg 1800
agccccctgc tcacctttgc cggcggcctg ccagatgtgc cggtagactc agcggcagtc 1860
accgcgttct accgcggctg catgacactg gagggtcaacc ggaggctgct ggacctggac 1920
gaggcgcgct acaagcacag cgacatcacg gccactcctt gccccccgtt ggagcccgc 1980
gcagccgact acaagacca tgacggtgat tataaagatc atgacatcga ctacaaggat 2040
gacgatgaca agtga 2055

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SEQ ID NO: 158
FEATURE
source
Location/Qualifiers
1..907
mol_type = protein
organism = synthetic construct

SIGNAL
1..21

SEQUENCE: 158
METDTLLLV LLLWVPGSTG DEVQLVESGG GVVQGRSLR LSAAAGFAF SSGMHVVRQ 60
APGKGLEWVA VIWFDGTKKY YTSVKGRFT ISRDNSKNTL YLQMNLTAE DTAVYYCARD 120
RGIGARRGPY YMDVWKGTTT VTSSASTKG PSVFPAPSS KSTSGGTAAL GCLVKDYFPE 180
PVTVSWNSGA LTSVGHVTPA VLQSSGLYSL SSVVTPSSS LGTQTYICNV NHKPSNTKVD 240
KKVEPKSCDK THTCPPCPAP ELLGGPSVFL FPPKPKDTLM ISRTPEVTCV VVDVSHEDPE 300
VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNGKEYKCK VSNKALPAPI 360
EKTISKAKGQ PREPQVYTL PPSRDELTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK 420
TTPPVLDSDG SFFLYSKLTV DKSRRQQGNV FSCVMHEAL HNHYTQKSL LSPGKGGGGS 480
GGGSDILPC VPFSVAKSVK SLYLGRMFSG TPVIRLRFKR LQPTRLVAEF DFRTFDPEGI 540
LLFAGGHQDS TWIVLALRAG RLELQLRYNG VGRVTSSGPV INHGMWQTIS VEELARNLVI 600
KVNRAVMKI AVAGDLFQPE RGLYHLNLTV GGIPFHEKDL VQPINRLDG CMRSWNWNLG 660
EDTTIQETVK VNTRMQCFSV TERGSFYPGS GFAPYSLDYM RTPLDVGTES TWEVEVVAHI 720
RPAADTGVLF ALWAPDLRAV PLSVALVDYH STKKLKKQLV VLAVEHTALA LMEIKVCDGQ 780
EHVTVSLRD GEATLEVDGT RGQSEVSAAQ LQERLAVLER HLRSPVLTF GGLPDVPVTS 840
APVTAFYRGC MTEVNRRL DLDEAAYKHS DITAHSCPPV EPAAADYKDH DGDYKDHDID 900
YKDDDDK 907

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SEQ ID NO: 159
FEATURE
source
Location/Qualifiers
1..2724
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 159
atggagacag acacactcct gctatgggta ctgctgctct gggttccagg ttccactggt 60
gacgaggttc agcttgtega gtctgggggg ggagtcgttc agccaggtag aagcctcaga 120
ctgagctgtg ccgcaagtgg gtttgctttt tcatcttacg gtatgcactg ggtgagacag 180
gctcctggca aaggactcga gtgggtcgct gtaatatggt tcgatggtag aaagaataac 240
tataccgata gtgtgaaagg aagattcacc atttcacgag acaacagtaa aaataccttg 300
taccttcaga tgaacacctt gagagcagaa gacacagccg tgtactactg cgccagagat 360
agaggtatcg gagcaaggcg tggtccttat tataatggatg tgtgggggaa gggacaacaaca 420
gtgactgtga gctctgcctc caccaagggc ccatcggtct tccccctggc accctcctcc 480

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aagagcacct ctggggggcac agcgggcccgt ggetgcctgg tcaaggacta cttccccgaa 540
ccgggtgacgg tgtcgtggaa ctacggcgcc ctgaccagcg gcgtgcacac cttccccggct 600
gtcctacagt cctcaggact ctactccctc agcagcgtgg tgactgtgcc cttcagcagc 660
ttggggcacc agacctacat ctgcaacgtg aatcacaaag ccagcaacac caaggtggac 720
aagaaagtgt agcccaaatc ttgtgacaaa actcacacat gccacacgtg cccagcacct 780
gaactcctgg ggggaccgtc agtcttctc tccccccaa aaccaagga caccctcatg 840
atctcccgga cccctgaggt cacatgcgtg gtggtggacg tgagccacga agaccctgag 900
gtcaagtcca actggtacgt ggacggcggt gaggtgcata atgccaagac aaagccgagg 960
gaggagcagt acaacagcac gtaccgtgtg gtcagcgtcc tcacgcctcc gcaccaggac 1020
tggtgaatg gcaaggagta caagtgcaag gtctccaaca agccctccc agcccccac 1080
gagaaaacca tctccaaagc caaaggcgag ccccgagAAC cacaggtga caccctgccc 1140
ccatcccggt atgagctgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 1200
tatcccgag acatcgccgt ggagtgagg agcaatgggc agccggagaa caactacaag 1260
accacgcctc ccgtgctgga ctccagcgcc tcttctctcc tctacagcaa gctcaccgtg 1320
gacaagagca ggtggcagca ggggaacgtc ttctcatgct ccgtgatgca tgaggctctg 1380
cacaaccact acacgcagaa gagcctctcc ctgtccccg gtaaaaggcg aggtggaagc 1440
ggagggcggt gaagcgacat ctgcccgtgc gtgcccttca gcgtggcaca gagtgtgaag 1500
tctctgtacc tgggcccggat gttcagtggt acccccgtga tccgactgag cttcaagagg 1560
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cggtcgagcg tgcagctgag ctacaacggt gtcggccgtg tcaccagcag cggcccggtc 1740
atcaaccatg gcatgtggca gacaatctct gttgaggagc tggcgcgga tctggtcatc 1800
aaggtcaaca gggatgctgt catgaaaaac gcggtggccg gggacttgtt ccaaccggag 1860
cgaggactgt atcatctgaa cctcaccgtg ggaggtattc ccttccatga gaaggacctc 1920
gtgcagccta taaaccctcg tctggatggc tgtatgagga gctggaactg gctgaacgga 1980
gaagacacca ccatccagga aacggtgaaa gtgaacacga ggtgagctg cttctcggtg 2040
acggagagag gctctttcta ccccgggagc ggcttcgcct tctacagcct ggactacatg 2100
cggacccctc tggacgtcgg gactgaatca acctgggaag tagaagtcgt ggctcacatc 2160
cgcccgagcg cagacacagg cgtgctgttt gcgctctggg ccccgacct ccgtgcccgtg 2220
cctctctctg tggcactggt agactatcac tccacgaaga aactcaagaa gcagctgggtg 2280
gtcctggccg tggagcatcac ggccttgccc ctaatggaga tcaaggtctg cgacggccaa 2340
gagcacgtgg tcacgctctc gctgagggac ggtgaggcca cctcgagggt ggacggcacc 2400
aggggcccga gcgaggtgag cgcgcgcgag ctgcaggaga ggctggccgt gctcgagagg 2460
cacctgcgga gcccgctgct cacccttgcc ggcggcctgc cagatgtgcc ggtgacttca 2520
gcgcagtcac ccgcgttcta ccgcggctgc atgacactgg aggtcaaacg gaggtgctg 2580
gacctggagc aggcggcgta caagcacagc gacatcacgg cccactcctg cccccccgtg 2640
gagcccgccg cagccgacta caaagaccat gacgggtgatt ataaagatca tgacatcgac 2700
tacaaggatg acgatgacaa gtga 2724

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SEQ ID NO: 160      moltype = AA length = 5
FEATURE            Location/Qualifiers
source              1..5
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 160
GGGS 5

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SEQ ID NO: 161      moltype = AA length = 232
FEATURE            Location/Qualifiers
source              1..232
                    mol_type = protein
                    organism = synthetic construct
SIGNAL              1..16

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SEQUENCE: 161
MGWSCILFL VATATGDIQM TQSPSSLSAS VGDRVITICR ASQSISSYLN WYQQKPGKAP 60
KLLIYAASSL QSGVPSRFSG SGSGTDFTLT ISSLPEDFA TYYCQSYST PLTFGGGTVK 120
EIKRKRTVAA PSVFIFFPSD EQLKSGTASV VCLLNIFYPR EAKVQWKVDN ALQSGNSQES 180
VTEQDSKDSY YSLSSLTILS KADYEKKHVV ACEVTHQGLS SPVTKSFNRG EC 232

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SEQ ID NO: 162      moltype = AA length = 914
FEATURE            Location/Qualifiers
source              1..914
                    mol_type = protein
                    organism = synthetic construct
SIGNAL              1..21

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SEQUENCE: 162
METDTLLLWV LLLVPGSTG DDIQMTQSPS SLSASVGDV TITCRASQSI SSYLNWYQQK 60
PGKAPKLLIY AASSLQSGVP SRFSGSGSGT DFTLTISLQ PEDFATYYCQ QSYSTPLTFG 120
GGTKVEIKGG GSGGGGSGG GSGEVQLVES GGGVQPGRS LRLSCAASGF AFSYGMHWV 180
RQAPGKGLEW VAVIWFDTGK KYITDSVKGR FTISRDNKSN TLYLQMNLT R AEDTAVYICA 240
RDRIGARRG PYYMDVWKGK TTVTVSSGGG GSGGGGSAPE FLGGPSVFLF PPKPKDTLYI 300
TREPEVTCVV VDVSDQEDPEV QFNWYVDGVE VHNATKPRE EQFNSTYRVS SVLTVLHQDW 360
LNGKEYKCKV SNKGLPSIE KTISKAKGQP REPQVYTFPP EQEEMTKNQV SLRCLVKGFY 420
PSDI AVEWES NGQPENNYKT TKPVLDSGGS FRLESRLTVD KSRWQEGNVF SCSVMHEACS 480
WHLCKSLSL S LGKGGGSGG GSGGGGSDI LPCVPFSVAK SVKSLYLGRM FSGTPVIRLR 540
FKRLQPTRLV AEFDFRTFDP EGILLFAGGH QDSTWIVLAL RAGRLEQLR YNGVGRVTSS 600
GPVINHGMMQ TISVEELARN LVIKVNRAV MKI A VAGDLF QPERGLYHLN LTVGGIPFHE 660
KDLVQPINPR LDGCMRSWNW LNGEDTIQE TVKVNTRMQC FSVTERGSFY PGSGFAFYSL 720
DYMRTPLDVG TESTWEVEVV AHIRPAADTG VLFALWAPDL RAVPLSVALV DYHSTKKLKK 780

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QLVVLAWEHT	ALALMEIKVC	DGQEHVTVS	LRDGEATLEV	DGTRGQSEVS	AAQLQERLAV	840
LERHLRSPVL	TFAGGLPDVP	VTSAPVTAFY	RGCMTLEVNR	RLLDLDEAAY	KHSDITAHSC	900
PPVEPAAAH	HHHH					914

SEQ ID NO: 163 moltype = AA length = 925
 FEATURE Location/Qualifiers
 source 1..925
 mol_type = protein
 organism = synthetic construct
 1..21

SIGNAL
 SEQUENCE: 163

METDTLLLVW	LLLWVPGSTG	DDIQMTQSPS	SLSASVGDRV	TITCRASQSI	SSYLNWYQQK	60
PGKAPKLLIY	AASSLQSGVP	SRFSGSGSGT	DFTLTISSLQ	PEDFATYYCQ	QSYSTPLTFG	120
GGTKVEIKGG	GGSGGGGSGG	GGSEVQLVES	GGGVVQPGRS	LRLSCAASGF	AFSSYGMHWV	180
RQAPGKGLEW	VAVIWFDTGK	KYYTDSVKGR	FTISRDN SKN	TLYLQMN TLR	AEDTAVYYCA	240
RDRGIGARRG	PYYMDVWGKG	TTVTVSSGGG	GSGGGGSDKT	HTCPPCPAPE	LLGGPSVFLF	300
PPKPKDTLYI	TREPEVTCVV	VDVSHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EQYASTYRVV	360
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	420
SLTCLVKGFY	PSDIAVEWES	NGQPENNYDT	TPPVLDSDGS	FFLYSDLTVD	KSRWQQGNVF	480
SCSVMEALH	NHYTQKSLSL	SPGKGGGSGG	GGSGGGGSD	ILPCVPFSVA	KSVKSLYLGR	540
MFSGTPVIRL	RFKRLQPTRL	VAEFDFTFTD	PEGILLFAGG	HQDSTWIVLA	LRAGRLELQL	600
RYNGVGRVTS	SGPVINHGMW	QTISVEELAR	NLVIKVN RDA	VMKIAVAGDL	FQPERGLYHL	660
NLTVGGIPPH	EKDLVQPINP	RLDGC MRSWN	WLNGEDTTIQ	ETVKVNTRMQ	CFSVTERGSF	720
YPGSGFAFYS	LDYMRTPLDV	GTESTWEVEV	VAHIRPAADT	GVLFALWAPD	LRAVPLSVAL	780
VDYHSTKKLK	KQLVVLAWEH	TALALMEIKV	CDGQEHVTV	SLRDGEATLE	VDGTRGQSEV	840
SAAQLQERLA	VLERHLRSPV	LTFAGGLPDV	PVTSAPVTAF	YRGCM TLEV N	RRLDLDEAA	900
YKHS DITAH S	CPPVEPAAAH	HHHHH				925

SEQ ID NO: 164 moltype = AA length = 504
 FEATURE Location/Qualifiers
 source 1..504
 mol_type = protein
 organism = synthetic construct
 1..21

SIGNAL
 SEQUENCE: 164

METDTLLLVW	LLLWVPGSTG	DDIQMTQSPS	SLSASVGDRV	TITCRASQSI	SSYLNWYQQK	60
PGKAPKLLIY	AASSLQSGVP	SRFSGSGSGT	DFTLTISSLQ	PEDFATYYCQ	QSYSTPLTFG	120
GGTKVEIKGG	GGSGGGGSGG	GGSEVQLVES	GGGVVQPGRS	LRLSCAASGF	AFSSYGMHWV	180
RQAPGKGLEW	VAVIWFDTGK	KYYTDSVKGR	FTISRDN SKN	TLYLQMN TLR	AEDTAVYYCA	240
RDRGIGARRG	PYYMDVWGKG	TTVTVSSGGG	GSGGGGSDKT	HTCPPCPAPE	LLGGPSVFLF	300
PPKPKDTLYI	TREPEVTCVV	VDVSHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EQYASTYRVV	360
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SRKEMTKNQV	420
SLTCLVKGFY	PSDIAVEWES	NGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWQQGNVF	480
SCSVMEALH	NHYTQKSLSL	SPGK				504

SEQ ID NO: 165 moltype = AA length = 925
 FEATURE Location/Qualifiers
 source 1..925
 mol_type = protein
 organism = synthetic construct
 1..21

SIGNAL
 SEQUENCE: 165

METDTLLLVW	LLLWVPGSTG	DDIQMTQSPS	SLSASVGDRV	TITCRASQSI	SSYLNWYQQK	60
PGKAPKLLIY	AASSLQSGVP	SRFSGSGSGT	DFTLTISSLQ	PEDFATYYCQ	QSYSTPLTFG	120
GGTKVEIKGG	GGSGGGGSGG	GGSEVQLVES	GGGVVQPGRS	LRLSCAASGF	AFSSYGMHWV	180
RQAPGKGLEW	VAVIWFDTGK	KYYTDSVKGR	FTISRDN SKN	TLYLQMN TLR	AEDTAVYYCA	240
RDRGIGARRG	PYYMDVWGKG	TTVTVSSGGG	GSGGGGSDKT	HTCPPCPAPE	LLGGPSVFLF	300
PPKPKDTLYI	TREPEVTCVV	VDVSHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EQYASTYRVV	360
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	420
SLTCLVKGFY	PSDIAVEWES	NGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWQQGNVF	480
SCSVMEALH	NHYTQKSLSL	SPGKGGGSGG	GGSGGGGSD	ILPCVPFSVA	KSVKSLYLGR	540
MFSGTPVIRL	RFKRLQPTRL	VAEFDFTFTD	PEGILLFAGG	HQDSTWIVLA	LRAGRLELQL	600
RYNGVGRVTS	SGPVINHGMW	QTISVEELAR	NLVIKVN RDA	VMKIAVAGDL	FQPERGLYHL	660
NLTVGGIPPH	EKDLVQPINP	RLDGC MRSWN	WLNGEDTTIQ	ETVKVNTRMQ	CFSVTERGSF	720
YPGSGFAFYS	LDYMRTPLDV	GTESTWEVEV	VAHIRPAADT	GVLFALWAPD	LRAVPLSVAL	780
VDYHSTKKLK	KQLVVLAWEH	TALALMEIKV	CDGQEHVTV	SLRDGEATLE	VDGTRGQSEV	840
SAAQLQERLA	VLERHLRSPV	LTFAGGLPDV	PVTSAPVTAF	YRGCM TLEV N	RRLDLDEAA	900
YKHS DITAH S	CPPVEPAAAH	HHHHH				925

SEQ ID NO: 166 moltype = AA length = 730
 FEATURE Location/Qualifiers
 source 1..730
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166

METDTLLLVW	LLLWVPGSTG	DEVQLVESGG	GVVQPGRSRL	LSCAASGF AF	SSYGMHWVRQ	60
APGKGLEWVA	VIWFDGTTKY	YTD SVKGRFT	ISRDN SKNTL	YLQMN TLR AE	DTAVYYCARD	120
RIGARRGPT	YMDVWVGKGP	VTSSASTKG	PSVFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	180
PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	240

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KKVEPKSCDK	THTCPPCPAP	ELGGPSVFL	FPPKPKDTLM	ISRTPEVTCV	VVDVSHEDPE	300
VKFNWYVDGV	EVHNAKT KPR	EEQYASTYRV	VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	360
EKTISKAKGQ	PREPQVYTL	PSRDELTKNQ	VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	420
TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMEAL	HNHYTQKSLS	LSPGKGGGGS	480
GGGGSEVKLV	ESGGDLVKPG	GSLKLSCAAS	GFTFSSYGMS	WVRQTPDKRL	EWVATISSGG	540
SYTYYPDSVK	GRFTISRDN	KNTLYLQMS	LKSEDTAMYY	CARHPIYYTY	DDTMDYWGQG	600
TSVTVSSGGG	GSGGGGSGGG	GSDIVLTQSP	AIMAASPGEK	VTMTCSASSS	VSSGNFHWYQ	660
QKPGTSPKLW	IYRTSNLASG	VPARFSGSGS	GTSYSLTISS	MEAEADAATY	CQQWSGYPW	720
FGGGTKLEIK						730

SEQ ID NO: 167 moltype = AA length = 730
 FEATURE Location/Qualifiers
 source 1..730
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 167

METDTLLLV	LLLWPGSTG	DEVQLVESGG	GVVQPGRSLR	LSCAASGF	SSYGMHWVRQ	60
APGKGLEWVA	VIWFDGTKKY	YTDVSVKGRFT	ISRDNKNTL	YLQMNTRLAE	DTAVYYCARD	120
RGIGARRGPY	YMDVWGKGT	VTSSASTKG	PSVFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	180
PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	SSVTVTPSSS	LGTQTYICNV	NHKPSNTKVD	240
KKVEPKSCDK	THTCPPCPAP	ELGGPSVFL	FPPKPKDTLM	ISRTPEVTCV	VVDVSHEDPE	300
VKFNWYVDGV	EVHNAKT KPR	EEQYASTYRV	VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	360
EKTISKAKGQ	PREPQVYTL	PSRDELTKNQ	VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	420
TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMEAL	HNHYTQKSLS	LSPGKGGGGS	480
GGGGSEVKLV	ESGGDLVKPG	GSLKLSCAAS	GFTFSSYGMS	WVRQTPDKRL	EWVATISSGG	540
SYTYYPDSVK	GRFTISRDN	KNTLYLQMS	LKSEDTAMYY	CARHPIYYTY	DDTMDYWGQG	600
TSVTVSSGGG	GSGGGGSGGG	GSDIVLTQSP	AIMAASPGEK	VTMTCSASSS	VSSGNFHWYQ	660
QKPGTSPKLW	IYRTSNLASG	VPARFSGSGS	GTSYSLTISS	MEAEADAATY	CQQWSGYPW	720
FGGGTKLEIK						730

SEQ ID NO: 168 moltype = AA length = 730
 FEATURE Location/Qualifiers
 source 1..730
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 168

METDTLLLV	LLLWPGSTG	DEVQLVESGG	GVVQPGRSLR	LSCAASGF	SSYGMHWVRQ	60
APGKGLEWVA	VIWFDGTKKY	YTDVSVKGRFT	ISRDNKNTL	YLQMNTRLAE	DTAVYYCARD	120
RGIGARRGPY	YMDVWGKGT	VTSSASTKG	PSVFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	180
PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	SSVTVTPSSS	LGTQTYICNV	NHKPSNTKVD	240
KKVEPKSCDK	THTCPPCPAP	ELGGPSVFL	FPPKPKDTLM	ISRTPEVTCV	VVDVSHEDPE	300
VKFNWYVDGV	EVHNAKT KPR	EEQYASTYRV	VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	360
EKTISKAKGQ	PREPQVYTL	PSRDELTKNQ	VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYD	420
TTPPVLDSDG	SFFLYDLTV	DKSRWQQGNV	FSCSVMEAL	HNHYTQKSLS	LSPGKGGGGS	480
GGGGSEVKLV	ESGGDLVKPG	GSLKLSCAAS	GFTFSSYGMS	WVRQTPDKRL	EWVATISSGG	540
SYTYYPDSVK	GRFTISRDN	KNTLYLQMS	LKSEDTAMYY	CARHPIYYTY	DDTMDYWGQG	600
TSVTVSSGGG	GSGGGGSGGG	GSDIVLTQSP	AIMAASPGEK	VTMTCSASSS	VSSGNFHWYQ	660
QKPGTSPKLW	IYRTSNLASG	VPARFSGSGS	GTSYSLTISS	MEAEADAATY	CQQWSGYPW	720
FGGGTKLEIK						730

SEQ ID NO: 169 moltype = AA length = 475
 FEATURE Location/Qualifiers
 source 1..475
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 169

METDTLLLV	LLLWPGSTG	DEVQLVESGG	GVVQPGRSLR	LSCAASGF	SSYGMHWVRQ	60
APGKGLEWVA	VIWFDGTKKY	YTDVSVKGRFT	ISRDNKNTL	YLQMNTRLAE	DTAVYYCARD	120
RGIGARRGPY	YMDVWGKGT	VTSSASTKG	PSVFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	180
PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	SSVTVTPSSS	LGTQTYICNV	NHKPSNTKVD	240
KKVEPKSCDK	THTCPPCPAP	ELGGPSVFL	FPPKPKDTLM	ISRTPEVTCV	VVDVSHEDPE	300
VKFNWYVDGV	EVHNAKT KPR	EEQYASTYRV	VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	360
EKTISKAKGQ	PREPQVYTL	PSRDELTKNQ	VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	420
TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMEAL	HNHYTQKSLS	LSPGK	475

The invention claimed is:

1. A binding molecule comprising a first region capable of binding to a TAM (Tyro3, Axl, MerTK) receptor and a second region capable of specifically binding to a target substance, said target substance being a substance of which aberrant accumulation in a living tissue is characteristic of or associated with a disease, wherein the first region and the second region are coupled to each other directly or via a linker,

wherein the first region comprises a Gas6 protein comprising (i) the sequence of SEQ ID NO: 5 or (ii) a

sequence having sequence identity of at least 85% to the sequence of SEQ ID NO: 5 and comprising a sex hormone-binding globulin (SHBG)-like domain of Gas6 protein,

wherein the first region does not comprise N-terminal gamma carboxyglutamic acid (GLA) domain and epidermal growth factor (EGF)-like domain, and wherein the binding molecule induces phagocytosis.

2. The binding molecule according to claim 1, wherein the Gas6 protein comprises the amino acid sequence of SEQ ID NO: 1 and the amino acid sequence of SEQ ID NO: 2.

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3. The binding molecule according to claim 1, wherein the Gas6 protein comprises one or more sequences selected from the group consisting of SEQ ID NO: 5, SEQ ID NO: 63, SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67, SEQ ID NO: 68, SEQ ID NO: 69, SEQ ID NO: 71, SEQ ID NO: 72, SEQ ID NO: 74, SEQ ID NO: 75, SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, SEQ ID NO: 80, SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, and SEQ ID NO: 87.

4. The binding molecule according to claim 1, which is a monomer or multimer.

5. The binding molecule according to claim 1, wherein the target substance is β -amyloid.

6. The binding molecule according to claim 1, wherein the target substance is soluble amyloid, oligomeric amyloid, aggregated amyloid, or combinations thereof.

7. The binding molecule according to claim 1, wherein the second region that specifically binds to the target substance is selected from the group consisting of an antibody or an antigen-binding fragment thereof, an antibody-like protein, a peptide, an aptamer, and a soluble receptor, which each specifically bind to the target substance.

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8. A pharmaceutical composition comprising the binding molecule of claim 1 and a pharmaceutically acceptable carrier.

9. The binding molecule according to claim 1, which further comprises an immunoglobulin Fc domain.

10. The binding molecule according to claim 9, wherein the immunoglobulin Fc domain has a reduced or abolished Fc receptor binding affinity, compared to wild-type immunoglobulin Fc domain.

11. The binding molecule according to claim 1, wherein the second region is an antibody or an antigen-binding fragment thereof that specifically binds to amyloid.

12. The binding molecule according to claim 1, wherein the first region comprises the sequence of SEQ ID NO: 5 or a sequence having at least 95% sequence identity thereto, and wherein the second region is an antibody or an antigen-binding fragment thereof that specifically binds to soluble amyloid, oligomeric amyloid, and/or aggregated amyloid.

13. The binding molecule according to claim 3, which further comprises an immunoglobulin Fc domain.

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